Senior Numeracy Short Course

Teacher's Resource Book

Senior Short Course: Numeracy

Teaching Guides · Worked Solutions · Marking Guides · Lesson Plans

For teacher use only. Not for distribution to students.

Version 1.0

Senior Numeracy Short Course — Teacher's Resource Book

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Introduction for Teachers

About This Book

This Teacher's Resource Book is the companion to the *Senior Numeracy Short Course Student Workbook*. It is designed to support teachers in delivering the Senior Numeracy Short Course in Queensland.

For each chapter of the Student Workbook, this book provides:

- A **Teaching Guide** — objectives, key vocabulary, and suggested delivery notes
- **Common Student Misconceptions** — errors observed in the key skills of each chapter, with suggested correction strategies
- **Fully Worked Solutions** — complete solutions to every question in the Practice Exercise, Application Exercise, Chapter Review, and Multi-Step Problems of the Student Workbook
- **Marking Guides** — instrument-specific standards for both formal assessment tasks, embedded at the relevant point in the course

Important Note on Solutions

The worked solutions in this book correspond exactly to the questions in the Student Workbook. No additional or alternative questions appear here. Teachers should not distribute this book to students.

Course Delivery: Six Months Across Two Terms

This course is designed as an **18-week teaching program**. In practice, most schools will deliver this course across **two 10-week terms**, providing a six-month timeframe. This additional two-week buffer accommodates school interruptions, including public holidays, camps, sports carnivals, and student-free days.

Recommendation: Treat the 18-week plan as your *core delivery timeline* and use the additional weeks in each term as flex time for consolidation, extension, or re-teaching.

Scheduling Advice

- **Term 1 (Weeks 1–9):** Topic 1 — Personal Identity and Community, concluding with Assessment Task 1 (Project)
- **Term 2 (Weeks 10–18):** Topic 2 — Workplace and Employment, concluding with Assessment Task 2 (Examination)
- Set AT1 due date no later than **Week 8 of Term 1** to allow marking time before end of term
- Schedule AT2 for **Week 8 of Term 2** so that Week 9 and the buffer weeks can be used for result return and review

18-Week Lesson Plan at a Glance

Wk	Term	Chapter	Focus	Assessment
1	T1	Ch 1: Numbers in Everyday Life	Whole numbers, place value, operations, BODMAS	—
2	T1	Ch 2: Fractions, Decimals & Percentages	Fractions, decimals, percentages, conversions	—
3	T1	Ch 3: Rates, Ratios & Comparisons	Rates, ratios, unit rates, comparisons	—
4	T1	Ch 4: Length, Perimeter & Area	Length, perimeter, area (rect., triangle)	—
5	T1	Ch 5: Shapes, Area & Maps	2D shapes, area units, maps, coordinates	—
6	T1	Ch 6: Time & Angles	12/24-hr time, timetables, angles	—
7	T1	Ch 7: Data & Graphs	Data collection, tables, bar & line graphs	—
8	T1	Ch 8: Chance & Probability	Probability language, calculation, expression	—
9	T1	AT1 Project Work	Review Topic 1; project completion	**AT1 Due**
10	T2	Ch 9: Numbers at Work	Whole numbers and technology in workplace	—
11	T2	Ch 10: Workplace Numbers	Fractions, decimals, % in workplace budgets	—
12	T2	Ch 11: 3D Shapes & Volume	3D shapes, volume of prisms, unit conversions	—
13	T2	Ch 12: Mass, Capacity & Temperature	Mass, capacity, temperature; unit conversions	—
14	T2	Ch 13: Workplace Measurement Mixed	Mixed M&S; volume units, maps, scale	—
15	T2	Ch 14: Data at Work	Workplace data, graphs, mean, median, mode	—
16	T2	Ch 15: Probability & Trends at Work	Workplace probability, data trends	—
17	T2	AT2 Exam Preparation	Review Topic 2; practice exam	—
18	T2	Examination + Wrap-Up	Open-book supervised examination	**AT2 Exam**

Assessment Overview

Assessment Task 1 — Project (Topic 1)

Type:	Individual project (QCAA Option A)
Context:	Personal Identity and Community
When:	Week 9 (Term 1)
Response modes:	Written ($\leq$600 words), multimodal ($\leq$6 A4 pages or 5 min), or spoken ($\leq$4 min)
Marking guide:	See AT1 Marking Guide section of this book
Grading:	A–E instrument-specific standards

Assessment Task 2 — Examination (Topic 2)

Type:	Individual supervised open-book examination (QCAA Option D)
Context:	Workplace and Employment
When:	Week 18 (Term 2)
Conditions:	Open-book; students may bring notes, exercise books, and student workbook
Time:	5 min perusal + 45 min working time
Marking guide:	See AT2 Marking Guide section of this book
Grading:	A–E with percentage cut-offs (A $>$80%, B $>$60%, C $>$40%, D $>$20%, E $\geq$0%)

General Teaching Notes

Key Principles for Delivering This Course

- **Real-world contexts first:** All topics should be introduced through real-life scenarios before moving to abstract skills. The student workbook models this approach in every chapter.
- **Open-book exam mindset:** Because AT2 is open-book, students should understand where to find information in their workbook. Train them to tab chapters and annotate examples throughout the course.
- **Calculator use:** Students should use calculators regularly. QCAA expects technology use to be embedded in everyday numeracy practice.
- **Aboriginal and Torres Strait Islander perspectives:** Embed Indigenous perspectives in examples where appropriate — land area, community data, economic contexts, and cultural storytelling all provide authentic connections.
- **Differentiation:** Chapter exercises progress from straightforward to applied. Use Practice Exercises for consolidation, Application Exercises for working students, and Chapter Reviews as formative assessment.
- **Worked examples in class:** Each chapter's Worked Examples section should be completed with the class, not as independent work. Model the thinking aloud.

Part I

Topic 1 — Personal Identity and Community

1. Numbers in Everyday Life

Chapter 1 Information

Week taught:	Week 1 (Term 1)
Topic:	Topic 1 — Personal Identity and Community
Skills area:	Number & Algebra
Lesson reference:	Lessons 1–4

Teaching Guide

Key concepts: Place value (ones to millions); reading and writing whole numbers in words and numerals; ordering and comparing whole numbers; four operations; BODMAS order of operations; rounding and estimation; using a calculator.

Suggested delivery sequence:

1. Lesson 1: Introduce numeracy and course overview; whole numbers in everyday contexts; place value chart.
2. Lesson 2: Reading, writing and ordering numbers; place value of digits.
3. Lesson 3: Four operations with whole numbers; BODMAS; estimation and rounding.
4. Lesson 4: Calculator skills; work through Worked Examples together as a class; students begin Practice Exercise.

Common Student Misconceptions

Common Misconceptions — Chapter 1

1. **Place value confusion with zeros:** Students write "one million, fifty thousand, and six" as 1,500,006 instead of 1,050,006. *Strategy:* Use a place value chart for every conversion until automaticity develops.

2. **BODMAS errors — ignoring left-to-right rule:** Students treat $12 \div 3 \times 2$ as $12 \div 6 = 2$ instead of $4 \times 2 = 8$. *Strategy:* Stress that division and multiplication are equal in priority; work left to right.

3. **Rounding in the wrong direction:** Students round 4,500 to the nearest thousand as 4,000. *Strategy:* Explicitly state: "5 or more rounds up — always."

4. **Estimation not attempted:** Students skip estimation and go straight to the calculator. *Strategy:* Make estimation a required step; check that the estimate and calculator answer are in the same order of magnitude.

5. **Multi-step problems — only one operation performed:** Students stop after the first step of a two-step word problem. *Strategy:* Teach students to re-read the question after each step.

Worked Solutions

Chapter 1 Overview for Teachers

Chapter:	1 — Numbers in Everyday Life
Week:	Week 1, Term 1 (Lessons 1–4)
Topic:	Topic 1 — Personal Identity and Community
Skills area:	Number & Algebra
Focus:	Place value; reading, ordering and comparing whole numbers; rounding to nearest ten, hundred and thousand; BODMAS order of operations; addition, subtraction, multiplication and division

Worked Solutions — Practice Exercise

Part A — Reading and Writing Whole Numbers

Question 1 Write 4,507 in words.

Split into groups: 4 | 507
4 = four thousand 507 = five hundred and seven

Answer: four thousand, five hundred and seven

Question 2 Write 1,030,060 in words.

Split into groups: 1 | 030 | 060
1 = one million 030 = thirty thousand 060 = sixty

Answer: one million, thirty thousand, and sixty

Teaching note: Students often write "one million, zero thirty thousand" or forget the zero-placeholder groups entirely. Remind them that groups with all zeros are not spoken.

Question 3 Write "sixty-two thousand three hundred and fifteen" as a numeral.

sixty-two thousand = 62,000 three hundred and fifteen = 315
Combined: 62,000 + 315

Answer: 62,315

Question 4 Write "nine million four hundred thousand and seven" as a numeral.

nine million = 9,000,000 four hundred thousand = 400,000 seven = 7
Zeros hold missing places: 9,400,007

Answer: 9,400,007

Teaching note: Common error is writing 9,400,700 or 9,4000,007. Use a place value chart.

Part B — Place Value

Question 5 Place value of digit 8 in 285,604.

Hundred Thous.	**Ten Thous.**	Thousands	Hundreds	Tens	Ones
2	**8**	5	6	0	4

Answer: ten thousands

Question 6 Value of digit 3 in 1,307,250.

Millions	**Hund. Thous.**	Ten Thous.	Thousand	Hundred	Tens	Ones
1	**3**	0	7	2	5	0

$3 \times 100{,}000 = 300{,}000$

Answer: 300,000

Question 7 Digits in 5,924,831.

Millions	Hund. Thous.	**Ten Thous.**	Thousan	**Hundre**	Tens	Ones
5	9	**2**	4	**8**	3	1

(a) Ten thousands place: **2**
(b) Hundreds place: **8**
(c) Millions place: **5**

Part C — Ordering Numbers

Question 8 Order 45,678 45,768 45,087 45,870 from smallest to largest.

All start with 45,___. Compare the hundreds digits:
45,**0**87 → 0 45,**6**78 → 6 45,**7**68 → 7 45,**8**70 → 8
Order by hundreds digit: $0 < 6 < 7 < 8$

Answer: 45,087 < 45,678 < 45,768 < 45,870

Question 9 **Order 1,200,000 1,020,000 1,002,000 1,200,200 from largest to smallest.**

All start with 1,___,___. Compare the hundred-thousands digits:
1,**2**00,000 → 2 1,**2**00,200 → 2 1,**0**20,000 → 0 1,**0**02,000 → 0
The two with 2: compare ten-thousands: 1,200,000 vs 1,200,200 → same to that point; compare thousands: 1,200,**0**00 vs 1,200,**2**00 → 200 > 000
The two with 0: 1,0**2**0,000 > 1,0**0**2,000

Answer: 1,200,200 > 1,200,000 > 1,020,000 > 1,002,000

Part D — Calculations

Question 10 3,456 + 2,897

3,456 + 2,897:
6 + 7 = 13 (write 3, carry 1) 5 + 9 + 1 = 15 (write 5, carry 1)
4 + 8 + 1 = 13 (write 3, carry 1) 3 + 2 + 1 = 6

Answer: 6,353

Question 11 12,500 − 4,836

12,500 − 4,836:
0 − 6: borrow → 10 − 6 = 4 0 − 1 − 3: borrow → 10 − 1 − 3 = 6
5 − 1 − 8: borrow → 15 − 1 − 8 = 6 2 − 1 − 4: borrow → 12 − 1 − 4 = 7
1 − 1 = 0

Answer: 7,664

Question 12 365 × 24

365 × 20 = 7,300
365 × 4 = 1,460
7,300 + 1,460 = 8,760

Answer: 8,760

Question 13 $2{,}808 \div 12$

$12 \times 200 = 2{,}400$ remainder $2{,}808 - 2{,}400 = 408$
$12 \times 30 = 360$ remainder $408 - 360 = 48$
$12 \times 4 = 48$ remainder 0
$200 + 30 + 4 = 234$

Answer: 234

Part E — Estimate Then Calculate

Question 14 **Estimate** $489 + 312$ **(nearest hundred) then find exact answer.**

Estimate: $489 \approx 500$ $312 \approx 300$ $500 + 300 = 800$
Exact: $489 + 312 = 801$

Estimate: 800 **Exact: 801**

Teaching note: Estimate and exact are only $1 apart — a good opportunity to show students that rounding can give a very close prediction.

Question 15 **Estimate** 67×48 **(nearest ten) then use calculator.**

Estimate: $67 \approx 70$ $48 \approx 50$ $70 \times 50 = 3{,}500$
Calculator: $67 \times 48 = 3{,}216$

Estimate: 3,500 **Calculator answer: 3,216**

Question 16 **Evaluate** $(12 + 8) \times 5 - 14 \div 2$ **using BODMAS.**

Step 1 — Brackets: $(12 + 8) = 20$
Expression becomes: $20 \times 5 - 14 \div 2$

Step 2 — Division and Multiplication (left to right):
$20 \times 5 = 100$ then $14 \div 2 = 7$
Expression becomes: $100 - 7$

Step 3 — Subtraction: $100 - 7 = 93$

Answer: 93

Worked Solutions — Application Exercise

Question 1 Mia's shopping: jacket \$89 T-shirt \$34 jeans \$67. Total spent?

$\$89 + \$34 + \$67$
$= \$123 + \$67 = \$190$

Answer: Mia spends \$190 in total.

Question 2 Jake: starts with \$245 earns \$320 spends \$198. How much left?

Opening balance + earnings: $\$245 + \$320 = \$565$
Less grocery spend: $\$565 - \$198 = \$367$

Answer: Jake has \$367 left.

Question 3 Brisbane population 2,582,000 Hobart 238,000. How many more in Brisbane?

$2{,}582{,}000 - 238{,}000 = 2{,}344{,}000$

Answer: 2,344,000 more people live in Brisbane than Hobart.

Question 4 Sydney to Melbourne 877 km. Sam drives there and back. Total distance?

Return trip $= 2 \times 877 = 1{,}754$ km

Answer: Sam drives a total of 1,754 km.

Question 5 Cinema: \$18 per ticket 264 people. Total ticket revenue?

$264 \times \$18$:
$264 \times 10 = 2{,}640$
$264 \times 8 = 2{,}112$
$2{,}640 + 2{,}112 = 4{,}752$

Answer: Total revenue is \$4,752.

Question 6 Gym class: 4 teams of 7 students then 3 more added.

(a) Original class: $4 \times 7 = 28$ students

(b) After addition: $28 + 3 = 31$ students

(a) 28 students **(b) 31 students**

Question 7 Jordan downloads 4 apps: three use 340 MB each one uses 1,200 MB. Total data?

$3 \times 340 = 1{,}020$ MB
$1{,}020 + 1{,}200 = 2{,}220$ MB

Answer: 2,220 MB of data is used.

Question 8 Athletics club: 252 members in 9 groups. Then 3 more groups added at the same size.

(a) Members per group: $252 \div 9 = 28$

(b) New total groups: $9 + 3 = 12$
Total members: $12 \times 28 = 336$

(a) 28 members per group **(b) 336 members total**

Question 9 Zara earns \$875/week saves \$125/week. How much saved after 8 weeks?

$8 \times \$125 = \$1{,}000$

Answer: Zara has saved \$1,000 after 8 weeks.

Teaching note: Some students may attempt to use \$875 in this calculation. Redirect: the question asks about savings only; the \$875 is context for where the money comes from.

Question 10 Council park: 3,200 m^2 divided into 8 equal sections. Area of each?

$3{,}200 \div 8 = 400$

Answer: Each section has an area of 400 m^2.

Question 11 AFL grand final attendance: 100,024 people.

(a) Nearest thousand: look at hundreds digit $= 0 \rightarrow$ round down
$100{,}024 \approx 100{,}000$

(b) Nearest ten thousand: look at thousands digit $= 0 \rightarrow$ round down
$100{,}024 \approx 100{,}000$

(a) 100,000 (b) 100,000

Teaching note: Both answers are the same here because the number is very close to 100,000. This is a good discussion point — it does not mean an error was made.

Question 12 Phone plan: 15,000 MB/month. Liam uses 11,350 MB (Month 1) and 2,840 MB (Month 2).

(a) Total data used across both months:
$11{,}350 + 2{,}840 = 14{,}190$ MB

(b) Monthly limit check:
Month 1: $11{,}350 < 15{,}000$ ✓ within limit
Month 2: $2{,}840 < 15{,}000$ ✓ within limit

(a) 14,190 MB (b) Yes — Liam stayed within his 15,000 MB monthly limit in both months. Month 1: 11,350 MB used; Month 2: 2,840 MB used.

Worked Solutions — Chapter Review

Question 1 Write 1,204,750 in words.

Groups: 1 | 204 | 750
1 = one million 204 = two hundred and four thousand 750 = seven hundred and fifty

Answer: one million, two hundred and four thousand, seven hundred and fifty

Question 2 Value of digit 6 in 2,637,100.

Millions	**Hund. Thous.**	Ten Thous.	Thousan	Hundred	Tens	Ones
2	**6**	3	7	1	0	0

$6 \times 100{,}000 = 600{,}000$

Answer: 600,000

Question 3 Order 303,040 330,400 300,340 303,400 from largest to smallest.

Compare ten-thousands digit:
3**3**0,400 → 3 3**0**3,040 → 0 3**0**0,340 → 0 3**0**3,400 → 0
Largest: 330,400
Remaining three all have 0 in ten-thousands. Compare thousands:
303,**4**00 → 4 303,**0**40 → 0 300,**3**40 → 3
Order: 303,400 > 300,340 > 303,040

Answer: 330,400 > 303,400 > 303,040 > 300,340

Question 4 5,874 + 3,649

$4 + 9 = 13$ (write 3, carry 1) $7 + 4 + 1 = 12$ (write 2, carry 1)
$8 + 6 + 1 = 15$ (write 5, carry 1) $5 + 3 + 1 = 9$

Answer: 9,523

Question 5 $20{,}000 - 8{,}375$

$20{,}000 - 8{,}375$:
Borrow across the zeros: $20{,}000 = 19{,}999 + 1$
$19{,}999 - 8{,}375 = 11{,}624$ then add back the 1: $11{,}624 + 1 = 11{,}625$

Answer: 11,625

Teaching note: Subtracting from zeros requires repeated borrowing. The "round down and add back" strategy shown above is one clean approach. Accept standard column subtraction with borrowing shown step by step.

Question 6 48×125

$48 \times 100 = 4{,}800$
$48 \times 25 = 48 \times 20 + 48 \times 5 = 960 + 240 = 1{,}200$
$4{,}800 + 1{,}200 = 6{,}000$

Answer: 6,000

Question 7 $4{,}995 \div 15$

$15 \times 300 = 4{,}500$ remainder $4{,}995 - 4{,}500 = 495$
$15 \times 30 = 450$ remainder $495 - 450 = 45$
$15 \times 3 = 45$ remainder 0
$300 + 30 + 3 = 333$

Answer: 333

Question 8 Estimate $923 + 478$ (nearest hundred) then find exact answer.

Estimate: $923 \approx 900$ $478 \approx 500$ $900 + 500 = 1{,}400$
Exact: $923 + 478 = 1{,}401$

Estimate: 1,400 **Exact: 1,401**

Question 9 Adelaide to Darwin 3,027 km. Family drives there and back. Total distance?

Return journey $= 2 \times 3{,}027 = 6{,}054$ km

Answer: The family travels 6,054 km in total.

Question 10 **Cricket club orders 15 sets of gear at \$285 each. Club has \$3,500 in account.**

(a) Total cost: $15 \times \$285$
$15 \times 200 = 3{,}000 \quad 15 \times 85 = 1{,}275$
$3{,}000 + 1{,}275 = \$4{,}275$

(b) Shortfall: $\$4{,}275 - \$3{,}500 = \$775$

(a) \$4,275 **(b) The club needs an extra \$775.**

Question 11 **Amira earns \$640/week pays \$185 rent \$120 food \$75 transport.**

(a) Total expenses: $\$185 + \$120 + \$75 = \$380$
Left over: $\$640 - \$380 = \$260$

(b) Over 12 weeks: $\$260 \times 12 = \$3{,}120$

Answer: Amira has \$260 left over each week, and \$3,120 over 12 weeks.

Question 12 **Fundraiser: 345 chocolate bars at \$3 each and 128 raffle tickets at \$5 each.**

Chocolate bar revenue: $345 \times \$3 = \$1{,}035$
Raffle ticket revenue: $128 \times \$5 = \$640$
Total raised: $\$1{,}035 + \$640 = \$1{,}675$

Answer: Total raised = \$1,675.

Chapter 1 Solutions Complete

Summary of answers for quick marking:

Practice Exercise: 1. four thousand, five hundred and seven 2. one million, thirty thousand, and sixty 3. 62,315 4. 9,400,007 5. ten thousands 6. 300,000 7. (a) 2 (b) 8 (c) 5 8. 45,087 < 45,678 < 45,768 < 45,870 9. 1,200,200 > 1,200,000 > 1,020,000 > 1,002,000 10. 6,353 11. 7,664 12. 8,760 13. 234 14. est. 800, exact 801 15. est. 3,500, calc. 3,216 16. 93

Application Exercise: 1. \$190 2. \$367 3. 2,344,000 4. 1,754 km 5. \$4,752 6. (a) 28 (b) 31 7. 2,220 MB 8. (a) 28 (b) 336 9. \$1,000 10. 400 m^2 11. (a) 100,000 (b) 100,000 12. (a) 14,190 MB (b) yes, within limit both months

Chapter Review: 1. one million, two hundred and four thousand, seven hundred and fifty 2. 600,000 3. 330,400 > 303,400 > 303,040 > 300,340 4. 9,523 5. 11,625 6. 6,000 7. 333 8. est. 1,400, exact 1,401 9. 6,054 km 10. (a) \$4,275 (b) \$775 11. (a) \$260 (b) \$3,120 12. \$1,675

2. Fractions, Decimals and Percentages

Chapter 2 Information

Week taught:	Week 2 (Term 1)
Topic:	Topic 1 — Personal Identity and Community
Skills area:	Number & Algebra
Lesson reference:	Lessons 5–8

Teaching Guide

Key concepts: Equivalent fractions; simplifying fractions; mixed numbers and improper fractions; converting between fractions, decimals and percentages; finding a percentage of a quantity; expressing one quantity as a percentage of another; percentage increase and decrease.

Suggested delivery sequence:

1. Lesson 5: Fractions — parts of a whole, equivalent fractions, simplifying, ordering.
2. Lesson 6: Converting between fractions, decimals and percentages.
3. Lesson 7: Percentages of amounts; percentage increase and decrease.
4. Lesson 8: Worked Examples as a class; Practice and Application Exercises.

Common Student Misconceptions

Common Misconceptions — Chapter 2

1. **Adding fractions with different denominators:** Students add numerators and denominators separately, e.g., $\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$. *Strategy:* Always find a common denominator first.

2. **Percentage of a quantity — dividing instead of multiplying:** Students compute 15% of \$80 as $80 \div 15$. *Strategy:* Convert % to decimal, then multiply.

3. **Confusing "percentage of" with "percentage change":** Students apply the wrong formula. *Strategy:* Explicitly identify problem type before calculating.

4. **Decimal place confusion when converting:** Students write 0.7 as 7% instead of 70%. *Strategy:* Practise the $\times 100$ rule; use a conversion triangle.

5. **Improper fraction conversion errors:** Students write $\frac{7}{3}$ as $2\frac{2}{3}$ with the wrong remainder. *Strategy:* Use division with remainder: $7 \div 3 = 2$ remainder 1, so $2\frac{1}{3}$.

Worked Solutions

Chapter 2 Overview for Teachers

Chapter:	2 — Fractions, Decimals and Percentages
Week:	Week 2, Term 1 (Lessons 5–8)
Topic:	Topic 1 — Personal Identity and Community
Skills area:	Number & Algebra
Focus:	Simplifying fractions using HCF; equivalent fractions; ordering fractions and decimals; converting between fractions, decimals and percentages; percentage of an amount; discount and GST

Conversion Reference Table — Completed

Students complete this table at the start of the chapter. The completed version below is for teacher reference.

Fraction	Decimal	Percentage
$\frac{1}{2}$	0.5	50%
$\frac{1}{4}$	0.25	25%
$\frac{3}{4}$	0.75	75%
$\frac{1}{10}$	0.1	10%
$\frac{1}{5}$	0.2	20%
$\frac{2}{5}$	0.4	40%
$\frac{1}{3}$	$0.\overline{3}$	$33.\overline{3}\%$
$\frac{3}{4}$	0.75	75%
$\frac{4}{5}$	0.8	80%
$\frac{7}{20}$	0.35	35%

Teaching note: The $\frac{1}{3}$ *row introduces recurring decimals. Acknowledge that* $0.\overline{3}$ *(or* $0.333\ldots$*) is exact and* $33.\overline{3}\%$ *is the precise percentage. For practical purposes students may write* 33.3% *(1 d.p.) and this is acceptable.*

Worked Solutions — Practice Exercise

Part A — Simplifying Fractions

Question 1 Simplify $\frac{6}{9}$. Show the HCF.

Factors of 6: 1, 2, **3**, 6 Factors of 9: 1, **3**, 9 HCF = 3

$$\frac{6}{9} = \frac{6 \div 3}{9 \div 3} = \frac{2}{3}$$

Answer: $\frac{2}{3}$

Question 2 Simplify $\frac{15}{25}$.

HCF of 15 and 25 = 5

$$\frac{15}{25} = \frac{15 \div 5}{25 \div 5} = \frac{3}{5}$$

Answer: $\frac{3}{5}$

Question 3 Simplify $\frac{18}{24}$.

Factors of 18: 1, 2, 3, **6**, 9, 18 Factors of 24: 1, 2, 3, 4, **6**, 8, 12, 24 HCF = 6

$$\frac{18}{24} = \frac{18 \div 6}{24 \div 6} = \frac{3}{4}$$

Answer: $\frac{3}{4}$

Part B — Equivalent Fractions

Question 4 $\frac{3}{5} = \frac{?}{20}$

Denominator: $5 \times 4 = 20$, so multiply numerator by 4: $3 \times 4 = 12$

Answer: $\frac{3}{5} = \frac{12}{20}$

Question 5 $\frac{2}{7} = \frac{6}{?}$

Numerator: $2 \times 3 = 6$, so multiply denominator by 3: $7 \times 3 = 21$

Answer: $\frac{2}{7} = \frac{6}{21}$

Question 6 **Are $\frac{4}{10}$ and $\frac{6}{15}$ equivalent?**

Simplify each fraction to its simplest form:

$\frac{4}{10}$: HCF $= 2 \rightarrow \frac{2}{5}$

$\frac{6}{15}$: HCF $= 3 \rightarrow \frac{2}{5}$

Both simplify to $\frac{2}{5}$.

Answer: Yes — $\frac{4}{10}$ and $\frac{6}{15}$ are equivalent. Both equal $\frac{2}{5}$.

Part C — Ordering Fractions and Decimals

Question 7 **Order $\frac{3}{4}$ $\frac{1}{2}$ $\frac{2}{3}$ $\frac{1}{4}$ from smallest to largest.**

Convert each to a decimal:

$\frac{3}{4} = 0.75$ $\frac{1}{2} = 0.5$ $\frac{2}{3} = 0.667$ $\frac{1}{4} = 0.25$

Order the decimals: $0.25 < 0.5 < 0.667 < 0.75$

Answer (smallest to largest): $\frac{1}{4} < \frac{1}{2} < \frac{2}{3} < \frac{3}{4}$

Part D — Conversions

Question 8 Complete the conversion table.

Fraction	Decimal	Percentage
$\frac{1}{2}$	0.5	50%
$\frac{3}{5}$	0.6	60%
$\frac{1}{4}$	0.25	25%
$\frac{2}{5}$	0.4	40%
$\frac{2}{25}$	0.08	8%

Working for row 5 (0.08): $0.08 = \frac{8}{100}$, *HCF = 4, simplifies to* $\frac{2}{25}$. *As percentage:* $0.08 \times 100 = 8\%$.

Part E — Calculating Percentages

Question 9 Calculate 15% of \$200.

15% of \$200 $= 0.15 \times 200 =$ \$30
Check via 10%: 10% = \$20; 5% = \$10; *total* = \$30 ✓

Answer: \$30

Question 10 Calculate 25% of \$80.

25% of \$80 $= 0.25 \times 80 =$ \$20
Alternatively: $25\% = \frac{1}{4}$, *so* \$80 ÷ 4 = \$20 ✓

Answer: \$20

Question 11 **Calculate 40% of 350 items.**

40% of $350 = 0.40 \times 350 = 140$

Answer: 140 items

Question 12 **Convert 0.6 to a fraction (simplest form) and a percentage.**

Fraction: $0.6 = \frac{6}{10}$. HCF $= 2$. Simplified: $\frac{3}{5}$

Percentage: $0.6 \times 100 = 60\%$

Answer: $\frac{3}{5}$ and 60%

Question 13 **Convert 85% to a decimal and a fraction in simplest form.**

Decimal: $85\% \div 100 = 0.85$

Fraction: $85\% = \frac{85}{100}$. HCF of 85 and 100 $= 5$.

$\frac{85}{100} = \frac{85 \div 5}{100 \div 5} = \frac{17}{20}$

Answer: 0.85 and $\frac{17}{20}$

Question 14 **Class of 28 students. $\frac{1}{4}$ are absent. How many absent?**

$\frac{1}{4}$ of $28 = 28 \div 4 = 7$

Answer: 7 students are absent.

Question 15 **Which is greater: 65% or $\frac{2}{3}$?**

Convert $\frac{2}{3}$ to a percentage: $2 \div 3 \times 100 = 66.\overline{6}\% \approx 66.7\%$

Compare: $66.7\% > 65\%$

Answer: $\frac{2}{3}$ is greater. $\frac{2}{3} \approx 66.7\%$, which is greater than 65%.

Worked Solutions — Application Exercise

Question 1 Woolworths: 25% off jacket originally \$48. What is the sale price?

Discount $= 25\% \times \$48 = 0.25 \times 48 = \$12$

Sale price $= \$48 - \$12 = \$36$

Quick method: 75% *of* $\$48 = 0.75 \times 48 = \$36$ ✓

Answer: The sale price is \$36.

Question 2 Water bottle holds 750 mL. Tom drinks $\frac{3}{5}$ of it. How much drunk? How much left?

Amount drunk: $\dfrac{3}{5} \times 750 = 3 \times 150 = 450$ mL

Amount left: $750 - 450 = 300$ mL

Answer: Tom has drunk 450 mL and has 300 mL left.

Question 3 Layla scored 34 out of 40. What percentage?

$$\frac{34}{40} \times 100 = 34 \div 40 \times 100 = 0.85 \times 100 = 85\%$$

Answer: Layla scored 85%.

Question 4 Phone plan 10 GB. Amara used 6.5 GB.

(a) Fraction used: $\dfrac{6.5}{10} = \dfrac{13}{20}$ (given in hint — multiply numerator and denominator by 2 to remove the decimal)

(b) Percentage used: $\dfrac{13}{20} \times 100 = 65\%$

(c) Percentage remaining: $100\% - 65\% = 35\%$

(a) $\dfrac{13}{20}$ **(b) 65%** **(c) 35% remains**

Question 5 Laptop \$1,299 with 20% off. What is the sale price?

Discount $= 20\% \times \$1{,}299 = 0.20 \times 1{,}299 = \$259.80$

Sale price $= \$1{,}299 - \$259.80 = \$1{,}039.20$
Quick method: 80% *of* $\$1{,}299 = 0.80 \times 1{,}299 = \$1{,}039.20$ ✓

Answer: The sale price is \$1,039.20.

Question 6 Cereal: 12 g fat per serve. Daily recommended

$$\frac{12}{65} \times 100 = 18.46\ldots\% \approx 18.5\%$$

Answer: One serve provides 18.5% of the recommended daily fat.

Teaching note: Accept 18.46% if the student rounds to 2 d.p. Ensure students are comfortable rounding percentage answers to 1 d.p. when required.

Question 7 Plumber charges \$340 before GST. GST

(a) GST amount $= 10\% \times \$340 = 0.10 \times 340 = \$34$

(b) Total price $= \$340 + \$34 = \$374$
Quick method: $\$340 \times 1.10 = \$374$ ✓

(a) GST = \$34 **(b) Total = \$374**

Question 8 Milo: science 45/50 maths 38/40. Which is the higher percentage?

Science: $\frac{45}{50} \times 100 = 90\%$

Maths: $\frac{38}{40} \times 100 = 95\%$

$95\% > 90\%$

Answer: The higher score is Milo's maths result (95% vs 90% in science).

Question 9 School fete: $\frac{3}{8}$ of 240 students bought a sausage sizzle.

(a) Students who bought: $\dfrac{3}{8} \times 240 = 3 \times 30 = 90$

(b) Percentage who did NOT buy:
$\dfrac{3}{8} = 37.5\%$ bought, so $100\% - 37.5\% = 62.5\%$ did not buy.
Alternative: $\frac{5}{8}$ *did not buy;* $\frac{5}{8} \times 100 = 62.5\%$ ✓

(a) 90 students **(b) 62.5% did not buy a sausage sizzle.**

Question 10 Cinema ticket \$22 including GST. Find pre-GST price and GST component.

(a) \$22 represents 110% of the pre-GST price.
Pre-GST price $= \$22 \div 1.10 = \$20$

(b) GST component $= \$22 - \$20 = \$2$
Check: 10% *of* $\$20 = \$2$ ✓

(a) Pre-GST price = \$20 **(b) GST component = \$2**

Teaching note: This is the reverse GST calculation — the most common error is students computing 10% of \$22 (giving \$2.20) instead of dividing by 1.10. Use the question hint to guide students: "\$22 = 110% of pre-GST" so pre-GST = \$22 ÷ 1.10.

Question 11 Everything 30% off. Dress \$85 shoes \$120. Find total after discount.

(a) Discount on dress $= 30\% \times \$85 = 0.30 \times 85 = \$25.50$

(b) Discount on shoes $= 30\% \times \$120 = 0.30 \times 120 = \$36$

(c) Sale price of dress $= \$85 - \$25.50 = \$59.50$
Sale price of shoes $= \$120 - \$36 = \$84$
Total $= \$59.50 + \$84 = \$143.50$
Quick method: 70% *of* $(\$85 + \$120) = 0.70 \times \$205 = \$143.50$ ✓

(a) \$25.50 **(b) \$36** **(c) Total cost = \$143.50**

Question 12 **Sports day: $\frac{2}{5}$ swimming 35% athletics rest cricket.**

(a) Swimming: $\dfrac{2}{5} = 0.4 = 40\%$

(b) Cricket: $100\% - 40\% - 35\% = 25\%$

(c) Cricket as a fraction: $25\% = \dfrac{25}{100}$, HCF $= 25$, simplifies to $\dfrac{1}{4}$

(a) 40% **(b) 25%** **(c)** $\dfrac{1}{4}$

Worked Solutions — Chapter Review

Question 1 **Simplify $\frac{24}{36}$.**

HCF of 24 and 36 = 12

$$\frac{24}{36} = \frac{24 \div 12}{36 \div 12} = \frac{2}{3}$$

Answer: $\frac{2}{3}$

Question 2 **Write $\frac{3}{8}$ as a decimal and as a percentage.**

Decimal: $3 \div 8 = 0.375$

Percentage: $0.375 \times 100 = 37.5\%$

Answer: 0.375 **and** 37.5%

Question 3 **Write 45% as a decimal and a fraction in simplest form.**

Decimal: $45\% \div 100 = 0.45$

Fraction: $\frac{45}{100}$. HCF of 45 and 100 = 5.

$$\frac{45}{100} = \frac{9}{20}$$

Answer: 0.45 **and** $\frac{9}{20}$

Question 4 $\frac{5}{8} = \frac{?}{40}$

Denominator: $8 \times 5 = 40$, so multiply numerator by 5: $5 \times 5 = 25$

Answer: $\frac{5}{8} = \frac{25}{40}$

Question 5 **Order** 0.4 $\frac{3}{8}$ 42% $\frac{1}{3}$ **from smallest to largest.**

Convert all to decimals:

$0.4 = 0.400$ $\frac{3}{8} = 0.375$ $42\% = 0.420$ $\frac{1}{3} = 0.\overline{3} \approx 0.333$

Order: $0.333 < 0.375 < 0.400 < 0.420$

Answer (smallest to largest): $\frac{1}{3} < \frac{3}{8} < 0.4 < 42\%$

Question 6 **Calculate 35% of $260.**

35% of $\$260 = 0.35 \times 260 = \$91$
Check: $10\% = \$26$; $30\% = \$78$; $5\% = \$13$; *total* $= \$91$ ✓

Answer: $91

Question 7 **Luca scored 27 out of 30. What percentage?**

$\frac{27}{30} \times 100 = 27 \div 30 \times 100 = 0.9 \times 100 = 90\%$

Answer: 90%

Question 8 **GST (10%) on a $340 item. Total price including GST?**

GST $= 10\% \times \$340 = \$34$

Total $= \$340 + \$34 = \$374$

GST = $34 **Total including GST = $374**

Question 9 **Phone plan $65/month with 15% loyalty discount. New monthly cost?**

Discount $= 15\% \times \$65 = 0.15 \times 65 = \$9.75$

New cost $= \$65 - \$9.75 = \$55.25$
Quick method: 85% *of* $\$65 = 0.85 \times 65 = \$55.25$ ✓

Answer: New monthly cost = $55.25

Question 10 Jacket \$180 25% off then 10% GST on the sale price.

(a) Sale price before GST:
Discount $= 25\% \times \$180 = \$45$
Sale price $= \$180 - \$45 = \$135$

(b) GST on sale price:
$10\% \times \$135 = \$13.50$

(c) Final price $= \$135 + \$13.50 = \$148.50$

(a) \$135 (b) \$13.50 (c) Final price = \$148.50

Question 11 Juice bottle 1,250 mL. Tom drinks $\frac{2}{5}$ in morning and 30% in afternoon.

(a) Morning: $\dfrac{2}{5} \times 1{,}250 = 2 \times 250 = 500$ mL

(b) Afternoon: 30% of $1{,}250 = 0.30 \times 1{,}250 = 375$ mL

(c) Total drunk $= 500 + 375 = 875$ mL
Left $= 1{,}250 - 875 = 375$ mL

(a) 500 mL (b) 375 mL (c) 375 mL of juice is left.

Teaching note: The afternoon drink is 30% of the original bottle — this is stated explicitly in the question. Some students may try to take 30% of the remaining amount after the morning drink. Redirect by re-reading the question together.

Question 12 Store A: TV \$799. Store B: TV \$849 with 10% off. Which is better value?

Store A price: \$799

Store B price after discount: $\$849 \times 0.90 = \$764.10$

Compare: $\$764.10 < \$799$

Answer: Store B gives the better price (\$764.10 after 10% discount vs \$799 at Store A).

Chapter 2 Solutions Complete

Summary of answers for quick marking:

Practice Exercise: 1. $\frac{2}{3}$ 2. $\frac{3}{5}$ 3. $\frac{3}{4}$ 4. $\frac{3}{5} = \frac{12}{20}$ 5. $\frac{2}{7} = \frac{6}{21}$ 6. Yes, both equal $\frac{2}{5}$ 7. $\frac{1}{4} < \frac{1}{2} < \frac{2}{3} < \frac{3}{4}$ 8. See table (row answers: 0.5/50%; 0.6/60%; $\frac{1}{4}$/25%; $\frac{2}{5}$/0.4; $\frac{2}{25}$/8%) 9. \$30 10. \$20 11. 140 items 12. $\frac{3}{5}$, 60% 13. 0.85, $\frac{17}{20}$ 14. 7 students 15. $\frac{2}{3}$ is greater ($\approx 66.7\% > 65\%$)

Application Exercise: 1. \$36 2. 450 mL drunk, 300 mL left 3. 85% 4. (a) $\frac{13}{20}$ (b) 65% (c) 35% 5. \$1,039.20 6. 18.5% 7. (a) \$34 (b) \$374 8. maths 95% (higher than science 90%) 9. (a) 90 (b) 62.5% 10. (a) \$20 (b) \$2 11. (a) \$25.50 (b) \$36 (c) \$143.50 12. (a) 40% (b) 25% (c) $\frac{1}{4}$

Chapter Review: 1. $\frac{2}{3}$ 2. 0.375, 37.5% 3. 0.45, $\frac{9}{20}$ 4. $\frac{25}{40}$ 5. $\frac{1}{3} < \frac{3}{8} < 0.4 <$ 42% 6. \$91 7. 90% 8. \$34 GST, \$374 total 9. \$55.25 10. (a) \$135 (b) \$13.50 (c) \$148.50 11. (a) 500 mL (b) 375 mL (c) 375 mL left 12. Store B (\$764.10)

3. Rates, Ratios and Comparisons

Chapter 3 Information

Week taught:	Week 3 (Term 1)
Topic:	Topic 1 — Personal Identity and Community
Skills area:	Number & Algebra
Lesson reference:	Lessons 9–12

Teaching Guide

Key concepts: Writing and simplifying ratios; dividing a quantity in a given ratio; unit rates (per hour, per kg, per km); comparing rates to find best value; distance–speed–time calculations.

Suggested delivery sequence:

1. Lesson 9: Ratios — writing, simplifying, equivalent ratios.
2. Lesson 10: Dividing a quantity in a ratio; real-life ratio problems.
3. Lesson 11: Rates — unit rates, best value, speed calculations.
4. Lesson 12: Worked Examples; Practice and Application Exercises.

Common Student Misconceptions

Common Misconceptions — Chapter 3

1. **Reversing the order of a ratio:** Students write ratio of boys to girls as girls:boys. *Strategy:* Underline the key words; always write the ratio in the order stated.
2. **Simplifying ratios — only simplifying one part:** Students simplify 6:9 as 2:9. *Strategy:* Both parts must be divided by the same common factor.
3. **Dividing in a ratio — wrong fraction of total:** For $2:3$, students compute $\frac{2}{3}$ of the total instead of $\frac{2}{5}$. *Strategy:* Always find the total number of parts first.
4. **Unit rate — dividing backwards:** For 300 km in 4 hours, students compute $4 \div 300$. *Strategy:* Read the unit: "km *per* hour" means km on top.
5. **Best value — comparing totals rather than unit price:** Students choose the larger pack without calculating unit price. *Strategy:* Always reduce to a common unit before comparing.

Worked Solutions

Chapter 3 Overview for Teachers

Chapter:	3 — Rates, Ratios and Comparisons
Week:	Week 3, Term 1 (Lessons 9–12)
Topic:	Topic 1 — Personal Identity and Community
Skills area:	Number & Algebra
Focus:	Unit rates and unit pricing; distance–speed–time formula; simplifying ratios; equivalent ratios; scaling recipes; sharing in a ratio

Worked Solutions — Practice Exercise

Part A — Rates

Question 1 Car travels 300 km in 4 hours. Average speed?

$$S = \frac{D}{T} = \frac{300}{4} = 75 \text{ km/h}$$

Answer: 75 km/h

Question 2 Emma reads 45 pages in 3 hours. Rate in pages per hour?

$$\text{Rate} = 45 \div 3 = 15 \text{ pages per hour}$$

Answer: 15 pages per hour

Question 3 Tap fills 180 L tank in 4 hours. Litres per hour?

$$\text{Rate} = 180 \div 4 = 45 \text{ L/h}$$

Answer: 45 litres per hour

Question 4 Cyclist rides at 24 km/h for 2.5 hours. Distance?

$$D = S \times T = 24 \times 2.5 = 60 \text{ km}$$

Answer: 60 km

Part B — Unit Price and Best Buy

Question 5 500 g yoghurt for \$3.50. Price per 100 g?

$$\frac{\$3.50}{500} \times 100 = \$0.70 \text{ per } 100\,\text{g}$$

Answer: \$0.70 per 100 g

Question 6 **2 L juice for \$4.40 vs 1.5 L for \$3.45. Better value?**

2 L option: $\frac{\$4.40}{2000} \times 100 = \$0.22$ per 100 mL

1.5 L option: $\frac{\$3.45}{1500} \times 100 = \$0.23$ per 100 mL

$\$0.22 < \$0.23$

Answer: The better buy is the 2 L bottle (\$0.22 per 100 mL vs \$0.23).

Part C — Writing and Simplifying Ratios

Question 7 **Bag has 5 red and 3 blue marbles. Ratio of red to blue?**

Answer: 5:3

Question 8 **Class of 25. 10 absent. Ratio of absent to present simplified.**

Present students $= 25 - 10 = 15$

Absent : Present $= 10 : 15$

HCF of 10 and 15 $= 5 \quad \rightarrow \quad 10 \div 5 : 15 \div 5 = 2 : 3$

Answer: 2:3

Question 9 **Simplify each ratio.**

(a) $15 : 25$ HCF $= 5 \quad \rightarrow$ **3:5**

(b) $24 : 36$ HCF $= 12 \quad \rightarrow$ **2:3**

(c) $9 : 12 : 15$ HCF $= 3 \quad \rightarrow$ **3:4:5**

Part D — Scaling Ratios

Question 10 **Ratio $3 : 4$. Scale so the first part equals 12.**

Scale factor $= 12 \div 3 = 4$

$3 \times 4 : 4 \times 4 = 12 : 16$

Answer: 12:16

Question 11 **Flour:sugar $= 3 : 1$. Using 12 cups of flour. How much sugar?**

Scale factor $= 12 \div 3 = 4$

Sugar $= 1 \times 4 = 4$ cups

Answer: 4 cups of sugar

Question 12 **Cordial:water $= 1 : 4$. Making 2.5 L total. How much of each?**

Total parts $= 1 + 4 = 5$

Value of 1 part $= 2.5 \div 5 = 0.5$ L

Cordial $= 1 \times 0.5 = 0.5$ L
Water $= 4 \times 0.5 = 2$ L

Check: $0.5 + 2 = 2.5$ L ✓

Answer: 0.5 L cordial and 2 L water

Part E — Conversion Review

Question 13 **Write 0.35 as a fraction (simplest form) and a percentage.**

Fraction: $0.35 = \frac{35}{100}$. HCF $= 5$. $\frac{7}{20}$

Percentage: $0.35 \times 100 = 35\%$

Answer: $\frac{7}{20}$ and 35%

Question 14 **Write $\frac{3}{8}$ as a decimal and a percentage.**

Decimal: $3 \div 8 = 0.375$

Percentage: $0.375 \times 100 = 37.5\%$

Answer: 0.375 and 37.5%

Worked Solutions — Application Exercise

Question 1 Chips: 400 g for \$3.60 vs 700 g for \$5.95. Better value per 100 g?

400 g bag: $\dfrac{\$3.60}{400} \times 100 = \$0.90$ per 100 g

700 g bag: $\dfrac{\$5.95}{700} \times 100 = \$0.85$ per 100 g

$\$0.85 < \$0.90$

Answer: The better buy is the 700 g bag at \$0.85 per 100 g.

Question 2 Cyclist completes 45 km in 1.5 hours. Average speed?

$$S = \frac{D}{T} = \frac{45}{1.5} = 30 \text{ km/h}$$

Answer: Her average speed is 30 km/h.

Question 3 Brisbane to Cairns 1,700 km at 100 km/h. Stops: 30 min + 15 min + 15 min.

(a) Driving time: $T = \dfrac{D}{S} = \dfrac{1700}{100} = 17$ hours

(b) Total stop time $= 30 + 15 + 15 = 60$ min $= 1$ hour

Total time $= 17 + 1 = 18$ hours

(a) 17 hours driving **(b) Total time = 18 hours**

Question 4 Pancake recipe for 4: 2 cups flour 1 cup milk 3 eggs. Scale to 12 people.

Scale factor $= 12 \div 4 = 3$

Flour: $2 \times 3 = 6$ cups
Milk: $1 \times 3 = 3$ cups
Eggs: $3 \times 3 = 9$ eggs

Answer (for 12 people): flour 6 cups, milk 3 cups, eggs 9.

Question 5 **White:blue paint = 5 : 2. Using 10 L of white.**

Scale factor $= 10 \div 5 = 2$

(a) Blue paint $= 2 \times 2 = 4$ L

(b) Total paint $= 10 + 4 = 14$ L

(a) 4 L of blue paint **(b) 14 L total**

Question 6 **Jess and Sam share \$600 in the ratio 3 : 2.**

Total parts $= 3 + 2 = 5$

1 part $= \$600 \div 5 = \$120$

Jess (3 parts) $= 3 \times \$120 = \$360$
Sam (2 parts) $= 2 \times \$120 = \$240$

Check: $\$360 + \$240 = \$600$ ✓

Answer: Jess receives \$360 and Sam receives \$240.

Question 7 **Concrete: cement:sand:gravel = 1 : 3 : 5. Builder needs 18 kg sand.**

Sand is 3 parts. Value of 1 part $= 18 \div 3 = 6$ kg

(a) Cement $= 1 \times 6 = 6$ kg
(b) Gravel $= 5 \times 6 = 30$ kg
(c) Total $= 6 + 18 + 30 = 54$ kg

(a) 6 kg cement **(b) 30 kg gravel** **(c) 54 kg total**

Question 8 **Plan A: 15 GB for \$45. Plan B: 20 GB for \$65. More data per dollar?**

Plan A: $15 \div 45 = 0.333$ GB per dollar

Plan B: $20 \div 65 = 0.308$ GB per dollar

$0.333 > 0.308$

Answer: Plan A gives more data per dollar (0.333 GB/\$ vs 0.308 GB/\$).

Teaching note: Students may compare cost per GB instead (Plan A: \$3.00/GB; Plan B: \$3.25/GB) — this gives the same conclusion. Accept either valid approach.

Question 9 Prize \$1,200 shared among three friends in ratio 2 : 3 : 4.

Total parts $= 2 + 3 + 4 = 9$

1 part $= \$1{,}200 \div 9 = \$133.33$ (recurring)

Person 1 (2 parts): $\$1{,}200 \times \frac{2}{9} = \$266.67$

Person 2 (3 parts): $\$1{,}200 \times \frac{3}{9} = \$400.00$

Person 3 (4 parts): $\$1{,}200 \times \frac{4}{9} = \$533.33$

Check: $\$266.67 + \$400.00 + \$533.33 = \$1{,}200$ ✓

Person 1: \$266.67 Person 2: \$400.00 Person 3: \$533.33

Teaching note: The answer involves a recurring decimal (\$133.3$\overline{3}$ per part). Using fractions of the total (\$1,200 $\times \frac{2}{9}$ etc.) avoids rounding errors. Ensure students check that the three shares sum to \$1,200.

Question 10 OJ: 250 mL for \$1.80 vs 600 mL for \$4.20.

(a) Unit price per 100 mL:

250 mL: $\frac{\$1.80}{250} \times 100 = \$0.72$ per 100 mL

600 mL: $\frac{\$4.20}{600} \times 100 = \$0.70$ per 100 mL

(b) Better buy: 600 mL (\$0.70 per 100 mL)

(c) Cost of 3 small bottles $= 3 \times \$1.80 = \$5.40$
Extra spent compared to 1 large bottle $= \$5.40 - \$4.20 = \$1.20$

(a) 250 mL: \$0.72/100 mL; 600 mL: \$0.70/100 mL (b) 600 mL is better value (c) \$1.20 more

Question 11 Apple juice:ginger ale = 3 : 5. Total punch = 4 L.

Total parts $= 3 + 5 = 8$

1 part $= 4 \div 8 = 0.5$ L

(a) Apple juice $= 3 \times 0.5 = 1.5$ L
(b) Ginger ale $= 5 \times 0.5 = 2.5$ L

(c) Fraction that is apple juice $= \frac{3}{8}$ (already in simplest form)

(a) 1.5 L apple juice **(b) 2.5 L ginger ale** **(c)** $\frac{3}{8}$

Question 12 QLD ≈ 5 million people NSW ≈ 8 million people.

(a) QLD : NSW = 5 : 8. HCF = 1. Already in simplest form.

(b) Combined population $= 5 + 8 = 13$ million

QLD fraction $= \frac{5}{13} = 0.3846\ldots \approx \mathbf{0.38}$ (2 d.p.)

(c) NSW percentage $= \frac{8}{13} \times 100 = 61.538\ldots\% \approx \mathbf{61.5}\%$ (1 d.p.)

(a) 5:8 **(b) 0.38** **(c) 61.5%**

Worked Solutions — Chapter Review

Question 1 Car travels 360 km in 4.5 hours. Average speed?

$$S = \frac{360}{4.5} = 80 \text{ km/h}$$

Answer: 80 km/h

Question 2 750 mL bottle of water costs \$2.85. Price per 100 mL?

$$\frac{\$2.85}{750} \times 100 = \$0.38 \text{ per } 100\,\text{mL}$$

Answer: \$0.38 per 100 mL

Question 3 1 kg coffee for \$6.20 vs 1.5 kg for \$8.70. Better value?

$$1\,\text{kg option: } \frac{\$6.20}{1000} \times 100 = \$0.62 \text{ per } 100\,\text{g}$$

$$1.5\,\text{kg option: } \frac{\$8.70}{1500} \times 100 = \$0.58 \text{ per } 100\,\text{g}$$

$$\$0.58 < \$0.62$$

Answer: The 1.5 kg bag is better value at \$0.58 per 100 g.

Question 4 Class has 18 adults and 12 children. Ratio adults:children simplified.

$18 : 12$. HCF $= 6$. $\rightarrow 3 : 2$

Answer: 3:2

Question 5 $4 : 7 = ? : 35$

Scale factor $= 35 \div 7 = 5$

First part $= 4 \times 5 = 20$

Answer: $20 : 35$

Question 6 Write 72% as a decimal and a fraction in simplest form.

Decimal: $72\% \div 100 = 0.72$

Fraction: $\frac{72}{100}$. HCF $= 4$. $\rightarrow \frac{18}{25}$

Answer: 0.72 **and** $\frac{18}{25}$

Question 7 Write $\frac{5}{6}$ as a decimal (3 d.p.) and a percentage.

Decimal: $5 \div 6 = 0.8333\ldots \approx 0.833$ (3 d.p.)

Percentage: $0.833 \times 100 = 83.3\%$

Answer: 0.833 **and** 83.3%

Teaching note: The exact value is $0.\overline{83}$ (recurring). The question specifies 3 d.p., so 0.833 is correct. Accept $83.\overline{3}\%$ or 83.3% for the percentage.

Question 8 Ratio $2:5$ total amount $= 70$. Find each part.

Total parts $= 2 + 5 = 7$

1 part $= 70 \div 7 = 10$

First part (2): $2 \times 10 = 20$
Second part (5): $5 \times 10 = 50$

Check: $20 + 50 = 70$ ✓

Answer: First part = 20, second part = 50

Question 9 Car uses 8 L per 100 km. Fuel needed for 450 km trip?

$$\text{Fuel} = \frac{8}{100} \times 450 = 8 \times 4.5 = 36 \text{ L}$$

Answer: 36 litres of fuel needed.

Question 10 **Recipe for 6: 3 cups rice 2 cups beans 1 cup spice. Scale to 15 people.**

Scale factor $= 15 \div 6 = 2.5$

Rice: $3 \times 2.5 = 7.5$ cups
Beans: $2 \times 2.5 = 5$ cups
Spice: $1 \times 2.5 = 2.5$ cups

Answer (for 15): rice 7.5 cups, beans 5 cups, spice 2.5 cups

Question 11 **Tom (3 hrs) and Lucy (5 hrs) share \$192 in ratio of hours worked.**

Ratio Tom : Lucy $= 3 : 5$

Total parts $= 3 + 5 = 8$

1 part $= \$192 \div 8 = \$24$

Tom $= 3 \times \$24 = \$72$
Lucy $= 5 \times \$24 = \$120$

Check: $\$72 + \$120 = \$192$ ✓

Answer: Tom earns \$72, Lucy earns \$120.

Question 12 **Sam earns \$720/week. Spends $\frac{1}{3}$ on rent 25% on food saves the rest.**

(a) Rent $= \dfrac{1}{3} \times \$720 = \$240$

(b) Food $= 25\% \times \$720 = 0.25 \times 720 = \$180$

(c) Savings $= \$720 - \$240 - \$180 = \$300$

(d) Savings as fraction of income $= \dfrac{300}{720}$. HCF $= 60$. $\rightarrow \dfrac{5}{12}$

(a) \$240 rent **(b) \$180 food** **(c) \$300 savings** **(d) $\dfrac{5}{12}$**

Teaching note: Some students may convert $\frac{1}{3}$ to $33.\overline{3}\%$ and add to 25% to get $58.\overline{3}\%$, then subtract from 100% for savings. This is valid but introduces rounding risk. The direct subtraction approach in part (c) is cleaner.

Question 13 **Plan X: 50 GB for \$60. Plan Y: 80 GB for \$95. More data per dollar?**

Plan X: $50 \div 60 = 0.833$ GB per dollar

Plan Y: $80 \div 95 = 0.842$ GB per dollar

$0.842 > 0.833$

Answer: Plan Y gives more data per dollar (0.842 GB/\$ vs 0.833 GB/\$).

Question 14 **Punch: lemonade:juice = 3 : 5. Total = 6.4 L. Juice at \$2.50/L lemonade at \$1.80/L.**

Total parts $= 3 + 5 = 8$

1 part $= 6.4 \div 8 = 0.8$ L

(a) Lemonade $= 3 \times 0.8 = 2.4$ L
(b) Juice $= 5 \times 0.8 = 4$ L

(c) Cost of juice $= 4 \times \$2.50 = \$10.00$
Cost of lemonade $= 2.4 \times \$1.80 = \$4.32$
Total cost $= \$10.00 + \$4.32 = \$14.32$

(a) 2.4 L lemonade **(b) 4 L juice** **(c) Total cost = \$14.32**

Chapter 3 Solutions Complete

Summary of answers for quick marking:

Practice Exercise: 1. 75 km/h 2. 15 pages/h 3. 45 L/h 4. 60 km 5. \$0.70/100 g 6. 2 L bottle (\$0.22/100 mL) 7. 5:3 8. 2:3 9. (a) 3:5 (b) 2:3 (c) 3:4:5 10. 12:16 11. 4 cups 12. 0.5 L cordial, 2 L water 13. $\frac{7}{20}$, 35% 14. 0.375, 37.5%

Application Exercise: 1. 700 g bag (\$0.85/100 g) 2. 30 km/h 3. (a) 17 h (b) 18 h total 4. 6 cups flour, 3 cups milk, 9 eggs 5. (a) 4 L (b) 14 L 6. Jess \$360, Sam \$240 7. (a) 6 kg (b) 30 kg (c) 54 kg 8. Plan A (0.333 GB/\$) 9. \$266.67, \$400, \$533.33 10. (a) \$0.72, \$0.70 (b) 600 mL (c) \$1.20 more 11. (a) 1.5 L (b) 2.5 L (c) $\frac{3}{8}$ 12. (a) 5:8 (b) 0.38 (c) 61.5%

Chapter Review: 1. 80 km/h 2. \$0.38/100 mL 3. 1.5 kg (\$0.58/100 g) 4. 3:2 5. 20:35 6. 0.72, $\frac{18}{25}$ 7. 0.833, 83.3% 8. 20 and 50 9. 36 L 10. 7.5 cups rice, 5 cups beans, 2.5 cups spice 11. Tom \$72, Lucy \$120 12. (a) \$240 (b) \$180 (c) \$300 (d) $\frac{5}{12}$ 13. Plan Y (0.842 GB/\$) 14. (a) 2.4 L (b) 4 L (c) \$14.32

4. Length, Perimeter and Area

Chapter 4 Information

Week taught:	Week 4 (Term 1)
Topic:	Topic 1 — Personal Identity and Community
Skills area:	Measurement & Space
Lesson reference:	Lessons 13–16

Teaching Guide

Key concepts: Length unit conversions (mm, cm, m, km); perimeter of rectangles, squares, triangles and composite shapes; area of rectangles, squares, and triangles; area units (cm^2, m^2).

Suggested delivery sequence:

1. Lesson 13: Length units and conversions; measuring in context.
2. Lesson 14: Perimeter — rectangles, squares, composite shapes.
3. Lesson 15: Area of rectangles and squares.
4. Lesson 16: Area of triangles; Worked Examples; Practice Exercise.

Common Student Misconceptions

Common Misconceptions — Chapter 4

1. **Unit conversion errors:** Students convert 500 cm to m by multiplying (500×100). *Strategy:* Use a conversion ladder; moving *up* the ladder means divide.
2. **Confusing perimeter and area:** Students use an area formula for a perimeter question. *Strategy:* Perimeter is a distance; area is a surface.
3. **Triangle area — forgetting to halve:** Students use $A = b \times h$ without halving. *Strategy:* A triangle is always half of a rectangle with the same base and height.
4. **Using slant side as height:** Students use the slant side of a triangle as the height. *Strategy:* The height must be perpendicular to the base.
5. **Composite shapes — not subtracting missing pieces:** Students add all parts rather than subtracting cut-out sections. *Strategy:* Clearly mark which regions are included.

Worked Solutions

Chapter 4 Overview for Teachers

Chapter:	4 — Length, Perimeter and Area
Week:	Week 4, Term 1 (Lessons 13–16)
Topic:	Topic 1 — Personal Identity and Community
Skills area:	Measurement & Space
Focus:	Measuring and estimating length; perimeter of squares, rectangles and triangles; area of squares, rectangles and triangles; practical problems involving tiling, carpeting and fencing

Worked Solutions — Practice Exercise

Part A — Length Conversions

Question 1 Convert each measurement.

(a) 3.5 km → m: multiply by 1,000 $3.5 \times 1{,}000 = 3{,}500$ **m**

(b) 450 cm → m: divide by 100 $450 \div 100 = 4.5$ **m**

(c) 82 mm → cm: divide by 10 $82 \div 10 = 8.2$ **cm**

(d) 2.4 m → cm: multiply by 100 $2.4 \times 100 = 240$ **cm**

(e) 0.6 km → m: multiply by 1,000 $0.6 \times 1{,}000 = 600$ **m**

(f) 1,250 m → km: divide by 1,000 $1{,}250 \div 1{,}000 = 1.25$ **km**

Part B — Perimeter

Question 2 Calculate the perimeter of each shape.

(a) Rectangle $L = 8$ m, $W = 5$ m:
$P = 2L + 2W = 2(8) + 2(5) = 16 + 10 = 26$ **m**

(b) Square side = 6.5 cm:
$P = 4S = 4 \times 6.5 = 26$ **cm**

(c) Triangle sides 7 cm, 9 cm, 11 cm:
$P = 7 + 9 + 11 = 27$ **cm**

(d) Rectangle $L = 12.5$ m, $W = 4$ m:
$P = 2(12.5) + 2(4) = 25 + 8 = 33$ **m**

Part C — Area

Question 3 Calculate the area of each shape.

(a) Rectangle $L = 9$ m, $W = 4$ m:
$A = L \times W = 9 \times 4 = 36$ **m**^{2}

(b) Square side $= 7$ cm:
$A = S^2 = 7^2 = 49$ **cm**^{2}

(c) Triangle $B = 10$ m, $H = 6$ m:
$A = \dfrac{B \times H}{2} = \dfrac{10 \times 6}{2} = \dfrac{60}{2} = 30$ **m**^{2}

(d) Triangle $B = 8.4$ cm, $H = 5$ cm:
$A = \dfrac{8.4 \times 5}{2} = \dfrac{42}{2} = 21$ **cm**^{2}

Part D — Counting Squares

Question 4 Find area by counting squares (each square

(a) L-shaped region (vertices at $(1,1), (7,1), (7,4), (4,4), (4,7), (1,7)$):

This is made of two rectangles — count directly:
Bottom section: $6 \times 3 = 18$ whole squares
Top section: $3 \times 3 = 9$ whole squares
Total whole squares $= 27$

Area = 27 cm^2 (exact — all whole squares, no partials)

(b) Irregular blob (counting estimate):

Whole squares $\approx$ **22** Partial squares $\approx$ **10** $\times \frac{1}{2} = 5$

Estimated area $\approx$ 27 cm^2

Accept any answer in the range 24–30 cm^2. The exact area (by geometry) is approximately 27 cm^2. Encourage students to be systematic — count all complete squares first, then go around the edge pairing up partials.

Worked Solutions — Application Exercise

Question 1 **Rectangular garden bed 8 m × 5 m. Timber border around edge.**

(a) Perimeter (length of timber):
$P = 2(8) + 2(5) = 16 + 10 = 26$ m

(b) Cost $= 26 \times \$12 = \$312$

(a) 26 m of timber needed **(b) Total cost = \$312**

Question 2 **Bedroom 4.5 m × 3.8 m. Carpet at \$45 per m^2.**

(a) Area $= 4.5 \times 3.8 = 17.1$ m^2

(b) Cost $= 17.1 \times \$45 = \$769.50$

(a) 17.1 m^2 **(b) Cost = \$769.50**

Question 3 **Triangular garden bed $B = 7$ m $H = 5$ m. Mulch at \$8 per m^2.**

(a) Area $= \dfrac{7 \times 5}{2} = \dfrac{35}{2} = 17.5$ m^2

(b) Cost $= 17.5 \times \$8 = \$140$

(a) 17.5 m^2 **(b) Cost = \$140**

Question 4 **School oval 110 m × 68 m.**

(a) Perimeter $= 2(110) + 2(68) = 220 + 136 = 356$ m

(b) Area $= 110 \times 68 = 7{,}480$ m^2

(a) 356 m **(b) 7,480 m^2**

Question 5 **Wall 4 m × 2.5 m with window 1.2 m × 0.9 m. Painter skips window.**

(a) Wall area $= 4 \times 2.5 = 10$ m^2

(b) Window area $= 1.2 \times 0.9 = 1.08$ m^2

(c) Paint area $= 10 - 1.08 = 8.92$ m^2

(a) 10 m^2 (b) 1.08 m^2 (c) Area to be painted = 8.92 m^2

Question 6 **Bathroom 3 m × 3 m. Tiles 20 cm × 20 cm at \$3.50 each.**

(a) Convert: 3 m = 300 cm. Tiles per side $= 300 \div 20 = 15$

(b) Total tiles $= 15 \times 15 = 225$

(c) Cost $= 225 \times \$3.50 = \$787.50$

(a) 15 tiles per side (b) 225 tiles needed (c) Total cost = \$787.50

Question 7 **Yard 15 m × 10 m. One 15 m side is wall. Gate 1.2 m in one 10 m side.**

Sides needing fencing: one 15 m side + two 10 m sides $= 15 + 10 + 10 = 35$ m

Less gate opening: $35 - 1.2 = 33.8$ m of fencing

(b) Fencing cost $= 33.8 \times \$18 = \$608.40$
Gate cost $= \$320$
Total $= \$608.40 + \$320 = \$928.40$

(a) 33.8 m of fencing (b) Total cost = \$928.40

Teaching note: Students often include the wall side in their perimeter calculation. Re-read the question: one 15 m side already has a wall, so only three sides need fencing. Sketching a diagram helps.

Question 8 Kitchen 3.6 m × 2.4 m. Tiles 30 cm × 30 cm at \$4.80 each.

(a) Floor area $= 3.6 \times 2.4 = 8.64\ \text{m}^2$

(b) Tile area $= 0.30 \times 0.30 = 0.09\ \text{m}^2$

(c) Number of tiles $= 8.64 \div 0.09 = 96$

(d) Cost $= 96 \times \$4.80 = \$460.80$

(a) 8.64 m^2 (b) 0.09 m^2 (c) 96 tiles (d) Total cost = \$460.80

Teaching note: Students must convert tile dimensions to metres before comparing to the floor area, or convert the floor to cm. Both approaches work; the key is consistency of units throughout.

Question 9 Triangular sail $B = 3.6$ m $H = 5.2$ m. Find area.

$$A = \frac{B \times H}{2} = \frac{3.6 \times 5.2}{2} = \frac{18.72}{2} = 9.36\ \text{m}^2$$

Answer: Area = 9.36 m^2

Question 10 Park 20 m × 15 m. Triangular flower bed ($B = 6$ m $H = 5$ m) is paved. Grass area?

Park area $= 20 \times 15 = 300\ \text{m}^2$

Flower bed area $= \dfrac{6 \times 5}{2} = 15\ \text{m}^2$

Grass area $= 300 - 15 = 285\ \text{m}^2$

Answer: Grass area = 285 m^2

Question 11 Square vegetable patch with perimeter 28 m.

(a) Side length $= 28 \div 4 = 7$ m

(b) Area $= 7^2 = 49\ \text{m}^2$

(a) 7 m (b) Area = 49 m^2

Question 12 Pool 25 m × 10 m. Tiled path 1 m wide all around outside.

(a) Outer dimensions: path adds 1 m on each side.
Length $= 25 + 1 + 1 = 27$ m Width $= 10 + 1 + 1 = 12$ m

(b) Outer area $= 27 \times 12 = 324$ m^2

(c) Pool area $= 25 \times 10 = 250$ m^2
Path area $= 324 - 250 = 74$ m^2

(a) 27 m × 12 m (b) 324 m^2 (c) Path area = 74 m^2

Teaching note: Common error is adding 1 m to only one end of each side, giving 26 m and 11 m. The path wraps all the way around, adding 1 m to both ends of each dimension.

Worked Solutions — Chapter Review

Question 1 Complete the conversion table.

mm	cm	m
450	45	**0.45**
75	**7.5**	**0.075**
2,800	**280**	2.8

m	km
2,800	**2.8**
350	0.35

Key working: mm ↔ cm: ÷10 or ×10; cm ↔ m: ÷100 or ×100; m ↔ km: ÷1,000 or ×1,000.

Question 2 Perimeter of rectangle $L = 7\,\text{m}$ $W = 4.5\,\text{m}$.

$P = 2(7) + 2(4.5) = 14 + 9 = 23$ m

Answer: 23 m

Question 3 Perimeter of square with side 9.5 cm.

$P = 4 \times 9.5 = 38$ cm

Answer: 38 cm

Question 4 Triangle sides 5 cm 7 cm 9 cm. Perimeter?

$P = 5 + 7 + 9 = 21$ cm

Answer: 21 cm

Question 5 Area of rectangle $L = 11\,\text{m}$ $W = 6\,\text{m}$.

$A = 11 \times 6 = 66$ m^2

Answer: 66 m^2

Question 6 Area of square with side 8.5 cm.

$A = 8.5^2 = 8.5 \times 8.5 = 72.25 \text{ cm}^2$

Answer: 72.25 cm^2

Question 7 Triangle $B = 14$ m $H = 9$ m. Area?

$$A = \frac{14 \times 9}{2} = \frac{126}{2} = 63 \text{ m}^2$$

Answer: 63 m^2

Question 8 Triangular road sign $B = 60$ cm $H = 52$ cm. Area?

$$A = \frac{60 \times 52}{2} = \frac{3{,}120}{2} = 1{,}560 \text{ cm}^2$$

Answer: 1,560 cm^2

Question 9 Living room 5.2 m × 4 m. Floorboards \$38/m^2 materials + \$12/m^2 installation.

(a) Area $= 5.2 \times 4 = 20.8 \text{ m}^2$

(b) Combined rate $= \$38 + \$12 = \$50$ per m^2
Total cost $= 20.8 \times \$50 = \$1{,}040$

(a) 20.8 m^2 (b) Total cost = \$1,040

Question 10 Backyard 18 m × 12 m. Lawn \$7/m^2 fencing \$22/m.

(a) Area $= 18 \times 12 = 216 \text{ m}^2$

(b) Lawn cost $= 216 \times \$7 = \$1{,}512$

(c) Perimeter $= 2(18) + 2(12) = 36 + 24 = 60$ m

(d) Fencing cost $= 60 \times \$22 = \$1{,}320$

(e) Total $= \$1{,}512 + \$1{,}320 = \$2{,}832$

(a) 216 m^2 (b) \$1,512 (c) 60 m (d) \$1,320 (e) Total = \$2,832

Question 11 Square frame outer side 40 cm inner (picture) side 32 cm. Frame border area?

Outer area $= 40^2 = 1{,}600$ cm^2

Inner area $= 32^2 = 1{,}024$ cm^2

Frame area $= 1{,}600 - 1{,}024 = 576$ cm^2

Answer: Frame area = 576 cm^2

Question 12 Bed A: rectangle 8 m × 3.5 m. Bed B: triangle $B = 10$ m $H = 6$ m. Which is larger?

Bed A (rectangle): $A = 8 \times 3.5 = 28$ m^2

Bed B (triangle): $A = \dfrac{10 \times 6}{2} = 30$ m^2

Difference $= 30 - 28 = 2$ m^2

Answer: Bed B has the larger area by 2 m^2.

Chapter 4 Solutions Complete

Summary of answers for quick marking:

Practice Exercise: 1. (a) 3,500 m (b) 4.5 m (c) 8.2 cm (d) 240 cm (e) 600 m (f) 1.25 km 2. (a) 26 m (b) 26 cm (c) 27 cm (d) 33 m 3. (a) 36 m^2 (b) 49 cm^2 (c) 30 m^2 (d) 21 cm^2 4. (a) 27 cm^2 (b) ≈27 cm^2 (accept 24–30)

Application Exercise: 1. (a) 26 m (b) \$312 2. (a) 17.1 m^2 (b) \$769.50 3. (a) 17.5 m^2 (b) \$140 4. (a) 356 m (b) 7,480 m^2 5. (a) 10 m^2 (b) 1.08 m^2 (c) 8.92 m^2 6. (a) 15 (b) 225 tiles (c) \$787.50 7. (a) 33.8 m (b) \$928.40 8. (a) 8.64 m^2 (b) 0.09 m^2 (c) 96 tiles (d) \$460.80 9. 9.36 m^2 10. 285 m^2 11. (a) 7 m (b) 49 m^2 12. (a) 27 m×12 m (b) 324 m^2 (c) 74 m^2

Chapter Review: 1. See table (450 mm/0.45 m; 7.5 cm/0.075 m; 2,800 mm/280 cm; 2.8 km; 350 m) 2. 23 m 3. 38 cm 4. 21 cm 5. 66 m^2 6. 72.25 cm^2 7. 63 m^2 8. 1,560 cm^2 9. (a) 20.8 m^2 (b) \$1,040 10. (a) 216 m^2 (b) \$1,512 (c) 60 m (d) \$1,320 (e) \$2,832 11. 576 cm^2 12. Bed B larger by 2 m^2

5. Shapes, Area and Maps

Chapter 5 Information

Week taught:	Week 5 (Term 1)
Topic:	Topic 1 — Personal Identity and Community
Skills area:	Measurement & Space
Lesson reference:	Lessons 17–20

Teaching Guide

Key concepts: Properties of 2D shapes; area unit conversions (mm^2, cm^2, m^2, km^2); reading maps with grid references and compass directions; using map scales to calculate real distances.

Suggested delivery sequence:

1. Lesson 17: Properties of 2D shapes; classifying triangles and quadrilaterals.
2. Lesson 18: Area unit conversions; counting squares for irregular areas.
3. Lesson 19: Maps — grid references, compass directions, scale.
4. Lesson 20: Worked Examples; Practice and Application Exercises.

Common Student Misconceptions

Common Misconceptions — Chapter 5

1. **Area unit conversions — using factor of 10:** Students convert 1 m^2 to cm^2 as 100 instead of 10,000. *Strategy:* 1 m = 100 cm, so 1 m^2 = 100^2 = 10,000 cm^2.

2. **Grid references — reversing row and column:** Students read (3, 5) as row 3, column 5. *Strategy:* "Along the corridor, then up the stairs."

3. **Map scale errors:** Students interpret scale as a fraction rather than multiplying. *Strategy:* 1 cm on map = scale factor cm in real life.

4. **Confusing types of triangles:** Students give only one classification for a triangle with multiple properties. *Strategy:* A triangle can have multiple classifications simultaneously.

5. **Counting squares for irregular shapes:** Students count only whole squares. *Strategy:* Pair up partial squares; two halves = one whole.

Worked Solutions

Chapter 5 Overview for Teachers

Chapter:	5 — Shapes, Area and Maps
Week:	Week 5, Term 1 (Lessons 17–20)
Topic:	Topic 1 — Personal Identity and Community
Skills area:	Measurement & Space
Focus:	Shape properties, area unit conversions, map skills (Springvale map)

Key concepts: Identifying 2D shapes by properties; converting mm^2, cm^2, m^2, km^2; reading grid references and compass directions; calculating real distances from a map scale.

Map used throughout: The Springvale Neighbourhood Map (scale 1:10,000; 1 cm = 100 m). Grid squares are 1.4 cm × 1.4 cm. Key locations:

A2	Train Station	B4	School (Springvale SC)
C2	Shopping Centre	C4	Café
D3	Riverside Park	D6	Swimming Pool
E5	Library	F4	Hospital

Note on map questions: Answers to distance questions depend on students measuring from the printed map. Solutions below use the grid-square size (1.4 cm) to calculate expected distances. Accept reasonable measurements (±0.2 cm).

Completed Shape Properties Reference Table

Shape	Sides	Equal sides	Right angles	Lines of sym.	Other
Square	4	**4**	**4**	**4**	All sides equal
Rectangle	**4**	2 pairs	4	**2**	Opp. sides parallel
Equilateral △	**3**	**3**	0	**3**	All angles 60°
Right-angled △	3	**0**	**1**	0	One angle = 90°
Isosceles △	3	**2**	0	**1**	2 equal angles
Circle	0	—	—	**infinite**	No straight sides

Note: Right-angled triangle has 0 equal sides in the general case. A right-angled isosceles triangle (45–45–90) would have 2 equal sides, but students are not expected to know this.

Practice Exercise

Part A — Shape Properties

Question 1 — Name the shape

(a) 4 sides, opposite sides equal and parallel, 4 right angles, not all sides equal.
⇒ **Rectangle**

(b) 3 sides, 2 equal sides, 2 equal angles.
⇒ **Isosceles triangle**

(c) 4 equal sides, no right angles (not a square).
⇒ **Rhombus**

(d) No sides, no angles, perfectly round boundary.
⇒ **Circle**

Question 2 — Lines of symmetry

(a) Square: **4** lines of symmetry (2 through midpoints of opposite sides, 2 through opposite corners).

(b) Rectangle (not a square): **2** lines of symmetry (through midpoints of opposite sides only).

(c) Equilateral triangle: **3** lines of symmetry (one from each vertex to the midpoint of the opposite side).

(d) Circle: **infinite** lines of symmetry (any diameter divides it into two identical halves).

Part B — Area Unit Conversions

Question 3 — Convert area units

Key rules: $m^2 \leftrightarrow cm^2$: $\times$ or $\div$ 10,000; $km^2 \leftrightarrow m^2$: $\times$ or $\div$ 1,000,000; $mm^2 \leftrightarrow cm^2$: $\times$ or $\div$ 100.

(a) $3\,\text{m}^2 \to \text{cm}^2$: multiply by 10,000.

$$3 \times 10{,}000 = 30{,}000\ \text{cm}^2$$

(b) $80{,}000\,\text{cm}^2 \to \text{m}^2$: divide by 10,000.

$$80{,}000 \div 10{,}000 = 8\ \text{m}^2$$

(c) $5\,\text{km}^2 \to \text{m}^2$: multiply by 1,000,000.

$$5 \times 1{,}000{,}000 = 5{,}000{,}000\ \text{m}^2$$

(d) $250\,\text{mm}^2 \to \text{cm}^2$: divide by 100.

$$250 \div 100 = 2.5\ \text{cm}^2$$

(e) $0.4\,\text{m}^2 \to \text{cm}^2$: multiply by 10,000.

$$0.4 \times 10{,}000 = 4{,}000\ \text{cm}^2$$

(f) $2{,}500{,}000\,\text{m}^2 \to \text{km}^2$: divide by 1,000,000.

$$2{,}500{,}000 \div 1{,}000{,}000 = 2.5\ \text{km}^2$$

Part C — Grid References (Springvale map)

Question 4 — Name the location at each grid reference

(a) A2 ⇒ **Train Station**
(b) C2 ⇒ **Shopping Centre**
(c) B4 ⇒ **Springvale School (SC)**
(d) D6 ⇒ **Swimming Pool**

Question 5 — Write the grid reference for each location

(a) Hospital ⇒ **F4**
(b) Riverside Park ⇒ **D3**
(c) Café ⇒ **C4**
(d) Swimming Pool ⇒ **D6**

Part D — Compass Directions (Springvale map)

Map orientation reminder

On the Springvale map: columns A–F run left (West) to right (East); row numbers increase going upward (North). So a higher row number means further North.

Question 6 — Compass direction of travel

(a) Café (C4) → Library (E5):
Column C → E = 2 steps right = **East**; Row 4 → 5 = 1 step up = **North**.
Direction: **East and North (Northeast)**

(b) Hospital (F4) → School (B4):
Column F → B = 4 steps left = **West**; Row stays at 4 = no change.
Direction: **West**

(c) Train Station (A2) → Swimming Pool (D6):
Column A → D = 3 steps right = **East**; Row 2 → 6 = 4 steps up = **North**.
Direction: **East and North (Northeast)**

(d) Library (E5) → Shopping Centre (C2):
Column E → C = 2 steps left = **West**; Row 5 → 2 = 3 steps down = **South**.
Direction: **West and South (Southwest)**

Part E — Map Scale Calculations

Question 7 — Map distance to real distance (scale 1 cm

(a) $1.5\,\text{cm} \times 100 = 150\ \text{m}$

(b) $3.2\,\text{cm} \times 100 = 320\ \text{m}$

(c) $0.8\,\text{cm} \times 100 = 80\ \text{m}$

Question 8 — Real distance to map distance

Step 1: Real distance $= 450\,\text{m}$. Scale: $1\,\text{cm} = 100\,\text{m}$.

Step 2: Map distance $= 450 \div 100 = 4.5\ \text{cm}$

Application Exercise

Question 1 — Mia's route: School (B4) → Café (C4) → Riverside Park (D3)

Leg 1: School (B4) → Café (C4)
Column B → C: 1 step right = East. Same row.
Travel East from the School to the Café.

Leg 2: Café (C4) → Riverside Park (D3)
Column C → D: 1 step right = East. Row 4 → 3: 1 step down = South.
Travel East and South from the Café to Riverside Park.

Question 2 — Train Station (A2) to Shopping Centre (C2); map distance 2 cm

(a) Actual distance:

$$2\,\text{cm} \times 100 = 200\text{ m}$$

(b) Time to walk at 80 m per minute:

$$200 \div 80 = 2.5\text{ minutes}$$

Question 3 — Hospital (F4) to Library (E5); map distance 1.4 cm

$$1.4\,\text{cm} \times 100 = 140\text{ m}$$

Question 4 — Jordan: Swimming Pool (D6) to Train Station (A2)

(a) Route using compass directions:
Column D → A = 3 steps left = West; Row 6 → 2 = 4 steps down = South.
Travel West and South from the Swimming Pool to the Train Station.

(Sample: head West along the top road, then turn South towards the station entrance.)

(b) Map route ≈ 4.5 cm:

$$4.5\,\text{cm} \times 100 = 450\text{ m}$$

Question 5 — Which two locations are furthest apart?

Identify candidates. Compare diagonal distances using grid squares (each 1.4 cm).

Train Station (A2) to Hospital (F4):
Horizontal: $5 \times 1.4 = 7.0$ cm; Vertical: $2 \times 1.4 = 2.8$ cm

$$\text{Distance} = \sqrt{7.0^2 + 2.8^2} = \sqrt{49.0 + 7.84} = \sqrt{56.84} \approx 7.5\,\text{cm}$$

Train Station (A2) to Swimming Pool (D6):
Horizontal: $3 \times 1.4 = 4.2$ cm; Vertical: $4 \times 1.4 = 5.6$ cm

$$\text{Distance} = \sqrt{4.2^2 + 5.6^2} = \sqrt{17.64 + 31.36} = \sqrt{49} = 7.0\,\text{cm}$$

Conclusion. The **Train Station (A2) and Hospital (F4)** are the furthest apart.
Map distance ≈ 7.5 cm; real-world distance $= 7.5 \times 100 =$ 750 m

Teaching note: Students are not expected to use Pythagoras — they should measure with a ruler. Accept any pair with a measured distance near 7–8 cm and a correctly calculated real distance.

Question 6 — Riverside Park (D3); grid square 1.4 cm × 1.4 cm

(a) Real side length:

$$1.4\,\text{cm} \times 100 = 140\ \text{m}$$

(b) Real area:

$$140 \times 140 = 19{,}600\ \text{m}^2$$

Question 7 — New café halfway between Train Station (A2) and School (B4)

Step 1: Find the midpoint.
Train Station: column A, row 2. School: column B, row 4.
Midpoint column: halfway between A and B (no exact grid letter — describe as "between A and B").
Midpoint row: $(2 + 4) \div 2 = 3$

Answer: The new café would appear **approximately halfway between grid squares A3 and B3**, or "at about column A.5, row 3."

Teaching note: Students should describe the location as clearly as possible using the grid — e.g., "between A3 and B3" or "just right of A, at row 3." Accept any reasonable description.

Question 8 — Bedroom floor plan; scale 1:100 (1 cm

(a) Actual dimensions:

$$4.5\,\text{cm} \times 1 = 4.5\,\text{m}; \quad 3\,\text{cm} \times 1 = 3\,\text{m}$$

Dimensions: **4.5 m by 3 m**

(b) Actual floor area:

$$4.5 \times 3 = 13.5\ \text{m}^2$$

Question 9 — Football field 7,140 m² to km²

Step 1: Divide by 1,000,000 to convert m^2 to km^2.

$$7{,}140 \div 1{,}000{,}000 = 0.007140\,\text{km}^2$$

Rounded to 4 decimal places: **0.0071 km²**

Question 10 — Square courtyard; area 625 m^2

(a) Side length of a square with area 625 m^2:

$$\text{side} = \sqrt{625} = 25 \text{ m}$$

(b) Convert 625 m^2 to cm^2:

$$625 \times 10{,}000 = 6{,}250{,}000 \text{ cm}^2$$

Question 11 — Room on floor plan; scale 1:50 (1 cm

(a) Actual dimensions:
Scale 1:50 means 1 cm on plan = 0.5 m in reality.

$$7\,\mathrm{cm} \times 0.5 = 3.5\,\mathrm{m}; \qquad 5\,\mathrm{cm} \times 0.5 = 2.5\,\mathrm{m}$$

Dimensions: **3.5 m by 2.5 m**

(b) Actual area:

$$3.5 \times 2.5 = 8.75 \text{ m}^2$$

Question 12 — Triangular garden; base 4 m; height 3 m; area in cm^2

Step 1: Calculate area in m^2.

$$A = \tfrac{1}{2} \times \text{base} \times \text{height} = \tfrac{1}{2} \times 4 \times 3 = 6\,\mathrm{m}^2$$

Step 2: Convert to cm^2.

$$6\,\mathrm{m}^2 \times 10{,}000 = 60{,}000 \text{ cm}^2$$

Chapter Review

Question 1 — Complete the sentence about a rectangle

A rectangle has **4** sides, **4** right angles and **2** lines of symmetry.

Question 2 — Name a 2D shape with exactly 1 line of symmetry

Isosceles triangle (the line of symmetry runs from the apex to the midpoint of the base).

Also accept: kite, or any scalene triangle with a special property — but isosceles triangle is the standard expected answer at this level.

Question 3 — Convert 6.2 m^2 to cm^2

$$6.2 \times 10{,}000 = 62{,}000 \text{ cm}^2$$

Question 4 — Convert 350,000 cm^2 to m^2

$$350{,}000 \div 10{,}000 = 35 \text{ m}^2$$

Question 5 — Convert 1.5 km^2 to m^2

$$1.5 \times 1{,}000{,}000 = 1{,}500{,}000 \text{ m}^2$$

Question 6 — What is at grid reference F4?

Column F, row 4 on the Springvale map ⇒ **Hospital**

Question 7 — Direction from Shopping Centre (C2) to Hospital (F4)

Column C → F = 3 steps right = **East**.
Row 2 → 4 = 2 steps up = **North**.
Direction: **East and North (or Northeast)**

Question 8 — Map distance 3.5 cm on 1:10,000 map

$$3.5\,\text{cm} \times 100 = 350\ \text{m}$$

Question 9 — Towns 4 cm apart on a 1:50,000 map; distance in km

Step 1: Apply the scale.

$$4\,\text{cm} \times 50{,}000 = 200{,}000\,\text{cm}$$

Step 2: Convert cm to km (divide by 100,000).

$$200{,}000\,\text{cm} \div 100{,}000 = 2\ \text{km}$$

Teaching note: Students often make errors converting cm to km. Reinforce the chain: cm ÷ 100 = m, then m ÷ 1000 = km. Or divide by 100,000 directly.

Question 10 — House block 18 m × 30 m

(a) Area in m^2:

$$18 \times 30 = 540\ \text{m}^2$$

(b) Convert to cm^2:

$$540 \times 10{,}000 = 5{,}400{,}000\ \text{cm}^2$$

Question 11 — Multi-step: Floor plan 1:100; 6 cm × 4 cm; flooring \$52 per m²

(a) Real dimensions (scale 1:100; 1 cm = 1 m):

$$6\,\text{cm} \to 6\,\text{m}; \qquad 4\,\text{cm} \to 4\,\text{m}$$

Real dimensions: **6 m by 4 m**

(b) Real floor area:

$$6 \times 4 = 24\ \text{m}^2$$

(c) Total cost of flooring:

$$24\,\text{m}^2 \times \$52 = \$1{,}248$$

Question 12 — Multi-step: Round trip School (B4) → Library (E5) → Swimming Pool (D6) → School (B4)

Compass directions for each leg:

Leg 1 School (B4) → Library (E5):
Column B → E = 3 steps right = East; Row 4 → 5 = 1 step up = North.
Direction: **East and North**

Leg 2 Library (E5) → Swimming Pool (D6):
Column E → D = 1 step left = West; Row 5 → 6 = 1 step up = North.
Direction: **West and North**

Leg 3 Swimming Pool (D6) → School (B4):
Column D → B = 2 steps left = West; Row 6 → 4 = 2 steps down = South.
Direction: **West and South**

Estimated map distances (grid squares = 1.4 cm each):

Leg 1: horizontal $3 \times 1.4 = 4.2$ cm, vertical $1 \times 1.4 = 1.4$ cm $\Rightarrow$ approx. 4.4 cm
Leg 2: horizontal $1 \times 1.4 = 1.4$ cm, vertical $1 \times 1.4 = 1.4$ cm $\Rightarrow$ approx. 2.0 cm
Leg 3: horizontal $2 \times 1.4 = 2.8$ cm, vertical $2 \times 1.4 = 2.8$ cm $\Rightarrow$ approx. 4.0 cm

Total map distance $\approx 4.4 + 2.0 + 4.0 = \mathbf{10.4}$ cm

Real-world total distance:

$$10.4\,\text{cm} \times 100 = \approx 1{,}040 \text{ m} \approx 1 \text{ km}$$

Teaching note: Students measure the distances on their printed map and will get slightly different values depending on their ruler. Accept total real distance in the range 950 m – 1,100 m. The key skill is applying the scale correctly, not matching a specific value.

Chapter 5 Quick-Reference Answer Summary

Shape Properties Reference Table (completed):
Square: 4 equal sides, 4 right angles, 4 lines of symmetry.
Rectangle: 4 sides, 4 right angles, 2 lines of symmetry.
Equilateral △: 3 sides, 3 equal, 3 lines of symmetry.
Right-angled △: 0 equal sides, 1 right angle, 0 lines of symmetry.
Isosceles △: 2 equal sides, 1 line of symmetry.
Circle: infinite lines of symmetry.

Practice Exercise:

- Q1: (a) Rectangle (b) Isosceles triangle (c) Rhombus (d) Circle
- Q2: (a) 4 (b) 2 (c) 3 (d) infinite
- Q3: (a) 30,000 cm^2 (b) 8 m^2 (c) 5,000,000 m^2 (d) 2.5 cm^2 (e) 4,000 cm^2 (f) 2.5 km^2
- Q4: (a) Train Station (b) Shopping Centre (c) School (d) Swimming Pool
- Q5: (a) F4 (b) D3 (c) C4 (d) D6
- Q6: (a) East and North (b) West (c) East and North (d) West and South
- Q7: (a) 150 m (b) 320 m (c) 80 m
- Q8: 4.5 cm

Application Exercise:

- Q1: Leg 1 East; Leg 2 East and South
- Q2: (a) 200 m (b) 2.5 min
- Q3: 140 m
- Q4: (a) West and South (b) 450 m
- Q5: Train Station and Hospital ≈ 750 m (accept any well-justified answer)
- Q6: (a) 140 m (b) 19,600 m^2
- Q7: Between A3 and B3 (halfway)
- Q8: (a) 4.5 m × 3 m (b) 13.5 m^2
- Q9: 0.0071 km^2
- Q10: (a) 25 m (b) 6,250,000 cm^2
- Q11: (a) 3.5 m × 2.5 m (b) 8.75 m^2
- Q12: 60,000 cm^2

Chapter Review:

- Q1: 4 sides, 4 right angles, 2 lines of symmetry
- Q2: Isosceles triangle (accept: kite)
- Q3: 62,000 cm^2
- Q4: 35 m^2
- Q5: 1,500,000 m^2
- Q6: Hospital
- Q7: East and North (Northeast)
- Q8: 350 m
- Q9: 2 km
- Q10: (a) 540 m^2 (b) 5,400,000 cm^2
- Q11: (a) 6 m × 4 m (b) 24 m^2 (c) $1,248
- Q12: Legs: East+North, West+North, West+South; total ≈ 1,040 m (accept 950–1,100 m)

6. Time and Angles

Chapter 6 Information

Week taught:	Week 6 (Term 1)
Topic:	Topic 1 — Personal Identity and Community
Skills area:	Measurement & Space
Lesson reference:	Lessons 21–24

Teaching Guide

Key concepts: 12-hour and 24-hour time; converting between time formats; calculating elapsed time; reading timetables; angle types; angle sums (straight line = 180°, revolution = 360°, triangle = 180°).

Suggested delivery sequence:

1. Lesson 21: 12-hr and 24-hr time; converting between formats.
2. Lesson 22: Elapsed time; reading timetables and schedules.
3. Lesson 23: Types of angles; measuring and drawing angles; angle relationships.
4. Lesson 24: Worked Examples; Practice and Application Exercises.

Common Student Misconceptions

Common Misconceptions — Chapter 6

1. **12-hr to 24-hr — adding 12 to AM times:** Students write 9:00 am as 21:00. *Strategy:* Only add 12 for PM times (except 12:00 pm); 12:00 am becomes 00:00.
2. **Elapsed time — not carrying when subtracting minutes:** Students compute 3:10 − 1:45 incorrectly. *Strategy:* Convert to total minutes first, subtract, then convert back.
3. **Timetable reading — confusing arrival and departure:** Students use the wrong column. *Strategy:* Highlight start and finish columns before calculating.
4. **Angle types — confusing obtuse and reflex:** Students classify 200° as obtuse. *Strategy:* Reflex is $> 180°$; obtuse is between 90° and 180°.
5. **Angles on a straight line — not recognising 180°:** Students do not identify the relationship. *Strategy:* Colour the straight line and write the equation before solving.

Worked Solutions

Chapter 6 Overview for Teachers

Chapter:	6 — Time and Angles
Week:	Week 6, Term 1 (Lessons 21–24)
Topic:	Topic 1 — Personal Identity and Community
Skills area:	Measurement & Space
Focus:	12-/24-hour time conversions; elapsed time (jump method); timetable and calendar reading; angle classification

Reference data used throughout:

Route 72 timetable (Springvale → City Central):

Service	Springvale	Park Rd	Shopping Ctr	Riverside	City Central
706	06:15	06:22	06:30	06:38	06:50
720	07:00	07:08	07:16	07:25	07:38
735	08:30	08:38	08:47	08:56	09:10
750	10:15	10:22	10:30	10:39	10:52
760	13:40	13:47	13:55	14:04	14:17
775	16:20	16:28	16:37	16:46	17:00
790	18:05	18:12	18:20	18:29	18:42

Route 72R timetable (City Central → Springvale):

Service	City Central	Riverside	Shopping Ctr	Park Rd	Springvale
801	07:20	07:33	07:41	07:50	07:57
815	09:00	09:13	09:21	09:30	09:37
830	12:30	12:43	12:51	13:00	13:07
845	15:00	15:13	15:21	15:30	15:37
860	17:30	17:43	17:51	18:00	18:07
875	19:15	19:28	19:36	19:45	19:52

June 2026 calendar: Mon 1, Tue 2, Wed 3, Thu 4, Fri 5, Sat 6, Sun 7; Mon 8…Sun 14; Mon 15…Sun 21; Mon 22…Sun 28; Mon 29, Tue 30.

Practice Exercise

Part A — Time Conversions

Question 1 — Convert to 24-hour format

Rule: am times → keep (write with 4 digits). pm times → add 12 to the hour. Midnight (12:00 am) → 00:00. Noon (12:00 pm) → 12:00.

(a) 7:30 am → am, keep as is → **07:30**

(b) 1:15 pm → $1 + 12 = 13$ → **13:15**

(c) 11:50 pm → $11 + 12 = 23$ → **23:50**

(d) Midnight (12:00 am) → **00:00**

(e) 12:30 pm (noon) → noon starts at 12:00 → **12:30**

Common errors: (d) Students often write 12:00 instead of 00:00 for midnight. Stress: midnight = start of the day = 00:00 in 24-hour time. (e) Students sometimes add 12 to noon: $12 + 12 = 24$*, which is wrong. Noon is already 12:xx in 24-hour time.*

Question 2 — Convert to 12-hour format

Rule: 00:xx → 12:xx am. 01:00–11:59 → am (as is). 12:xx → 12:xx pm. 13:00–23:59 → subtract 12 and add pm.

(a) 06:45 → before noon, am → **6:45 am**

(b) 14:30 → $14 - 12 = 2$ → **2:30 pm**

(c) 21:05 → $21 - 12 = 9$ → **9:05 pm**

(d) 00:20 → 00 = midnight hour → **12:20 am**

(e) 12:00 → noon → **12:00 pm**

Common error: (d) Students write "0:20 am" instead of "12:20 am". Explain: 00:xx in 24-hour time is always 12:xx am in 12-hour time.

Part B — Elapsed Time

Question 3 — Elapsed time using the jump method

(a) Start 09:15 → End 11:45

09:15 → 10:00: $60 - 15 = 45$ min
10:00 → 11:00: 1 hour
11:00 → 11:45: 45 min

Total: 45 min + 1 hr + 45 min = 1 hr 90 min = **2 hours 30 minutes**

(b) Start 07:50 → End 10:20

07:50 → 08:00: 10 min
08:00 → 10:00: 2 hours
10:00 → 10:20: 20 min

Total: 10 min + 2 hr + 20 min = **2 hours 30 minutes**

(c) Start 13:35 → End 16:10

13:35 → 14:00: 25 min
14:00 → 16:00: 2 hours
16:00 → 16:10: 10 min

Total: 25 min + 2 hr + 10 min = **2 hours 35 minutes**

(d) Start 22:40 → End 02:15 (next day)

22:40 → 23:00: 20 min
23:00 → 00:00 (midnight): 1 hour
00:00 → 02:00: 2 hours
02:00 → 02:15: 15 min

Total: 20 min + 1 hr + 2 hr + 15 min = **3 hours 35 minutes**

Teaching note (d): Students often forget to count through midnight. Emphasise: from 23:00 to 00:00 is exactly 1 hour. After midnight the hours start again from 00:00.

Part C — Timetable Reading

Question 4 — Service 760 (Springvale → City Central)

From the timetable, Service 760:
Departs Springvale: **13:40** Arrives City Central: **14:17**

Question 5 — Service 790 duration

Service 790: Springvale 18:05 → City Central 18:42

18:05 → 18:30: 25 min 18:30 → 18:42: 12 min

Total: $25 + 12 =$ **37 minutes**

Question 6 — First service from City Central to Springvale after 3:00 pm

3:00 pm = 15:00. Checking Route 72R departures from City Central:

Service 845 departs 15:00 (exactly 3:00 pm ✓)

Service 860 departs 17:30

Service 845 departs at exactly 15:00 (3:00 pm). If a student arrives at the stop at 3:00 pm they can board this service.

Service 845, departing City Central at **15:00 (3:00 pm)**

Note: "After 3:00 pm" is ambiguous — Service 845 departs at exactly 15:00. In a practical transport context, if you are at the stop at 3:00 pm you can board. Accept Service 860 (17:30) if a student interprets "after" as strictly later than 15:00.

Question 7 — Service 830 duration (City Central → Springvale)

Service 830: City Central 12:30 → Springvale 13:07

12:30 → 13:00: 30 min 13:00 → 13:07: 7 min

Total: $30 + 7 =$ **37 minutes**

Part D — Date Calculations

Question 8 — Days from June 3 to June 19

$19 - 3 =$ **16 days**

(Count the days that pass — do not include the start date.)

Question 9 — Project due June 25; today June 10

Days remaining: $25 - 10 =$ **15 days**

Question 10 — Camp starts June 12; runs 5 days

Day 1 = June 12 (first day of camp)
Day 2 = June 13, Day 3 = June 14, Day 4 = June 15, Day 5 = June 16

Camp ends: **June 16**

Teaching note: "Runs for 5 days starting June 12" includes June 12 as day 1. The last day is June 12 + 4 = June 16. Students who add 5 directly get June 17 — a common error. Encourage students to list the days explicitly if unsure.

Question 11 — Day of the week for June 17, 2026

From the June 2026 calendar: Row Mon 15, Tue 16, **Wed 17**, Thu 18, Fri 19...

June 17 is a **Wednesday**

Part E — Angle Classification

Question 12 — Classify each angle

(a) 75°	$0° < 75° < 90°$	**Acute**
(b) 180°	Exactly 180°	**Straight angle**
(c) 210°	$180° < 210° < 360°$	**Reflex**
(d) 90°	Exactly 90°	**Right angle**
(e) 115°	$90° < 115° < 180°$	**Obtuse**
(f) 360°	Complete turn	**Full rotation**

Question 13 — Draw angle types

This is a drawing exercise. Check that students' sketches match the descriptions:

Acute:	Two lines meeting with a small opening (less than a right angle)
Right:	Two lines meeting at 90° with a small square symbol
Obtuse:	Two lines meeting with an opening between 90° and 180°
Straight:	A single straight line (two lines forming a flat angle)
Reflex:	Two lines with the measured angle going the "long way round" ($> 180°$)
Full rotation:	A complete circle (360°) drawn around a centre point

Application Exercise

Question 1 — Mia's school day: 8:45 am to 3:10 pm

Convert to 24-hour: 8:45 am = 08:45; 3:10 pm = 15:10

Jump method:
08:45 → 09:00: 15 min
09:00 → 15:00: 6 hours
15:00 → 15:10: 10 min

Total: $15 + 6 \text{ hr} + 10 =$ **6 hours 25 minutes**

Question 2 — Jake works 09:30 to 17:15; earns $24.50/hr

(a) Duration:
09:30 → 10:00: 30 min
10:00 → 17:00: 7 hours
17:00 → 17:15: 15 min

Total: **7 hours 45 minutes**

(b) Pay calculation:
7 hours 45 minutes = 7.75 hours (already on a 15-minute boundary, no rounding needed).

$$7.75 \times \$24.50 = \$189.875 \approx \$189.88$$

Working: $7 \times \$24.50 = \$171.50$; $0.75 \times \$24.50 = \$18.375$; total $= \$189.875$.

Teaching note: 7 hr 45 min = $7\frac{3}{4}$ hr = 7.75 hr. Students who convert 45 min to a decimal often write 7.45 (treating minutes as hundredths of an hour) — a very common error. Stress: 45 min ÷ 60 min = 0.75 hr.

Question 3 — Priya needs to reach Shopping Centre before 10:45

(a) Shopping Centre arrival times (Route 72):

Service 706	arrives 06:30 ✓
Service 720	arrives 07:16 ✓
Service 735	arrives 08:47 ✓
Service 750	arrives 10:30 ✓
Service 760	arrives 13:55 × (too late)

The *latest* service arriving before 10:45 is **Service 750**, departing Springvale at 10:15.

Answer: **Service 750** (departs Springvale 10:15, arrives Shopping Centre 10:30)

(b) Minutes early: 10:45 − 10:30 = **15 minutes early**

Question 4 — Sam: Service 775 to City Central, returns on Service 860

(a) Service 775 arrives City Central: **17:00**

(b) Service 860 departs City Central: **17:30**

(c) Time for shopping: 17:30 − 17:00 = **30 minutes**

Teaching note: This question highlights the importance of checking BOTH the outbound and return timetables. Students sometimes read from the wrong timetable or use the wrong row. Encourage students to underline the relevant row in each timetable.

Question 5 — Soccer match starts 14:30, lasts 1 hr 50 min

Step 1: Add the duration to the start time.

14:30 + 1 hr = 15:30
15:30 + 50 min = 16:20

24-hour finish time: **16:20**
12-hour: 16 − 12 = 4 → **4:20 pm**

Question 6 — Zara: 3-week challenge starting June 2

(a) 3 weeks = 21 days. Day 1 = June 2, Day 21 = June 2 +20 = June 22.
Challenge ends: **June 22**

(b) From the calendar: Mon 22 → June 22 is a **Monday**

Teaching note: "Starts on June 2 and runs for 3 weeks" means June 2 is day 1 and June 22 is day 21. Students who add 21 directly get June 23 — watch for this. To check: count 3 weeks (Mon 2 → Mon 9 → Mon 16 → Mon 23... that is 3 full weeks after June 2, ending June 23). The answer depends on interpretation. Accept June 22 (if June 2 = day 1) or June 23 (if 3 weeks after June 2). Mark both as correct; consistency of method is key.

Question 7 — School holidays June 27 to July 14 (June has 30 days)

Step 1: Count days remaining in June (from June 27 inclusive, to June 30).
$30 - 27 + 1 = 4$ days in June (27, 28, 29, 30).
Step 2: Count days in July up to and including July 13 (students return July 14, so the last holiday is July 13).
13 days in July.
Step 3: Total $= 4 + 13 =$ **17 days**

Teaching note: Whether to include the last day depends on whether July 14 is a "return" day (not a holiday). Here, "students return on July 14" means July 14 is not a holiday day. The holiday period is June 27–July 13 = 17 days. Alternatively, students may count June 27 to July 14 directly: days left in June from 27 = 3 (28, 29, 30), + 14 days in July = 17 days. Both approaches give 17.

Question 8 — Library book borrowed June 5; due in 21 days

June 5 + 21 days:

Days left in June from June 5: $30 - 5 = 25$ days remain (so 21 days later is still in June).

$5 + 21 = 26$ Return by: **June 26**

Question 9 — Clock angles

Key fact: Clock face $= 360° \div 12 = 30°$ per section (one number to the next).

(a) At 6:00, the minute hand points to 12. Angle from 12 to 6 = 6 sections:

$$6 \times 30° = 180° \quad \text{– a } \textbf{straight angle}$$

(b) At 3:00, minute hand at 12, hour hand at 3. Angle from 12 to 3 = 3 sections (clockwise):

$$3 \times 30° = 90° \quad \text{– a } \textbf{right angle}$$

Question 10 — Carpenter's 45° cuts

(a) 45°: since $0° < 45° < 90° \Rightarrow$ **Acute angle**

(b) Two 45° cuts joined at the corner:

$$45° + 45° = 90° \quad \text{(a right angle)}$$

This is why picture frame corners make a perfect right-angle corner.

Question 11 — Bus route: two right turns of 90°

(a) Total angle after two right turns:

$$90° + 90° = 180°$$

(b) After a total turn of 180°, the bus is facing in the **opposite direction** to where it started.

Explanation: A 180° turn is a U-turn — a straight angle. The bus has effectively reversed its heading.

Question 12 — Jayden: Service 720 to Riverside, then 830R to Park Road

(a) Service 720 arrives at Riverside: **07:25**

(b) Wait time: 07:25 to 12:43

Jump method:
07:25 → 08:00: 35 min
08:00 → 12:00: 4 hours
12:00 → 12:43: 43 min

Total: 35 min + 4 hr + 43 min = 4 hr 78 min = **5 hours 18 minutes**

Teaching note: $35 + 43 = 78$ *minutes* $= 1$ *hr* 18 *min. Add that to 4 hr → 5 hr 18 min. Students who don't convert 78 min to hours and minutes will write "4 hr 78 min" — a common error. Stress that minutes must be less than 60.*

Chapter Review

Questions 1–4 — Time conversions

Q1. 4:20 pm → 24-hour: $4 + 12 = 16$ → **16:20**

Q2. 12:15 am (midnight) → 24-hour: → **00:15**

Q3. 17:45 → 12-hour: $17 - 12 = 5$ → **5:45 pm**

Q4. 03:30 → 12-hour: before noon, am → **3:30 am**

Question 5 — Elapsed time 10:40 to 13:15

10:40 → 11:00: 20 min
11:00 → 13:00: 2 hours
13:00 → 13:15: 15 min

Total: 20 min + 2 hr + 15 min = **2 hours 35 minutes**

Question 6 — Movie starts 7:15 pm; runs 2 hr 35 min

Step 1: Convert to 24-hour: 7:15 pm = 19:15.

Step 2: Add duration.
19:15 + 2 hr = 21:15
21:15 + 35 min = 21:50

24-hour finish: **21:50**
12-hour: $21 - 12 = 9$ → **9:50 pm**

Question 7 — Route 72 service arriving City Central closest to 7:30 am

Check Route 72 City Central arrivals:

Service 706	arrives City Central 06:50	$\lvert 07{:}30 - 06{:}50 \rvert = 40$ min away
Service 720	arrives City Central 07:38	$\lvert 07{:}38 - 07{:}30 \rvert = 8$ min away
Service 735	arrives City Central 09:10	far too late

Service 720 arrives at 07:38 — only 8 minutes after 07:30. Closest to 07:30.

Question 8 — Service 845 duration (City Central → Springvale)

Service 845: departs City Central 15:00, arrives Springvale 15:37.

15:00 → 15:37: = **37 minutes**

Question 9 — How many Mondays in June 2026?

From calendar: Mondays are June 1, 8, 15, 22, 29.

Total: **5 Mondays**

Question 10 — Assignment given June 4; due June 22

$22 - 4 =$ **18 days**

Question 11 — Classify angles with real-world examples

(a) 35°: **Acute** Example: *a wheelchair ramp, an escalator incline, or the angle of a book propped open slightly*

(b) 90°: **Right angle** Example: *the corner of a room, a square table top, a door frame*

(c) 200°: **Reflex** Example: *the larger angle swept by a door opened past horizontal, or the angle behind the hands of a clock at 1:00*

Accept any sensible real-world example. Mark the classification strictly; the example is for open-ended extension.

Question 12 — Multi-step: Lena on Service 720 from Springvale, returns on Service 830

(a) Service 720 (07:00 from Springvale) arrives City Central: **07:38**

(b) Service 830 departs City Central: **12:30**

(c) Time at City Central: 07:38 to 12:30

07:38 → 08:00: 22 min
08:00 → 12:00: 4 hours
12:00 → 12:30: 30 min

Total: $22 + 4$ hr $+ 30 = 4$ hr 52 min → **4 hours 52 minutes**

Question 13 — Multi-step: Excursion 08:45 to 15:20; 45-min activity booklet

(a) Total excursion duration: 08:45 to 15:20

08:45 → 09:00: 15 min
09:00 → 15:00: 6 hours
15:00 → 15:20: 20 min

Total: $15 + 6 \text{ hr} + 20 =$ **6 hours 35 minutes**

(b) Time remaining after booklet (45 min):

$$6 \text{ hr } 35 \text{ min} - 45 \text{ min} = 6 \text{ hr } (35 - 45) \text{ min}$$

Borrow 1 hour: $5 \text{ hr } (60 + 35) \text{ min} - 45 \text{ min} = 5 \text{ hr } 95 \text{ min} - 45 \text{ min} =$ **5 hours 50 minutes**

Chapter 6 Quick-Reference Answer Summary

Practice Exercise:

- Q1: (a) 07:30 (b) 13:15 (c) 23:50 (d) 00:00 (e) 12:30
- Q2: (a) 6:45 am (b) 2:30 pm (c) 9:05 pm (d) 12:20 am (e) 12:00 pm
- Q3: (a) 2 hr 30 min (b) 2 hr 30 min (c) 2 hr 35 min (d) 3 hr 35 min
- Q4: Departs 13:40; arrives 14:17
- Q5: 37 minutes
- Q6: Service 845 at 15:00 (accept Service 860 if "strictly after" 15:00)
- Q7: 37 minutes
- Q8: 16 days
- Q9: 15 days
- Q10: June 16
- Q11: Wednesday
- Q12: (a) Acute (b) Straight (c) Reflex (d) Right (e) Obtuse (f) Full rotation
- Q13: Drawing exercise — see angle descriptions above

Application Exercise:

- Q1: 6 hr 25 min
- Q2: (a) 7 hr 45 min (b) $189.88
- Q3: (a) Service 750 (departs 10:15) (b) 15 minutes early
- Q4: (a) 17:00 (b) 17:30 (c) 30 minutes
- Q5: 16:20 (24-hr); 4:20 pm (12-hr)
- Q6: (a) June 22 (b) Monday
- Q7: 17 days
- Q8: June 26
- Q9: (a) 180° (straight) (b) 90° (right)
- Q10: (a) Acute (b) 90° (right angle)
- Q11: (a) 180° (b) Yes, facing opposite direction (180° = U-turn)
- Q12: (a) 07:25 (b) 5 hr 18 min

Chapter Review:

- Q1: 16:20 Q2: 00:15 Q3: 5:45 pm Q4: 3:30 am
- Q5: 2 hr 35 min
- Q6: 9:50 pm / 21:50
- Q7: Service 720 (arrives 07:38)
- Q8: 37 minutes
- Q9: 5 Mondays
- Q10: 18 days
- Q11: (a) Acute (b) Right (c) Reflex
- Q12: (a) 07:38 (b) 12:30 (c) 4 hr 52 min
- Q13: (a) 6 hr 35 min (b) 5 hr 50 min

7. Data and Graphs

Chapter 7 Information

Week taught:	Week 7 (Term 1)
Topic:	Topic 1 — Personal Identity and Community
Skills area:	Statistics & Probability
Lesson reference:	Lessons 25–28

Teaching Guide

Key concepts: Types of data (categorical vs. numerical); collecting and organising data; constructing and interpreting bar graphs and line graphs; mean, median and mode; identifying misleading graphs.

Suggested delivery sequence:

1. Lesson 25: Types of data; tally charts and frequency tables.
2. Lesson 26: Constructing and reading bar graphs; misleading graphs.
3. Lesson 27: Line graphs; mean, median and mode.
4. Lesson 28: Worked Examples; Practice and Application Exercises.

Common Student Misconceptions

Common Misconceptions — Chapter 7

1. **Mean — dividing by the highest value:** Students add all values and divide by the maximum. *Strategy:* Mean = sum ÷ *number of values*.
2. **Median — not ordering data first:** Students find the middle value of an unordered list. *Strategy:* Always order from smallest to largest first.
3. **Mode — missing multiple modes:** Students report only one mode when data is bimodal. *Strategy:* Mode is the most frequent value; there can be more than one.
4. **Bar graphs — gaps for continuous data:** Students draw gaps in charts for continuous data. *Strategy:* Categorical data has gaps; continuous (histogram-style) data does not.
5. **Line graphs — connecting non-continuous data:** Students connect points where intermediate values are not meaningful. *Strategy:* Discuss whether values between data points make sense.

Worked Solutions

Chapter 7 Overview for Teachers

Chapter:	7 — Data and Graphs
Week:	Week 7, Term 1 (Lessons 25–28)
Topic:	Topic 1 — Statistics & Probability
Skills area:	Statistics & Probability
Focus:	Tally charts; frequency tables; reading/drawing bar and line graphs; mean; trend language

Key data sets used in questions:

- Practice Ex Q1 text messages: 3,7,2,5,8,3,7,10,2,5,7,3,8,5,7,2,10,3,5,7 (n=20)
- Practice Ex bar graph: Soccer 48, Netball 35, Basketball 52, Swimming 28, AFL 42, Tennis 22 (total 227)
- Practice Ex line graph: Sports store monthly sales (Jan $42k – Dec $48k)
- Application Ex Q1 café drinks: Coffee 11, Tea 5, Juice 5, Smoothie 3 (n=24)
- Application Ex Q6 soccer clubs: Riverside FC 85/90/102/115/128; Northside FC 130/118/110/105/98
- Chapter Review Q2 pets: 0×6, 1×7, 2×4, 3×2, 4×1 (n=20)
- Chapter Review Q9 football club: Juniors 64/78/85/96/110; Seniors 42/45/48/50/55
- Chapter Review Q10 Brisbane temps: 32,35,30,28,33,36,34,29,31,32 (n=10)

Note on drawing exercises: Several questions ask students to draw graphs on blank grids. These cannot be reproduced exactly in a solutions book. This file provides: the complete data to plot, the correct scale and labels, and a description of what a correctly drawn graph looks like. Use these to mark student work.

Practice Exercise

Part A — Tally Charts and Frequency Tables

Question 1 — Frequency table for text messages

Data: 3, 7, 2, 5, 8, 3, 7, 10, 2, 5, 7, 3, 8, 5, 7, 2, 10, 3, 5, 7 (n = 20)

Count each value:

Messages	Tally	Frequency
2	|||	**3**
3	||||	**4**
5	||||	**4**
7	~~||||~~ |||	**5**
8	||	**2**
10	||	**2**
Total		**20**

Check: $3 + 4 + 4 + 5 + 2 + 2 = 20$ ✓

Question 2 — Most common number of messages

Highest frequency is **5**, for value **7**.

The most common number of text messages sent was 7.

Question 3 — Fraction who sent 7 or more messages

7 or more: frequency 7 (5 students) + frequency 8 (2 students) + frequency 10 (2 students) = 9 students.

$$\text{Fraction} = \frac{9}{20}$$

9 and 20 share no common factor ⇒ already in simplest form: $\frac{9}{20}$

Part B — Reading Bar Graphs

Bar graph data — Sports Participation at Springvale High

Soccer: 48 Netball: 35 Basketball: 52 Swimming: 28 AFL: 42 Tennis: 22
Total: 227

Question 4 — Sport with most participants

Highest bar = Basketball (52 students)
Basketball has the most participants.

Question 5 — Basketball minus Tennis

$52 - 22 = 30$ more students play Basketball than Tennis.

Question 6 — Total students in the graph

$48 + 35 + 52 + 28 + 42 + 22 = 227$ students

Question 7 — Percentage playing Soccer

$$\frac{48}{227} \times 100 = 21.14\ldots\% \approx 21\%$$

Part C — Reading Line Graphs

Line graph data — Springvale Sports Store Monthly Sales ($000s)

Jan: 42, Feb: 38, Mar: 30, Apr: 22, May: 18, Jun: 15, Jul: 14, Aug: 16, Sep: 20, Oct: 26, Nov: 35, Dec: 48

Question 8 — Lowest sales month

Lowest point on the graph: **July** ($14,000)

Question 9 — November sales value

The axis is in \$000s. November reads as 35 on the axis.
November sales = 35 × \$1,000 = \$35,000

Common error: Students read 35 and write \$35, forgetting the axis is in thousands. Remind them to always check axis labels and units.

Question 10 — Trend from January to July

Sales decreased steadily from \$42,000 in January to \$14,000 in July, showing a consistent downward trend.

Question 11 — Why higher sales in December than July

Accept any well-reasoned seasonal explanation, for example:

December is the Christmas/holiday season, when people buy more sporting goods as gifts or for summer activities.

Also accept: "Summer sport seasons begin in December" or "People are on holiday and more active."

Part D — Drawing Graphs

Question 12 — Line graph for netball goals (drawing exercise)

Data to plot:

Game	1	2	3	4	5	6	7	8
Goals	32	28	35	30	38	42	36	44

A correct graph must include:

- Title (e.g., "Netball Team Goals per Game")
- Horizontal axis labelled "Game" with values 1–8
- Vertical axis labelled "Goals" with a suitable scale (e.g., 0–50 in steps of 5 or 10)
- Points plotted at (1,32), (2,28), (3,35), (4,30), (5,38), (6,42), (7,36), (8,44)
- Points joined with straight lines

Question 13 — Describe the trend

The team's goal scoring shows an overall upward trend across the season, despite some fluctuation from game to game. The team generally improved, scoring 32 goals in Game 1 and 44 in Game 8.

Accept any response that correctly identifies: (1) the general increasing trend, and (2) acknowledges the variation/fluctuation between games.

Part E — Calculating the Mean

Question 14 — Mean km walked

Data: 8, 12, 7, 15, 8 $\quad (n = 5)$

$$\text{Sum} = 8 + 12 + 7 + 15 + 8 = 50$$

$$\text{Mean} = \frac{50}{5} = 10 \text{ km}$$

Question 15 — Quiz scores: 14, 18, 12, 16, 15, 17

(a) Sum $= 14 + 18 + 12 + 16 + 15 + 17 = 92; \quad n = 6$

$$\text{Mean} = \frac{92}{6} = 15.3\overline{3} \approx 15.3 \text{ (out of 20)}$$

(b) The seventh score is 19. Since $19 > 15.3$, adding this score will **raise the mean**.

$$\text{New mean} = \frac{92 + 19}{7} = \frac{111}{7} = 15.86 \approx 15.9$$

Explanation: Any new score above the current mean pulls the mean upward.

Application Exercise

Question 1 — Café drink orders (n

Count each drink from the list:

Row 1 (10): Coffee, Tea, Coffee, Juice, Coffee, Smoothie, Tea, Coffee, Juice, Coffee
Row 2 (10): Coffee, Tea, Smoothie, Coffee, Juice, Tea, Coffee, Coffee, Smoothie, Juice
Row 3 (4): Tea, Coffee, Juice, Coffee

Drink	Tally	Frequency
Coffee	𝍸 𝍸 |	**11**
Tea	𝍸	**5**
Juice	𝍸	**5**
Smoothie	|||	**3**
Total		**24**

Check: $11 + 5 + 5 + 3 = 24$ ✓

Teaching note: Encourage students to work through the list systematically, crossing off each item as they tally it. A common error is miscounting — always verify the total equals the number of items listed.

Question 2 — Coffee fraction and juice percentage

(a) Fraction of drinks that were coffee:

$$\frac{11}{24}$$

11 and 24 share no common factor (11 is prime) $\Rightarrow$ already simplest: $\frac{11}{24}$

(b) Juice percentage:

$$\frac{5}{24} \times 100 = 20.83\ldots\% \approx 20.8\% \quad \text{(accept 21\%)}$$

Question 3 — Bar graph for café drinks (drawing exercise)

Data to plot:
Coffee: 11, Tea: 5, Juice: 5, Smoothie: 3

A correct graph must include:

- Title (e.g., "Springvale Café — Morning Drink Orders")
- Horizontal axis labelled "Drink type" with four categories
- Vertical axis labelled "Frequency (number of orders)" with scale 0–12 (steps of 2)
- Bars of equal width with gaps between them
- Bars at correct heights: Coffee 11, Tea 5, Juice 5, Smoothie 3

Question 4 — Queensland tourism visitor numbers

Data: Jan:420, Feb:390, Mar:350, Apr:310, May:280, Jun:260, Jul:265, Aug:290, Sep:330, Oct:370, Nov:400, Dec:445 (thousands)

(a) Drawing exercise. Correct graph:

- Title (e.g., "Queensland Monthly Tourism Visitors 2025")
- Horizontal axis: Months Jan–Dec
- Vertical axis: "Visitors (000s)"; scale 0–500 in steps of 50 or 100
- Points plotted and joined; axis labelled with units

(b) Lowest visitor month: **June** (260,000 visitors)

(c) Trend June to December:
Visitor numbers increase steadily from June (260,000) to December (445,000), showing a consistent upward trend over the second half of the year.

(d) Marketing campaigns in May/June:
Yes, the data supports the suggestion. May and June show the lowest visitor numbers of the year (280,000 and 260,000), so targeted marketing during these months could help attract more visitors during the quietest period. The data identifies a clear seasonal dip that marketing could aim to address.

Question 5 — Mango crates: Mon 18, Tue 24, Wed 21, Thu 27, Fri 22, Sat 30

(a) Mean crates per day:

$$\text{Sum} = 18 + 24 + 21 + 27 + 22 + 30 = 142$$

$$\text{Mean} = \frac{142}{6} = 23.6\overline{6} \approx 23.7 \text{ crates per day}$$

(b) Target: at least 24 crates/day on average.
Actual mean: ≈ 23.7 crates/day.
$23.7 < 24 \Rightarrow$ **No, this week's mean does not meet the target of 24 crates per day.**
The farmer is short by approximately 0.3 crates per day on average.

Question 6 — Soccer club membership comparison

Data: Riverside FC: 85, 90, 102, 115, 128 (2021–2025)
Northside FC: 130, 118, 110, 105, 98 (2021–2025)

(a) Drawing exercise. Correct graph:

- Title (e.g., "Soccer Club Membership 2021–2025")
- Horizontal axis: Years 2021–2025
- Vertical axis: "Members"; scale 0–150 (steps of 10 or 20)
- Two clearly labelled lines (one for each club)
- A legend identifying Riverside FC and Northside FC

(b) Mean membership:

$$\text{Riverside FC: } \frac{85+90+102+115+128}{5} = \frac{520}{5} = 104$$

$$\text{Northside FC: } \frac{130+118+110+105+98}{5} = \frac{561}{5} = 112.2$$

(c) Difference in trends:
Riverside FC shows a consistent upward trend, growing from 85 to 128 members. Northside FC shows a consistent downward trend, declining from 130 to 98 members. By 2025, Riverside FC has surpassed Northside FC in membership.

(d) Prediction for 2026:
Riverside FC, which is continuing to grow steadily (gaining about 10–13 members per year). Northside FC is declining and has already fallen below Riverside's 2025 total.

Question 7 — Canteen sandwiches: Mon 42, Tue 48, Wed 55, Thu 50, Fri 68

(a) Mean:

$$\text{Sum} = 42 + 48 + 55 + 50 + 68 = 263$$

$$\text{Mean} = \frac{263}{5} = 52.6 \text{ sandwiches per day}$$

(b) Prepare 10% more than the mean:

$$52.6 \times 1.10 = 57.86 \approx 58 \text{ sandwiches}$$

(c) With 50 fewer students on Thursday due to an excursion:
Expect Thursday sales to be below the mean. Fewer students at school means fewer potential customers at the canteen, so sandwich sales would likely be lower than the typical Thursday figure (50).

Chapter Review

Question 1 — Tally mark counts

(a) One group of five + three additional marks $= 5 + 3 = 8$

(b) Two groups of five + one additional mark $= 5 + 5 + 1 = 11$

Question 2 — Frequency table for pets per household

Data: 0, 1, 2, 0, 3, 1, 0, 2, 1, 1, 0, 2, 3, 1, 0, 4, 1, 2, 0, 1 $\quad (n = 20)$

Pets	Tally	Frequency
0	𝍸 |	**6**
1	𝍸 ||	**7**
2	||||	**4**
3	||	**2**
4	|	**1**
Total		**20**

Check: $6 + 7 + 4 + 2 + 1 = 20$ ✓

Question 3 — Analysis of pets data

(a) Most common number of pets: **1 pet** (frequency 7 — the highest in the table)

(b) Households with 2 or more pets: $4 + 2 + 1 = 7$ out of 20.

$$\frac{7}{20}$$

7 and 20 share no common factor $\Rightarrow$ simplest form: $\frac{7}{20}$

Question 4 — New total if 15 more students join Tennis

Original total (from Practice Ex Q6): 227 students.
15 new students join Tennis:

$$227 + 15 = 242 \text{ students}$$

Question 5 — Cairns temperatures: 28, 30, 27, 32, 31, 29, 33

(a) Mean:

$$\text{Sum} = 28 + 30 + 27 + 32 + 31 + 29 + 33 = 210$$

$$\text{Mean} = \frac{210}{7} = 30°\text{C}$$

(b) Following week mean: 31.5°C. Compare: $31.5 > 30$.
The second week was hotter on average (31.5°C vs 30°C).

Question 6 — Upward trend on a line graph

A — The website had more visitors each day as the week went on.

An upward trend means values are generally increasing over time.

Question 7 — Bar graph or line graph?

(a) School total enrolments each year 2015–2025:
Line graph. Enrolments are measured over time (year by year). A line graph shows change over time clearly.

(b) Number of students preferring each of five subjects:
Bar graph. The five subjects are separate categories (not time). A bar graph compares distinct categories.

Question 8 — Monthly electricity usage (kWh)

Data: Jan:380, Feb:360, Mar:340, Apr:300, May:280, Jun:265, Jul:270, Aug:285, Sep:295, Oct:315, Nov:345, Dec:375

(a) Drawing exercise. Correct graph:

- Title (e.g., "Monthly Electricity Usage — Queensland Home 2025")
- Horizontal axis: Months Jan–Dec
- Vertical axis: "Usage (kWh)"; scale 0–400 (steps of 50)
- 12 points plotted and joined with straight lines

(b) Lowest usage month: **June** (265 kWh)

(c) Overall trend and seasonal pattern:
Electricity usage decreases from January to June, then increases from July to December. There is a clear U-shaped seasonal pattern: usage is highest in summer (Jan–Mar) and late spring/early summer (Nov–Dec), likely due to air conditioning, and lowest in autumn/winter (Jun–Jul). This is consistent with Queensland's climate where summer is hot and air conditioning usage is highest.

(d) Mean monthly usage (sum given as 3,810 kWh):

$$\text{Mean} = \frac{3{,}810}{12} = 317.5 \text{ kWh per month}$$

(e) Cost in January:

$$380\,\text{kWh} \times \$0.28 = \$106.40$$

Multi-Step Problem 9 — Springvale Junior Football Club

Data: Juniors: 64, 78, 85, 96, 110 (2021–2025); Seniors: 42, 45, 48, 50, 55 (2021–2025)

(a) Drawing exercise — grouped (side-by-side) bar graph:

- Title: e.g., "Springvale Junior Football Club Membership 2021–2025"
- Horizontal axis: Years 2021–2025
- Vertical axis: "Members"; scale 0–120 (steps of 20)
- Two bars per year (different shading), with a legend: Juniors / Seniors

(b) Mean junior membership:

$$\text{Sum} = 64 + 78 + 85 + 96 + 110 = 433$$

$$\text{Mean} = \frac{433}{5} = 86.6 \text{ junior members per year}$$

(c) Total members in 2025:

$$110 + 55 = 165 \text{ members}$$

(d) Has junior membership grown by more than 50% since 2021?

2021 juniors: 64. 2025 juniors: 110.

$$\text{Percentage increase} = \frac{110 - 64}{64} \times 100 = \frac{46}{64} \times 100 = 71.875\% \approx 71.9\%$$

$71.9\% > 50\% \Rightarrow$ **Yes, the claim is correct.** Junior membership has grown by approximately 72%, which is more than 50%.

(e) Sample sentences for the grant application:

"Junior membership has increased by approximately 72% since 2021, from 64 to 110 members, demonstrating a strong and consistent upward trend. The mean junior membership over the past five years has been 86.6, and the club now has a total of 165 members in 2025."

Mark for: use of trend language (increasing/upward trend), at least one specific statistic (mean, total, percentage), correct values, and clear full sentences.

Multi-Step Problem 10 — Brisbane February temperatures

Data: 32, 35, 30, 28, 33, 36, 34, 29, 31, 32 $(n = 10)$

(a) Frequency table (groups 28–30, 31–33, 34–36):

Temperature Group (°C)	Tally	Frequency	
28–30	|||	**3**	(28, 30, 29)
31–33	||||	**4**	(32, 33, 31, 32)
34–36	|||	**3**	(35, 36, 34)
Total		**10**	

Check: $3 + 4 + 3 = 10$ ✓

(b) Mean temperature:

$$\text{Sum} = 32 + 35 + 30 + 28 + 33 + 36 + 34 + 29 + 31 + 32 = 320$$

$$\text{Mean} = \frac{320}{10} = 32°\text{C}$$

(c) Long-term average: 29.8°C. This February: 32°C.

$$32 - 29.8 = 2.2°\text{C}$$

This February was above the long-term average by 2.2°C.

(d) Drawing exercise — line graph (Day 1 to Day 10):

- Title: e.g., "Daily Maximum Temperature — Brisbane, February"
- Horizontal axis: Day 1–10
- Vertical axis: "Temperature (°C)"; scale 25–40 (steps of 5)
- Points: (1,32), (2,35), (3,30), (4,28), (5,33), (6,36), (7,34), (8,29), (9,31), (10,32)

(e) Pattern description:
The temperature pattern is fluctuating — it goes up and down with no clear overall upward or downward trend. For example, the temperature rises from Day 1 (32°C) to Day 2 (35°C), then drops to Day 4 (28°C), rises again to Day 6 (36°C), and drops again to Day 8 (29°C). The data alternates between warm and cooler days without a consistent direction.

Chapter 7 Quick-Reference Answer Summary

Practice Exercise:

- Q1 Frequency table: 2×3, 3×4, 5×4, 7×5, 8×2, 10×2 (total 20)
- Q2: 7 messages (most common)
- Q3: 9/20
- Q4: Basketball Q5: 30 more Q6: 227 students Q7: 21%
- Q8: July Q9: $35,000 Q10: Steadily decreasing Jan–Jul Q11: Christmas/holiday season
- Q12: Drawing (points: 32,28,35,30,38,42,36,44)
- Q13: Overall upward trend with fluctuation
- Q14: 10 km Q15: (a) 15.3 (b) raises the mean (new mean ≈ 15.9)

Application Exercise:

- Q1: Coffee 11, Tea 5, Juice 5, Smoothie 3 (total 24)
- Q2: (a) 11/24 (b) 20.8% (accept 21%)
- Q3: Drawing (bars: Coffee 11, Tea 5, Juice 5, Smoothie 3)
- Q4: (b) June (c) Increasing trend Jun–Dec (d) Yes, data supports winter campaigns
- Q5: (a) 23.7 crates/day (b) No, does not meet 24-crate target
- Q6: (b) Riverside mean 104, Northside mean 112.2 (c) Riverside increasing, Northside decreasing (d) Riverside FC
- Q7: (a) 52.6 (b) 58 sandwiches (c) Below mean (fewer students)

Chapter Review:

- Q1: (a) 8 (b) 11
- Q2: 0×6, 1×7, 2×4, 3×2, 4×1 (total 20)
- Q3: (a) 1 pet (b) 7/20
- Q4: 242 students
- Q5: (a) 30°C (b) Second week hotter (31.5°C vs 30°C)
- Q6: Answer A
- Q7: (a) Line graph (b) Bar graph
- Q8: (b) June (c) U-shape seasonal pattern, highest in summer (d) 317.5 kWh/-month (e) $106.40
- Q9: (b) Mean juniors 86.6 (c) 165 total (d) Yes — grew by ≈72% > 50%
- Q10: (a) 28–30: 3; 31–33: 4; 34–36: 3 (b) 32°C (c) 2.2°C above average (e) Fluctuating

8. Chance and Probability

Chapter 8 Information

Week taught:	Week 8 (Term 1)
Topic:	Topic 1 — Personal Identity and Community
Skills area:	Statistics & Probability
Lesson reference:	Lessons 29–32

Teaching Guide

Key concepts: Probability language scale (impossible through certain); listing outcomes and sample spaces; calculating $\Pr(E) = \text{favourable outcomes} \div \text{total outcomes}$; expressing probability as a fraction, decimal, and percentage; complementary events.

Suggested delivery sequence:

1. Lesson 29: Probability language; placing events on the 0–1 scale.
2. Lesson 30: Sample spaces; listing outcomes; calculating probability.
3. Lesson 31: Expressing probability in multiple forms; complementary events.
4. Lesson 32: Worked Examples; Practice and Application Exercises; begin AT1 preparation.

Common Student Misconceptions

Common Misconceptions — Chapter 8

1. **Probability greater than 1:** Students write $\Pr(E) = 3/2$ using unfavourable outcomes as denominator. *Strategy:* Probability is always between 0 and 1; the denominator is the *total* number of outcomes.

2. **Not listing all outcomes:** Students miss outcomes from a sample space. *Strategy:* Use a table or tree diagram to organise outcomes systematically.

3. **Complementary events — subtracting from 100:** Students compute $P(\text{not A}) = 100 - 0.3 = 99.7$. *Strategy:* $P(A) + P(\text{not A}) = 1$; ensure both are in the same form.

4. **Confusing theoretical and experimental probability:** Students expect exactly 5 heads in 10 flips. *Strategy:* Theoretical probability predicts the *long-run* pattern, not every trial.

5. **Expressing probability as a percentage without converting:** Students write $\frac{3}{4}$ as "3/4%". *Strategy:* Practise all three conversions: fraction $\rightarrow$ decimal $\rightarrow$ percentage.

Worked Solutions

Chapter 8 Overview for Teachers

Chapter:	8 — Chance and Probability
Week:	Week 8, Term 1 (Lessons 29–32)
Topic:	Topic 1 — Statistics & Probability
Focus:	Probability language; sample space; theoretical and experimental probability; fraction/decimal/percentage conversions

Key formula:

$$\Pr(E) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$$

Probability language benchmarks:

Impossible	$\Pr = 0$	Cannot happen
Unlikely	$0 < \Pr < \frac{1}{2}$	Less likely than not
Even chance	$\Pr = \frac{1}{2}$	Equally likely either way
Likely	$\frac{1}{2} < \Pr < 1$	More likely than not
Certain	$\Pr = 1$	Will definitely happen

Chapter 8 is also the AT1 Project checkpoint (end of Topic 1). This chapter completes the probability strand before the assessment task.

Practice Exercise

Part A — Probability Language

Question 1 — Probability word for each event

(a) The sun rises tomorrow.
This will definitely happen. $\Pr = 1$ **Certain**

(b) You roll a 9 on a standard six-sided die.
A standard die has faces 1–6 only; 9 is not possible. $\Pr = 0$ **Impossible**

(c) You flip a fair coin and get Tails.
$\Pr(\text{Tails}) = \frac{1}{2}$ **Even chance**

(d) A bag has 9 red and 1 blue ball; you pick red.
$\Pr(\text{red}) = \frac{9}{10} = 0.9 > 0.5$ **Likely**

(e) You randomly pick a day in July that is *both* a Monday and a Friday.
A single day cannot simultaneously be Monday and Friday. $\Pr = 0$ **Impossible**

(f) You randomly pick a vowel from the 5 vowels A, E, I, O, U.
You are selecting from {A, E, I, O, U}; every option is a vowel. $\Pr = \frac{5}{5} = 1$ **Certain**

Teaching note: Q1(e) is a useful discussion point — students sometimes say "unlikely" because Monday and Friday are both real days. The key insight is that any single randomly chosen day has ONE name; it cannot be two different days simultaneously, making this impossible. Q1(f) tests whether students read carefully: since the selection is made from *the vowels, a vowel is guaranteed.*

Question 2 — Plot events on the probability scale

Positions on the scale (0 = left end, 1 = right end):

Event	Probability	Position
(a) Sun rises	1	far right
(b) Roll a 9	0	far left
(c) Coin Tails	0.5	centre
(d) Pick red (9/10)	0.9	near right
(e) Mon & Fri	0	far left (same as b)
(f) Pick vowel	1	far right (same as a)

Mark b and e at 0; c at 0.5; d at 0.9; a and f at 1.

Part B — Sample Space

Question 3 — List the sample space

(a) Four-sided die (faces 1–4):

$$S = \{1,\ 2,\ 3,\ 4\} \quad \text{(4 outcomes)}$$

(b) One day from the school working week:

$$S = \{\text{Monday},\ \text{Tuesday},\ \text{Wednesday},\ \text{Thursday},\ \text{Friday}\} \quad \text{(5 outcomes)}$$

(c) One letter from the word MATHS:

$$S = \{\text{M},\ \text{A},\ \text{T},\ \text{H},\ \text{S}\} \quad \text{(5 outcomes)}$$

(d) One season from Summer, Autumn, Winter, Spring:

$$S = \{\text{Summer},\ \text{Autumn},\ \text{Winter},\ \text{Spring}\} \quad \text{(4 outcomes)}$$

Question 4 — Spinner with 8 equal sections numbered 1–8

$S = \{1, 2, 3, 4, 5, 6, 7, 8\}$ **8 outcomes** in the sample space.

Part C — Calculating Probability

Question 5 — Box: 4 green, 6 yellow, 2 purple counters

(a) Total outcomes: $4 + 6 + 2 = 12$

(b) $\Pr(\text{green}) = \frac{4}{12} = \frac{1}{3}$

(c) $\Pr(\text{yellow}) = \frac{6}{12} = \frac{1}{2} = 0.5$

(d) $\Pr(\text{purple}) = \frac{2}{12} = \frac{1}{6} = 0.1\overline{6} \approx 16.7\%$

(e) $\Pr(\text{orange}) = 0$ There are no orange counters in the box. This event is **impossible**.

(f) $\Pr(\text{not yellow}) = 1 - \frac{6}{12} = \frac{6}{12} = \frac{1}{2}$
(Green + Purple $= 4 + 2 = 6$ favourable outcomes out of 12.)

Question 6 — Standard six-sided die

Sample space: $\{1, 2, 3, 4, 5, 6\}$ Total outcomes $= 6$

(a) $\Pr(\text{rolling a } 4) = \frac{1}{6}$

(b) Even numbers: $\{2, 4, 6\}$ — 3 favourable outcomes.

$$\Pr(\text{even}) = \frac{3}{6} = \frac{1}{2}$$

(c) Greater than 4: $\{5, 6\}$ — 2 favourable outcomes.

$$\Pr(\text{greater than } 4) = \frac{2}{6} = \frac{1}{3}$$

(d) Less than 7: $\{1, 2, 3, 4, 5, 6\}$ — all 6 outcomes qualify.

$$\Pr(\text{less than } 7) = \frac{6}{6} = 1 \quad \textbf{(certain)}$$

Every number on a standard die is less than 7, so this event must occur.

Part D — Converting and Comparing

Question 7 — Conversion table

Given value	Fraction	Decimal	Percentage
$\frac{1}{4}$	$\frac{1}{4}$	**0.25**	**25%**
0.6	$\frac{3}{5}$	0.6	**60%**
35%	$\frac{7}{20}$	**0.35**	35%
$\frac{7}{20}$	$\frac{7}{20}$	**0.35**	**35%**
0.08	$\frac{2}{25}$	0.08	**8%**

Teaching note: $0.6 = \frac{6}{10} = \frac{3}{5}$; $35\% \div 100 = 0.35 = \frac{35}{100} = \frac{7}{20}$; $0.08 = \frac{8}{100} = \frac{2}{25}$.

Question 8 — Order from least to most likely

Convert all to decimals:

$\Pr(A) = \frac{2}{5} = 0.40$ $\Pr(B) = 0.65$ $\Pr(C) = 48\% = 0.48$ $\Pr(D) = \frac{3}{4} = 0.75$

Comparing: $0.40 < 0.48 < 0.65 < 0.75$

$$\text{Least likely: } A \longrightarrow C \longrightarrow B \longrightarrow D : \text{Most likely}$$

Part E — Experimental Probability

Question 9 — Drawing pin: 50 trials, 32 point-up, 18 point-down

(a) $\Pr(\text{point-up}) = \dfrac{32}{50} = \dfrac{16}{25} = 0.64 = 64\%$

(b) $\Pr(\text{point-down}) = \dfrac{18}{50} = \dfrac{9}{25} = 0.36 = 36\%$

(c) $0.64 + 0.36 = 1.00$ ✓

Yes, they add to 1. On every trial the pin must land either point-up or point-down — these are the only two possible outcomes. Because they are mutually exclusive (can't both happen) and exhaustive (one must happen), their probabilities must sum to 1.

Question 10 — Die rolled 60 times; rolled a 6 exactly 8 times

(a) Experimental $\Pr(\text{rolling a } 6) = \dfrac{8}{60} = \dfrac{2}{15} \approx 0.133 = 13.3\%$

(b) Theoretical $\Pr(\text{rolling a } 6) = \dfrac{1}{6} \approx 0.167 = 16.7\%$

(c) The experimental result (13.3%) is reasonably close to but not identical to the theoretical value (16.7%). **This is expected** with only 60 trials — random variation causes differences in small samples. With thousands of rolls, the experimental probability would get closer and closer to $\frac{1}{6}$ (Law of Large Numbers).

Application Exercise

Question 1 — Weather forecast for Townsville

Rain probabilities: Mon 20%, Tue 45%, Wed 60%, Thu 80%, Fri 55%, Sat 10%

(a) Highest probability of rain: **Thursday** (80%)

(b) More likely than not means Pr > 50%:
Wednesday (60%), Thursday (80%) and Friday (55%) all exceed 50%.

(c) Wednesday's probability as a simplified fraction:

$$60\% = \frac{60}{100} = \frac{6}{10} = \frac{3}{5}$$

(d) Saturday: $\Pr(\text{rain}) = 10\% = 0.10 < 0.5 \Rightarrow$ **Unlikely.**
With only a 10% chance of rain, there is a 90% chance of no rain. Mara's outdoor party is very likely to have fine weather.

Question 2 — School raffle: 200 tickets, 1 prize

(a) Lachlan buys 5 tickets:

$$\Pr(\text{win}) = \frac{5}{200} = \frac{1}{40} = 2.5\%$$

(b) Sofia buys 20 tickets:

$$\Pr(\text{win}) = \frac{20}{200} = \frac{1}{10} = 10\%$$

(c) **Sofia** has the better chance.
Difference: $10\% - 2.5\% = 7.5\%$
(d) Sofia's cost: $20 \times \$2 = \$40$

Question 3 — Spinner: 10 sections (4 Red, 3 Blue, 2 Green, 1 Gold)

(a) Drawing exercise. The circle should be divided into 10 equal sections (each 36°), labelled: 4 sections Red, 3 sections Blue, 2 sections Green, 1 section Gold.

(b) $\Pr(\text{Gold}) = \frac{1}{10} = 0.1 = 10\%$

(c) $\Pr(\text{Red}) = \frac{4}{10} = \frac{2}{5}$

(d) $\Pr(\text{not Blue}) = 1 - \frac{3}{10} = \frac{7}{10} = 0.7 = 70\%$

(e) Expected Green in 200 spins:

$\Pr(\text{Green}) = \frac{2}{10} = \frac{1}{5}$ Expected count $= 200 \times \frac{1}{5} = 40$ times

Question 4 — Standard deck of 52 cards

Recall: 52 cards, 4 suits of 13 each; Hearts & Diamonds = red (26 cards); Clubs & Spades = black (26 cards).

(a) $\Pr(\text{Heart}) = \frac{13}{52} = \frac{1}{4}$

(b) $\Pr(\text{King}) = \frac{4}{52} = \frac{1}{13}$ (one King in each of the 4 suits)

(c) $\Pr(\text{red card}) = \frac{26}{52} = \frac{1}{2}$

(d) $\Pr(\text{Ace of Spades}) = \frac{1}{52}$ (exactly one such card in the deck)

(e) $\Pr(\text{red}) = \frac{26}{52} = 0.5$ $\Pr(\text{Club}) = \frac{13}{52} = \frac{1}{4} = 0.25$

$0.5 > 0.25$ **Drawing a red card is more likely.** It has twice the probability of drawing a Club (50% vs 25%).

Question 5 — Quality control: 400 jars/hour, 6 faulty

(a) $\Pr(\text{faulty}) = \frac{6}{400} = \frac{3}{200} = 0.015 = 1.5\%$

(b) Faulty jars in 8-hour shift:

$$6 \times 8 = 48 \text{ faulty jars}$$

(c) Target: fewer than 2% faulty. Current rate: 1.5%.

$1.5\% < 2\%$ **Yes, the current rate is within target.**

Question 6 — Experimental probability: Chloe 20 flips, Tom 200 flips

(a) Chloe: 14 Heads from 20 flips.

$$\Pr(\text{Heads}) = \frac{14}{20} = \frac{7}{10} = 0.70 = 70\%$$

(b) Tom: 128 Heads from 200 flips.

$$\Pr(\text{Heads}) = \frac{128}{200} = \frac{16}{25} = 0.64 = 64\%$$

(c) **Tom's results** give a better estimate. With 200 trials (vs Chloe's 20), Tom's larger sample is less affected by random variation and will be closer to the true probability (Law of Large Numbers).

(d) A fair coin has $\Pr(\text{Heads}) = 0.5 = 50\%$. Tom's result is 64%, which is notably higher than 50%. **The coin does not appear to be fair** — it seems biased towards Heads. However, even 200 trials may not be completely conclusive; a much larger experiment would give more certainty.

Chapter Review

Question 1 — Match event with probability word

(a) Rolling a 3 on a standard die: $\Pr = \frac{1}{6} \approx 0.17 < 0.5$ **Unlikely**

(b) Picking red from a bag of only red balls: $\Pr = 1$ **Certain**

(c) Flipping a fair coin and getting Tails: $\Pr = \frac{1}{2}$ **Even chance**

(d) Snowing in Darwin in summer: Darwin is tropical; snow is climatically impossible. **Impossible**

(e) Bag has 7 red, 3 blue; picking red: $\Pr = \frac{7}{10} = 0.7 > 0.5$ **Likely**

Question 2 — Bag: 2 red, 4 blue, 3 green, 1 yellow marble

(a) Total: $2 + 4 + 3 + 1 = 10$

(b) $\Pr(\text{blue}) = \dfrac{4}{10} = \dfrac{2}{5}$

(c) $\Pr(\text{red}) = \dfrac{2}{10} = 0.2$

(d) $\Pr(\text{green}) = \dfrac{3}{10} = 30\%$

(e) $\Pr(\text{not red}) = \dfrac{10-2}{10} = \dfrac{8}{10} = \dfrac{4}{5}$

Question 3 — Spinner: 5 equal sections numbered 1–5

(a) Sample space: $S = \{1,\ 2,\ 3,\ 4,\ 5\}$ (5 outcomes)

(b) $\Pr(\text{spinning a 3}) = \dfrac{1}{5}$

(c) Even numbers from $\{1, 2, 3, 4, 5\}$: $\{2, 4\}$ — 2 favourable.

$$\Pr(\text{even}) = \frac{2}{5}$$

(d) Greater than 2: $\{3, 4, 5\}$ — 3 favourable.

$$\Pr(\text{greater than 2}) = \frac{3}{5}$$

Question 4 — Order from least to most likely

Convert all to decimals:

$45\% = 0.45 \qquad \frac{1}{3} \approx 0.333 \qquad 0.70 \qquad \frac{9}{10} = 0.90 \qquad 20\% = 0.20$

Order: $0.20 < 0.333 < 0.45 < 0.70 < 0.90$

$$20\% \longrightarrow \tfrac{1}{3} \longrightarrow 45\% \longrightarrow 0.70 \longrightarrow \tfrac{9}{10}$$

Question 5 — Thunderstorm probability 0.35

(a) $0.35 = 35\%$

(b) $0.35 < 0.5 \Rightarrow$ **Unlikely.** A thunderstorm is less likely than not — there is a 65% chance of no storm.

(c) $\Pr(\text{no storm}) = 1 - 0.35 = 0.65 = 65\%$

(Because the storm either occurs or doesn't, these two complementary probabilities must sum to 1.)

Question 6 — Die rolled 30 times (results: 1×6, 2×3, 3×7, 4×4, 5×5, 6×5)

(a) Experimental Pr(rolling a 3): rolled 3 seven times in 30 trials.

$$\frac{7}{30} \approx 0.233 = 23.3\%$$

(b) Theoretical $\Pr(\text{rolling a } 3) = \frac{1}{6} \approx 0.167 = 16.7\%$

(c) Experimental Pr(greater than 4): rolled 5 or $6 = 5 + 5 = 10$ times.

$$\frac{10}{30} = \frac{1}{3} \approx 0.333 = 33.3\%$$

(d) Expected rolls of 1 in 30 (theoretical):

$$30 \times \frac{1}{6} = 5 \text{ times}$$

Question 7 — Lollies: 6 snakes, 4 bears, 5 frogs, 5 worms (total 20)

Total: $6 + 4 + 5 + 5 = 20$

(a) $\Pr(\text{snake or worm}) = \dfrac{6+5}{20} = \dfrac{11}{20} = 0.55 = 55\%$

(b) $\Pr(\text{not bear}) = \dfrac{20-4}{20} = \dfrac{16}{20} = \dfrac{4}{5} = 0.8 = 80\%$

(c) $\Pr(\text{frog}) = \dfrac{5}{20} = \dfrac{1}{4} = 0.25 \quad \Pr(\text{bear}) = \dfrac{4}{20} = \dfrac{1}{5} = 0.20$

$0.25 > 0.20$ **Picking a frog is more likely than picking a bear** (25% vs 20%).

Multi-Step Problem 8 — Springvale Primary Lucky Dip

Contents: 10 small + 6 medium + 3 large + 1 grand prize. Tickets cost \$3 each.

(a) Total prizes: $10 + 6 + 3 + 1 = 20$

(b) $\Pr(\text{grand prize}) = \dfrac{1}{20} = 5.0\%$

(c) $\Pr(\text{large prize}) = \dfrac{3}{20} = 15\%$

(d) $\Pr(\text{small or medium}) = \dfrac{10+6}{20} = \dfrac{16}{20} = \dfrac{4}{5} = 80\%$

(e) After 5 small and 2 medium drawn: $20 - 5 - 2 = 13$ prizes remain

(f) New $\Pr(\text{grand prize}) = \dfrac{1}{13} \approx 7.7\%$

It has increased. The grand prize is still in the box, but the total number of prizes is now smaller (13 instead of 20). With fewer prizes to compete against, the chance of drawing the grand prize is higher.

(g) Revenue from 200 tickets: $200 \times \$3 = \$600$

Multi-Step Problem 9 — Thumbtack experiment

(a) Experimental Pr(point-up):

Student	Calculation	Experimental Pr(point-up)
Aiden	$6 \div 10$	**0.60**
Bella	$28 \div 50$	**0.56**
Carlos	$118 \div 200$	**0.59**

(b) **Carlos's result is most reliable.** He conducted 200 trials — far more than Aiden (10) or Bella (50). A larger number of trials reduces the effect of random variation and gives a more accurate estimate of the true probability.

(c) Carlos's estimate: $118 \div 200 = 0.59$.

$\frac{1}{2} = 0.50 \quad \frac{3}{5} = 0.60$

$|0.59 - 0.60| = 0.01$ vs $|0.59 - 0.50| = 0.09$ **Closer to $\frac{3}{5}$ (0.60).**

(d) Expected point-up in 500 flips using Pr = 0.59:

$$500 \times 0.59 = 295 \text{ times}$$

(e) A fair coin has Pr(heads) = 0.50. This thumbtack gives Pr(point-up) ≈ 0.59 across all three students' results. **The thumbtack does not behave like a fair coin** — it is consistently more likely to land point-up ($\approx 59\%$) than point-down ($\approx 41\%$). The evidence from all three students supports this conclusion, with Carlos's 200-trial result being the most reliable.

Chapter 8 Quick-Reference Answer Summary

Practice Exercise:

- Q1: (a) Certain (b) Impossible (c) Even chance (d) Likely (e) Impossible (f) Certain
- Q2: b,e at 0; c at 0.5; d at 0.9; a,f at 1
- Q3: (a) {1,2,3,4} (b) {Mon–Fri} (c) {M,A,T,H,S} (d) {Summer,Autumn,Winter,Spring}
- Q4: 8 outcomes
- Q5: (a) 12 (b) 1/3 (c) 0.5 (d) 16.7% (e) 0 (impossible) (f) 1/2
- Q6: (a) 1/6 (b) 1/2 (c) 1/3 (d) 1 (certain)
- Q7: Row 1: 0.25, 25% Row 2: 3/5, 60% Row 3: 7/20, 0.35 Row 4: 0.35, 35% Row 5: 2/25, 8%
- Q8: A (0.40) → C (0.48) → B (0.65) → D (0.75)
- Q9: (a) 0.64 (b) 0.36 (c) Yes, exhaustive outcomes sum to 1
- Q10: (a) 2/15 ≈ 13.3% (b) 1/6 ≈ 16.7% (c) Close but not exact — normal for 60 trials

Application Exercise:

- Q1: (a) Thursday (b) Wed, Thu, Fri (c) 3/5 (d) Unlikely (10%)
- Q2: (a) 1/40 = 2.5% (b) 1/10 = 10% (c) Sofia by 7.5% (d) $40
- Q3: (b) 1/10 = 10% (c) 2/5 (d) 7/10 = 70% (e) 40 times
- Q4: (a) 1/4 (b) 1/13 (c) 1/2 (d) 1/52 (e) Red card more likely (1/2 vs 1/4)
- Q5: (a) 3/200 = 1.5% (b) 48 faulty jars (c) Yes, within 2% target
- Q6: (a) 70% (b) 64% (c) Tom (200 trials) (d) Coin appears biased towards Heads

Chapter Review:

- Q1: (a) Unlikely (b) Certain (c) Even chance (d) Impossible (e) Likely
- Q2: (a) 10 (b) 2/5 (c) 0.2 (d) 30% (e) 4/5
- Q3: (a) {1,2,3,4,5} (b) 1/5 (c) 2/5 (d) 3/5
- Q4: 20% → 1/3 → 45% → 0.70 → 9/10
- Q5: (a) 35% (b) Unlikely (c) 0.65
- Q6: (a) 7/30 ≈ 23.3% (b) 1/6 ≈ 16.7% (c) 10/30 = 1/3 (d) 5 times
- Q7: (a) 11/20 = 55% (b) 4/5 = 80% (c) Frog more likely (25% vs 20%)
- MS8: (a) 20 (b) 5% (c) 15% (d) 80% (e) 13 remain (f) 1/13 ≈ 7.7%, increased (g) $600
- MS9: (a) Aiden 0.60, Bella 0.56, Carlos 0.59 (b) Carlos (c) Closer to 3/5 (d) 295 times (e) Not fair — biased towards point-up

Assessment Task 1 — Marking Guide

Assessment Task 1 — Project (Week 9)

The full Assessment Task 1 instrument is located in the following source files, which should be compiled and appended here:

- **Project Scenario and Student Instructions:** `Stage3_Task1_ProjectScenario.tex`
- **Marking Guide (A–E instrument-specific standards):** `Stage3_Task1_MarkingGuide.tex`

This marking guide is for teacher use only and must not be distributed to students in advance of the task.

Key Assessment Standards Summary (AT1 — Project):

Grade	Standard descriptor
A	Demonstrates a thorough understanding of numeracy concepts; all calculations accurate; clear, well-organised presentation; all required features present; strong real-world connection evident.
B	Demonstrates a sound understanding; most calculations accurate with minor errors; well-organised; required features largely present; good real-world connection.
C	Demonstrates a satisfactory understanding; calculations attempted with some errors; generally organised; most required features present; some real-world connection.
D	Demonstrates a limited understanding; several calculation errors; partially organised; some required features missing; limited real-world connection.
E	Demonstrates a minimal understanding; significant errors or omissions; limited organisation; required features largely absent.

Part II

Topic 2 — Workplace and Employment

9. Numbers at Work

Chapter 9 Information

Week taught:	Week 10 (Term 2)
Topic:	Topic 2 — Workplace and Employment
Skills area:	Number & Algebra
Lesson reference:	Lessons 37–40

Teaching Guide

Key concepts: Whole numbers in workplace contexts; reading payslips and invoices; using technology for workplace calculations; order of operations in workplace scenarios; estimating and checking workplace figures.

Suggested delivery sequence:

1. Lesson 37: Welcome to Topic 2; review of whole number operations; workplace number contexts.
2. Lesson 38: Reading and interpreting payslips and invoices.
3. Lesson 39: Using calculators and technology for workplace calculations.
4. Lesson 40: Worked Examples; Practice and Application Exercises.

Common Student Misconceptions

Common Misconceptions — Chapter 9

1. **Gross vs. net pay:** Students confuse gross income (before tax) with take-home (net) pay. *Strategy:* Gross = before deductions; net = what goes into your bank account.

2. **Order of operations in multi-line invoices:** Students add all items including tax before identifying the subtotal. *Strategy:* Identify the document structure (subtotal → tax → total) before calculating.

3. **Calculator errors — not checking answers:** Students trust an obviously wrong calculator answer. *Strategy:* Estimate first; if the calculator answer differs from the estimate by more than 10%, re-enter.

4. **Overtime — applying overtime rate to all hours:** Students apply the 1.5× rate to all hours worked. *Strategy:* Separate ordinary hours and overtime hours before applying rates.

5. **Rounding money — to nearest dollar:** Students write \$47.8 rather than \$47.80. *Strategy:* Money always uses two decimal places.

Worked Solutions

Chapter 9 Overview for Teachers

Chapter:	9 — Numbers at Work
Week:	Week 10, Term 1 (Lessons 37–40)
Topic:	Topic 2 — Workplace and Employment
Focus:	Calculating gross pay; reading and completing timesheets; interpreting pay slips; fortnightly pay; paper payroll spreadsheets

Key formulas:

$$\text{Gross pay} = \text{hours worked} \times \text{hourly rate}$$

$$\text{Fortnightly pay} = 2 \times \text{weekly pay} \qquad \text{Net pay} = \text{Gross pay} - \text{Deductions}$$

Common student errors to watch for:

- Treating minutes as decimal hundredths when converting time (e.g. writing 4 h 30 min as 4.30 h instead of 4.5 h).
- Forgetting to subtract break time from total shift duration.
- Confusing gross pay with net pay.
- Calculating superannuation on net pay rather than gross pay.

Practice Exercise Solutions

Part A — Single Shift Pay

Question 1 — Gross Pay Table

	Employee	Hours	Rate	Gross pay
(a)	Aisha	3 h	\$27.40/hr	**\$82.20**
(b)	Ben	5 h	\$26.80/hr	**\$134.00**
(c)	Chloe	6.5 h	\$29.20/hr	**\$189.80**
(d)	Dan	8 h	\$34.50/hr	**\$276.00**
(e)	Eva	4.5 h	\$24.10/hr	**\$108.45**

Workings:

(a) $3 \times \$27.40 = \$82.20$
(b) $5 \times \$26.80 = \$134.00$
(c) $6.5 \times \$29.20 = \$189.80$
(d) $8 \times \$34.50 = \$276.00$
(e) $4.5 \times \$24.10 = \$108.45$

Question 2 — Maya's Pay

$$7 \times \$25.60 = \$179.20$$

Question 3 — Liam's Public Holiday Pay

$$3.5 \times \$49.20 = \$172.20$$

Note for teachers: Liam's rate already includes the double-time penalty loading — students simply multiply hours by the given rate. No further loading calculation is required.

Part B — Timesheet

Question 4 — Hours Worked per Shift

Day	Start	Finish	Break	Total	Hours worked
Monday	08:00	13:00	30 min	5 h 00 min	**4 h 30 min**
Tuesday	12:00	17:30	30 min	5 h 30 min	**5 h 00 min**
Wednesday	09:00	14:00	0 min	5 h 00 min	**5 h 00 min**
Thursday	15:00	21:00	30 min	6 h 00 min	**5 h 30 min**
Friday	08:30	13:30	0 min	5 h 00 min	**5 h 00 min**
				Total hours:	**25 h 00 min**

Total: $4.5 + 5.0 + 5.0 + 5.5 + 5.0 = 25$ h

Question 5 — Weekly Gross Pay

$$25 \times \$24.80 = \$620.00$$

Question 6 — Fortnightly Pay

$$2 \times \$620.00 = \$1{,}240.00$$

Part C — Completing a Timesheet

Question 7 — Zara's Bakery Timesheet

Day	Start	Finish	Break	Hours worked
Mon 7 Jul	06:00	12:00	30 min	**5 h 30 min**
Tue 8 Jul	not rostered			—
Wed 9 Jul	06:00	10:30	0 min	**4 h 30 min**
Thu 10 Jul	06:00	12:00	30 min	**5 h 30 min**
Fri 11 Jul	not rostered			—
Sat 12 Jul	06:00	11:00	30 min	**4 h 30 min**
Sun 13 Jul	not rostered			—
			Total hours:	**20 h 00 min**

$$\begin{aligned}
\text{Mon: } 12{:}00 - 06{:}00 &= 6\text{ h } - 30\text{ min break} = 5\text{ h } 30\text{ min} \\
\text{Wed: } 10{:}30 - 06{:}00 &= 4\text{ h } 30\text{ min (no break)} \\
\text{Thu: } 12{:}00 - 06{:}00 &= 6\text{ h } - 30\text{ min break} = 5\text{ h } 30\text{ min} \\
\text{Sat: } 11{:}00 - 06{:}00 &= 5\text{ h } - 30\text{ min break} = 4\text{ h } 30\text{ min} \\
\text{Total: } 5.5 + 4.5 + 5.5 + 4.5 &= 20\text{ h}
\end{aligned}$$

Part D — Pay Slips

Question 8 — Rosa's Pay Slip (Springvale Hardware)

(a) Gross pay: **$386.40**

(b) Net pay: **$358.40**

(c) Tax withheld: **$28.00**

(d) Superannuation:

$$11.5\% \times \$386.40 = 0.115 \times \$386.40 = \$44.44$$

(e) Fortnightly gross pay:

$$2 \times \$386.40 = \$772.80$$

Note for teachers: (d) — super is paid by the employer on top of gross pay and does not reduce Rosa's net pay. Students should note the pay slip does not list super as a deduction from take-home pay.

Part E — Spreadsheet Table

Question 9 — Payroll Spreadsheet ($26.40/hr)

Row	A: Employee	B: Hours	C: Rate	D: Gross Pay
2	Hamish	4.5	$26.40	**$118.80**
3	Jess	6.0	$26.40	**$158.40**
4	Kobi	3.5	$26.40	**$92.40**
5	Layla	5.0	$26.40	**$132.00**
6	**Total**			**$501.60**

$4.5 + 6.0 + 3.5 + 5.0 = 19.0$ h total $\quad \$118.80 + \$158.40 + \$92.40 + \$132.00 = \$501.60$

Question 10 — Total Gross Pay

$501.60 (read directly from the SUM row above)

Application Exercise Solutions

Question 1 — Callum at JB Hi-Fi

(a) Pay per shift:

Day	Hours × Rate	Pay
Tuesday	5.0 × \$28.20	**\$141.00**
Thursday	5.0 × \$28.20	**\$141.00**
Saturday	6.0 × \$35.25	**\$211.50**

(b) Total gross pay:

$$\$141.00 + \$141.00 + \$211.50 = \$493.50$$

(c) Net pay:

$$\$493.50 - \$52.00 = \$441.50$$

(d) Fortnightly gross pay:

$$2 \times \$493.50 = \$987.00$$

Question 2 — Mia Comparing Jobs

(a) Weekly gross pay:

$$\begin{aligned} \text{Job A (Coles):} \quad & 20 \times \$24.80 = \$496.00 \\ \text{Job B (Café):} \quad & 16 \times \$26.40 = \$422.40 \end{aligned}$$

(b) **Job A** pays more per week, by **\$73.60** (\$496.00 − \$422.40).

(c) **Job B** has the higher hourly rate (\$26.40 vs \$24.80/hr).

(d) **Disagree.** Although Job B pays a higher hourly rate, Mia would work 4 fewer hours per week. Job A pays \$73.60 more per week in total. A higher hourly rate does not always mean higher weekly earnings — hours worked also matter.

Question 3 — Noah's Timesheet (Gold Coast Resort)

(a) Completed timesheet:

Day	Start	Finish	Break	Hours worked
Monday	11:00	15:00	0 min	**4 h 00 min**
Wednesday	17:00	22:30	30 min	**5 h 00 min**
Friday	17:00	23:00	30 min	**5 h 30 min**
Saturday	12:00	18:00	30 min	**5 h 30 min**
Sunday	11:00	16:00	0 min	**5 h 00 min**
			Total hours:	**25 h 00 min**

(b) Weekday gross (Mon 4 h + Wed 5 h + Fri 5.5 h = 14.5 h):

$$14.5 \times \$26.80 = \$388.60$$

(c) Weekend gross (Sat 5.5 h + Sun 5 h = 10.5 h):

$$10.5 \times \$33.50 = \$351.75$$

(d) Total gross pay:

$$\$388.60 + \$351.75 = \$740.35$$

Question 4 — Construction Crew Payroll

(a) Completed payroll spreadsheet:

Row	A: Employee	B: Hours	C: Rate	D: Gross Pay
2	Riley	38	$36.80	**$1,398.40**
3	Sam	32	$36.80	**$1,177.60**
4	Taylor	40	$36.80	**$1,472.00**
5	**Total**			**$4,048.00**

(b) Total gross pay: **$4,048.00**

(c) Mean gross pay per worker:

$$\$4{,}048.00 \div 3 = \$1{,}349.33$$

Question 5 — Priya's Aged-Care Pay Slip

Gross fortnightly pay: \$742.80

(a) Weekly gross pay:

$$\$742.80 \div 2 = \$371.40$$

(b) Hours worked in the fortnight:

$$\$742.80 \div \$29.10 \approx 25.5 \text{ hours}$$

(c) Fortnightly net pay:

$$\$742.80 - \$68.00 = \$674.80$$

Question 6 — Springvale Nursery Payroll Table

(a) Completed totals row:

Employee	Hours	Rate	Gross Pay	Tax
Finn	15	\$24.80	\$372.00	\$30.00
Georgia	20	\$27.60	\$552.00	\$52.00
Harry	12.5	\$24.80	\$310.00	\$22.00
Isla	25	\$27.60	\$690.00	\$70.00
Total	**72.5**		**\$1,924.00**	**\$174.00**

(b) **Isla** had the highest gross pay (\$690.00).

(c) Finn's net pay: \$372.00 − \$30.00 = \$342.00

(d) Total tax withheld: **\$174.00**

(e) Georgia and Isla earn \$27.60/hr compared to \$24.80/hr for Finn and Harry because they likely have more experience, higher skills or qualifications, or are classified under a different award rate for their role.

Chapter Review Solutions

Question 1 — Gross Pay Calculations

(a) $6 \times \$25.40 = \$152.40$

(b) $4.5 \times \$32.00 = \$144.00$

(c) $8 \times \$24.10 = \$192.80$

Question 2 — Hours Worked per Shift

Start	Finish	Break	Hours worked
07:00	13:30	30 min	**6 h 00 min**
14:00	20:00	0 min	**6 h 00 min**
09:30	16:00	30 min	**6 h 00 min**

Row 1: 13:30 − 07:00 = 6 h 30 min −30 min break = 6 h 00 min Row 3: 16:00 − 09:30 = 6 h 30 min −30 min break = 6 h 00 min

Question 3 — Sam's Weekly Pay Table

Day	Hours	Rate	Pay
Tuesday	5.0 h	\$27.20	**\$136.00**
Thursday	4.5 h	\$27.20	**\$122.40**
Saturday	6.0 h	\$27.20	**\$163.20**
Sunday	5.0 h	\$34.00	**\$170.00**
		Total gross pay:	**\$591.60**

$\$136.00 + \$122.40 + \$163.20 + \$170.00 = \$591.60$

Question 4 — Sam's Net Pay

$$\$591.60 - \$56.00 = \$535.60$$

Question 5 — Sam's Fortnightly Gross Pay

$$2 \times \$591.60 = \$1{,}183.20$$

Question 6 — Sam's Superannuation

$$11.5\% \times \$591.60 = 0.115 \times \$591.60 = \$68.03$$

Question 7 — Net Pay from Pay Slip

$$\$534.00 - \$47.00 = \$487.00$$

Question 8 — Hours Worked (Back-Calculation)

$$\$504.00 \div \$31.50 = 16 \text{ hours}$$

Question 9 — Gross Pay vs Net Pay (Written Response)

Sample answer: Gross pay is the total amount you earn from working, before any money is taken out. Net pay (also called take-home pay) is what is left after deductions — such as income tax and union fees — are subtracted. Net pay is the amount that actually goes into your bank account.

Multi-Step Problem Solutions

Multi-Step Problem 10 — Ava's Week at Work

Pay rates: Weekday \$24.80/hr Saturday \$31.00/hr Sunday \$37.20/hr

(a) **Completed timesheet:**

Day	Start → Finish	Break	Hours worked
Monday	15:30 → 19:00	0 min	**3 h 30 min**
Wednesday	15:30 → 19:00	0 min	**3 h 30 min**
Friday	15:30 → 19:30	0 min	**4 h 00 min**
Saturday	09:00 → 14:00	30 min	**4 h 30 min**
Sunday	10:00 → 14:00	0 min	**4 h 00 min**
		Total:	**19 h 30 min**

(b) **Pay by group:**

$$\begin{aligned} \text{Weekday (Mon + Wed): } & 2 \times 3.5 \times \$24.80 = \$173.60 \\ \text{Weekday (Fri): } & 4.0 \times \$24.80 = \$99.20 \\ \text{Weekday total: } & \$173.60 + \$99.20 = \$272.80 \\ \text{Saturday: } & 4.5 \times \$31.00 = \$139.50 \\ \text{Sunday: } & 4.0 \times \$37.20 = \$148.80 \end{aligned}$$

(c) **Total gross pay:**

$$\$272.80 + \$139.50 + \$148.80 = \$561.10$$

(d) **Net pay:**

$$\$561.10 - \$38.00 = \$523.10$$

(e) **Annual gross pay:**

$$52 \times \$561.10 = \$29{,}177.20$$

(f) **Weekly savings (30% of net pay):**

$$0.30 \times \$523.10 = \$156.93$$

(g) **Will Ava afford the \$500 laptop after 10 weeks?**

$$10 \times \$156.93 = \$1{,}569.30$$

Yes. \$1,569.30 > \$500, so Ava will have more than enough money after 10 weeks of saving.

Multi-Step Problem 11 — Springvale Café Team Payroll

Rates: Weekday \$26.80/hr Saturday \$33.50/hr

(a) **Completed payroll table:**

Employee	Wkday hrs	Sat hrs	Wkday pay	Sat pay	Gross pay
Ben	16	5	**\$428.80**	**\$167.50**	**\$596.30**
Cleo	20	6	**\$536.00**	**\$201.00**	**\$737.00**
Dom	12	0	**\$321.60**	**\$0.00**	**\$321.60**
				Total gross pay:	**\$1,654.90**

Ben: $16 \times \$26.80 = \$428.80$; $5 \times \$33.50 = \$167.50$; Gross $= \$596.30$
Cleo: $20 \times \$26.80 = \$536.00$; $6 \times \$33.50 = \$201.00$; Gross $= \$737.00$
Dom: $12 \times \$26.80 = \$321.60$; $0 \times \$33.50 = \$0.00$; Gross $= \$321.60$
Total: $\$596.30 + \$737.00 + \$321.60 = \$1{,}654.90$

(b) **Superannuation (11.5% of gross pay):**

$$\text{Ben: } 0.115 \times \$596.30 = \$68.57$$
$$\text{Cleo: } 0.115 \times \$737.00 = \$84.76$$
$$\text{Dom: } 0.115 \times \$321.60 = \$36.98$$
$$\text{Total super: } \$68.57 + \$84.76 + \$36.98 = \$190.31$$

(c) **Total weekly cost to the café (gross pay + super):**

$$\$1{,}654.90 + \$190.31 = \$1{,}845.21$$

(d) **Cleo's extra earnings (4 extra weekday hours):**

$$4 \times \$26.80 = \$107.20$$

(e) **Dom's pay rise over 12 weeks:**

$$\text{Current weekly pay: } 12 \times \$26.80 = \$321.60$$
$$\text{New weekly pay: } 12 \times \$28.50 = \$342.00$$
$$\text{Extra per week: } \$342.00 - \$321.60 = \$20.40$$
$$\text{Over 12 weeks: } 12 \times \$20.40 = \$244.80$$

Chapter 9 Quick Reference — Numbers at Work

Single shift gross pay

$$\text{Gross} = \text{hours} \times \text{rate}$$

Fortnightly pay

$$\text{Fortnightly} = 2 \times \text{weekly}$$

Net pay

$$\text{Net} = \text{Gross} - \text{Deductions}$$

Superannuation (2025)

$$\text{Super} = 11.5\% \times \text{Gross}$$

Key reminder: Convert time correctly — 30 min = 0.5 h, 15 min = 0.25 h, 45 min = 0.75 h. Never treat minutes as decimal hundredths.

10. Workplace Numbers

Chapter 10 Information

Week taught:	Week 11 (Term 2)
Topic:	Topic 2 — Workplace and Employment
Skills area:	Number & Algebra
Lesson reference:	Lessons 41–44

Teaching Guide

Key concepts: Fractions and percentages in workplace settings; calculating discounts, GST, and profit/loss; budgeting with fractions; percentage increase and decrease in employment contexts.

Suggested delivery sequence:

1. Lesson 41: Fractions in budgeting; allocating income as fractions.
2. Lesson 42: GST (10%) calculations; discounts; percentage of a price.
3. Lesson 43: Profit and loss; percentage change in workplace contexts.
4. Lesson 44: Worked Examples; Practice and Application Exercises.

Common Student Misconceptions

Common Misconceptions — Chapter 10

1. **GST — confusing adding GST with finding pre-GST price:** Students divide by 10 and subtract. *Strategy:* Separate "add GST" ($\times 1.1$) from "find pre-GST" ($\div 1.1$).
2. **Discount — subtracting the percentage number from the price:** Method breaks down for non-round numbers. *Strategy:* Always calculate the discount amount first, then subtract.
3. **Profit confused with revenue:** Students report total selling price as profit. *Strategy:* Profit = Revenue − Cost; identify the cost price explicitly.
4. **Percentage change — using the new value as base:** Students calculate 15% of the new wage. *Strategy:* Percentage change is always on the *original* value.
5. **Budget fractions summing to more than 1:** Students don't check that allocated fractions add to 1 or less. *Strategy:* Always verify the total equals 1 before finalising.

Worked Solutions

Chapter 10 Overview for Teachers

Chapter:	10 — Workplace Numbers
Week:	Week 11, Term 1 (Lessons 41–44)
Topic:	Topic 2 — Workplace and Employment
Focus:	Ratios; GST (10%); markup and discount; completing invoices; budget planning

Key formulas:

$$\text{Price incl. GST} = \text{price} \times 1.10 \qquad \text{Price excl. GST} = \text{total} \div 1.10$$

$$\text{Markup price} = \text{original} \times (1 + r) \qquad \text{Sale price} = \text{original} \times (1 - r)$$

Common errors:

- Removing GST by subtracting 10% (wrong: $\times 0.90$). Correct method: divide by 1.10.
- Mixing up excl./incl. GST when comparing quotes from different suppliers.
- Applying markup then discount sequentially without recalculating the base each time.

Practice Exercise Solutions

Part A — Ratio Mixing and Scaling

Question 1 — Fertiliser 1:5, make 12 litres

(a) Total parts: $1 + 5 = 6$ parts

(b) Litres per part: $12 \div 6 = 2$ litres per part

(c) Fertiliser: $1 \times 2 = 2$ litres

(d) Water: $5 \times 2 = 10$ litres

(e) Check: $2 + 10 = 12$ litres ✓

Question 2 — Marinade ratio 2:3:1, different batch sizes

Total parts: $2 + 3 + 1 = 6$

Total volume	Soy sauce	Honey	Vinegar
6 cups	**2 cups**	**3 cups**	**1 cup**
12 cups	**4 cups**	**6 cups**	**2 cups**
18 cups	**6 cups**	**9 cups**	**3 cups**

For each row: qty per part = total ÷6, then multiply by 2, 3 and 1 respectively.

Question 3 — Paint 3 white:1 blue, 20 litres

Total parts = 4; qty per part = $20 \div 4 = 5$ litres
White: $3 \times 5 = 15$ litres Blue: $1 \times 5 = 5$ litres
Check: $15 + 5 = 20$ ✓

Part B — GST Calculations

Question 4 — Adding GST (×1.10)

Item	Price excl. GST	Price incl. GST
Toolbox	$65.00	**$71.50**
Safety boots	$148.00	**$162.80**
Work gloves	$22.50	**$24.75**
First aid kit	$84.00	**$92.40**
Hi-vis vest	$36.80	**$40.48**

Each price ×1.10, e.g. $\$22.50 \times 1.10 = \$24.75$

Question 5 — Removing GST (÷1.10)

Price incl. GST	Price excl. GST	GST amount
$44.00	**$40.00**	**$4.00**
$99.00	**$90.00**	**$9.00**
$132.00	**$120.00**	**$12.00**
$275.00	**$250.00**	**$25.00**

Pre-GST price = total ÷1.10; GST = total − pre-GST price.
Common error: subtracting 10% gives the wrong answer (e.g. $\$44 \times 0.90 = \$39.60 \neq \$40.00$).

Part C — Markup and Discount

Question 6 — Markup and Discount Table

Original price	Change	Multiplier	New price
\$60.00	20% markup	×1.20	**\$72.00**
\$250.00	30% discount	×0.70	**\$175.00**
\$45.00	15% markup	×1.15	**\$51.75**
\$180.00	10% discount	×0.90	**\$162.00**
\$320.00	50% discount	×0.50	**\$160.00**

Question 7 — Headphones: cost \$54, 60% markup

(a) Selling price (excl. GST):

$$\$54.00 \times 1.60 = \$86.40$$

(b) Selling price including GST:

$$\$86.40 \times 1.10 = \$95.04$$

Part D — Completing the Invoice

Question 8 — Springvale Office Supplies Invoice

Description	Qty	Unit price	Line total
A4 copy paper (ream)	10	$6.80	**$68.00**
Ballpoint pens (box 12)	4	$9.50	**$38.00**
Sticky notes (pack)	6	$3.40	**$20.40**
Whiteboard markers (set)	3	$12.60	**$37.80**
Stapler	2	$8.90	**$17.80**
		Subtotal (excl. GST)	**$182.00**
		GST (10%)	**$18.20**
		TOTAL (incl. GST)	**$200.20**

$\$68.00 + \$38.00 + \$20.40 + \$37.80 + \$17.80 = \$182.00$ GST $= \$182.00 \times 0.10 = \$18.20$ Total $= \$182.00 \times 1.10 = \$200.20$

Part E — Budget Planning

Question 9 — Tradie's Purchase List (budget $150 excl. GST)

Sample answer (students' items will vary; any selection totalling $\leq$ $150 excl. GST is correct):

Item	Qty	Unit price	Line total
Paintbrush set	1	$14.50	$14.50
Drop sheet	2	$9.80	$19.60
Safety glasses	2	$11.00	$22.00
Wood screws	3	$7.20	$21.60
Masking tape	3	$3.90	$11.70
Filler (tube)	2	$6.40	$12.80
		Subtotal (excl. GST)	**$102.20**
		GST (10%)	**$10.22**
		Total (incl. GST)	**$112.42**

Yes, the order is within the $150 budget (excl. GST): $102.20 < $150.00.

Question 10 — Catering Budget Remaining

$500.00 − $312.40 = $187.60 remaining (excl. GST)

Application Exercise Solutions

Question 1 — Painter's Ratio (4:1 first coat, 9:1 second coat)

(a) First coat (4:1), 15 litres:

Total parts = 5; per part = $15 \div 5 = 3$ L

Paint: $4 \times 3 = 12$ litres Water: $1 \times 3 = 3$ litres

(b) Second coat (9:1), 20 litres:

Total parts = 10; per part = $20 \div 10 = 2$ L

Paint: $9 \times 2 = 18$ litres Water: $1 \times 2 = 2$ litres

(c) Cost of paint for second coat (18 litres at \$28.50/litre excl. GST):

$$18 \times \$28.50 = \$513.00 \text{ (excl. GST)}$$

Question 2 — Commercial Cleaning Solutions

(a) Concentrate needed for each solution:

Solution	Ratio	Total parts	L/part	Total	Concentrate
Floor cleaner	1:7	8	2 L	16 L	**2 L**
Window spray	1:3	4	2 L	8 L	**2 L**
Sanitiser	1:9	10	2 L	20 L	**2 L**

(b) Total concentrate = $2 + 2 + 2 = 6$ litres at \$18.40/litre:

$$6 \times \$18.40 = \$110.40 \text{ (excl. GST)}$$

(c) Total including GST:

$$\$110.40 \times 1.10 = \$121.44$$

Question 3 — Springvale Sports & Outdoors Invoice

Description	Qty	Unit price	Line total
Football (size 5)	12	\$28.00	**\$336.00**
Training bibs (pack of 10)	3	\$22.50	**\$67.50**
Drink bottles (750 mL)	20	\$6.80	**\$136.00**
Mesh bag (for balls)	2	\$18.40	**\$36.80**
First aid kit (junior)	1	\$54.00	**\$54.00**
		Subtotal (excl. GST)	**\$630.30**
		GST (10%)	**\$63.03**
		TOTAL (incl. GST)	**\$693.33**

(a) Club budget \$600 (incl. GST); invoice total \$693.33. **No** — the invoice exceeds the budget by \$93.33.

(b) Reduce drink bottles to 10 (save $10 \times \$6.80 = \$68.00$ excl. GST):

$$\begin{aligned} \text{New subtotal} &= \$630.30 - \$68.00 = \$562.30 \\ \text{New GST} &= \$56.23 \\ \text{New total} &= \$618.53 \end{aligned}$$

Question 4 — Comparing Quotes: 50 Folding Chairs

(a) **Supplier A** (excl. GST): $50 \times \$38.00 + \$25.00 = \$1{,}925.00$ (excl. GST)

$$\$1{,}925.00 \times 1.10 = \$2{,}117.50 \text{ (incl. GST)}$$

(b) **Supplier B** (already incl. GST, free delivery):

$$50 \times \$44.00 = \$2{,}200.00 \text{ (incl. GST)}$$

(c) **Supplier A** is cheaper, by $\$2{,}200.00 - \$2{,}117.50 = \$82.50$.

Question 5 — Boutique: 60% Markup then 20% Off

Item	Wholesale	After 60% markup ($\times 1.60$)	Sale price (20% off, $\times 0.80$)
Dress	\$45.00	**\$72.00**	**\$57.60**
Jeans	\$38.00	**\$60.80**	**\$48.64**
Jacket	\$72.00	**\$115.20**	**\$92.16**

(b) Jacket sale price including GST:

$$\$92.16 \times 1.10 = \$101.38$$

(c) **Yes, the boutique still makes a profit.**
Sale price (excl. GST) = \$92.16 > \$72.00 (wholesale).
Profit = \$92.16 − \$72.00 = \$20.16.

Question 6 — Catering Budget \$120 excl. GST for 20 People

Sample purchase list (answers will vary; any selection within \$120 excl. GST that feeds 20 people is valid):

Item	Qty	Unit price	Line total
Sausage rolls (12-pack)	2	\$8.40	\$16.80
Mini quiches (12-pack)	2	\$9.60	\$19.20
Fruit platter (serves 10)	2	\$18.50	\$37.00
Juice bottles (6-pack)	2	\$7.20	\$14.40
Coffee pods (10-pack)	1	\$6.80	\$6.80
Tea bags (25-pack)	1	\$4.20	\$4.20
Disposable cups (50-pack)	1	\$5.60	\$5.60
		Subtotal (excl. GST)	**\$104.00**
		GST (10%)	**\$10.40**
		Total (incl. GST)	**\$114.40**

(b) **Yes** — subtotal \$104.00 < \$120.00. Within budget.

(c) Cost per person: \$114.40 ÷ 20 = \$5.72

Chapter Review Solutions

Question 1 — Ratio 1:3, make 8 litres

Total parts = 4; per part = $8 \div 4 = 2$ L
Concentrate: **2 litres** Water: **6 litres**

Question 2 — Concrete 1:3:5, scale 9 bags to 45 bags

Multiplier = $45 \div 9 = 5$

Ingredient	Original	×5	New quantity
Cement	1	×5	**5 bags**
Sand	3	×5	**15 bags**
Gravel	5	×5	**25 bags**
Total	9		**45 bags**

Question 3 — GST-Inclusive Prices (×1.10)

(a) $\$75.00 \times 1.10 = \$82.50$

(b) $\$240.00 \times 1.10 = \$264.00$

(c) $\$18.50 \times 1.10 = \$20.35$

Question 4 — Finding Pre-GST Price (÷1.10)

(a) $\$55.00 \div 1.10 = \$50.00$ (excl. GST); GST = $\$5.00$

(b) $\$363.00 \div 1.10 = \$330.00$ (excl. GST); GST = $\$33.00$

Question 5 — Camera $320: 35% Markup then 10% Discount then GST

(a) Price after 35% markup:

$$\$320.00 \times 1.35 = \$432.00 \text{ (excl. GST)}$$

(b) Sale price after 10% discount:

$$\$432.00 \times 0.90 = \$388.80 \text{ (excl. GST)}$$

(c) Final price including GST:

$$\$388.80 \times 1.10 = \$427.68$$

Question 6 — Springvale Garden Centre Invoice

Description	Qty	Unit price	Line total
Potting mix (25 L bag)	6	$12.80	**$76.80**
Fertiliser (1 kg)	4	$9.40	**$37.60**
Garden gloves (pair)	3	$8.60	**$25.80**
Seedling trays (4-pack)	5	$6.20	**$31.00**
		Subtotal (excl. GST)	**$171.20**
		GST (10%)	**$17.12**
		TOTAL (incl. GST)	**$188.32**

Question 7 — GST Check: total $407.00 incl. GST

Pre-GST price = $407.00 ÷ 1.10 = $370.00 GST = $407.00 − $370.00 = $37.00

No, the colleague is incorrect. The GST is $37.00, not $40.00.

Common error: $407 × 0.10 = $40.70 (also wrong). To find GST within a GST-inclusive total, always divide by 1.10 first, then subtract.

Multi-Step Problem Solutions

Multi-Step Problem 8 — Flynn's Tiling Job

Areas: Floor = 12 m^2; two feature walls = $2 \times 6 = 12$ m^2; each box covers 1.5 m^2.

(a) **Floor tiles** (including 10% extra):

$$\text{Boxes needed (exact)} = 12 \div 1.5 = 8 \text{ boxes}$$
$$\text{With 10\% extra} = 8 \times 1.10 = 8.8 \rightarrow \text{round up} = 9 \text{ boxes}$$

(b) **Feature wall tiles** (two walls × 6 m^2 each = 12 m^2, plus 10% extra):

$$\text{Boxes needed (exact)} = 12 \div 1.5 = 8 \text{ boxes}$$
$$\text{With 10\% extra} = 8 \times 1.10 = 8.8 \rightarrow \text{round up} = 9 \text{ boxes}$$

(c) **Adhesive powder** (ratio 3 powder:1 water, 8 litres total):

$$\text{Total parts} = 4; \quad \text{per part} = 8 \div 4 = 2 \text{ L}$$
$$\text{Powder needed} = 3 \times 2 = 6 \text{ litres}$$

(d) **Invoice:**

Description	Qty	Unit price	Line total
Floor tiles (boxes)	9	$48.00	**$432.00**
Feature wall tiles (boxes)	9	$62.00	**$558.00**
Tile adhesive powder (litres)	6	$4.50	**$27.00**
Tile grout (bags)	3	$18.00	**$54.00**
		Subtotal (excl. GST)	**$1,071.00**
		GST (10%)	**$107.10**
		TOTAL (incl. GST)	**$1,178.10**

(e) Flynn's budget is $900 (incl. GST). Total = $1,178.10 > $900.
No — the total exceeds the budget by $1,178.10 − $900.00 = $278.10.

Multi-Step Problem 9 — School Canteen: Comparing Quotes

(a) **Subtotals (excl. GST):**

Item	Qty	Quote A	Quote B	A total / B total
Bread rolls (bags)	10	\$3.60	\$3.90	\$36.00 / \$39.00
Cheese (packs)	8	\$5.40	\$4.80	\$43.20 / \$38.40
Ham (packs)	10	\$4.80	\$5.20	\$48.00 / \$52.00
Juice (6-packs)	5	\$6.80	\$6.40	\$34.00 / \$32.00
Containers (packs)	4	\$8.60	\$9.20	\$34.40 / \$36.80
Subtotal				**\$195.60 / \$198.20**

(b) **Quote A** is cheaper overall (excl. GST) by \$198.20 − \$195.60 = \$2.60.

(c) Quote A including GST:

$$\$195.60 \times 1.10 = \$215.16$$

(d) Is Quote A within the \$400 budget (excl. GST)? \$195.60 < \$400.00. **Yes.**

(e) Add 2 extra bags of bread rolls (Quote A rate \$3.60/bag):

$$\text{New subtotal} = \$195.60 + 2 \times \$3.60 = \$195.60 + \$7.20 = \$202.80$$
$$\text{New total (incl. GST)} = \$202.80 \times 1.10 = \$223.08$$

Chapter 10 Quick Reference — Workplace Numbers

GST

$$\text{Incl. GST} = \text{price} \times 1.10$$
$$\text{Excl. GST} = \text{total} \div 1.10$$

Markup/Discount

$$\text{Markup} = \text{original} \times (1 + r)$$
$$\text{Discount} = \text{original} \times (1 - r)$$

Ratio mixing

Total parts = sum of all parts
Qty per part = total volume ÷ total parts

Invoice flow

Line total = qty × unit price
Subtotal → GST → Total

11. 3D Shapes and Volume

Chapter 11 Information

Week taught:	Week 12 (Term 2)
Topic:	Topic 2 — Workplace and Employment
Skills area:	Measurement & Space
Lesson reference:	Lessons 45–48

Teaching Guide

Key concepts: Identifying and describing 3D shapes (prisms, pyramids, cylinders, cones, spheres); faces, edges and vertices; volume of rectangular prisms and other prisms using $V = A \times H$; volume unit conversions.

Suggested delivery sequence:

1. Lesson 45: 3D shapes — identification, properties, nets.
2. Lesson 46: Volume of rectangular prisms; $V = l \times w \times h$.
3. Lesson 47: Volume of other prisms: $V = A_{\text{cross-section}} \times H$.
4. Lesson 48: Worked Examples; Practice and Application Exercises.

Common Student Misconceptions

Common Misconceptions — Chapter 11

1. **Confusing prisms and pyramids:** Students call any 3D shape a "prism." *Strategy:* A prism has two identical parallel end-faces; a pyramid has one base and triangular faces meeting at a point.

2. **Triangular prism volume — not halving:** Students use $V = b \times h \times H$ without halving. *Strategy:* First calculate cross-sectional area ($A = \frac{1}{2}bh$), then multiply by length.

3. **Volume unit conversions — using linear factor:** Students divide cm^3 to m^3 by 100 instead of 10^6. *Strategy:* Cube the linear factor: $(100)^3 = 1{,}000{,}000$.

4. **Dimension identification from diagrams:** Students swap height and length from a tilted diagram. *Strategy:* Redraw in standard upright orientation before applying the formula.

5. **Counting faces, edges, vertices — double-counting:** Students count shared edges twice. *Strategy:* Count systematically; use Euler's formula $F + V - E = 2$ to check.

Worked Solutions

Chapter 11 Overview for Teachers

Chapter:	11 — 3D Shapes and Volume
Week:	Week 12, Term 1 (Lessons 45–48)
Topic:	Topic 2 — Workplace and Employment
Focus:	Identifying 3D shapes; faces/edges/vertices; volume of rectangular and triangular prisms; unit conversions; workplace applications

Key formulas:

$$V_{\text{rectangular prism}} = L \times W \times H$$

$$V_{\text{triangular prism}} = \frac{b \times h}{2} \times L \qquad 1\ \text{cm}^3 = 1\ \text{mL}; \quad 1000\ \text{cm}^3 = 1\ \text{L}; \quad 1\ \text{m}^3 = 1000\ \text{L}$$

Pedagogical notes:

- Emphasise the two-step method: (1) find base area, (2) multiply by height/length.
- Students often confuse the prism "height" with the triangle "height" — label each clearly.
- Unit conversions: $1\ \text{m}^3 = 10^6\ \text{cm}^3$ (multiply all three dimensions by 100).
- Volume method for fitting boxes gives a theoretical maximum; practical packing is always less.

Practice Exercise — Worked Solutions

Part A — Shape Identification and Properties

Question 1 — Name each 3D shape

(a) A shape with 6 rectangular faces, 12 edges and 8 vertices.
Rectangular prism (cuboid)
(b) A shape with 5 faces (2 triangles and 3 rectangles), 9 edges and 6 vertices.
Triangular prism
(c) A shape with 2 circular faces and 1 curved surface.
Cylinder
(d) A special rectangular prism where all 6 faces are identical squares.
Cube (square prism)

Question 2 — Properties table

Shape	Faces	Edges	Vertices
Rectangular prism	**6**	**12**	**8**
Triangular prism	**5**	**9**	**6**
Cube	**6**	**12**	**8**

Teacher note: Use the general rule for a prism with an n-sided base:

$$\text{Faces} = n + 2, \qquad \text{Edges} = 3n, \qquad \text{Vertices} = 2n$$

For a rectangle ($n = 4$): $4 + 2 = 6$ faces, 12 edges, 8 vertices. For a triangle ($n = 3$): 5 faces, 9 edges, 6 vertices.

Question 3 — Hexagonal prism properties

A hexagon has $n = 6$ sides.

$\text{Faces} = n + 2 = 6 + 2 = 8$ faces
$\text{Edges} = 3n = 3 \times 6 = 18$ edges
$\text{Vertices} = 2n = 2 \times 6 = 12$ vertices

Answer: 8 faces, 18 edges, 12 vertices

Part B — Volume of Rectangular Prisms

Question 4 — Volume table

Using $V = L \times W \times H$:

	L	W	H	Volume
(a)	5 cm	4 cm	3 cm	**60 cm^3**
(b)	8 m	3 m	2 m	**48 m^3**
(c)	12 cm	7 cm	5 cm	**420 cm^3**
(d)	2.5 m	1.5 m	0.2 m	**0.75 m^3**
(e)	40 cm	25 cm	15 cm	**15 000 cm^3**

Selected workings:
(d) $V = 2.5 \times 1.5 \times 0.2 = 3.75 \times 0.2 = 0.75$ m^3
(e) $V = 40 \times 25 \times 15 = 1000 \times 15 = 15\,000$ cm^3

Question 5 — Find the height of the storage box

Given: $V = 3000\ \text{cm}^3$, $L = 25$ cm, $W = 12$ cm.

Rearrange $V = L \times W \times H$:

$$H = \frac{V}{L \times W} = \frac{3000}{25 \times 12} = \frac{3000}{300} = 10\ \text{cm}$$

The box is 10 cm tall.

Part C — Volume of Triangular Prisms

Question 6 — Volume of triangular prisms

(a) $b = 6$ cm, $h = 4$ cm, $L = 10$ cm

Step 1: Base area $= \dfrac{b \times h}{2} = \dfrac{6 \times 4}{2} = \dfrac{24}{2} = 12\ \text{cm}^2$

Step 2: $V = A \times L = 12 \times 10 = 120\ \text{cm}^3$

(b) $b = 9$ m, $h = 5$ m, $L = 8$ m

Step 1: Base area $= \dfrac{9 \times 5}{2} = \dfrac{45}{2} = 22.5\ \text{m}^2$

Step 2: $V = 22.5 \times 8 = 180\ \text{m}^3$

(c) $b = 14$ cm, $h = 8$ cm, $L = 20$ cm

Step 1: Base area $= \dfrac{14 \times 8}{2} = \dfrac{112}{2} = 56\ \text{cm}^2$

Step 2: $V = 56 \times 20 = 1120\ \text{cm}^3$

Part D — Volume Unit Conversions

Question 7 — Unit conversions

(a) $5\ \text{m}^3 = 5 \times 10^6\ \text{cm}^3 = 5\,000\,000\ \text{cm}^3$

(b) $8000\ \text{cm}^3 = 8000 \div 1000 = 8$ L

(c) $2500\ \text{cm}^3 = 2500$ mL (since $1\ \text{cm}^3 = 1$ mL)

(d) $3\ \text{m}^3 = 3 \times 1000 = 3000$ L

(e) $450\,000\ \text{cm}^3 = 450\,000 \div 1\,000\,000 = 0.45\ \text{m}^3$

Question 8 — Rectangular tank capacity in litres

Tank dimensions: $L = 80$ cm, $W = 50$ cm, $H = 40$ cm.

Step 1: Volume $= 80 \times 50 \times 40 = 160\,000$ cm^3

Step 2: Convert to litres: $160\,000 \div 1000 =$ 160 L

The tank holds 160 litres.

Application Exercise — Worked Solutions

Question 1 — Concrete slab (driveway)

Dimensions: $L = 12$ m, $W = 3.5$ m, thickness $= 0.12$ m.
(a) Volume of concrete:

$V = L \times W \times H = 12 \times 3.5 \times 0.12 = 42 \times 0.12 = 5.04$ m^3

(b) Cost excluding GST:

Cost $= 5.04 \times \$185 = \$932.40$

(c) Total cost including GST (10%):

Total $= \$932.40 \times 1.10 = \$1025.64$

Question 2 — Shipping container

Container: $L = 5.9$ m, $W = 2.35$ m, $H = 2.39$ m.
(a) Internal volume:

$V = 5.9 \times 2.35 \times 2.39$
$= 5.9 \times 5.6165$
$= 33.137\ldots \approx 33.14$ m^3

(b) Volume of one pallet (1.2 m × 1.0 m × 1.8 m):

$V_{\text{pallet}} = 1.2 \times 1.0 \times 1.8 = 2.16$ m^3

(c) Number of pallets (round down):

Pallets $= 33.14 \div 2.16 = 15.34\ldots \rightarrow$ 15 pallets

Teacher note: Volume method gives the theoretical maximum. In practice fewer pallets fit due to the container door, stacking stability, and aisle access.

Question 3 — Triangular concrete ramp

Ramp: triangle base = 2.4 m, triangle height = 0.3 m, prism width = 1.5 m.
(a) Labelled diagram: a triangular prism lying on its rectangular face, with the triangle at one end showing $b = 2.4$ m (horizontal) and $h = 0.3$ m (vertical rise), and the width of the ramp $L = 1.5$ m shown along the depth.
(b) Volume of concrete:

Step 1: Base area $= \dfrac{b \times h}{2} = \dfrac{2.4 \times 0.3}{2} = \dfrac{0.72}{2} = 0.36\ \text{m}^2$

Step 2: $V = A \times L = 0.36 \times 1.5 = 0.54\ \text{m}^3$

(c) Bags needed (0.2 m^3 each; round up):

$0.54 \div 0.2 = 2.7 \rightarrow$ round up $\rightarrow$ 3 bags

Question 4 — Gift box packaging

Small box: 20 cm × 15 cm × 10 cm. Large box: 35 cm × 25 cm × 15 cm.
(a) Volume of each box:

$V_{\text{small}} = 20 \times 15 \times 10 = 3\,000\ \text{cm}^3$

$V_{\text{large}} = 35 \times 25 \times 15 = 13\,125\ \text{cm}^3$

(b) Small boxes fitting inside one large box (volume method; round down):

$13\,125 \div 3\,000 = 4.375 \rightarrow$ 4 small boxes

(c) Can 500 small boxes fit in the storage room (4 m × 3 m × 2.4 m)?

Room volume $= 4 \times 3 \times 2.4 = 28.8\ \text{m}^3 = 28\,800\,000\ \text{cm}^3$

500 small boxes $= 500 \times 3\,000 = 1\,500\,000\ \text{cm}^3$

$28\,800\,000 > 1\,500\,000$ ✓

Yes — 500 small boxes require only about 5.2% of the room's volume.

Question 5 — Rectangular water tank (farm)

Tank: $L = 3$ m, $W = 2$ m, $H = 1.5$ m.
(a) Volume:

$$V = 3 \times 2 \times 1.5 = 9 \text{ m}^3$$

(b) Convert to litres:

$$9 \times 1000 = 9\,000 \text{ L}$$

(c) Tank currently 40% full:

$$0.40 \times 9\,000 = 3\,600 \text{ L currently in the tank}$$

(d) Time to fill from current level at 50 L/min:

Water needed $= 9\,000 - 3\,600 = 5\,400$ L

Time $= 5\,400 \div 50 = 108$ minutes

Question 6 — Comparing storage containers

Option A: 6 m × 3 m × 2.5 m, \$85/week. Option B: 5 m × 4 m × 2.8 m, \$90/week.
(a) Volume of each container:

$V_A = 6 \times 3 \times 2.5 = 45 \text{ m}^3$

$V_B = 5 \times 4 \times 2.8 = 56 \text{ m}^3$

(b) Greater volume: **Option B (56 m^3)**
(c) Cost per cubic metre per week:

Option A: \$85 ÷ 45 ≈ \$1.89 per m^3/week

Option B: \$90 ÷ 56 ≈ \$1.61 per m^3/week

(d) Better value: **Option B** — lower cost per cubic metre (\$1.61 vs \$1.89) and larger total volume. Unless the business only needs 45 m^3 or less, Option B is the better choice.

Chapter Review — Worked Solutions

Review Question 1 — Identify the shape

5 faces, 9 edges, 6 vertices.
Triangular prism — base shape is a **triangle**.

Review Question 2 — Octagonal prism faces

An octagon has $n = 8$ sides.

$$\text{Faces} = n + 2 = 8 + 2 = 10 \text{ faces}$$

(2 octagonal bases + 8 rectangular side faces)

Review Question 3 — Calculate volumes

(a) Rectangular prism: $L = 9$ cm, $W = 6$ cm, $H = 4$ cm

$$V = 9 \times 6 \times 4 = 216 \text{ cm}^3$$

(b) Rectangular prism: $L = 5$ m, $W = 2.5$ m, $H = 0.15$ m

$$V = 5 \times 2.5 \times 0.15 = 12.5 \times 0.15 = 1.875 \text{ m}^3$$

(c) Triangular prism: $b = 10$ cm, $h = 6$ cm, $L = 15$ cm

$$\text{Base area} = \frac{10 \times 6}{2} = 30 \text{ cm}^2$$

$$V = 30 \times 15 = 450 \text{ cm}^3$$

(d) Triangular prism: $b = 4$ m, $h = 3$ m, $L = 7$ m

$$\text{Base area} = \frac{4 \times 3}{2} = 6 \text{ m}^2$$

$$V = 6 \times 7 = 42 \text{ m}^3$$

Review Question 4 — Unit conversions

$$2.4 \text{ m}^3 = 2.4 \times 1\,000\,000 = 2\,400\,000 \text{ cm}^3$$

$$15\,000 \text{ cm}^3 = 15\,000 \div 1000 = 15 \text{ L}$$

Review Question 5 — Rectangular fish tank

Tank: 90 cm × 40 cm × 45 cm.

(a) Volume:

$$V = 90 \times 40 \times 45 = 3600 \times 45 = 162\,000 \text{ cm}^3$$

(b) Convert to litres:

$$162\,000 \div 1000 = 162 \text{ L}$$

(c) Fill to 80%:

$$0.80 \times 162 = 129.6 \text{ L}$$

Review Question 6 — Volume vs. Capacity

Volume is the amount of three-dimensional space a solid object occupies, measured in cubic units (cm^3, m^3). *Workplace example:* the volume of a concrete block tells you how much material is in it.
Capacity is the amount a hollow container can hold, measured in litres or millilitres. *Workplace example:* the capacity of a chemical storage drum tells you how much liquid it can safely contain.
Key link: For a container, its capacity equals its internal volume ($1\ cm^3 = 1$ mL; 1 L $= 1000\ cm^3$).

Multi-Step Problems — Worked Solutions

Multi-Step Problem 7 — Concrete pour (warehouse floor)

Floor: 20 m × 15 m, slab thickness = 120 mm.

(a) Convert thickness to metres:

$$120 \text{ mm} = 0.12 \text{ m}$$

(b) Volume of concrete:

$V = 20 \times 15 \times 0.12 = 300 \times 0.12 = 36 \text{ m}^3$

(c) Truck loads (6 m^3 each; round up):

$36 \div 6 = 6$ truck loads (exactly 6 — no rounding needed)

(d) Cost excluding GST (\$195/m^3):

$36 \times \$195 = \$7\,020$

(e) Total cost including GST:

$\$7\,020 \times 1.10 = \$7\,722.00$

(f) Time to pour at 3 m^3/hour:

$36 \div 3 = 12$ hours

(g) Total labour cost (4 workers, 12 hours, \$38/hr each):

Labour $= 4 \times 12 \times \$38 = 48 \times \$38 = \$1\,824.00$

Summary:

Item	Amount
Volume of concrete	36 m^3
Truck loads	6
Concrete cost (excl. GST)	\$7,020.00
Concrete cost (incl. GST)	\$7,722.00
Time for pour	12 hours
Labour cost	\$1,824.00

Multi-Step Problem 8 — Warehouse loading

Storage space: 12 m × 8 m × 4 m.
Type X box: 60 cm × 50 cm × 40 cm.
Type Y box: 80 cm × 60 cm × 50 cm.

(a) Volume of storage space:

$V_{\text{room}} = 12 \times 8 \times 4 = 384 \text{ m}^3$

(b) Volume of one Type X box:

$V_X = 60 \times 50 \times 40 = 120\,000 \text{ cm}^3$
Convert: $120\,000 \div 1\,000\,000 = 0.12 \text{ m}^3$

(c) Volume of one Type Y box:

$0.8 \text{ m} \times 0.6 \text{ m} \times 0.5 \text{ m} = 0.24 \text{ m}^3$

(d) Maximum Type X boxes (full storage):

$384 \div 0.12 = 3\,200$ Type X boxes

(e) Maximum Type Y boxes (full storage):

$384 \div 0.24 = 1\,600$ Type Y boxes

(f) Mixed scenario — half each (192 m^3 each):

Type X in half: $192 \div 0.12 = 1\,600$ Type X
Type Y in half: $192 \div 0.24 = 800$ Type Y

(g) Total value of stock in mixed scenario:

Type X value: $1\,600 \times \$240 = \$384\,000$
Type Y value: $800 \times \$185 = \$148\,000$
Total: \$532 000

Summary:

Item	Type X	Type Y
Box volume	0.12 m^3	0.24 m^3
Max (full room)	3,200 boxes	1,600 boxes
Mixed (half each)	1,600 boxes	800 boxes
Value per box	\$240	\$185
Value (mixed)	\$384,000	\$148,000
Total value (mixed)	**\$532,000**	

Chapter 11 — Key Answers at a Glance

Practice — Part A
Q1: (a) Rect. prism (b) Tri. prism (c) Cylinder (d) Cube
Q2: Rect. prism 6/12/8; Tri. prism 5/9/6; Cube 6/12/8
Q3: 8 faces, 18 edges, 12 vertices

Practice — Part B
Q4: 60 cm^3; 48 m^3; 420 cm^3; 0.75 m^3; 15 000 cm^3
Q5: Height = 10 cm

Practice — Part C
Q6(a): 120 cm^3 (b): 180 m^3 (c): 1120 cm^3

Practice — Part D
Q7: 5 000 000 cm^3; 8 L; 2500 mL; 3000 L; 0.45 m^3
Q8: 160 L

Application
Q1: 5.04 m^3; \$932.40; \$1025.64
Q2: 33.14 m^3; 2.16 m^3; 15 pallets
Q3: 0.54 m^3; 3 bags
Q4: 3000/13125 cm^3; 4 small; Yes fits
Q5: 9 m^3; 9000 L; 3600 L; 108 min
Q6: 45/56 m^3; B larger; A \$1.89; B \$1.61; B better value

Chapter Review
Q1: Triangular prism, triangle base
Q2: 10 faces
Q3: 216 cm^3; 1.875 m^3; 450 cm^3; 42 m^3
Q4: 2 400 000 cm^3; 15 L
Q5: 162 000 cm^3; 162 L; 129.6 L

Multi-Step 7: 36 m^3; 6 trucks; \$7020; \$7722; 12 h; \$1824
Multi-Step 8: 384 m^3; X=0.12 m^3, Y=0.24 m^3;
X max 3200, Y max 1600;
Mixed: 1600 X + 800 Y = \$532 000

12. Mass, Capacity and Temperature

Chapter 12 Information

Week taught:	Week 13 (Term 2)
Topic:	Topic 2 — Workplace and Employment
Skills area:	Measurement & Space
Lesson reference:	Lessons 49–52

Teaching Guide

Key concepts: Mass unit conversions (mg, g, kg, t); capacity unit conversions (mL, L, kL, ML); temperature (reading thermometers, interpreting temperature data in workplace contexts including food safety).

Suggested delivery sequence:

1. Lesson 49: Mass conversions; weighing in workplace contexts.
2. Lesson 50: Capacity conversions; liquid measurements in industry.
3. Lesson 51: Temperature; reading and comparing temperatures; workplace applications.
4. Lesson 52: Worked Examples; Practice and Application Exercises.

Common Student Misconceptions

Common Misconceptions — Chapter 12

1. **Mass vs. weight:** Students treat mass and weight as identical. *Strategy:* In everyday Australian usage these are interchangeable; note mass is measured in kg.

2. **Metric prefix confusion — kilo vs. mega:** Students convert kL to L by multiplying by 1,000,000. *Strategy:* Use a consistent prefix chart: kilo = ×1,000, mega = ×1,000,000.

3. **Capacity in formulas — using wrong unit:** Students quote 1.5 L correctly but then use 1.5 where mL is needed. *Strategy:* Identify the required unit before substituting.

4. **Negative temperatures:** Students write negative Celsius values as impossible. *Strategy:* Use a thermometer diagram showing below-zero; connect to cold-storage contexts.

5. **Unit conversion direction:** Students convert g to kg by multiplying by 1,000. *Strategy:* Moving up the size ladder = divide; down = multiply.

Worked Solutions

Chapter 12 Overview for Teachers

Chapter:	12 — Mass, Capacity and Temperature
Week:	Week 13, Term 1 (Lessons 49–52)
Topic:	Topic 2 — Workplace and Employment
Focus:	Metric mass (mg/g/kg/t); metric capacity (mL/L/kL/ML); reading thermometers; Australian food safety temperature guidelines

Key rules:

Moving **down** the ladder (to a smaller unit) $\Rightarrow \times 1000$

Moving **up** the ladder (to a larger unit) $\Rightarrow \div 1000$

Cold zone: $<5\,°\text{C}$ Danger zone: $5\,°\text{C} - 60\,°\text{C}$ Hot zone: $>60\,°\text{C}$

Capacity links: $1\text{ cm}^3 = 1\text{ mL}$; $1000\text{ cm}^3 = 1\text{ L}$; $1\text{ m}^3 = 1\text{ kL} = 1000\text{ L}$

Pedagogical notes:

- The conversion ladder applies identically to both mass and capacity — reinforce this parallel structure.
- Two-step conversions (e.g. kg → mg) should be shown with intermediate working.
- For food safety, students must know the exact boundaries: $< 5\,°\text{C}$ and $> 60\,°\text{C}$. "Approximately" is not acceptable in the exam.
- The 2-hour rule: food in the danger zone for more than 2 hours must be discarded.

Practice Exercise — Worked Solutions

Part A — Mass Conversions

Question 1 — Mass conversion table

Complete the table — each row has one known value; derive the rest using ×1000 or ÷1000.

	mg	g	kg	t
(a)	**500 000**	500	**0.5**	**0.0005**
(b)	**3 200 000**	**3200**	3.2	**0.0032**
(c)	2000	**2**	**0.002**	**0.000002**
(d)	**500 000 000**	**500 000**	**500**	0.5
(e)	**8 500 000**	8500	**8.5**	**0.0085**
(f)	**250 000**	**250**	0.25	**0.00025**

Teacher note: For rows (a) and (f), expect students to focus on the two adjacent columns. Award full marks for any correctly derived pair.

Question 2 — Convert each mass

(a) 4.7 kg × 1000 = 4700 g

(b) 12 500 g ÷ 1000 = 12.5 kg

(c) 3 t × 1000 = 3000 kg

(d) 650 mg ÷ 1000 = 0.65 g

(e) Two steps: 2.1 t $\xrightarrow{\times 1000}$ 2100 kg $\xrightarrow{\times 1000}$ 2 100 000 g

(f) Two steps: 0.008 kg $\xrightarrow{\times 1000}$ 8 g $\xrightarrow{\times 1000}$ 8000 mg

Part B — Capacity Conversions

Question 3 — Capacity conversion table

	mL	L	kL	ML
(a)	**4500**	4.5	**0.0045**	**0.0000045**
(b)	750	**0.75**	**0.00075**	**0.00000075**
(c)	**2 800 000**	**2800**	2.8	**0.0028**
(d)	**3 000 000**	**3000**	**3**	0.003
(e)	**12 000 000**	12 000	**12**	**0.012**
(f)	6 500 000	**6500**	**6.5**	**0.0065**

Question 4 — Convert each capacity

(a) 3.6 L × 1000 = 3600 mL

(b) 4200 mL ÷ 1000 = 4.2 L

(c) 5 kL × 1000 = 5000 L

(d) 0.75 L × 1000 = 750 mL

(e) 8 ML × 1000 = 8000 kL

(f) 3500 L ÷ 1000 = 3.5 kL

Part C — Choosing Appropriate Units

Question 5 — Appropriate units

(a) Tablet of paracetamol **mg**

(b) Fuel tank of a family car **L**

(c) Loaded semi-trailer truck **t**

(d) Volume of water in a swimming pool **kL**

(e) Bag of potatoes at the supermarket **kg**

(f) Amount of medicine in a syringe **mL**

(g) Water storage in a large dam **ML**

(h) Single grain of rice **mg**

Part D — Reading Thermometers

Question 6 — Thermometer readings

The mercury column height in each diagram indicates the temperature:

(a)	(b)	(c)
45°C	**2°C**	**65°C**

Question 7 — Temperature zones

(a) 45 °C — **danger zone** (5 °C to 60 °C). Food is unsafe at this temperature.

(b) 2 °C — **cold zone** (below 5 °C). Cold food is safe.

(c) 65 °C — **hot zone** (above 60 °C). Hot food is safe.

Part E — Food Safety

Question 8 — Food safety table

Food item	Recorded	Required	Safe?
Hot chips (takeaway display)	58°C	> 60°C	**No**
Yoghurt (fridge display)	4°C	< 5°C	**Yes**
Roast chicken (warming oven)	65°C	> 60°C	**Yes**
Cream cake (counter display)	12°C	< 5°C	**No**

Question 9 — Actions for unsafe items

Hot chips (58°C): **Reheat immediately to above 60°C before serving. Do not serve until the correct temperature is confirmed.**
Cream cake (12°C): **Remove from display and return to the refrigerator immediately. Do not serve until temperature drops below 5°C. If it has been above 5°C for more than 2 hours, discard it.**

Question 10 — The danger zone explained

The danger zone is the temperature range between 5°C and 60°C. In this range, bacteria multiply rapidly — some types can double in number every 20 minutes. Food left in the danger zone for more than 2 hours may contain dangerous levels of bacteria and must be discarded. Keeping food either below 5°C (cold storage) or above 60°C (hot holding) prevents bacterial growth.

Application Exercise — Worked Solutions

Question 1 — Pharmacy dosage

Patient: 500 mg twice a day for 10 days. Each tablet = 250 mg.

(a) Daily dose:

$$500 \times 2 = 1000 \text{ mg per day}$$

(b) Total over 10 days:

$$1000 \times 10 = 10\,000 \text{ mg}$$

(c) Convert to grams:

$$10\,000 \div 1000 = 10 \text{ g}$$

(d) Number of tablets:

$$10\,000 \div 250 = 40 \text{ tablets}$$

Question 2 — Freight payload

Maximum payload: 4.5 t. Load: 36 boxes at 85 kg each.

(a) Convert maximum to kg:

$$4.5 \times 1000 = 4500 \text{ kg}$$

(b) Total load:

$$36 \times 85 = 3060 \text{ kg}$$

(c) Is total load within limit?

$4500 - 3060 = 1440$ kg remaining capacity

Yes — the load is within the payload limit with **1440 kg** to spare.

(d) Extra boxes at 65 kg each:

$$1440 \div 65 = 22.15\ldots \rightarrow \text{round down} \rightarrow 22 \text{ additional boxes}$$

Question 3 — Catering: scaling a recipe

Original: 4 servings, 600 g pumpkin, 800 mL stock. Target: 60 servings.

(a) Scaling factor:

$$60 \div 4 = 15$$

(b) Pumpkin needed:

$$600 \times 15 = 9000 \text{ g}$$

(c) Convert to kilograms:

$$9000 \div 1000 = 9 \text{ kg}$$

(d) Stock needed:

$$800 \times 15 = 12\,000 \text{ mL}$$

(e) Convert to litres:

$$12\,000 \div 1000 = 12 \text{ L}$$

(f) 2-litre cartons needed (round up):

$$12 \div 2 = 6 \text{ cartons (exactly 6 — no rounding required)}$$

Question 4 — Irrigation

Pump rate: 850 L per minute.

(a) Litres per hour:

$$850 \times 60 = 51\,000 \text{ L per hour}$$

(b) Convert to kilolitres:

$$51\,000 \div 1000 = 51 \text{ kL}$$

(c) Minutes to pump 12 kL:

$12 \text{ kL} = 12\,000 \text{ L}$

$12\,000 \div 850 = 14.1 \text{ min} \approx 14.1 \text{ minutes}$

(d) 45 ML in litres:

$$45 \times 1\,000\,000 = 45\,000\,000 \text{ L}$$

Question 5 — Food safety inspection

Item	Temp.	Assessment and action
Prawn cocktail (display)	9°C	**Unsafe** — in danger zone. Return to fridge immediately; do not serve until below 5°C.
Beef stew (bain-marie)	63°C	**Safe** — above 60°C.
Chocolate mousse (fridge)	3°C	**Safe** — below 5°C.
Fried rice (warming lamp)	52°C	**Unsafe** — in danger zone. Reheat to above 60°C immediately.
Grilled fish (service pass)	74°C	**Safe** — above 60°C.

Question 6 — Cold chain temperature log

(a) Completed table:

Time	Truck temp.	Within safe cold range (< 5°C)?
06:00 (departure)	2°C	**Yes**
08:30	4°C	**Yes**
10:15	7°C	**No**
11:45	6°C	**No**
13:00 (arrival)	4°C	**Yes**

(b) Times outside safe cold range: **10:15 and 11:45** (approximately 10:15 am to 13:00 — about 2 hours 45 minutes above 5°C).
(c) Since the goods were in the danger zone for approximately 2.5 hours, which exceeds the 2-hour limit under the Australian Food Standards Code, **the delivery must be discarded**. The supermarket should refuse the delivery and the driver must report the temperature breach to the supplier.

Chapter Review — Worked Solutions

Review Question 1 — Mass conversions

(a) 6.4 kg × 1000 = 6400 g

(b) 900 mg ÷ 1000 = 0.9 g

(c) 7500 kg ÷ 1000 = 7.5 t

(d) 0.35 t × 1000 = 350 kg

(e) Two steps: 2.5 kg $\xrightarrow{\times 1000}$ 2500 g $\xrightarrow{\times 1000}$ 2 500 000 mg

Review Question 2 — Capacity conversions

(a) 2.8 L × 1000 = 2800 mL

(b) 5600 mL ÷ 1000 = 5.6 L

(c) 4 kL × 1000 = 4000 L

(d) 0.6 ML × 1000 = 600 kL

(e) 3200 L ÷ 1000 = 3.2 kL

Review Question 3 — Appropriate units

(a) Volume of a household water tank **kL**

(b) Mass of a vitamin C tablet **mg**

(c) Capacity of an IV drip bag in hospital **mL**

(d) Mass of a shipping container of steel **t**

Review Question 4 — Coolroom temperature (14°C)

(a) Is 14°C in the danger zone?
Yes — 14 °C is between 5 °C and 60 °C, so it is in the danger zone.
(b) Safe temperature for cold food:
Below 5°C
(c) Action required:
The coolroom must be checked and repaired immediately. The cooked chicken should be moved to a functioning refrigerator below 5°C. If the chicken has been above 5°C for more than 2 hours, it must be discarded.

Review Question 5 — Baker: flour for 24 loaves

750 g per loaf; 5 kg bags.

(a) Total flour (grams):

$$750 \times 24 = 18\,000 \text{ g}$$

(b) Convert to kilograms:

$$18\,000 \div 1000 = 18 \text{ kg}$$

(c) Bags needed (round up):

$$18 \div 5 = 3.6 \rightarrow \text{round up} \rightarrow 4 \text{ bags}$$

Review Question 6 — Sports drink: 18 players

750 mL per player; drinks sold in 1.5 L bottles.

(a) Total mL:

$$750 \times 18 = 13\,500 \text{ mL}$$

(b) Convert to litres:

$$13\,500 \div 1000 = 13.5 \text{ L}$$

(c) Bottles needed:

$$13.5 \div 1.5 = 9 \text{ bottles (exactly 9 — no rounding)}$$

Multi-Step Problems — Worked Solutions

Multi-Step Problem 7 — Hospital kitchen (120 patients)

Each patient: 200 mL soup, 150 g bread roll.

(a) Total soup (mL):

$$120 \times 200 = 24\,000 \text{ mL}$$

(b) Convert to litres:

$$24\,000 \div 1000 = 24 \text{ L}$$

(c) Pots needed (8 L each; round up):

$$24 \div 8 = 3 \text{ pots (exactly 3)}$$

(d) Total mass of bread rolls (grams):

$$120 \times 150 = 18\,000 \text{ g}$$

(e) Convert to kilograms:

$$18\,000 \div 1000 = 18 \text{ kg}$$

(f) Trays needed (12 rolls per tray):

$$120 \div 12 = 10 \text{ trays}$$

(g) Soup at 68°C:

Yes, safe. 68 °C is above 60 °C (the hot zone minimum), so the soup is being held at a safe temperature.

(h) Leftover soup: 3.5 L at 45°C:

The soup is currently in the danger zone (45°C is between 5°C and 60°C). To safely refrigerate and reuse it, it must first be rapidly cooled to below 5°C (using an ice bath or blast chiller). If the soup has already been in the danger zone for more than 2 hours in total (including service time), it must be discarded.

Summary:

Item	Amount
Total soup volume	24,000 mL = 24 L
Pots required	3 pots (8 L each)
Total bread mass	18,000 g = 18 kg
Bread trays	10 trays (12 rolls each)
Soup temperature	68°C — safe (hot zone)

Multi-Step Problem 8 — Water tank management (farm)

Tank A: 4 m × 2.5 m × 2 m. Tank B: 3 m × 3 m × 1.8 m.

(a) Volume of each tank:

$V_A = 4 \times 2.5 \times 2 = 20\ \text{m}^3$

$V_B = 3 \times 3 \times 1.8 = 16.2\ \text{m}^3$

(b) Convert to litres (1 m^3 = 1000 L):

$$V_A = 20\,000\ \text{L} \qquad V_B = 16\,200\ \text{L}$$

(c) Combined capacity in kilolitres:

$(20\,000 + 16\,200) = 36\,200\ \text{L} \div 1000 = 36.2\ \text{kL}$

(d) Current water levels:

Tank A (60% full): $0.60 \times 20\,000 = 12\,000$ L

Tank B (35% full): $0.35 \times 16\,200 = 5\,670$ L

(e) Time to fill both tanks from current levels (pump: 120 L/min):

Tank A needs: $20\,000 - 12\,000 = 8\,000$ L

Tank B needs: $16\,200 - 5\,670 = 10\,530$ L

Total needed: $8\,000 + 10\,530 = 18\,530$ L

Time $= 18\,530 \div 120 = 154.4$ min

$= 154$ min = 2 hours 34 minutes (round up to ensure full)

(f) Days current supply lasts (4500 L/day):

Current total: $12\,000 + 5\,670 = 17\,670$ L

$17\,670 \div 4\,500 = 3.93$ days $\approx$ 3 full days (with partial 4th day)

Teacher note: Accept 3 days or 3.9 days; emphasise that on the 4th day the tanks will not have a full day's supply.

Summary:

Item	Tank A	Tank B
Volume (m^3)	20 m^3	16.2 m^3
Volume (litres)	20,000 L	16,200 L
Current level	12,000 L (60%)	5,670 L (35%)
Litres to fill	8,000 L	10,530 L
Combined capacity	36.2 kL	
Time to fill	2 h 34 min	
Days supply remains	$\approx$ 3.9 days	

Chapter 12 — Key Answers at a Glance

Practice — Part A (Mass)
Q1: See completed table above
Q2: 4700 g; 12.5 kg; 3000 kg; 0.65 g; 2 100 000 g; 8000 mg

Practice — Part B (Capacity)
Q3: See completed table above
Q4: 3600 mL; 4.2 L; 5000 L; 750 mL; 8000 kL; 3.5 kL

Practice — Part C (Units)
Q5: mg; L; t; kL; kg; mL; ML; mg

Practice — Part D (Thermometers)
Q6: 45°C; 2°C; 65°C
Q7: Danger; Cold; Hot

Practice — Part E (Food safety)
Q8: No; Yes; Yes; No
Q9: Reheat chips >60°C; Return cake to fridge

Application
Q1: 1000 mg/day; 10 000 mg; 10 g; 40 tablets
Q2: 4500 kg; 3060 kg; Yes, 1440 kg spare; 22 more boxes
Q3: Factor 15; 9000 g; 9 kg; 12 000 mL; 12 L; 6 cartons
Q4: 51 000 L/hr; 51 kL; 14.1 min; 45 000 000 L
Q5: Prawns unsafe; Stew safe; Mousse safe; Rice unsafe; Fish safe
Q6: Unsafe at 10:15 & 11:45; discard delivery

Chapter Review
Q1: 6400 g; 0.9 g; 7.5 t; 350 kg; 2 500 000 mg
Q2: 2800 mL; 5.6 L; 4000 L; 600 kL; 3.2 kL
Q3: kL; mg; mL; t
Q4: Yes danger zone; <5°C; cool/discard if >2 h
Q5: 18 000 g; 18 kg; 4 bags
Q6: 13 500 mL; 13.5 L; 9 bottles

MS7: 24 000 mL; 24 L; 3 pots; 18 000 g; 18 kg; 10 trays
MS8: A 20 m^3/B 16.2 m^3; 36.2 kL combined;
A 12 000 L; B 5 670 L; fill in 2 h 34 min; 3.9 days supply

13. Workplace Measurement Mixed

Chapter 13 Information

Week taught:	Week 14 (Term 2)
Topic:	Topic 2 — Workplace and Employment
Skills area:	Measurement & Space
Lesson reference:	Lessons 53–56

Teaching Guide

Key concepts: Mixed measurement in workplace scenarios; volume unit conversions revisited; reading maps, coordinates and scale in workplace contexts; selecting appropriate units for workplace tasks.

Suggested delivery sequence:

1. Lesson 53: Review of all measurement topics from Chapters 11–12; mixed conversion practice.
2. Lesson 54: Volume unit conversions extended; capacity and volume relationships (1 L = 1000 cm^3).
3. Lesson 55: Workplace maps — scale, coordinates, distances in workplace settings.
4. Lesson 56: Worked Examples; mixed Practice and Application Exercises.

Common Student Misconceptions

Common Misconceptions — Chapter 13

1. **Volume–capacity relationship:** Students do not connect that 1 cm^3 = 1 mL. *Strategy:* Physically demonstrate with a 1 cm cube and 1 mL of water.

2. **Wrong formula for wrong quantity:** Students apply a volume formula to an area question. *Strategy:* Identify what is being asked (flat surface → area; 3D space → volume) before selecting a formula.

3. **Map scale — not converting:** Students report the ruler measurement as the answer. *Strategy:* Always multiply map distance by the scale factor to get real distance.

4. **Inappropriate units:** Students express a 2 km distance as 2,000,000 mm. *Strategy:* Choose the unit that gives a number between 1 and 10,000 for practical purposes.

5. **Multi-step substitution errors:** Students substitute the wrong value when multiple measurements are given. *Strategy:* List given values and label them before substituting.

Worked Solutions

Chapter 13 Overview for Teachers

Chapter:	13 — Workplace Measurement
Week:	Week 14, Term 1 (Lessons 53–56)
Topic:	Topic 2 — Workplace and Employment
Focus:	Volume unit conversions ($mm^3/cm^3/m^3$); reading floor plans using scale; interpreting delivery maps using compass directions and distance

Key formulas:

$$1\ cm^3 = 1000\ mm^3 \qquad 1\ m^3 = 1\,000\,000\ cm^3 \qquad 1\ m^3 = 1000\ L$$

$$\text{Real length} = \text{plan measurement} \times \text{scale factor}$$

Pedagogical notes:

- Volume conversions use $\times/\div$ 1000 ($mm^3 \leftrightarrow cm^3$) and $\times/\div$ 1 000 000 ($cm^3 \leftrightarrow m^3$). Students frequently confuse these with the length ladder ($\times 10$, $\times 100$).
- Floor plan and map questions require the student to measure the printed diagram. The answers below are based on the stated plan dimensions; accept values within $\pm 5\%$ of the printed measurement.
- Route planning has no single correct answer — mark on efficiency (no unnecessary backtracking) and correct total distance.

Practice Exercise — Worked Solutions

Part A — Volume Unit Conversions

Question 1 — Convert each volume

(a) $3\ \text{cm}^3 \times 1000 = 3000\ \text{mm}^3$

(b) $12\,000\ \text{mm}^3 \div 1000 = 12\ \text{cm}^3$

(c) $2.5\ \text{m}^3 \times 1\,000\,000 = 2\,500\,000\ \text{cm}^3$

(d) $3\,400\,000\ \text{cm}^3 \div 1\,000\,000 = 3.4\ \text{m}^3$

(e) $0.008\ \text{m}^3 \times 1\,000\,000 = 8000\ \text{cm}^3$

(f) $750\ \text{cm}^3 \times 1000 = 750\,000\ \text{mm}^3$

Question 2 — Plastic component (25 mm × 15 mm × 8 mm)

(a) Volume in mm^3:

$$25 \times 15 \times 8 = 3000\ \text{mm}^3$$

(b) Convert to cm^3:

$$3000 \div 1000 = 3\ \text{cm}^3$$

Part B — Floor Plan Reading

Floor plan questions — teacher guidance

Questions 3–7 require students to refer to the **Springvale Warehouse floor plan** (Scale: 1 cm = 4 m). Answers below match the plan as drawn. Accept reasonable variations where measurement is required (±0.5 cm on the plan = ±2 m real).

Questions 3–5 — Floor plan identification

Q3. Room in the north-east section: **Staff Room**
Q4. Number of doors shown on the plan: **5 doors** (main entrance, loading dock door, Cold Store door, office corridor door, WC door — accept 4–6 depending on print quality)
Q5. Room completely enclosed within the Main Warehouse: **Supervisor Office**

Question 6 — Supervisor Office dimensions (plan 5 cm × 2 cm, scale 1 cm = 4 m)

Real width $= 5 \times 4 = 20$ m

Real depth $= 2 \times 4 = 8$ m

Floor area $= 20 \times 8 = 160\ \text{m}^2$

Question 7 — Main Office floor area (plan 5 cm × 4 cm, scale 1 cm = 4 m)

Real length = $5 \times 4 = 20$ m

Real width = $4 \times 4 = 16$ m

Floor area = $20 \times 16 = 320$ m^2

Part C — Delivery Map

Map questions — teacher guidance

Questions 8–12 require the **Springvale Delivery Map** (Scale: 1 cm = 200 m). Compass direction answers are definitive; distance answers are approximate — accept ±10%.

Question 8 — Furthest south delivery stop

School (F) — located in the south-east area of the map, furthest from the north boundary.

Question 9 — Compass directions from Warehouse (W)

(a) W → Supermarket (B): **North-west**

(b) W → Pharmacy (C): **North** (or north-east)

(c) W → School (F): **South** (or south-east)

Question 10 — Map distance table (Scale: 1 cm = 200 m)

From → To	Map dist. (cm)	Real distance (m)
W → Pharmacy (C)	≈ 3.5 **cm**	≈ 700 **m**
W → Hardware (E)	≈ 3.0 **cm**	≈ 600 **m**
Supermarket (B) → Pharmacy (C)	≈ 3.3 **cm**	≈ 660 **m**

Teacher note: Values depend on the printed map. Multiply student's measured cm by 200 to check.

Question 11 — Efficient route: W → C → E → W

Most efficient route: W → C → E → W (visit both stops on the northern/eastern side before returning)

W → C (Pharmacy): ≈ 700 m
C → E (Hardware): ≈ 800 m
E → W: ≈ 600 m
Total ≈ 2100 m

Accept any efficient two-stop route; deduct marks only if the route involves clear unnecessary backtracking.

Question 12 — Route: Café (D) to School (F)

From the Café (D), head east along Oak Street to the Warehouse intersection, then turn south-east along Commerce Avenue to reach the School (F).
Accept any route description that correctly names streets and uses appropriate compass directions. Mark for logical sequence and correct directional language.

Application Exercise — Worked Solutions

Question 1 — Volume conversions (construction)

(a) $4800\ \text{mm}^3 \div 1000 = 4.8\ \text{cm}^3$
(b) $0.015\ \text{m}^3 \times 1\,000\,000 = 15\,000\ \text{cm}^3$
(c) $3.6\ \text{m}^3 \times 1000 = 3600\ \text{L}$
(d) $750\,000\ \text{cm}^3 \div 1\,000\,000 = 0.75\ \text{m}^3$

Question 2 — Floor plan: material quantities (Scale 1 cm = 4 m)

(a) Cold Store floor area (plan 4 cm × 4 cm):

Real: $4 \times 4 = 16$ m each side
Area $= 16 \times 16 =$ 256 m^2

(b) Floor tiles (30 cm × 30 cm):

Convert area: $256\ \text{m}^2 = 256 \times 10\,000 = 2\,560\,000\ \text{cm}^2$
Each tile: $30 \times 30 = 900\ \text{cm}^2$
Tiles needed: $2\,560\,000 \div 900 = 2844.4\ldots \rightarrow$ round up $\rightarrow$ 2845 tiles

(c) Rack sections across Main Warehouse width (real 44 m, each rack 2.4 m wide):

$44 \div 2.4 = 18.3\ldots \rightarrow$ round down $\rightarrow$ 18 rack sections

(d) Building perimeter (plan ≈ 16 cm × 12 cm):

Real: $16 \times 4 = 64$ m; $12 \times 4 = 48$ m
Perimeter $= 2 \times (64 + 48) = 2 \times 112 =$ 224 m

Question 3 — Full delivery route (Scale 1 cm = 200 m)

(a) Sample efficient route (no backtracking):
W → D → B → C → E → F → W
(Accept any route that visits all 5 stops without unnecessary retracing.)
(b) Total distance estimate:

Leg	Map (cm)	Real (m)
W → D (Café)	≈ 4.2	≈ 840
D → B (Supermarket)	≈ 3.3	≈ 660
B → C (Pharmacy)	≈ 3.3	≈ 660
C → E (Hardware)	≈ 4.0	≈ 800
E → F (School)	≈ 5.0	≈ 1000
F → W (return)	≈ 4.2	≈ 840
Total	≈ 24.0	≈ 4800 m

(c) Convert to kilometres:

$$4800 \div 1000 = 4.8 \text{ km}$$

(d) Driving time at 30 km/h:

$$\text{Time} = \frac{4.8 \text{ km}}{30 \text{ km/h}} = 0.16 \text{ h} = 0.16 \times 60 = 9.6 \text{ min} \approx 10 \text{ minutes}$$

(e) Total run time including delivery stops (5 stops × 15 min):

Delivery time: $5 \times 15 = 75$ min
Driving time: ≈ 10 min
Total ≈ 85 minutes

Question 4 — Scale drawing: office layout (Scale 1:50)

Office: 8 m × 5 m real.

(a) 8 m wall on plan (1:50 ⇒ divide real by 50):

$$8 \text{ m} = 800 \text{ cm} \div 50 = 16 \text{ cm}$$

(b) 5 m wall on plan:

$$5 \text{ m} = 500 \text{ cm} \div 50 = 10 \text{ cm}$$

(c) Real floor area:

$$8 \times 5 = 40 \text{ m}^2$$

(d) Desk: plan 4 cm × 2 cm; real dimensions (×50):

$$4 \times 50 = 200 \text{ cm} = 2.0 \text{ m} \qquad 2 \times 50 = 100 \text{ cm} = 1.0 \text{ m}$$

(e) Desk as percentage of office floor area:

Desk area $= 2.0 \times 1.0 = 2 \text{ m}^2$

$$\frac{2}{40} \times 100 = 5\%$$

Question 5 — Warehouse pallet storage (44 m × 38 m)

Each pallet: 1.2 m × 1.0 m.

(a) Pallets along 44 m (each 1.2 m wide):

$$44 \div 1.2 = 36.67\ldots \rightarrow \text{round down} \rightarrow 36 \text{ pallets}$$

(b) Pallets along 38 m (each 1.0 m deep):

$$38 \div 1.0 = 38 \text{ pallets}$$

(c) Maximum ground-floor pallets:

$$36 \times 38 = 1368 \text{ pallets}$$

(d) After reserving 25% for access aisles (75% usable):

$$1368 \times 0.75 = 1026 \text{ pallets}$$

Chapter Review — Worked Solutions

Section A — 3D Shapes and Volume (Chapter 11 revision)

Review Question 1 — Volume calculations

(a) Rectangular prism: $L = 7$ m, $W = 4$ m, $H = 2.5$ m

$$V = 7 \times 4 \times 2.5 = 28 \times 2.5 = 70 \text{ m}^3$$

(b) Triangular prism: $b = 6$ cm, $h = 4$ cm, $L = 9$ cm

Base area $= \dfrac{6 \times 4}{2} = 12 \text{ cm}^2$

$V = 12 \times 9 = 108 \text{ cm}^3$

Review Question 2 — Shipping box volume in cm^3

Box: $0.8 \text{ m} \times 0.6 \text{ m} \times 0.4 \text{ m}$

$V = 0.8 \times 0.6 \times 0.4 = 0.192 \text{ m}^3$

Convert: $0.192 \times 1\,000\,000 = 192\,000 \text{ cm}^3$

Section B — Mass, Capacity and Temperature (Chapter 12 revision)

Review Question 3 — Conversions

(a) $3.5 \text{ kg} \times 1000 = 3500 \text{ g}$

(b) $2800 \text{ mL} \div 1000 = 2.8 \text{ L}$

(c) $4 \text{ kL} \times 1000 = 4000 \text{ L}$

(d) $600 \text{ mg} \div 1000 = 0.6 \text{ g}$

(e) $1.2 \text{ t} \times 1000 = 1200 \text{ kg}$

Review Question 4 — Coolroom at 28°C (cooked ham)

Yes — 28 °C is between 5 °C and 60 °C, so it is in the **danger zone**.
Action required: Move the ham immediately to a functioning refrigerator below 5 °C. If the ham has been above 5 °C for more than 2 hours, it must be discarded. The coolroom must be inspected and repaired before further use.

Section C — Volume Unit Conversions

Review Question 5 — Conversions

(a) $4\ \text{cm}^3 \times 1000 = 4000\ \text{mm}^3$

(b) $9\ \text{m}^3 \times 1\,000\,000 = 9\,000\,000\ \text{cm}^3$

(c) $500\,000\ \text{cm}^3 \div 1\,000\,000 = 0.5\ \text{m}^3$

Section D — Floor Plan and Map

Review Question 6 — Staff Room area (plan 2 cm × 6 cm, scale 1 cm = 4 m)

Real: $2 \times 4 = 8$ m; $6 \times 4 = 24$ m

Area $= 8 \times 24 = 192\ \text{m}^2$

Review Question 7 — Loading dock wall

The south wall. Vehicles enter from the road on the south side of the building.

Review Question 8 — Café (D) to Supermarket (B): map distance

From the Delivery Map, D → B:

Map measurement: ≈ 3.3 cm

Real distance $= 3.3 \times 200 = \approx 660$ m

Review Question 9 — Route directions: W → F → E → C → W

(1) W → F (School): **south** (or south-east)

(2) F → E (Hardware): **north** (or north-west)

(3) E → C (Pharmacy): **north-west** (or west)

(4) C → W: **south** (or south-west)

Multi-Step Problems — Worked Solutions

Multi-Step Problem 10 — Cold Store fit-out (16 m × 16 m)

(a) Floor area:

$$16 \times 16 = 256 \text{ m}^2$$

(b) Shelving units along the 16 m wall (each unit 1.5 m wide):

$16 \div 1.5 = 10.67\ldots \rightarrow$ round down $\rightarrow$ 10 shelving units

Note: The access aisle (0.9 m) runs in front of each row — it affects the depth used, not the number along the wall. The question asks how many fit along the 16 m wall at 1.5 m each.

(c) Volume of one box (40 cm × 30 cm × 25 cm):

$$40 \times 30 \times 25 = 30\,000 \text{ cm}^3$$

(d) Total boxes in Cold Store:

10 units × 8 shelves × 12 boxes = 10 × 96 = 960 boxes

(e) Temperature log:

Time	Temperature	Safe (< 5°C)?
08:00	3°C	**Yes**
10:00	4.5°C	**Yes**
12:00	6°C	**No — danger zone**
14:00	5.5°C	**No — danger zone**

Temperatures were in the danger zone at 12:00 (6°C) and 14:00 (5.5°C). Action: Immediately check and repair the cold store refrigeration. Move all perishable goods to a working cold store below 5°C. Any food that has been in the danger zone for more than 2 hours must be discarded.

Summary:

Item	Answer
Floor area	256 m^2
Shelving units	10
Volume per box	30,000 cm^3
Total boxes	960
Danger zone times	12:00 and 14:00

Multi-Step Problem 11 — Full delivery run (all 5 stops)

(a) Planned route and total distance:
Sample route: **W → D → B → C → E → F → W**
Total estimated distance: ≈ 4800 **m** = 4.8 **km** (based on map measurements; see Application Q3)
(b) Cargo weights — does total exceed 1 tonne (1000 kg)?

Stop	Boxes	kg/box	Total (kg)
B — Supermarket	24	18	**432**
C — Pharmacy	6	8	**48**
D — Café	10	12	**120**
E — Hardware	15	22	**330**
F — School	8	6	**48**
Total			**978 kg**

978 kg < 1000 kg **No, the load does not exceed 1 tonne. It is 22 kg under the 1-tonne limit.**
(c) Time to complete the run:

Route distance: ≈ 4.8 km at 25 km/h

Driving time $= \frac{4.8}{25} \times 60 = 11.5$ min ≈ 12 min

Unloading time: 5 stops × 20 min = 100 min

Total time = 100 + 12 = 112 min = 1 hour 52 min

Departs 08:00 → returns ≈ **09:52 AM**

Chapter 13 — Key Answers at a Glance

Practice — Part A (Volume conversions)
Q1: 3000 mm^3; 12 cm^3; 2 500 000 cm^3; 3.4 m^3; 8000 cm^3; 750 000 mm^3
Q2: 3000 mm^3; 3 cm^3

Practice — Part B (Floor plan)
Q3: Staff Room
Q4: ≈ 5 doors
Q5: Supervisor Office
Q6: 20 m × 8 m = 160 m^2
Q7: 320 m^2

Practice — Part C (Delivery map)
Q8: School (F) — furthest south
Q9: NW; N; S
Q10: ≈700 m; ≈600 m; ≈660 m
Q11: ≈2100 m total (W→C→E→W)

Application
Q1: 4.8 cm^3; 15 000 cm^3; 3600 L; 0.75 m^3
Q2: 256 m^2; 2845 tiles; 18 racks; 224 m perimeter
Q3: ≈4800 m; 4.8 km; ≈10 min driving; 85 min total
Q4: 16 cm; 10 cm; 40 m^2; 2.0 m × 1.0 m; 5%
Q5: 36 along; 38 deep; 1368 max; 1026 with aisles

Chapter Review
Q1: 70 m^3; 108 cm^3
Q2: 192 000 cm^3
Q3: 3500 g; 2.8 L; 4000 L; 0.6 g; 1200 kg
Q4: Yes danger zone; move below 5°C / discard if >2 h
Q5: 4000 mm^3; 9 000 000 cm^3; 0.5 m^3
Q6: 192 m^2; Q7: South wall; Q8: ≈660 m

MS10: 256 m^2; 10 units; 30 000 cm^3/box; 960 boxes;
danger zone at 12:00 and 14:00
MS11: 978 kg (under 1 t by 22 kg); returns ≈09:52

14. Data at Work

Chapter 14 Information

Week taught:	Week 15 (Term 2)
Topic:	Topic 2 — Workplace and Employment
Skills area:	Statistics & Probability
Lesson reference:	Lessons 57–60

Teaching Guide

Key concepts: Collecting and displaying workplace data; constructing and interpreting bar graphs, line graphs and tables; calculating mean, median and mode for workplace data; identifying trends; critically evaluating data representations; paper spreadsheets.

Suggested delivery sequence:

1. Lesson 57: Workplace data collection; frequency tables and two-way tables.
2. Lesson 58: Constructing and interpreting graphs for workplace data.
3. Lesson 59: Mean, median, mode; which average is most appropriate?
4. Lesson 60: Worked Examples; Practice and Application Exercises.

Common Student Misconceptions

Common Misconceptions — Chapter 14

1. **Mean — omitting zeros from count:** Students treat zero values as non-existent. *Strategy:* Zero is a value; it counts toward the total number of data points.

2. **Always using the mean:** Students report the mean even when data is heavily skewed. *Strategy:* Mean for symmetric data; median when outliers are present; mode for categorical.

3. **Two-way tables — wrong row or column:** Students sum the wrong row/column for a total. *Strategy:* Label rows and columns clearly; trace before reading.

4. **Trend vs. highest point:** Students describe a trend as "it peaked at Week 6" rather than "it increased over Weeks 1–6." *Strategy:* A trend describes the *direction* of change over a period.

5. **Truncated y-axes:** Students accept a graph with a truncated y-axis as showing a large change. *Strategy:* Always check whether the y-axis starts at zero.

Worked Solutions

Chapter 14 Overview for Teachers

Chapter:	14 — Data at Work
Week:	Week 15, Term 2 (Lessons 57–60)
Topic:	Topic 2 — Workplace and Employment
Focus:	Recording data in tables; constructing and reading bar graphs; comparing supplier tables; paper spreadsheets with totals

Key skills:

- Two-column table: title, column headers with units, accurate data, totals row
- Bar graph checklist — **SALT**: Scale, Axes labelled, Legend, Title
- Spreadsheet: `=B#*C#` for row totals; `=SUM(D2:D#)` for column totals
- Supplier comparison: calculate item totals, sum, compare, write a recommendation
- Trend identification: increasing / decreasing / stable; use figures to justify

Pedagogical notes:

- Graph-drawing questions (Q8, App Q2, Review Q2, MS7d) are assessed on SALT compliance and scale choice, not artistic precision. Award full marks if bars are correctly proportioned and all four SALT elements are present.
- Mean calculations should be rounded to 1 decimal place unless a whole number results exactly.
- Supplier comparison conclusions must cite specific dollar figures to earn full marks on written justification questions.

Practice Exercise — Worked Solutions

Part A — Completing Tables

Question 1 — Supermarket customers per hour

Completed table:

Springvale Supermarket — Customers Served per Hour

Hour	Customers served
8 am	**24**
9 am	**38**
10 am	**45**
11 am	**52**
12 pm	**61**
1 pm	**48**
2 pm	**36**
Total	**304**

(a) Busiest hour: **12 pm** (61 customers)

(b) Quietest hour: **8 am** (24 customers)

(c) Difference: $61 - 24 = 37$ more customers at 12 pm than 8 am

(d) Mean: $304 \div 7 = 43.4$ customers per hour (1 d.p.)

Question 2 — Springvale Bakery weekly sales revenue

Completed table:

Springvale Bakery — Weekly Sales Revenue

Day	Food sales ($)	Drink sales ($)
Monday	620	185
Tuesday	540	162
Wednesday	680	204
Thursday	595	178
Friday	820	246
Saturday	1140	342
Sunday	980	294
Total	**$5375**	**$1611**

(a) Food total: $620 + 540 + 680 + 595 + 820 + 1140 + 980 = \$5375$
Drink total: $185 + 162 + 204 + 178 + 246 + 342 + 294 = \$1611$

(b) Highest food sales: **Saturday** ($1140)

(c) Saturday drink sales as fraction of Saturday food sales:

$$\frac{342}{1140} = \frac{3}{10}$$

$\frac{3}{10}$ (simplify: $342 \div 114 = 3$; $1140 \div 114 = 10$)

(d) Drinks as percentage of total weekly revenue:

Total revenue $= \$5375 + \$1611 = \$6986$
$\frac{1611}{6986} \times 100 = 23.1\%$ (1 d.p.)

Part B — Reading a Bar Graph

Questions 3–7 — Call Centre daily call volume (Mon 148, Tue 162, Wed 135, Thu 175, Fri 190)

Q3. Day with most calls: **Friday** (190 calls)
Q4. Total across the week:

$$148 + 162 + 135 + 175 + 190 = 810 \text{ calls}$$

Q5. Friday minus Wednesday:

$$190 - 135 = 55 \text{ more calls}$$

Q6. Mean calls per day:

$$810 \div 5 = 162 \text{ calls/day}$$

Q7. Total call time on busiest day (Friday: 190 calls, 4 min each):

$$190 \times 4 = 760 \text{ minutes}$$

Part C — Drawing a Bar Graph

Question 8 — Bar graph of bakery drink sales

Teacher guidance: Students draw a bar graph of the 7-day drink sales data (Mon \$185, Tue \$162, Wed \$204, Thu \$178, Fri \$246, Sat \$342, Sun \$294).
SALT marking guide:

- **S**cale — vertical axis from \$0 to at least \$350, consistent intervals (e.g. \$50 or \$100)
- **A**xes labelled — horizontal: "Day"; vertical: "Drink Sales (\$)"
- **L**egend — not required for a single-variable graph
- **T**itle — e.g. "Springvale Bakery — Weekly Drink Sales"

Bar heights (answers): Mon \$185, Tue \$162, Wed \$204, Thu \$178, Fri \$246, Sat \$342, Sun \$294. Saturday should be the tallest bar; Tuesday the shortest.

Part D — Spreadsheet

Question 9 — Café weekly stock order spreadsheet

Row	A: Item	B: Qty	C: Unit price ($)	D: Line total ($)
2	Coffee beans (kg)	4	28.50	**$114.00**
3	Full cream milk (2L)	15	3.40	**$51.00**
4	Almond milk (L)	8	3.80	**$30.40**
5	Sugar (kg)	3	2.60	**$7.80**
6	Paper cups (50-pk)	6	9.20	**$55.20**
7	**TOTAL**	—	—	**$258.40**

Working:

$4 \times \$28.50 = \$114.00$; $15 \times \$3.40 = \$51.00$; $8 \times \$3.80 = \$30.40$

$3 \times \$2.60 = \$7.80$; $6 \times \$9.20 = \$55.20$

Total $= 114.00 + 51.00 + 30.40 + 7.80 + 55.20 = \$258.40$

Question 10 — Cost per cup of coffee

Coffee bean cost (from Q9): $114.00 for 4 kg, which makes 300 coffees

Cost per cup $= \$114.00 \div 300 = \$0.38$ per cup

Application Exercise — Worked Solutions

Question 1 — Pharmacy weekly stock count

Completed table:

Springvale Pharmacy — Weekly Stock Count

Item	Units in stock
Paracetamol 500mg (boxes)	**48**
Ibuprofen 200mg (boxes)	**32**
Antihistamine tablets (boxes)	**19**
Antiseptic cream (tubes)	**27**
Bandages (rolls)	**55**
Hand sanitiser (bottles)	**63**
Total	**244**

(a) Lowest stock: **Antihistamine tablets** (19 units)

(b) Items below reorder threshold of 20 units: **Antihistamine tablets** (19 units — only item below 20)

(c) Hand sanitiser stock value:

$$63 \times \$6.80 = \$428.40$$

Question 2 — School canteen sandwich sales bar graph

Data: Mon 38, Tue 44, Wed 52, Thu 48, Fri 65
Teacher guidance: Students draw a bar graph from this data.
SALT guide: Scale 0–70 (intervals of 10); horizontal axis "Day"; vertical axis "Sandwiches Sold"; title "Daily Sandwich Sales — School Canteen"; no legend needed.

(a) Total sandwiches: $38 + 44 + 52 + 48 + 65 = 247$ sandwiches

(b) Mean daily sales:

$$247 \div 5 = 49.4 \text{ sandwiches/day}$$

(c) Recommendation: **Prepare the most sandwiches on Friday.** Friday had the highest sales (65), which is 25% more than the weekly mean. The data shows a clear upward trend through the week, so Fridays consistently see the highest demand.

Question 3 — Supplier comparison: Fresh Direct QLD vs Metro Produce Co.

Item	Supplier A total	Supplier B total
Tomatoes (10 kg)	$10 \times \$3.20 = \$32.00$	$10 \times \$3.60 = \$36.00$
Lettuce (8 heads)	$8 \times \$2.80 = \$22.40$	$8 \times \$2.50 = \$20.00$
Capsicum (5 kg)	$5 \times \$4.50 = \$22.50$	$5 \times \$4.20 = \$21.00$
Mushrooms (4 kg)	$4 \times \$8.40 = \$33.60$	$4 \times \$7.80 = \$31.20$
Avocados (12)	$12 \times \$1.90 = \$22.80$	$12 \times \$2.10 = \$25.20$
Total	**\$133.30**	**\$133.40**

(a) See calculations in table above.

(b) **Supplier A (Fresh Direct QLD) is cheaper by \$0.10 per order.**

(c) **The owner might still choose Supplier B for reasons such as fresher produce, faster delivery, better reliability, an established relationship, or superior quality on specific items (e.g. lettuce and avocados are individually cheaper from Supplier B).** Accept any reasonable, specific reason.

Question 4 — Springvale Café staffing spreadsheet

Completed spreadsheet (column D — Gross pay):

Row	A: Staff member	B: Hours	C: Rate ($/hr)	D: Gross pay ($)
2	Anika	18	26.80	**$482.40**
3	Brennan	22	26.80	**$589.60**
4	Carla	15	24.80	**$372.00**
5	Diego	25	24.80	**$620.00**
6	Elena	12	26.80	**$321.60**
7	**TOTAL**	**92**	—	**$2385.60**

Working:

$18 \times 26.80 = \$482.40$; $22 \times 26.80 = \$589.60$; $15 \times 24.80 = \$372.00$

$25 \times 24.80 = \$620.00$; $12 \times 26.80 = \$321.60$

Total hours $= 18 + 22 + 15 + 25 + 12 = 92$

Total pay $= 482.40 + 589.60 + 372.00 + 620.00 + 321.60 = \$2385.60$

(c) Most hours worked: **Diego** (25 hours)

(d) Total weekly wages bill: **$2385.60**

(e) Mean hours per staff member: $92 \div 5 = 18.4$ hours

Question 5 — Coolroom temperature log

Completed table:

Springvale Restaurant — Coolroom Temperature Log

Time	Temperature (°C)	Within safe range (< 5°C)?
06:00	3	**Yes**
08:00	4	**Yes**
10:00	6	**No — danger zone**
12:00	5	**No — not below 5°C**
14:00	7	**No — danger zone**
16:00	4	**Yes**
18:00	3	**Yes**

(b) Times outside safe range: **10:00 (6°C), 12:00 (5°C), and 14:00 (7°C).**

(c) **The supervisor should: move all perishable items to a working cold store below 5°C; investigate and repair the coolroom refrigeration; discard any food that has been above 5°C for more than 2 hours; log the incident.**

(d) **Line graph guidance:** Horizontal axis "Time" (06:00–18:00, 2-hour intervals); vertical axis "Temperature (°C)" (0–10°C); plot points and join with lines; draw a dashed horizontal line at 5°C labelled "Safety limit". Title: "Coolroom Temperature Log".

Question 6 — Product X and Y sales trends

Data: Product X: 240, 255, 270, 285, 298, 310 (Jan–Jun)
Product Y: 180, 165, 148, 132, 120, 105 (Jan–Jun)

(a) Trend for Product X: **Steadily increasing.** Sales rise from 240 in January to 310 in June — an increase of 70 units (about 29%) over 6 months.

(b) Trend for Product Y: **Steadily decreasing.** Sales fall from 180 in January to 105 in June — a decline of 75 units (about 42%) over 6 months.

(c) Recommendation: **Order more of Product X in July.** Product X has increased every month and shows a consistent upward trend. Product Y has declined every month and at its current rate will continue to fall. The manager should increase Product X stock and consider reducing or discontinuing Product Y unless there is a seasonal explanation for the decline.

Chapter Review — Worked Solutions

Review Question 1 — Maintenance requests

Completed table:

Day	Requests
Monday	**8**
Tuesday	**12**
Wednesday	**6**
Thursday	**15**
Friday	**11**
Total	**52**

(b) Day with most requests: **Thursday** (15 requests)

(c) Mean requests per day: $52 \div 5 = 10.4$ requests/day

Review Question 2 — Bar graph of maintenance requests

Teacher guidance: Students draw a bar graph from the data in Q1 (Mon 8, Tue 12, Wed 6, Thu 15, Fri 11).
SALT guide: Scale 0–16 (intervals of 2 or 4); horizontal axis "Day"; vertical axis "Number of Requests"; title "Daily Maintenance Requests"; Thursday bar should be tallest, Wednesday shortest.

Review Question 3 — Safety stock value spreadsheet

Row	A: Product	B: Units	C: Unit price ($)	D: Total value ($)
2	Safety helmets	25	18.50	**$462.50**
3	Hi-vis vests	40	12.80	**$512.00**
4	Safety boots	15	85.00	**$1275.00**
5	Work gloves	60	9.40	**$564.00**
6	**TOTAL**	**140**	—	**$2813.50**

Working:

$25 \times 18.50 = \$462.50; \quad 40 \times 12.80 = \$512.00$

$15 \times 85.00 = \$1275.00; \quad 60 \times 9.40 = \$564.00$

Total units $= 25 + 40 + 15 + 60 = 140$

Total value $= 462.50 + 512.00 + 1275.00 + 564.00 = \$2813.50$

Review Question 4 — Total stock value

$2813.50 (from Q3 spreadsheet total)

Review Question 5 — Item with highest total stock value

Safety boots — total value $1275.00. Although work gloves have the highest unit count (60), safety boots have the highest total value because each pair costs $85.00.

Review Question 6 — Office supply quotes comparison

Item	Quote A total	Quote B total
A4 paper (5 reams)	$5 \times \$6.40 = \$32.00$	$5 \times \$5.90 = \$29.50$
Pens (3 boxes)	$3 \times \$8.80 = \$26.40$	$3 \times \$9.60 = \$28.80$
Folders (10)	$10 \times \$2.20 = \$22.00$	$10 \times \$2.50 = \$25.00$
Staples (4 boxes)	$4 \times \$4.60 = \$18.40$	$4 \times \$4.10 = \$16.40$
Total	**$98.80**	**$99.70**

(a) See calculations in table above.

(b) **Quote A is cheaper by $0.90.** (Quote B − Quote A = $99.70−$98.80 = $0.90)

Multi-Step Problems — Worked Solutions

Multi-Step Problem 7 — Sports store two-week performance

Completed totals row:

Day	Week 1 ($)	Week 2 ($)
Monday	1240	1380
Tuesday	980	1050
Wednesday	1120	1290
Thursday	1350	1420
Friday	1680	1750
Saturday	2450	2680
Sunday	2180	2340
Total	**$11 000**	**$11 910**

(a) See totals in table above.

(b) Week 2 had higher total revenue:

$$\$11\,910 - \$11\,000 = \$910 \text{ more in Week 2}$$

(c) Mean daily revenue:

Week 1: $\$11\,000 \div 7 = \$1571.43/\text{day}$
Week 2: $\$11\,910 \div 7 = \$1701.43/\text{day}$

(d) **Grouped bar graph guidance:** Horizontal axis "Day" (Mon–Sun); vertical axis "Revenue ($)" (0–3000, intervals of $500); draw two bars per day (different shading); legend showing Week 1 and Week 2. Week 2 bar should be slightly taller than Week 1 on every day.

(e) Overall trend: **Week 2 revenue was higher than Week 1 on every day of the week. The grouped bar graph shows a consistent upward shift across all seven days.**

(f) **Yes, the data supports the manager's claim. Week 2 total revenue ($11 910) exceeded Week 1 ($11 000) by $910 (an 8.3% increase). Every single day in Week 2 recorded higher revenue than the same day in Week 1, with Saturday showing the largest gain ($230) and Tuesday the smallest ($70). The consistent day-by-day improvement suggests genuine revenue growth rather than a one-off spike.**

Multi-Step Problem 8 — Café monthly performance report

Completed Table 1 — Revenue by category:

Category	Week 1	Week 2	Week 3	Week 4
Hot drinks	1240	1380	1190	1450
Cold drinks	620	750	580	820
Food	2840	3120	2680	3380
Weekly total	**\$4700**	**\$5250**	**\$4450**	**\$5650**

Completed Table 2 — Staff wages:

Staff	Hours/week	Rate (\$/hr)	Weekly wage (\$)
Manager	38	34.50	**\$1311.00**
Barista	30	27.80	**\$834.00**
Wait staff (3 people)	20 each	24.80	**\$1488.00**
Total weekly wages			**\$3633.00**

Wait staff working: $3 \times 20 = 60$ hours; $60 \times \$24.80 = \$1488.00$

(a) See completed tables above.

(b) Total monthly revenue:

$$\$4700 + \$5250 + \$4450 + \$5650 = \$20\,050$$

(c) Total monthly wages (4 weeks):

$$4 \times \$3633.00 = \$14\,532.00$$

(d) Monthly profit:

Total costs $= \$14\,532 + \$3200 = \$17\,732$

Profit $= \$20\,050 - \$17\,732 = \$2318.00$

(e) Highest food revenue: **Week 4 (\$3380).** This may be explained by a weekend event, a public holiday falling in that week, or end-of-month spending behaviour. Accept any reasonable contextual explanation.

(f) **Sample monthly summary:**

The café generated \$20 050 in total revenue for the month, with food sales contributing the largest share in all four weeks. Week 4 was the strongest week (\$5650), driven by the highest food revenue

recorded ($3380). Total wages for the month were $14 532, and after adding $3200 in other costs, the café made a monthly profit of $2318. Revenue varied between weeks — Week 3 was the weakest ($4450) — suggesting the café should investigate whether staffing levels and stock ordering can be adjusted to reflect these fluctuations.

Chapter 14 — Key Answers at a Glance

Practice — Part A

Q1: Total 304; busiest 12 pm; quietest 8 am; 37 more; mean 43.4

Q2: Food $5375; drinks $1611; Sat highest; $\frac{3}{10}$; drinks 23.1%

Practice — Part B (Call centre)

Q3: Friday (190)

Q4: 810 calls

Q5: 55 more on Friday

Q6: 162 calls/day

Q7: 760 minutes

Practice — Part D

Q9: $114.00; $51.00; $30.40; $7.80; $55.20; Total $258.40

Q10: $0.38 per cup

Application

Q1: Total 244; Antihistamine lowest (19); reorder Antihistamine; $428.40

Q2: 247 sandwiches; mean 49.4; recommend Friday

Q3: Supplier A $133.30; Supplier B $133.40; A cheaper by $0.10

Q4: $482.40; $589.60; $372.00; $620.00; $321.60; total $2385.60; Diego most hrs; mean 18.4

Q5: Danger zone at 10:00, 12:00, 14:00

Q6: X increasing; Y decreasing; recommend order more X

Chapter Review

Q1: Total 52; Thursday most (15); mean 10.4

Q3: $462.50; $512.00; $1275.00; $564.00; total $2813.50

Q4: $2813.50; Q5: Safety boots

Q6: Quote A $98.80; Quote B $99.70; A cheaper by $0.90

MS7: W1 $11 000; W2 $11 910; diff $910; means $1571/$1701

MS8: Revenue $20 050; wages $14 532; profit $2318

15. Probability and Trends at Work

Chapter 15 Information

Week taught:	Week 16 (Term 2)
Topic:	Topic 2 — Workplace and Employment
Skills area:	Statistics & Probability
Lesson reference:	Lessons 61–64

Teaching Guide

Key concepts: Probability in workplace contexts (risk, quality control, defect rates); calculating probability using the standard formula; expressing probability in fractions, decimals and percentages; identifying and interpreting trends; making predictions from data trends.

Suggested delivery sequence:

1. Lesson 61: Probability in the workplace — risk assessment, quality control, insurance contexts.
2. Lesson 62: Calculating probability; listing outcomes in workplace scenarios.
3. Lesson 63: Trends in data — identifying, describing, and predicting from trends.
4. Lesson 64: Worked Examples; Practice and Application Exercises; begin AT2 preparation.

Common Student Misconceptions

Common Misconceptions — Chapter 15

1. **Probability vs. certainty:** Students treat "30% probability" as meaning exactly 30 events in every 100 must occur. *Strategy:* Probability is a measure of *likelihood*, not a guarantee.

2. **Over-extrapolation of trends:** Students extend a trend line far beyond the data range without qualification. *Strategy:* Predictions become less reliable the further they are from the data; note assumptions made.

3. **Single dip interpreted as trend reversal:** Students see one data point drop and conclude the trend has changed direction. *Strategy:* Look at the overall direction across all data points.

4. **Counting order-independent outcomes as distinct:** Students list "A fails, B passes" and "B passes, A fails" as two separate outcomes. *Strategy:* Clarify whether order matters before listing outcomes.

5. **Percentage rate written as probability without converting:** Students write probability of failure as 5 (from "5% failure rate") instead of 0.05. *Strategy:* Always divide percentage by 100 before using in a probability context.

Worked Solutions

Chapter 15 Overview for Teachers

Chapter:	15 — Probability and Trends at Work
Week:	Week 16, Term 2 (Lessons 61–64)
Topic:	Topic 2 — Workplace and Employment
Focus:	Defect-rate probability; expected number of defects; random selection; reading and describing line graph trends; writing conclusions; making recommendations

Key formulas:

$$\Pr(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}} \qquad \text{Expected defects} = \Pr(\text{defect}) \times \text{total produced}$$

Pedagogical notes:

- Probability answers should be given in all three forms (fraction, decimal, percentage) unless the question specifies otherwise.
- "Expected" means the long-run average — stress that actual results in a single trial will vary. See Review Q6.
- Line graph conclusions must cite specific values from the graph. Penalise vague statements such as "it went up" without quoting figures.
- Multi-step written conclusions are assessed on: correct identification of trend, at least one specific figure quoted, and a logical recommendation.

Practice Exercise — Worked Solutions

Part A — Probability Calculations

Question 1 — Coastal Cannery (250 tins checked; 10 faulty seals)

(a)

$$\Pr(\text{faulty seal}) = \frac{10}{250} = \frac{1}{25}$$

Decimal: $1 \div 25 = 0.04$ Percentage: $0.04 \times 100 = 4\%$

Fraction (simplified): $\frac{1}{25}$ **Decimal: 0.04** **Percentage: 4%**

(b) On average, 4 out of every 100 tins selected at random will have a faulty seal. In other words, there is a 4% chance of picking a defective tin.

Question 2 — Expected faulty tins in 2000-tin shift

Expected defects = Pr(defect) × total tins

$= 0.04 \times 2000 = 80$ faulty tins

Expected faulty tins: 80

Question 3 — Brisbane call centre (12 staff: 5 full-time, 7 part-time)

(a) Pr(full-time selected):

$$\Pr(\text{full-time}) = \frac{5}{12} \approx 0.42 \approx 41.7\%$$

Fraction: $\frac{5}{12}$ **Decimal (2 d.p.): 0.42** **Percentage: 41.7%**

(b) Pr(NOT full-time selected):

$$\Pr(\text{not full-time}) = \frac{7}{12} \approx 0.58 \approx 58.3\%$$

Fraction: $\frac{7}{12}$ **Decimal (2 d.p.): 0.58** **Percentage: 58.3%**

(c) Check: $\frac{5}{12} + \frac{7}{12} = \frac{12}{12} = 1$ ✓

Question 4 — Café casual roster (5 weekdays)

$$\Pr(\text{Wednesday}) = \frac{1}{5} = 0.2 = 20\%$$

Fraction: $\frac{1}{5}$ **Decimal: 0.2** **Percentage: 20%**

Question 5 — Quality audit bag (20 items: 8 pass, 7 minor repair, 5 scrap)

(a) $\Pr(\text{pass}) = \frac{8}{20} = \frac{2}{5}$

(b) $\Pr(\text{scrap}) = \frac{5}{20} = \frac{1}{4}$ Percentage: 25%

(c) $\Pr(\text{NOT scrapped}) = \frac{20-5}{20} = \frac{15}{20} = \frac{3}{4}$

Part B — Reading a Line Graph

Questions 6–10 — Metro Call Centre monthly calls 2024

Graph data: Jan 420, Feb 440, Mar 480, Apr 460, May 500, Jun 520, Jul 490, Aug 510, Sep 530, Oct 550, Nov 490, Dec 460
Q6. Highest call volume: **October** (550 calls)
Q7. Lowest call volume: **January** (420 calls)
Q8. Calls in March: **480 calls**
Q9. Increase from January to October:

$$550 - 420 = 130 \text{ calls}$$

Q10. Complete the sentence: **generally increasing** (call volumes generally trended upward over 2024 despite month-to-month variation; the year ends lower than the October peak but still above January)

*"Overall, the number of customer calls at Metro Call Centre during 2024 was **generally increasing**, with some variation from month to month."*

Application Exercise — Worked Solutions

Question 1 — Northside Bakery quality control (480 loaves/day, 12 defective)

(a) Defect rate:

$$\Pr(\text{defect}) = \frac{12}{480} = \frac{1}{40}$$

Decimal: $1 \div 40 = 0.025$ Percentage: $0.025 \times 100 = 2.5\%$

Fraction: $\frac{1}{40}$ **Decimal:** **0.025** **Percentage:** **2.5%**

(b) Expected defects at 720 loaves:

$0.025 \times 720 =$ 18 defective loaves

(c) Weekly defect cost (normal production 480/day, 5-day week):

Daily defects: $480 \times 0.025 = 12$ loaves per day

Weekly defects: $12 \times 5 = 60$ loaves

Weekly cost: $60 \times \$3.50 = \$210.00$

Question 2 — Sunrise Café random rostering (15 casuals: 6 Fri, 5 Sat, 4 Sun)

(a) Pr(Saturday staff member selected):

$$\Pr(\text{Saturday}) = \frac{5}{15} = \frac{1}{3} \approx 0.33 \approx 33.3\%$$

Fraction: $\frac{1}{3}$ **Decimal (2 d.p.):** **0.33** **Percentage:** **33.3%**

(b) Pr(selected staff member is NOT rostered on Sunday):

$$\Pr(\text{not Sunday}) = \frac{15-4}{15} = \frac{11}{15} \approx 0.73 \approx 73.3\%$$

Fraction: $\frac{11}{15}$ **Decimal (2 d.p.):** **0.73** **Percentage:** **73.3%**

Question 3 — Greenthumb Garden Centre seasonal sales trend

Graph data: Jan \$22k, Feb \$20k, Mar \$28k, Apr \$38k, May \$52k, Jun \$65k, Jul \$70k, Aug \$68k, Sep \$60k, Oct \$45k, Nov \$32k, Dec \$25k

(a) Peak revenue: **July — \$70 000**

(b) Lowest revenue: **February — \$20 000**

(c) The data shows a clear seasonal pattern. Revenue is lowest in summer (Dec–Feb) and rises steadily through autumn, peaking in winter (June–July at \$65k–\$70k). This is typical for a Queensland garden centre — the cooler winter months are the main gardening season, while the hot, wet summer reduces outdoor activity. Revenue then falls again through spring and into summer.

(d) Model recommendation: The data shows that revenue peaks in June and July, so I recommend the garden centre begin significantly increasing stock orders in April to ensure full shelves during the winter peak. Additional casual staff should be rostered from May through August. Stock levels and staffing can be reduced in October to manage costs during the low-revenue summer months.

Question 4 — Medical centre waiting times line graph

Data: Jan 24, Feb 22, Mar 19, Apr 15, May 13, Jun 11 (minutes)

(a) Line graph guidance: Horizontal axis "Month" (Jan–Jun); vertical axis "Average Wait Time (min)", scale 0–25 (intervals of 5); plot 6 points and join with straight lines; title "Eastside Medical Centre — Average Patient Waiting Time (Jan–Jun)". The line should show a clear downward slope from upper-left to lower-right.

(b) Trend: **Steadily decreasing.** Waiting times fell consistently every month from 24 minutes in January to 11 minutes in June — a reduction of 13 minutes.

(c) Model conclusion:

The data shows that average patient waiting times at Eastside Medical Centre have decreased steadily over the six months since the new appointment system was introduced. Between January and June, waiting times fell from 24 minutes to 11 minutes — a reduction of 13 minutes (approximately 54%). The lowest waiting time was recorded in June (11 minutes), suggesting the system's benefits continue to improve. This indicates the new appointment system is working effectively, and I recommend that management continue with the system and set a target of reducing waiting times below 10 minutes by the end of the year.

Question 5 — Garden hose stock ordering (Aug–Nov, 2023 vs 2024)

Sales data:

Month	Aug	Sep	Oct	Nov
2023 Sales (units)	45	78	120	145
2024 Sales (units)	52	90	138	162
Increase (units)	**7**	**12**	**18**	**17**

(a) November increase from 2023 to 2024: $162 - 145 = 17$ units

(b) **Predicted November 2025 sales: approximately 175–180 units.** The year-on-year increase in November has been 17 units (2023 to 2024). Projecting the same growth forward: $162 + 17 \approx 179$ units. Accept any answer in the range 175–185 with a valid explanation.

(c) **I recommend the manager place a stock order of at least 180 garden hoses before November 2025, based on the consistent year-on-year growth of approximately 17 units each November. Ordering early (by mid-October) will prevent stockouts during the peak demand period.**

Question 6 — Electronics factory defect rate trend

Completed table:

Month	Total Produced	Defective	Defect Rate (fraction)	Defect %
July	1200	36	$\frac{3}{100}$	**3%**
August	1500	30	$\frac{1}{50}$	**2%**
September	1800	18	$\frac{1}{100}$	**1%**

Working: $36 \div 1200 = 0.03 = 3\%$; $30 \div 1500 = 0.02 = 2\%$; $18 \div 1800 = 0.01 = 1\%$

(b) Trend: **The defect rate has decreased by 1 percentage point each month — from 3% in July to 2% in August to 1% in September. Quality control is improving consistently.**

(c) Predicted October defect rate: **Approximately 0% to 0.5%.** If the 1 percentage point monthly decrease continues, October would reach 0%. In practice, a floor will be reached, so 0.5% is a realistic estimate.

(d) **I recommend the factory manager continue the process improvements that drove the reduction from July to September, as they are clearly working. A formal investigation should identify which specific changes caused the improvement so they can be made permanent and applied to other production lines.**

Chapter Review — Worked Solutions

Part A — Data and Tables (Chapter 14 Revision)

Review Question 1 — Retail morning shift sales

Completed table: Clothing 24, Footwear 12, Accessories 18, Electronics 6 → **Total: 60**

(a) Total items sold: **60**

(b) Highest sales category: **Clothing** (24 items)

(c) Electronics as fraction of total: $\frac{6}{60} = \frac{1}{10}$

Review Question 2 — Workplace incidents mean

Incidents per week: 5, 3, 4, 2, 4, 3

(a) Mean over 6 weeks:

Total $= 5 + 3 + 4 + 2 + 4 + 3 = 21$

Mean $= 21 \div 6 =$ 3.5 incidents/week

(b) Mean including Week 7 (0 incidents):

New total $= 21 + 0 = 21$

New mean $= 21 \div 7 =$ 3.0 incidents/week

Part B — Probability (Chapter 15)

Review Question 3 — Distribution centre (600 orders/day, 18 picking errors)

(a) Probability of a picking error:

$$\Pr(\text{error}) = \frac{18}{600} = \frac{3}{100} = 0.03 = 3\%$$

Fraction: $\frac{3}{100}$ **Decimal:** **0.03** **Percentage:** **3%**

(b) Expected errors in 1000 orders:

$$0.03 \times 1000 = 30 \text{ expected picking errors}$$

Review Question 4 — Café team (10 employees: 4 baristas, 3 wait staff, 3 kitchen hands)

(a) Pr(barista selected):

$$\Pr(\text{barista}) = \frac{4}{10} = \frac{2}{5} = 0.4 = 40\%$$

Fraction: $\frac{2}{5}$ **Decimal:** **0.4** **Percentage:** **40%**

(b) Pr(kitchen hand is NOT selected):

$$\Pr(\text{not kitchen hand}) = \frac{10-3}{10} = \frac{7}{10} = \frac{7}{10}$$

Part C — Line Graph Trends (Chapter 15)

Review Question 5 — Sporting goods store (Jul–Dec: 120, 145, 175, 200, 230, 290)

(a) Line graph guidance: Horizontal axis "Month" (Jul–Dec); vertical axis "Units Sold", scale 0–300 (intervals of 50); title "Monthly Units Sold — Sporting Goods Store". Line should rise steeply, especially from October to December.
(b) Trend: **Strongly and consistently increasing.** Units sold rose every single month from July (120) to December (290) without any decrease.
(c) Increase from July to December: $290 - 120 = 170$ units

Review Question 6 — "Exactly 9 defects" statement

The statement is not correct. $3\% \times 300 = 9$ gives the *expected* (average) number of defects, not a guaranteed exact count. In any one specific batch, the actual number of defects could be more or fewer than 9 — it might be 7, or 11, or even 0. The calculation gives a long-run average: over many batches of 300, the average number of defects would be approximately 9. The manager should say "*we expect approximately 9 defects*", not "*we will get exactly 9.*"

Review Question 7 — Percentage vs fraction for workplace probability

Percentages are more useful in workplace reports because they are immediately understood by most workers and managers without needing to calculate. For example, saying "our defect rate is 3%" is instantly clear, whereas "$\frac{3}{100}$" requires an extra mental step to interpret. Percentages also align naturally with other business metrics such as profit margins, waste targets, and performance KPIs, making comparisons straightforward.

Multi-Step Problems — Worked Solutions

Multi-Step Problem 1 — Retail store two-year trend analysis (Jan–Jun)

Revenue data:

Month	Jan	Feb	Mar	Apr	May	Jun
2023 ($'000)	48	45	52	58	60	62
2024 ($'000)	55	52	60	66	70	75

(a) Total revenue Jan–Jun:

2023: $48 + 45 + 52 + 58 + 60 + 62 = \$325\,000$

2024: $55 + 52 + 60 + 66 + 70 + 75 = \$378\,000$

(b) Increase from 2023 to 2024: $\$378\,000 - \$325\,000 = \$53\,000$

(c) **Dual-line graph guidance:** Two lines on the same axes, Jan–Jun, revenue \$40k–\$80k; one line for 2023 (e.g. dashed), one for 2024 (solid); both lines slope upward; legend required showing which line is which; title "Retail Store Revenue Jan–Jun: 2023 vs 2024".

(d) Comparison: **Both years show a similar increasing trend from January to June. The 2024 line runs consistently above the 2023 line in every month, showing that the business is growing year-on-year. The shape of both trends is similar (both rise at roughly the same rate), but 2024 values are approximately \$7k–\$13k higher in each month.**

(e) **Model recommendation:**

The data shows that revenue has increased in both 2023 and 2024, with the January-to-June total growing from \$325,000 to \$378,000 — an increase of \$53,000 (16.3%). Both years show a consistent upward trend through the first half of the year, with the strongest month being June in both cases (\$62k rising to \$75k). This suggests strong business momentum heading into the second half of 2024. I recommend maintaining current marketing strategies and planning for higher stock levels through the second half of the year to capitalise on the growth trend.

Multi-Step Problem 2 — Sports drink bottles quality control and cost analysis

Audit: 15 defective bottles in a sample of 500.

(a) Defect rate:

$$\Pr(\text{defect}) = \frac{15}{500} = \frac{3}{100} = 0.03 = 3\%$$

Fraction: $\frac{3}{100}$ **Decimal:** **0.03** **Percentage:** **3%**

(b) Expected daily defects (3 shifts × 1200 bottles = 3600/day):

Total daily production: $3 \times 1200 = 3600$ bottles

Expected defects: $0.03 \times 3600 =$ 108 defective bottles/day

(c) Daily disposal cost:

$$108 \times \$1.85 = \$199.80 \text{ per day}$$

(d) New defect rate 1.5%; reduction in defects:

New expected defects: $0.015 \times 3600 = 54$ bottles/day

Reduction: $108 - 54 =$ 54 fewer defective bottles per day

Daily cost saving: $54 \times \$1.85 =$ \$99.90 saved per day

(e) **Model conclusion:**

The new quality control process has halved the defect rate from 3% to 1.5%, resulting in 54 fewer defective bottles per day across the three production shifts. This represents a daily cost saving of \$99.90 in disposal costs — approximately \$499.50 per standard 5-day working week. Over a year, this saving would amount to approximately \$25,974. The data strongly supports continuing and expanding the new process, and I recommend the company formally document the changes made, apply them to all production lines, and set a target of reducing the defect rate further to below 1% within the next quarter.

Chapter 15 — Key Answers at a Glance

Practice — Part A (Probability)

Q1: $\frac{1}{25}$; 0.04; 4%
Q2: 80 faulty tins
Q3: $\frac{5}{12}$, 0.42, 41.7%; $\frac{7}{12}$, 0.58, 58.3%; adds to 1
Q4: $\frac{1}{5}$; 0.2; 20%
Q5: $\frac{2}{5}$; $\frac{1}{4}$ / 25%; $\frac{3}{4}$

Practice — Part B (Line graph)

Q6: October (550) Q7: January (420)
Q8: 480 Q9: 130 calls
Q10: generally increasing

Application

Q1: $\frac{1}{40}$; 0.025; 2.5%; 18 loaves; \$210/wk
Q2: $\frac{1}{3}$, 0.33, 33.3%; $\frac{11}{15}$, 0.73, 73.3%
Q3: July peak \$70k; Feb lowest \$20k
Q4: Decreasing 24→11 min; −13 min (54%)
Q5: Nov increase 17 units; predict ≈179 (2025)
Q6: Jul 3%, Aug 2%, Sep 1%; predict Oct ≈0.5%

Chapter Review

Q1: 60 total; Clothing highest; $\frac{1}{10}$ Electronics
Q2: mean 3.5; new mean 3.0
Q3: $\frac{3}{100}$; 0.03; 3%; 30 expected errors
Q4: $\frac{2}{5}$; 0.4; 40%; $\frac{7}{10}$ not kitchen hand
Q5: 120→290; +170 units; strongly increasing
Q6: Not correct — 9 is expected, not guaranteed
Q7: Percentages easier to understand; align with business KPIs

MS1: 2023 \$325k; 2024 \$378k; increase \$53k
MS2: $\frac{3}{100}$; 0.03; 3%; 108 defects/day; \$199.80/day;
new rate 1.5% → 54 defects/day; save \$99.90/day

Assessment Task 2 — Examination

AT2 Administration Checklist

- Open-book: students may bring their student workbook, notes, and exercise books
- Time: 5 minutes perusal (no writing) + 45 minutes working time
- Calculators are permitted
- Grading: A >80%, B >60%, C >40%, D >20%, E ≥0%
- Distribute examination papers face-down; begin perusal time simultaneously for all students
- Collect all materials at the conclusion; students retain no copies

Secure Assessment Instrument

Teacher use only. The examination paper on the following pages is a secure assessment instrument. Do not distribute to students before the scheduled examination. Photocopy one paper per student immediately before the examination session.

Examination Paper

ASSESSMENT TASK 2
EXAMINATION

Senior Numeracy Short Course

Senior Short Course: Numeracy

Topic 1: Personal Identity and Community
Topic 2: Workplace and Employment

Student Name: _____ **Class:** _____

Teacher: _____ **Date:** _____

Examination Conditions and Instructions

- **Time allowed:** 5 minutes perusal + 45 minutes working time
- This is an **OPEN-BOOK** supervised examination
- **Permitted materials:** Student workbook, exercise books, personal notes, QCAA Formula Book, calculator
- **NO** assistance from other students or the teacher during working time
- Write all answers in the spaces provided in this paper
- Show **ALL** working — answers alone may not receive full marks
- If you make a mistake, cross it out neatly and rewrite your answer
- Questions sample subject matter from **BOTH** Topic 1 and Topic 2

Total: 30 marks | 20 questions | Approximate time: 1.5 minutes per mark

During perusal time: Read all questions carefully. You may **NOT** write during perusal. Plan your approach — identify questions you can answer confidently and those you should return to.

Grade cut-offs: A > 80% (25–30) | B > 60% (19–24) | C > 40% (13–18) | D > 20% (7–12) | E ≥ 0% (0–6)

Question 1 [1 mark]

A family's electricity bill shows they used 1,245 kilowatt-hours last quarter. Round this number to the nearest hundred.

Question 2 [1 mark]

Convert the fraction $\frac{3}{8}$ to a decimal.

Question 3 [2 marks]

A phone plan costs \$65 per month. During a promotional sale, the price is reduced by 20%.

(a) Calculate the discount amount.

(b) Calculate the new monthly cost.

Question 4 [2 marks]

Orange juice is sold in two sizes:

- 2 L bottle for \$3.80
- 3 L bottle for \$5.40

Which bottle is better value for money per litre? Show all working.

Question 5 [1 mark]

Convert 85 mm to centimetres.

Question 6 [2 marks]

A rectangular park is 120 m long and 85 m wide. A path runs all the way around the outside of the park. Calculate the total length of the path using the formula $P = 2L + 2W$.

Question 7 [1 mark]

A map has a scale of 1:25,000. Two locations are 6 cm apart on the map. What is the actual distance between them in kilometres?

Question 8 [2 marks]

A concert starts at 7:45 pm and ends at 10:20 pm.

(a) How long does the concert last? Give your answer in hours and minutes.

(b) Write the start time in 24-hour format.

Question 9 [1 mark]

A survey of 28 students found: 14 prefer streaming music, 8 prefer radio and 6 prefer downloading. What percentage of students prefer streaming?

Question 10 [2 marks]

A bag contains 4 red, 6 blue and 2 yellow tokens. One token is drawn at random.

(a) What is the probability of drawing a blue token? Write as a simplified fraction.

(b) What is the probability of NOT drawing a red token? Write as a simplified fraction.

Question 11 [2 marks]

Jordan works the following hours this week: Monday 4 hours, Wednesday 5.5 hours, Saturday 7 hours. His pay rate is \$23.40 per hour. Calculate his total gross pay for the week. Show all working.

Question 12 [1 mark]

A supplier charges \$840 for materials (excluding GST). Calculate the GST (10%) payable on this invoice.

Question 13 [2 marks]

A builder mixes concrete using cement, sand and gravel in the ratio 1:2:4. The project requires 14 buckets of mix in total. How many buckets of each component are needed? Show all working.

Question 14 [1 mark]

A rectangular storage container is 2.5 m long, 1.2 m wide and 1 m high. Calculate its volume using $V = L \times W \times H$.

Question 15 [1 mark]

Convert 4.8 kg to grams.

Question 16 [2 marks]

A restaurant serves 240 customers on a Saturday. Each customer drinks an average of 350 mL of water.

(a) How many litres of water are needed in total?

(b) Water is supplied in 10 L containers. How many containers must be ordered?

Question 17 [1 mark]

A warehouse floor plan has a scale of 1:100. A shelf unit measures 4.5 cm on the plan. What is the actual length of the shelf unit? Give your answer in metres.

Question 18 [2 marks]

The table below shows a retail store's daily sales for one week.

Day	**Sales**
Monday	\$2,100
Tuesday	\$1,850
Wednesday	\$2,400
Thursday	\$1,600
Friday	\$3,200
Saturday	\$3,800
Total	

(a) Complete the table by calculating the total weekly sales.

(b) On which day were sales the highest?

Question 19 [1 mark]

A factory produces 600 items per hour. Quality control data shows that 1 in every 50 items is defective. How many defective items are expected per hour?

Question 20 [2 marks]

A line graph shows a café's monthly revenue over one year. Revenue is \$12,000 in January, rises steadily through the year, reaches \$28,000 in December, with a noticeable dip to \$15,000 in July.

(a) Describe the overall trend shown in the data.

(b) The café owner wants to prepare for the July dip. Write ONE specific recommendation, and explain how the data supports it.

— END OF EXAMINATION —

Check your working. If time permits, review your answers.

Extra Working Space

ASSESSMENT TASK 2 — MARKING GUIDE EXAMINATION

Senior Numeracy Short Course

Senior Short Course: Numeracy

Topic 1: Personal Identity and Community
Topic 2: Workplace and Employment

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Mark Allocation Summary

Question	Skill / Topic	Marks	Topic
Q1	Rounding whole numbers	1	Topic 1
Q2	Converting fractions to decimals	1	Topic 1
Q3	Percentage discount (2 parts)	2	Topic 1
Q4	Unit rate / best buy comparison	2	Topic 1
Q5	Unit conversion (mm to cm)	1	Topic 1
Q6	Perimeter of rectangle	2	Topic 1
Q7	Scale and map reading	1	Topic 1
Q8	Time duration and 24-hour clock (2 parts)	2	Topic 1
Q9	Percentage from a survey	1	Topic 1
Q10	Probability as a fraction (2 parts)	2	Topic 1
	Topic 1 Subtotal	**15**	
Q11	Total hours and gross pay calculation	2	Topic 2
Q12	GST calculation	1	Topic 2
Q13	Mixing ratio problems	2	Topic 2
Q14	Volume of rectangular prism	1	Topic 2
Q15	Unit conversion (kg to g)	1	Topic 2
Q16	Capacity calculations (2 parts)	2	Topic 2
Q17	Scale drawing (workplace plan)	1	Topic 2
Q18	Reading and totalling a data table (2 parts)	2	Topic 2
Q19	Expected frequency / probability	1	Topic 2
Q20	Interpreting trends and making recommendations	2	Topic 2
	Topic 2 Subtotal	**15**	
	TOTAL	**30**	

Grade Cut-offs: A > 80% **(25–30 marks)** | B > 60% **(19–24 marks)** | C > 40% **(13–18 marks)** | D > 20% **(7–12 marks)** | E ≥ 0% **(0–6 marks)**

General Marking Notes

- This is an open-book examination — students may reference their workbooks.
- **Show all working:** Award method marks where a student demonstrates correct process even if the final answer contains an arithmetic error.
- **Rounding:** Accept answers rounded to a reasonable number of decimal places unless otherwise specified.
- **Units:** Deduct 1 mark (up to the mark limit for that part) if units are entirely absent from a measurement answer, but only penalise once per question.
- **Q20(b):** Accept any well-reasoned recommendation that is supported by the given data — see acceptable responses list.

WORKED ANSWERS

Question 1 1 mark

Round 1,245 to the nearest hundred.

The hundreds digit is 2. The tens digit is 4 (less than 5), so round **down**.

$$1{,}245 \approx 1{,}200$$

Mark: 1 mark for correct answer.

Question 2 1 mark

Convert $\frac{3}{8}$ to a decimal.

$$\frac{3}{8} = 3 \div 8 = 0.375$$

Mark: 1 mark for **0.375**.

Question 3 2 marks

Phone plan: \$65/month; promotional discount: 20%.

(a) Discount amount:

$$20\% \times \$65 = 0.20 \times 65 = \$13.00$$

1 mark for \$13.

(b) New monthly cost:

$$\$65 - \$13 = \$52.00$$

1 mark for \$52. *(Accept follow-through from (a): \$65− student's discount.)*

Question 4 2 marks

Compare price per litre:

$$2\,\text{L bottle:} \quad \frac{\$3.80}{2} = \$1.90 \text{ per litre}$$

$$3\,\text{L bottle:} \quad \frac{\$5.40}{3} = \$1.80 \text{ per litre}$$

The 3 L bottle is better value for money (\$1.80/L < \$1.90/L).

Marks: 1 mark for correctly calculating both unit rates; 1 mark for correct conclusion (3 L bottle).

Question 5 1 mark

Convert 85 mm to centimetres:

$$85 \div 10 = 8.5 \text{ cm}$$

Mark: 1 mark for **8.5 cm**.

Question 6 2 marks

Rectangular park: $L = 120\,\text{m}$, $W = 85\,\text{m}$.

$$P = 2L + 2W = 2(120) + 2(85) = 240 + 170 = 410 \text{ m}$$

Marks: 1 mark for correct substitution into formula; 1 mark for correct answer 410 m.

Question 7 1 mark

Scale 1:25,000; map distance = 6 cm.

$$\text{Actual distance} = 6 \times 25{,}000 = 150{,}000 \text{ cm} = 1{,}500 \text{ m} = 1.5 \text{ km}$$

Mark: 1 mark for **1.5 km** (or 1,500 m).

Question 8 2 marks

Concert: 7:45 pm to 10:20 pm.

(a) Duration:

$$10{:}20 - 7{:}45 = 2 \text{ hours } 35 \text{ minutes}$$

Method: 7:45 → 8:45 (1 hr) → 9:45 (2 hr) → 10:20 (+35 min). **1 mark.**

(b) Start time in 24-hour format:

$$7{:}45\,\text{pm} = 19{:}45$$

1 mark for **19:45**.

Question 9 1 mark

28 students; 14 prefer streaming.

$$\frac{14}{28} \times 100 = 50\%$$

Mark: 1 mark for **50%**.

Question 10 2 marks

Bag: 4 red, 6 blue, 2 yellow. Total = 12 tokens.

(a) P(blue):

$$P(\text{blue}) = \frac{6}{12} = \frac{1}{2}$$

1 mark for $\frac{1}{2}$ (or $\frac{6}{12}$; accept unsimplified).

(b) P(not red):

$$P(\text{not red}) = \frac{6+2}{12} = \frac{8}{12} = \frac{2}{3}$$

1 mark for $\frac{2}{3}$ (or $\frac{8}{12}$; accept unsimplified).

Topic 2 Questions begin below (Q11–Q20)

Question 11 2 marks

Jordan's hours: Mon 4 h + Wed 5.5 h + Sat 7 h. Rate: \$23.40/h.

$$\text{Total hours} = 4 + 5.5 + 7 = 16.5 \text{ hours}$$

$$\text{Gross pay} = 16.5 \times \$23.40 = \$386.10$$

Marks: 1 mark for correctly totalling hours (16.5); 1 mark for correct gross pay (\$386.10).

Question 12 1 mark

GST (10%) on \$840:

$$10\% \times \$840 = 0.10 \times 840 = \$84.00$$

Mark: 1 mark for **\$84**.

Question 13 2 marks

Concrete mix ratio: cement : sand : gravel = 1 : 2 : 4. Total mix = 14 buckets.

$$\text{Total parts} = 1 + 2 + 4 = 7 \qquad \text{Each part} = \frac{14}{7} = 2 \text{ buckets}$$

Component	Parts	Buckets
Cement	1	**2**
Sand	2	**4**
Gravel	4	**8**
Total	7	**14**

Marks: 1 mark for correct method (dividing total by 7); 1 mark for all three correct quantities (2, 4, 8).

Question 14 1 mark

Container: $L = 2.5\,\text{m}$, $W = 1.2\,\text{m}$, $H = 1\,\text{m}$.

$$V = L \times W \times H = 2.5 \times 1.2 \times 1 = 3.0\ \text{m}^3$$

Mark: 1 mark for **3 m^3** (or 3.0 m^3).

Question 15 1 mark

Convert 4.8 kg to grams:

$$4.8 \times 1{,}000 = 4{,}800\ \text{g}$$

Mark: 1 mark for **4,800 g**.

Question 16 2 marks

Restaurant: 240 customers, 350 mL each.

(a) Total water:

$$240 \times 350 = 84{,}000\,\text{mL} = 84 \text{ litres}$$

1 mark for 84 L.

(b) Containers needed (10 L each):

$$\frac{84}{10} = 8.4 \rightarrow \text{must round } \textbf{up} \rightarrow 9 \text{ containers}$$

1 mark for **9** (must round up, not down). *Accept follow-through from (a).*

Question 17 1 mark

Warehouse plan scale 1:100; shelf unit = 4.5 cm on plan.

$$4.5 \times 100 = 450\,\text{cm} = 4.5\ \text{m}$$

Mark: 1 mark for **4.5 m** (or 450 cm).

Question 18 2 marks

Retail store daily sales:

Day	**Sales**
Monday	\$2,100
Tuesday	\$1,850
Wednesday	\$2,400
Thursday	\$1,600
Friday	\$3,200
Saturday	\$3,800
Total	**\$14,950**

(a) 2,100 + 1,850 + 2,400 + 1,600 + 3,200 + 3,800 = \$14,950 **1 mark.**

(b) Highest sales day: **Saturday** (\$3,800) **1 mark.**

Question 19 1 mark

Factory produces 600 items/hour; 1 in 50 is defective.

$$\frac{600}{50} = 12 \text{ defective items per hour}$$

Mark: 1 mark for **12**.

Question 20 2 marks

Café monthly revenue: $12,000 (Jan) rising to $28,000 (Dec), with a dip to $15,000 in July.

(a) Overall trend:

The revenue shows a general upward (increasing) trend across the year, rising from $12,000 in January to $28,000 in December, with an exception (dip) in July where revenue drops to $15,000.

1 mark for identifying the overall upward trend AND acknowledging the July dip.
Accept: "The trend is generally increasing with a dip in July." Do not award mark for describing only one feature.

(b) Recommendation (1 mark for a specific, data-supported recommendation):

Acceptable responses include:

- *"Reduce stock orders for June/July as revenue drops significantly, based on the graph showing the July dip to $15,000."*
- *"Run a winter promotion or special offer in July to attract more customers, as the data shows revenue dips to its lowest point mid-year."*
- *"Save a portion of the high-revenue months (Oct–Dec) to cover operating costs during the July slowdown, as the data shows consistent mid-year dips."*
- *"Reduce casual staff hours in July to lower wage costs during the slow period, supported by the July revenue dip shown in the data."*

Do NOT award mark for: vague responses with no data reference, or responses that contradict the data.

Student Mark Recording Sheet

Q1	Q2	Q3	Q4	Q5	Q6	Q7
/1	/1	/2	/2	/1	/2	/1
Q8	**Q9**	**Q10**	**Q11**	**Q12**	**Q13**	**Q14**
/2	/1	/2	/2	/1	/2	/1
Q15	**Q16**	**Q17**	**Q18**	**Q19**	**Q20**	
/1	/2	/1	/2	/1	/2	

Topic 1 (Q1–Q10):	/15	**Topic 2 (Q11–Q20):**	/15
TOTAL MARKS:	/30		
Grade Awarded:	A B C D E		

Grade Descriptors — Assessment Task 2 (Examination)

Grade	Marks	Descriptor
A	25–30	Demonstrates thorough understanding of numeracy skills from both topics. Applies correct methods consistently, shows clear working, and communicates mathematical reasoning effectively. Minor errors only.
B	19–24	Demonstrates sound understanding of most skills across both topics. Generally applies correct methods with clear working. Some errors in more complex questions or in communicating reasoning.
C	13–18	Demonstrates satisfactory understanding of basic numeracy skills. Correct responses to straightforward questions; errors increase with question complexity. Working is sometimes incomplete or unclear.
D	7–12	Demonstrates limited understanding of numeracy skills. Correct responses to only the most straightforward questions. Working is frequently missing or incorrect.
E	0–6	Demonstrates very limited understanding. Responses are mostly incorrect or missing. Little evidence of applying numeracy skills to the given contexts.

Exam Administration Notes

- **Perusal time:** 5 minutes. Students may *not* write during perusal. Monitor closely.
- **Working time:** 45 minutes. Begin timing only after perusal ends.
- **Open-book:** Students may use their workbooks, exercise books, personal notes, and the QCAA Formula Book. Calculators are permitted. Students may *not* share materials or communicate with others.
- **Supervision:** Remain in the room for the full duration. No student should leave without submitting their paper.
- **Absence:** If a student is absent on the exam day, follow your school's official assessment policy for missed supervised assessments.
- **After marking:** Record marks in your class grade book. Return papers to students with feedback. This marking guide should **not** be distributed to students.
- **Moderation:** Keep a sample of marked papers (one per grade boundary) for internal moderation purposes as required by QCAA.

— END OF MARKING GUIDE —

A. 18-Week Lesson Plans

The complete set of 72 lesson plans (Lessons 1–72, four lessons per week across 18 weeks) is reproduced below.

Source File: Stage1_Program_Overview.tex

This appendix reproduces the full lesson plan content from `Stage1_Program_Overview.tex`. Each lesson plan includes:

- Duration and topic reference
- Lesson objectives
- Required resources
- Timed lesson outline (50-minute sessions)
- Assessment and observation notes

To compile this section: ensure `Stage1_Program_Overview.tex` is in the same directory and the `docmute` package is installed.

Contents

Program Overview

Course at a Glance

Detail	Information
Course	Numeracy Short Course (QCAA 2025 v1.2)
Duration	18 weeks — 4 lessons × 50 minutes per week (72 lessons / ≈60 hours)
Topics	Topic 1: Personal Identity and Community (Weeks 1–9) Topic 2: Workplace and Employment (Weeks 10–18)
Assessment	Task 1: Project — Topic 1 (Option A) — completed during Week 9 Task 2: Examination — Topic 2 (Option D) — conducted in Week 18
Grading	A–E scale (no ATAR contribution)
Resources	QCAA Formula Book; Student Workbook (this course's textbook); calculator

Skills Coverage — Number & Algebra (N&A)

Skill	Topic 1	Topic 2
Whole numbers, fractions, decimals, percentages	Weeks 1–2	Review in Week 10
Technology calculations	Week 1	Week 10
Rates and ratios	Week 3	Week 11

Skills Coverage — Measurement & Space (M&S)

Skill	Topic 1	Topic 2
Length conversions (mm, cm, m, km)	Week 4	—
2D shapes — recognition and properties	Week 5	—
Area (rectangle, square, triangle; counting squares)	Weeks 4–5	—
Area unit conversions (mm^2, cm^2, m^2, km^2)	Week 5	—
Time (12-hr, 24-hr, dates, timetables)	Week 6	—
Maps, direction, coordinates, scales	Week 5	Week 14
Angles (360°, 180°, 90°)	Week 6	—
3D shapes — recognition and properties	—	Week 12
Volume of prisms ($V = A \times H$)	—	Week 12
Volume unit conversions (mm^3, cm^3, m^3, km^3)	—	Week 14
Mass conversions (mg, g, kg, t)	—	Week 13
Capacity conversions (mL, L, kL, ML)	—	Week 13
Temperature	—	Week 13

Skills Coverage — Statistics & Probability (S&P)

Skill	Topic 1	Topic 2
Data collection, tables, bar graphs, line graphs	Week 7	Week 15
Probability language (certain, likely, impossible)	Week 8	Week 16
Calculating probability: $\Pr(E) = \text{favourable} \div \text{total}$	Week 8	Week 16
Expressing probability (fractions, decimals, %)	Week 8	Week 16

Assessment Summary

Assessment Task 1 — Project (Topic 1: Personal Identity and Community)

- **Type:** Individual project (Option A)
- **Context:** Students respond to a real-world scenario related to personal identity and community
- **Class time:** ≈4 lessons (Week 9) plus additional time in own time

- **Response format:** Written (up to 600 words) OR multimodal (up to 6 A4 pages / 5 min) OR spoken (up to 4 min)
- **Grading:** A–E using instrument-specific standards

Assessment Task 2 — Examination (Topic 2: Workplace and Employment)

- **Type:** Individual supervised open-book examination (Option D)
- **Conditions:** Open-book; students may bring notes, exercise books, and student workbook
- **Time:** 5 min perusal + 45 min working time
- **Formula book:** QCAA Numeracy Short Course Formula Book provided
- **Content:** Short-response, real-context questions sampling subject matter from both topics
- **Grading:** A–E with percentage cut-offs (A >80%, B >60%, C >40%, D >20%, E ≥0%)

18-Week Program at a Glance

Week	Topic	Focus	Key Skills	Assessment
1	T1	Introduction + Whole Numbers	N&A	—
2	T1	Fractions, Decimals, Percentages	N&A	—
3	T1	Rates and Ratios	N&A	—
4	T1	Length, Perimeter and Area	M&S	—
5	T1	2D Shapes, Area Units, Maps	M&S	—
6	T1	Time and Angles	M&S	—
7	T1	Data Collection and Graphs	S&P	—
8	T1	Probability	S&P	—
9	T1	Project Work	All	Task 1 Due
10	T2	Workplace Introduction + N&A	N&A	—
11	T2	Ratios, Percentages and Budgets	N&A	—
12	T2	3D Shapes and Volume	M&S	—
13	T2	Mass, Capacity, Temperature	M&S	—
14	T2	Volume Units, Maps, Mixed M&S	M&S	—
15	T2	Workplace Data and Graphs	S&P	—
16	T2	Probability and Data Trends	S&P	—
17	T2	Exam Preparation	All	—
18	T2	Examination + Course Wrap-Up	All	Task 2 Exam

Notes for Teachers

- **Open-book exam:** Because Task 2 is open-book and students may bring their workbooks, lessons should emphasise understanding and application rather than memorisation. Students should know where to find formulas and worked examples in their workbook.
- **Workbook reference:** Each lesson plan references the relevant chapter(s) in the Student Workbook. Students should complete workbook exercises as directed.
- **Calculators:** Students should use a calculator (phone or scientific) regularly throughout

the course. Technology use is explicitly required in assessment.

- **Real contexts:** All lessons should be grounded in real-world scenarios. Avoid abstract mathematics — always connect to personal life or workplace situations.
- **Aboriginal and Torres Strait Islander perspectives:** Where appropriate, embed these perspectives in examples and contexts.

B. Topic 1: Personal Identity and Community

Weeks 1–9

Week 1 — Introduction and Whole Numbers

Topic Focus: Number and Algebra — Whole Numbers in personal contexts
Workbook Chapter: Chapter 1 — Numbers in Everyday Life

Lesson 1 — What is Numeracy?

Duration: 50 minutes

Lesson Objectives:

1. Understand what numeracy means and why it matters
2. Identify maths used in everyday personal life
3. Understand the structure of this course

Resources:

- Student Workbook Chapter 1 (Introduction)
- Whiteboard / projector
- Sticky notes

Lesson Outline:

Time	Activity
0–5 min	Welcome and introductions. Teacher explains course structure and what students will produce.
5–15 min	**Discussion:** "Where do you use maths every day?" Students brainstorm on sticky notes (shopping, sport scores, travel time, phone data, cooking). Share with class.
15–25 min	Teacher presents: what is numeracy? Difference between numeracy and maths. Real-life examples. Reference to workbook p.1–3.
25–40 min	Students complete Workbook Ch.1 Activity 1.1: "Maths in My Day" — list 10 everyday tasks that involve numbers.
40–48 min	Class share and discussion.
48–50 min	**Exit ticket:** "Name one thing you used maths for this week."

Assessment note: Observe student engagement. Informal only.

Lesson 2 — Place Value and Whole Numbers

Duration: 50 minutes

Lesson Objectives:

1. Read and write whole numbers up to millions
2. Identify place value of digits
3. Order and compare whole numbers

Resources:

- Student Workbook Chapter 1
- Place value chart (on board)
- Calculator

Lesson Outline:

Time	Activity
0–5 min	**Starter:** Show a supermarket receipt. "What are these numbers? What do they tell us?"
5–20 min	**Direct instruction:** Place value (ones, tens, hundreds, thousands, millions). Reading and writing numbers in words. Workbook p.4–6 worked examples.
20–35 min	Students complete Workbook Ch.1 Exercises 1.2 and 1.3: reading, writing and ordering whole numbers.
35–45 min	**Real context:** Read numbers from a real context (e.g. population statistics, sporting scores, distances between cities).
45–50 min	Review answers together. Correct misconceptions.

Lesson 3 — Operations with Whole Numbers

Duration: 50 minutes

Lesson Objectives:

1. Add, subtract, multiply and divide whole numbers
2. Solve simple word problems involving whole numbers
3. Use a calculator to check answers

Resources:

- Student Workbook Chapter 1
- Calculators
- Printed real-world scenarios (receipts, timetables)

Lesson Outline:

Time	Activity
0–8 min	**Starter warm-up:** mental maths — quick-fire addition and subtraction questions on the board.
8–20 min	**Direct instruction:** BODMAS/order of operations brief overview. Addition, subtraction, multiplication, division with whole numbers. Workbook p.7–9 worked examples.
20–40 min	Students complete Workbook Ch.1 Exercises 1.4–1.6: word problems in personal contexts (e.g. "You have $450. You spend $123 on shoes and $87 on clothes. How much do you have left?").
40–48 min	**Pair-check:** students swap books and check each other's answers using a calculator.
48–50 min	**Summary:** "When do we add? When do we multiply?" Quick class recap.

Lesson 4 — Technology and Calculations

Duration: 50 minutes

Lesson Objectives:

1. Use a calculator to perform calculations with whole numbers
2. Estimate answers before calculating
3. Interpret calculator results in context

Resources:

- Student Workbook Chapter 1
- Calculators (phone calculator is acceptable)
- Real-world problem cards

Lesson Outline:

Time	Activity
0–5 min	**Starter:** "If a cinema ticket is $18 and you go 4 times a month, how much do you spend in a year?" Students estimate first.
5–15 min	**Direct instruction:** Using a calculator correctly — entering numbers, order of operations, interpreting the display. Common mistakes.
15–35 min	Students complete Workbook Ch.1 Exercises 1.7–1.8: calculator-based problems in personal contexts. All answers must include working (estimation + calculator result).
35–45 min	**Activity:** "Shopping Budget" — students are given a $200 budget and a list of 10 items with prices. Using a calculator, they select items to stay within budget.
45–50 min	**Class discussion:** how did you decide what to buy? What maths did you use?

Week 2 — Fractions, Decimals and Percentages

Topic Focus: Number and Algebra — Converting between forms; personal contexts
Workbook Chapter: Chapter 2 — Fractions, Decimals and Percentages

Lesson 5 — Fractions

Duration: 50 minutes

Lesson Objectives:

1. Recognise and use common fractions (halves, thirds, quarters, etc.)
2. Find equivalent fractions

3. Compare and order fractions

Resources:

- Student Workbook Chapter 2
- Fraction wall visual (on board)
- Calculators

Lesson Outline:

Time	Activity
0–8 min	**Starter:** "A pizza is cut into 8 slices. You eat 3. What fraction did you eat?" Brief discussion.
8–22 min	**Direct instruction:** Numerator and denominator. Equivalent fractions (e.g. $\frac{1}{2} = \frac{2}{4} = \frac{4}{8}$). Simplifying fractions. Workbook p.2–4 worked examples.
22–38 min	Students complete Workbook Ch.2 Exercises 2.1–2.2: identifying, simplifying and comparing fractions in personal contexts (food, shopping, time).
38–48 min	**Real context activity:** Students look at sale signs (e.g. "$\frac{1}{3}$ off") and calculate what $\frac{1}{3}$ of various prices is.
48–50 min	**Exit question:** "What is $\frac{3}{4}$ of 24?"

Lesson 6 — Decimals

Duration: 50 minutes

Lesson Objectives:

1. Read and write decimals to two decimal places
2. Convert between fractions and decimals
3. Compare and order decimals

Resources:

- Student Workbook Chapter 2
- Calculators
- Petrol price display image / grocery price tags

Lesson Outline:

Time	Activity
0–5 min	**Starter:** Show a petrol price sign (e.g. 179.9 c/L). "What does this number mean? How do we read it?"
5–20 min	**Direct instruction:** Decimal place value (tenths, hundredths). Reading and writing decimals. Converting fractions to decimals using division. Workbook p.5–7 worked examples.
20–38 min	Students complete Workbook Ch.2 Exercises 2.3–2.4: decimals in personal contexts (prices, measurements, sports results).
38–48 min	**Comparison activity:** Order a set of 8 prices from cheapest to dearest (using decimals).
48–50 min	**Quick recap:** how do you convert $\frac{3}{4}$ to a decimal?

Lesson 7 — Percentages

Duration: 50 minutes

Lesson Objectives:

1. Understand what a percentage means
2. Convert between percentages, decimals and fractions
3. Recognise percentages in everyday life

Resources:

- Student Workbook Chapter 2
- Calculators
- Sample sale brochures / GST examples

Lesson Outline:

Time	Activity
0–8 min	**Starter:** "A shop says 20% off. What does that mean?" Class discussion.
8–22 min	**Direct instruction:** Percent = per hundred. Converting: % $\leftrightarrow$ decimal $\leftrightarrow$ fraction. Key equivalents: $50\% = 0.5 = \frac{1}{2}$, $25\% = 0.25 = \frac{1}{4}$, $10\% = 0.1 = \frac{1}{10}$, $20\% = 0.2 = \frac{1}{5}$. Workbook p.8–10 worked examples.
22–40 min	Students complete Workbook Ch.2 Exercises 2.5–2.6: conversion tables and real-life contexts (battery percentage, sale discounts, test scores).
40–48 min	**Conversion race:** Students race to complete a conversion table (%, fraction, decimal) in pairs.
48–50 min	**Exit ticket:** "Convert $\frac{3}{5}$ to a decimal and a percentage."

Lesson 8 — Calculating Percentages

Duration: 50 minutes

Lesson Objectives:

1. Calculate a percentage of an amount
2. Find discounted prices
3. Use technology to calculate percentages

Resources:

- Student Workbook Chapter 2
- Calculators
- Printed shopping scenarios

Lesson Outline:

Time	Activity
0–5 min	**Warm-up:** "10% of $80?" "50% of $120?" Mental maths.
5–20 min	**Direct instruction:** Calculating % of amount (method 1: convert to decimal then multiply; method 2: calculator). Examples: 20% of $250 = 0.2 × $250 = $50. GST (10%): adding 10% to a price. Workbook p.11–13 worked examples.
20–40 min	Students complete Workbook Ch.2 Exercises 2.7–2.9: calculating discounts, sale prices, tips, data usage.
40–48 min	**Shopping scenario:** Students receive a catalogue and must calculate the sale price of 5 items at various discount percentages.
48–50 min	Review key methods. Remind students: calculators can be used in the exam.

Week 3 — Rates and Ratios

Topic Focus: Number and Algebra — Rates and ratios in personal contexts
Workbook Chapter: Chapter 3 — Rates, Ratios and Comparisons

Lesson 9 — Rates

Duration: 50 minutes

Lesson Objectives:

1. Understand what a rate is and how it is expressed
2. Read and calculate rates (km/hr, $/kg, $/m)
3. Use rates to compare value

Resources:

- Student Workbook Chapter 3
- Calculators
- Supermarket unit pricing examples

Lesson Outline:

Time	Activity
0–8 min	**Starter:** Show two supermarket products — "500g of rice for $2.40 vs 1kg of rice for $4.10. Which is better value?"
8–22 min	**Direct instruction:** What is a rate? Rate = quantity per unit. Common rates: km/hr (speed), $/kg (unit price), $/m (price per metre). Reading rates. Calculating rates. Workbook p.2–5 worked examples.
22–40 min	Students complete Workbook Ch.3 Exercises 3.1–3.2: calculating and comparing rates in personal contexts (travel, shopping, data usage).
40–48 min	**Best buy activity:** Given 4 products with different sizes and prices, students calculate unit price and rank from best to worst value.
48–50 min	**Discussion:** "Does cheapest always mean best value?"

Lesson 10 — Ratios

Duration: 50 minutes

Lesson Objectives:

1. Understand what a ratio is
2. Simplify ratios
3. Apply ratios in personal contexts

Resources:

- Student Workbook Chapter 3
- Calculators
- Mixing/recipe examples

Lesson Outline:

Time	Activity
0–8 min	**Starter:** "A recipe needs 2 cups of flour for every 1 cup of sugar. If I use 6 cups of flour, how much sugar?"
8–22 min	**Direct instruction:** What is a ratio? Writing ratios (a:b). Simplifying ratios (like simplifying fractions). Scaling ratios up and down. Workbook p.6–8 worked examples.
22–40 min	Students complete Workbook Ch.3 Exercises 3.3–3.4: ratios in personal contexts (recipes, mixing paint, sharing money).
40–48 min	**Recipe activity:** Students scale a recipe (originally serves 4) to serve their whole class.
48–50 min	**Exit ticket:** "Simplify the ratio 12:8."

Lesson 11 — Converting Between Equivalent Forms

Duration: 50 minutes

Lesson Objectives:

1. Fluently convert between fractions, decimals and percentages
2. Recognise equivalent forms in real contexts
3. Use conversion in practical situations

Resources:

- Student Workbook Chapter 3
- Calculators
- Conversion reference card (from workbook)

Lesson Outline:

Time	Activity
0–10 min	**Warm-up revision:** class completes a quick conversion table together (%, decimal, fraction).
10–25 min	**Direct instruction:** When to use each form. Reading contexts that use different forms (e.g. sale = 20% off, nutrition label = 0.5g, recipe = $\frac{1}{4}$ cup). Workbook p.9–11 worked examples.
25–42 min	Students complete Workbook Ch.3 Exercises 3.5–3.7: mixed conversion problems in real contexts.
42–48 min	**Group activity:** Students create a "Conversion Cheat Sheet" for their workbook — best forms to use in different situations.
48–50 min	Students share their cheat sheets.

Lesson 12 — Mixed N&A Practice (Personal Contexts)

Duration: 50 minutes

Lesson Objectives:

1. Apply all N&A skills in combined personal context scenarios
2. Identify which skill to use for a given situation
3. Communicate mathematical results clearly

Resources:

- Student Workbook Chapter 3
- Calculators
- Mixed scenario cards

Lesson Outline:

Time	Activity
0–5 min	**Review:** class recap of all N&A skills covered. Teacher writes list on board.
5–15 min	**Worked example:** A multi-step problem combining whole numbers, percentages and rates. Walk through together.
15–40 min	Students complete Workbook Ch.3 Exercise 3.8 (Extended Practice): 6–8 multi-step problems set in personal contexts.
40–48 min	**Peer-marking:** students mark each other's work and provide written feedback.
48–50 min	Teacher addresses common errors.

Week 4 — Length, Perimeter and Area

Topic Focus: Measurement and Space — Length and area of 2D shapes
Workbook Chapter: Chapter 4 — Length, Perimeter and Area

Lesson 13 — Metric Length Units and Conversions

Duration: 50 minutes

Lesson Objectives:

1. Identify and use metric units for length (mm, cm, m, km)
2. Convert between metric length units
3. Estimate lengths in everyday situations

Resources:

- Student Workbook Chapter 4
- Rulers (30 cm)
- Conversion reference (in formula book and workbook)

Lesson Outline:

Time	Activity
0–8 min	**Starter:** "What unit would you use to measure: your height? a room? the distance to school?" Quick class discussion.
8–22 min	**Direct instruction:** The metric system for length. mm, cm, m, km — relationships (10mm=1cm, 100cm=1m, 1000m=1km). Converting up (÷) and down (×). Workbook p.2–4 worked examples.
22–40 min	Students complete Workbook Ch.4 Exercises 4.1–4.2: unit identification and conversion in personal contexts.
40–48 min	**Measurement activity:** Students measure 5 objects in the classroom using a ruler (in cm and mm), then convert to metres.
48–50 min	**Exit ticket:** "Convert 3.5 m to cm and mm."

Lesson 14 — Perimeter of 2D Shapes

Duration: 50 minutes

Lesson Objectives:

1. Calculate perimeter of rectangles, squares and triangles
2. Estimate perimeter in real contexts
3. Solve perimeter problems in personal contexts

Resources:

- Student Workbook Chapter 4
- Rulers
- Calculators
- Grid paper

Lesson Outline:

Time	Activity
0–5 min	**Starter:** "You want to put a fence around a garden that is 6m × 4m. How much fencing do you need?"
5–20 min	**Direct instruction:** Perimeter = sum of all sides. Formulas for rectangle ($P = 2L + 2W$) and square ($P = 4S$). Triangle (P = sum of three sides). Workbook p.5–7 worked examples.
20–38 min	Students complete Workbook Ch.4 Exercises 4.3–4.4: perimeter problems in personal contexts (fencing, framing, edging garden beds).
38–48 min	**Design activity:** Students design a rectangular bedroom on grid paper with a given perimeter constraint.
48–50 min	Share designs. Quick recap of perimeter formula.

Lesson 15 — Area of Rectangles and Squares

Duration: 50 minutes

Lesson Objectives:

1. Calculate area of rectangles using $A = L \times W$
2. Calculate area of squares using $A = S \times S$
3. Distinguish between area and perimeter

Resources:

- Student Workbook Chapter 4
- Grid paper
- Calculators
- Real-world examples (floor plans, turf quotes)

Lesson Outline:

Time	Activity
0–8 min	**Starter:** "What's the difference between perimeter and area? Use the classroom as an example."
8–22 min	**Direct instruction:** Area = space inside a shape. Rectangle: $A = L \times W$. Square: $A = S \times S$. Units: cm^2, m^2. Workbook p.8–10 worked examples.
22–40 min	Students complete Workbook Ch.4 Exercises 4.5–4.6: area calculations in personal contexts (tiles for a bathroom, grass for a lawn, carpet for a room).
40–48 min	**Real-world problem:** Given a room plan (rectangular), students calculate how many square metres of carpet are needed and estimate cost at $\$35/m^2$.
48–50 min	**Class discussion:** "Why is area important in real life?"

Lesson 16 — Area of Triangles and Counting Squares

Duration: 50 minutes

Lesson Objectives:

1. Calculate area of triangles using $A = (B \times H) \div 2$
2. Find area by counting grid squares
3. Apply area skills to irregular shapes

Resources:

- Student Workbook Chapter 4
- Grid paper / dot paper
- Calculators
- Formula book (triangle formula)

Lesson Outline:

Time	Activity
0–8 min	**Starter:** Review rectangle area. Then: "A triangular garden patch has a base of 5m and height of 3m. How do we find its area?"
8–22 min	**Direct instruction:** Triangle area: $A = \frac{B \times H}{2}$. Identifying base and perpendicular height. Counting squares method for irregular shapes. Workbook p.11–13 worked examples.
22–40 min	Students complete Workbook Ch.4 Exercises 4.7–4.9: triangle area calculations and counting-square activities.
40–48 min	**Design challenge:** Students draw an irregular garden shape on grid paper and calculate its area by counting squares.
48–50 min	**Exit ticket:** "Calculate the area of a triangle with base 8 cm and height 5 cm."

Week 5 — 2D Shapes, Area Units and Maps

Topic Focus: Measurement and Space — 2D shapes, area conversions, maps
Workbook Chapter: Chapter 5 — Shapes, Area and Maps

Lesson 17 — 2D Shapes: Recognition and Properties

Duration: 50 minutes

Lesson Objectives:

1. Identify triangles, rectangles (including squares) and circles
2. Describe properties of 2D shapes (sides, angles, symmetry)
3. Recognise 2D shapes in everyday objects

Resources:

- Student Workbook Chapter 5
- Shape cut-outs or images
- Rulers, protractors (optional)

Lesson Outline:

Time	Activity
0–8 min	**Starter:** "Look around the room. Can you spot a triangle? A rectangle? A circle? Where?"
8–22 min	**Direct instruction:** Key 2D shapes — triangles (3 sides), rectangles (4 sides, right angles), squares (4 equal sides, right angles), circles. Properties: number of sides, equal sides, angles, lines of symmetry. Workbook p.2–5 worked examples.
22–40 min	Students complete Workbook Ch.5 Exercises 5.1–5.2: identifying and describing shapes; matching shapes to real objects.
40–48 min	**Shape hunt:** Students photograph or sketch 8 real-world objects and label the 2D shape in each.
48–50 min	Share findings.

Lesson 18 — Using Shape Properties to Describe Objects

Duration: 50 minutes

Lesson Objectives:

1. Use properties of 2D shapes to describe and draw everyday objects
2. Identify shapes within composite designs
3. Communicate shape descriptions clearly

Resources:

- Student Workbook Chapter 5
- Rulers, pencils
- Grid paper

Lesson Outline:

Time	Activity
0–5 min	**Warm-up:** display an image of a house floor plan. "What shapes can you see?"
5–20 min	**Direct instruction:** Describing shapes by properties (e.g. "This shape has 3 sides, one of which is the base"). Drawing shapes from descriptions. Workbook p.6–7 worked examples.
20–38 min	Students complete Workbook Ch.5 Exercises 5.3–5.4: shape descriptions and drawing tasks.
38–48 min	**Floor plan activity:** Students design a simple room layout on grid paper using only named 2D shapes. They label each shape and describe its properties.
48–50 min	Share floor plans.

Lesson 19 — Area Unit Conversions

Duration: 50 minutes

Lesson Objectives:

1. Recognise and use area units (mm^2, cm^2, m^2, km^2)
2. Convert between area units
3. Choose appropriate units for given situations

Resources:

- Student Workbook Chapter 5
- Calculators
- Conversion reference (formula book / workbook)

Lesson Outline:

Time	Activity
0–8 min	**Starter:** "Would you measure the area of a postage stamp in m^2 or cm^2? What about a farm?" Discussion.
8–22 min	**Direct instruction:** Area unit conversions. 1 cm^2 = 100 mm^2. 1 m^2 = 10,000 cm^2. 1 km^2 = 1,000,000 m^2. Workbook p.8–10 worked examples.
22–40 min	Students complete Workbook Ch.5 Exercises 5.5–5.6: unit selection and conversion problems.
40–48 min	**Estimation activity:** Students estimate the area of the classroom floor in m^2, then check using measurement.
48–50 min	Quick class review of conversions.

Lesson 20 — Maps, Direction, Coordinates and Scales

Duration: 50 minutes

Lesson Objectives:

1. Read and use a map with a scale and key
2. Use direction (N, S, E, W) and coordinates to describe location
3. Estimate distances using map scales

Resources:

- Student Workbook Chapter 5
- Printed local area map (or Google Maps screenshot)
- Rulers
- Calculators

Lesson Outline:

Time	Activity
0–8 min	**Starter:** Display a map of local area. "How would you describe how to get from here to the nearest shops?"
8–22 min	**Direct instruction:** Map components: scale, key (symbols, labels), compass directions, grid coordinates. Using scale to calculate real distance. Workbook p.11–14 worked examples.
22–40 min	Students complete Workbook Ch.5 Exercises 5.7–5.8: locating positions, describing routes, calculating distances using scales.
40–48 min	**Route planning activity:** Using a provided map, students plan the shortest route from school to two landmarks, describe directions (N/S/E/W) and calculate distances.
48–50 min	Students share routes. Discuss: "Why do we need maps?"

Week 6 — Time and Angles

Topic Focus: Measurement and Space — Time, 24-hour clocks, angles
Workbook Chapter: Chapter 6 — Time and Angles

Lesson 21 — Reading and Calculating Time

Duration: 50 minutes

Lesson Objectives:

1. Read and write time in 12-hour and 24-hour formats
2. Calculate elapsed time and durations
3. Read and interpret simple timetables

Resources:

- Student Workbook Chapter 6
- Bus/train timetable (printed)
- Calculators

Lesson Outline:

Time	Activity
0–8 min	**Starter:** "Your bus departs at 08:35. You arrive at the stop at 8:20 am. How long do you wait?"
8–22 min	**Direct instruction:** 12-hour vs 24-hour time. Converting between them (pm times: add 12 to hours). Calculating elapsed time (end time – start time). Workbook p.2–5 worked examples.
22–40 min	Students complete Workbook Ch.6 Exercises 6.1–6.2: time reading and calculation in personal contexts.
40–48 min	**Timetable activity:** Students use a real or fictional bus timetable to answer questions about journey times, waiting times, and comparing routes.
48–50 min	**Exit ticket:** "Write 3:45 pm in 24-hour time."

Lesson 22 — Dates and Timetable Comparisons

Duration: 50 minutes

Lesson Objectives:

1. Work with dates, months and years
2. Calculate the number of days between dates
3. Compare journey options using timetables

Resources:

- Student Workbook Chapter 6
- Calendar
- Printed timetables

Lesson Outline:

Time	Activity
0–8 min	**Starter:** "How many days until your birthday? How many weeks is that?"
8–20 min	**Direct instruction:** Reading dates. Calculating time intervals using a calendar (days between dates). Comparing timetables (cost, time, convenience). Workbook p.6–8 worked examples.
20–38 min	Students complete Workbook Ch.6 Exercises 6.3–6.4: date calculations and timetable comparisons.
38–48 min	**Planning activity:** Students plan a 3-day weekend trip, listing dates, travel times, and comparing two transport options using timetables.
48–50 min	**Discussion:** "What factors other than time affect your transport choice?"

Lesson 23 — Angles

Duration: 50 minutes

Lesson Objectives:

1. Identify and measure right angles (90°), half rotations (180°) and full rotations (360°)
2. Recognise angles in everyday contexts
3. Use angle knowledge to describe direction and shape

Resources:

- Student Workbook Chapter 6
- Protractors
- Printed angle diagrams

Lesson Outline:

Time	Activity
0–8 min	**Starter:** "What angle does a clock make at 3:00? At 6:00? At 12:00?" Discuss.
8–22 min	**Direct instruction:** Types of angles. Right angle = 90°. Straight angle = 180°. Full rotation = 360°. How angles relate to direction changes (turn left = 90°). Workbook p.9–11 worked examples.
22–38 min	Students complete Workbook Ch.6 Exercises 6.5–6.6: identifying and classifying angles; angle problems in personal contexts (navigation, construction).
38–48 min	**Angles in the real world:** Students find and photograph 5 examples of angles in the school environment and classify them.
48–50 min	**Exit ticket:** "What is the angle of a full rotation? A right angle?"

Lesson 24 — Mixed M&S Practice (Personal Contexts)

Duration: 50 minutes

Lesson Objectives:

1. Apply all M&S skills in combined personal context scenarios
2. Select the correct formula and units for a given problem
3. Communicate solutions clearly

Resources:

- Student Workbook Chapter 6
- Calculators
- Formula book

- Mixed M&S scenario worksheet

Lesson Outline:

Time	Activity
0–5 min	**Recap:** teacher quickly reviews all M&S skills (length, area, shapes, time, maps, angles).
5–15 min	**Worked example:** A multi-step M&S problem — e.g. "Plan a community garden: calculate area, perimeter, label map, estimate time to build." Walk through together.
15–40 min	Students complete Workbook Ch.6 Exercise 6.8 (Mixed M&S Practice): 6 multi-step problems in personal contexts.
40–48 min	**Peer-review:** students compare answers with a partner and discuss any differences.
48–50 min	Teacher addresses top 3 errors.

Week 7 — Data Collection and Graphs

Topic Focus: Statistics and Probability — Collecting, organising and representing data
Workbook Chapter: Chapter 7 — Data and Graphs

Lesson 25 — Collecting and Organising Data in Tables

Duration: 50 minutes

Lesson Objectives:

1. Collect and record data in a tally chart and table
2. Organise data into categories
3. Read and interpret simple data tables

Resources:

- Student Workbook Chapter 7
- Tally chart templates
- Classroom survey activity

Lesson Outline:

Time	Activity
0–8 min	**Starter:** "How many students in this class walk to school? Take public transport? Get dropped off?" Quick class survey.
8–20 min	**Direct instruction:** Types of data (categorical, numerical). Tally charts. Frequency tables. Reading and interpreting tables. Workbook p.2–4 worked examples.
20–40 min	Students conduct a mini-survey (chosen from a list: favourite sport, preferred lunch, transport to school), collect data, record in a tally chart and build a frequency table. Workbook Ch.7 Exercise 7.1.
40–48 min	Share tables with class. Discuss: "What does the data tell us?"
48–50 min	**Exit question:** "What is the most common response in your survey?"

Lesson 26 — Bar Graphs

Duration: 50 minutes

Lesson Objectives:

1. Draw a bar graph from a data table
2. Label axes correctly (title, axis labels, scale)
3. Interpret information from a bar graph

Resources:

- Student Workbook Chapter 7
- Rulers, pencils, graph paper OR computer/tablet with spreadsheet
- Students' data from Lesson 25

Lesson Outline:

Time	Activity
0–5 min	Display an example bar graph. "What information can you read from this graph?"
5–20 min	**Direct instruction:** Components of a bar graph (title, x-axis, y-axis, scale, bars). How to choose a scale. Drawing bars accurately. Workbook p.5–8 worked examples.
20–40 min	Students create a bar graph from their survey data (Lesson 25). Must include: title, labelled axes, appropriate scale, even bar widths. Workbook Ch.7 Exercise 7.2.
40–48 min	**Interpretation questions:** Students write 3 statements about their bar graph ("The most popular... was...").
48–50 min	**Gallery walk:** students view each other's graphs.

Lesson 27 — Line Graphs

Duration: 50 minutes

Lesson Objectives:

1. Draw a line graph from a data table showing change over time
2. Interpret trends from a line graph
3. Use line graphs in personal contexts (petrol prices, temperature)

Resources:

- Student Workbook Chapter 7
- Graph paper / digital tools
- Printed data sets (monthly petrol prices, temperature data)

Lesson Outline:

Time	Activity
0–8 min	**Starter:** Display a line graph of monthly temperatures. "What does this graph show? What months are hottest? Coldest?"
8–22 min	**Direct instruction:** When to use a line graph (data over time). Plotting points, connecting with a line. Reading trends (increasing, decreasing, stable). Workbook p.9–11 worked examples.
22–40 min	Students use a provided data set (petrol prices over 6 months) to draw a line graph, then answer interpretation questions. Workbook Ch.7 Exercise 7.3.
40–48 min	**Discussion:** "Why do petrol prices change? What could cause a sudden rise or drop?"
48–50 min	**Exit question:** "What does an upward trend on a line graph tell us?"

Lesson 28 — Reading Charts and Tables from Media

Duration: 50 minutes

Lesson Objectives:

1. Read and interpret data from newspaper/online charts and tables
2. Make simple statements about what data shows
3. Identify trends and make basic comparisons

Resources:

- Student Workbook Chapter 7
- Printed newspaper/online graphs and tables (weather, sport, population)
- Calculators

Lesson Outline:

Time	Activity
0–5 min	Show a real newspaper page with a chart. "What story is this chart telling?"
5–15 min	**Direct instruction:** Reading graphs critically — checking axes, scale, title. Simple vs misleading graphs. Workbook p.12–14 worked examples.
15–40 min	Students rotate through 4 "stations" — each has a different chart/table from real media. Students answer 3–4 interpretation questions per station. Workbook Ch.7 Exercise 7.4.
40–48 min	**Class discussion:** share one interesting finding per station.
48–50 min	**Reflection:** "Where did you see data in the news this week?"

Week 8 — Probability

Topic Focus: Statistics and Probability — Probability language and calculations
Workbook Chapter: Chapter 8 — Chance and Probability

Lesson 29 — Probability Language

Duration: 50 minutes

Lesson Objectives:

1. Describe likelihood using everyday language (certain, likely, even chance, unlikely, impossible)
2. Place events on a probability scale
3. Recognise probability language in everyday contexts

Resources:

- Student Workbook Chapter 8
- Probability scale poster (0 to 1, or impossible to certain)
- Event cards

Lesson Outline:

Time	Activity
0–8 min	**Starter:** "Is it certain you'll eat today? Is it impossible it will snow here in summer? Is it likely or unlikely you'll win the lottery?"
8–20 min	**Direct instruction:** Probability language: impossible (0), unlikely, even chance ($\frac{1}{2}$), likely, certain (1). The probability scale 0 to 1. Workbook p.2–4 worked examples.
20–38 min	Students complete Workbook Ch.8 Exercises 8.1–8.2: classifying events using probability language; placing events on a probability scale.
38–48 min	**Event card activity:** In pairs, students sort 15 event cards from "impossible" to "certain" and justify their choices.
48–50 min	**Discussion:** "Can two people disagree on whether something is 'likely'? Why?"

Lesson 30 — Calculating Probability

Duration: 50 minutes

Lesson Objectives:

1. List all possible outcomes of a simple experiment
2. Calculate probability using $\Pr(E) = \text{favourable outcomes} \div \text{total outcomes}$
3. Apply probability to everyday chance events

Resources:

- Student Workbook Chapter 8
- Dice, coins, coloured cards
- Calculators

Lesson Outline:

Time	Activity
0–8 min	**Starter:** Roll a die. "What is the chance of rolling a 6?" "What is the chance of rolling an even number?" Students estimate.
8–22 min	**Direct instruction:** Sample space (all possible outcomes). Favourable outcomes. Formula: $\Pr(E) = \dfrac{\text{favourable outcomes}}{\text{total outcomes}}$. Examples: dice, coins, coloured bags. Workbook p.5–8 worked examples.
22–40 min	Students complete Workbook Ch.8 Exercises 8.3–8.4: calculating probability for coins, dice, spinners, and coloured-card draws.
40–48 min	**Experiment:** Students flip a coin 20 times and record results. Compare experimental frequency to theoretical probability ($\frac{1}{2}$).
48–50 min	**Discussion:** "Why didn't we get exactly 10 heads?" Introduction to idea of experimental vs theoretical probability.

Lesson 31 — Expressing Probability as Fractions, Decimals and Percentages

Duration: 50 minutes

Lesson Objectives:

1. Express probability as a fraction, decimal and percentage
2. Convert between forms
3. Compare probabilities expressed in different forms

Resources:

- Student Workbook Chapter 8
- Calculators

Lesson Outline:

Time	Activity
0–8 min	**Warm-up:** "What is Pr(rolling a 6)? Write it as a fraction." Then: "How do you write that as a decimal? A percentage?"
8–22 min	**Direct instruction:** Converting probability: fraction → decimal (÷) → percentage (×100). Range: 0 to 1 (or 0% to 100%). Workbook p.9–11 worked examples.
22–40 min	Students complete Workbook Ch.8 Exercises 8.5–8.6: expressing probabilities in all three forms; comparison problems.
40–48 min	**Probability matching activity:** Students are given probabilities in different forms and must match equivalent expressions ($\frac{1}{4} = 0.25 = 25\%$).
48–50 min	**Exit ticket:** "A bag has 3 red, 2 blue, 5 green balls. What is Pr(blue)? Write as a fraction, decimal and percentage."

Lesson 32 — Mixed S&P Practice and Project Introduction

Duration: 50 minutes

Lesson Objectives:

1. Apply all S&P skills in combined scenarios
2. Understand the requirements for Assessment Task 1 (Project)
3. Begin planning project response

Resources:

- Student Workbook Chapter 8
- Task 1 assessment sheet / scenario
- Calculators

Lesson Outline:

Time	Activity
0–5 min	**Recap:** class reviews all S&P skills.
5–20 min	Students complete Workbook Ch.8 Exercise 8.7 (Mixed S&P Practice): 5 multi-step problems combining data and probability in personal contexts.
20–38 min	**Assessment Task 1 introduction:** Teacher distributes and explains the project scenario. Students read the scenario, identify the mathematical information they will need, and list the numeracy skills that apply.
38–48 min	Students begin planning their project response. Workbook: Project Planning Page (Task 1).
48–50 min	Questions and answers about the project task.

Week 9 — Assessment Task 1: Project

Topic Focus: Personal Identity and Community — Project completion
Assessment: Task 1 due end of Week 9

Lesson 33 — Project Work: Understanding the Scenario and Planning

Duration: 50 minutes

Lesson Outline:

Time	Activity
0–5 min	Re-read the project scenario. Any clarifying questions answered by teacher.
5–15 min	**Planning:** Students map out the mathematical steps needed: what information do they have? What do they need to calculate? What format will their response take?
15–45 min	Students work on project — identifying mathematical information and selecting procedures. Teacher circulates and provides individual support.
45–50 min	Students note what they have completed and what still needs to be done.

Lesson 34 — Project Work: Applying Skills and Calculations

Duration: 50 minutes

Lesson Outline:

Time	Activity
0–5 min	**Brief check-in:** students share one challenge they faced in previous session.
5–45 min	Students continue project — performing calculations, applying numeracy skills to the scenario. Teacher provides one-on-one assistance.
45–50 min	**Progress checkpoint:** students self-assess against the task criteria (A–E standards sheet).

Lesson 35 — Project Work: Communicating Results

Duration: 50 minutes

Lesson Outline:

Time	Activity
0–5 min	Teacher models how to present findings clearly — using correct mathematical language, diagrams, labels.
5–45 min	Students finalise their response — writing up findings and recommendations in their chosen format (written, multimodal or spoken preparation).
45–50 min	Review format requirements (600 words / 6 pages / 4 minutes).

Lesson 36 — Project Completion and Submission

Duration: 50 minutes

Lesson Outline:

Time	Activity
0–30 min	Students complete any final additions to their project response. Teacher collects completed work.
30–45 min	**Reflection:** Students complete a self-reflection form — what went well? What did I find difficult? What would I do differently?
45–50 min	Brief preview of Topic 2: Workplace and Employment.

C. Topic 2: Workplace and Employment

Weeks 10–18

Week 10 — Introduction to the Workplace and N&A Review

Topic Focus: Number and Algebra — Rates and technology in workplace contexts
Workbook Chapter: Chapter 9 — Numbers at Work

Lesson 37 — What Maths Do We Use at Work?

Duration: 50 minutes

Lesson Objectives:

1. Identify numeracy skills used in various workplaces
2. Connect personal numeracy skills to workplace contexts
3. Understand Topic 2 structure and assessment

Resources:

- Student Workbook Chapter 9 (Introduction)
- Job-card activity (various occupations)

Lesson Outline:

Time	Activity
0–8 min	**Starter:** "What jobs are you interested in? What maths would you use in that job?" Quick poll.
8–20 min	**Discussion/activity:** Students receive a "job card" describing a role (barista, builder, nurse, retail worker, delivery driver). They list 5 maths tasks that person does daily.
20–35 min	Share with class. Teacher uses responses to introduce Topic 2 subject matter and connect to workbook chapters.
35–48 min	Students read Workbook Ch.9 Introduction and complete Activity 9.1: "My Future Workplace — Maths I'll Need."
48–50 min	Preview: what we'll learn in Topic 2.

Lesson 38 — Pay Rates and Time Records

Duration: 50 minutes

Lesson Objectives:

1. Interpret a pay rate ($/hour)
2. Calculate gross pay from hours worked
3. Record hours on a timesheet

Resources:

- Student Workbook Chapter 9
- Sample timesheet (printed)
- Calculators

Lesson Outline:

Time	Activity
0–8 min	**Starter:** "If you earn $19.50/hr and work 6 hours, how much do you earn?"
8–22 min	**Direct instruction:** Casual vs part-time pay. Gross pay = hours × rate. Reading a timesheet. Workbook p.4–7 worked examples.
22–40 min	Students complete Workbook Ch.9 Exercises 9.1–9.2: completing timesheets and calculating weekly pay in workplace scenarios.
40–48 min	**Scenario:** "You worked Mon 4h, Tue 3h, Thu 5h, Sat 6h at $21.50/hr. What is your gross pay? Create a timesheet."
48–50 min	Class review.

Lesson 39 — Technology Calculations in the Workplace

Duration: 50 minutes

Lesson Objectives:

1. Use a calculator for multi-step workplace calculations
2. Use a spreadsheet (or table) to record and total figures
3. Interpret calculator results in workplace contexts

Resources:

- Student Workbook Chapter 9
- Calculators
- Access to spreadsheet software (Excel / Google Sheets) if available

Lesson Outline:

Time	Activity
0–5 min	**Starter:** "At work, when would you use a calculator? When would you use a spreadsheet?"
5–18 min	**Direct instruction:** Spreadsheet basics (if available) — entering data, SUM formula. Otherwise: multi-step calculator calculations. Workbook p.8–10 worked examples.
18–40 min	Students complete Workbook Ch.9 Exercises 9.3–9.4: entering sales data, calculating totals and averages using technology.
40–48 min	**Data entry task:** Students enter a week of sales figures into a table/spreadsheet and calculate totals for each day and the weekly total.
48–50 min	Review: what does the data tell the business?

Lesson 40 — Calculating Gross Pay: Extended Practice

Duration: 50 minutes

Lesson Objectives:

1. Apply pay rate and time calculations in varied scenarios
2. Interpret a simple pay slip
3. Calculate total earnings over a week/fortnight

Resources:

- Student Workbook Chapter 9
- Sample pay slip (printed)
- Calculators

Lesson Outline:

Time	Activity
0–5 min	**Recap:** key formula for gross pay?
5–15 min	**Direct instruction:** Reading a pay slip — gross pay, tax withheld, net pay (note: students need to know these exist; full tax calculation not required). Workbook p.11–13 worked examples.
15–38 min	Students complete Workbook Ch.9 Exercise 9.5 (Extended Pay Scenarios): 5 different workplace scenarios involving pay rates, hours, and basic deductions.
38–48 min	**Role play:** "You are the payroll officer. Calculate the weekly pay for 3 employees using the timesheet provided."
48–50 min	Review answers.

Week 11 — Ratios, Percentages and Budgets at Work

Topic Focus: Number and Algebra — Ratios and percentages in workplace contexts
Workbook Chapter: Chapter 10 — Workplace Numbers

Lesson 41 — Ratios in the Workplace

Duration: 50 minutes

Lesson Objectives:

1. Apply ratios to mixing and dividing quantities at work
2. Scale ratios for workplace quantities
3. Use ratios to share costs or materials

Resources:

- Student Workbook Chapter 10
- Calculators
- Mixing scenario cards (paint, concrete, cleaning products)

Lesson Outline:

Time	Activity
0–8 min	**Starter:** "A cleaning solution is mixed in ratio 1 part concentrate : 10 parts water. If you need 22 litres total, how much concentrate?"
8–20 min	**Direct instruction:** Ratios in the workplace — mixing (concrete: 1:2:4), dividing profit, scaling recipes. Workbook p.2–4 worked examples.
20–40 min	Students complete Workbook Ch.10 Exercises 10.1–10.2: ratio problems in workplace contexts.
40–48 min	**Trade scenarios:** Students work through 3 trade-specific ratio problems (baker scaling recipe, painter mixing colours, gardener mixing fertiliser).
48–50 min	Quick recap of ratio method.

Lesson 42 — Percentages in the Workplace

Duration: 50 minutes

Lesson Objectives:

1. Calculate GST (10%) on goods and services
2. Apply discounts and markups in retail contexts
3. Interpret percentage-based information in workplace documents

Resources:

- Student Workbook Chapter 10
- Calculators
- Retail receipts / invoices (printed samples)

Lesson Outline:

Time	Activity
0–8 min	**Starter:** "A plumber charges $350 for labour + 10% GST. What is the total cost?"
8–22 min	**Direct instruction:** GST (Goods and Services Tax) = 10%. Calculating GST: price × 0.10. Total including GST: price × 1.10. Markups and discounts in business. Workbook p.5–7 worked examples.
22–40 min	Students complete Workbook Ch.10 Exercises 10.3–10.4: GST calculations, sale prices, markups in workplace contexts.
40–48 min	**Invoice task:** Students complete a simple invoice for a small business (list items, calculate subtotal, add GST, find total).
48–50 min	Share invoices.

Lesson 43 — Purchase Lists, Costs and Budgets

Duration: 50 minutes

Lesson Objectives:

1. Create a purchase list and calculate total cost
2. Work within a budget constraint
3. Compare costs of different options

Resources:

- Student Workbook Chapter 10
- Calculators
- Catalogue excerpts / price lists

Lesson Outline:

Time	Activity
0–5 min	**Starter:** "You have $500 to set up a small office. What would you need to buy?" Brainstorm.
5–18 min	**Direct instruction:** Writing a purchase list. Adding costs. Staying within budget. Comparing alternatives. Workbook p.8–10 worked examples.
18–40 min	Students complete Workbook Ch.10 Exercises 10.5–10.6: budgeting scenarios in workplace contexts (ordering stock, setting up equipment, catering for an event).
40–48 min	**Scenario:** "Your supervisor has given you $750 to buy supplies for the office kitchen. Using the provided price list, select items and calculate the total cost without exceeding the budget."
48–50 min	Present purchase lists.

Lesson 44 — Mixed N&A Workplace Practice

Duration: 50 minutes

Lesson Objectives:

1. Apply all N&A skills in combined workplace scenarios
2. Solve multi-step problems using appropriate operations and technology
3. Present results clearly

Resources:

- Student Workbook Chapter 10
- Calculators
- Mixed workplace scenarios

Lesson Outline:

Time	Activity
0–5 min	**Recap:** list all N&A skills. Which have we applied to workplace contexts?
5–45 min	Students work through Workbook Ch.10 Exercise 10.7 (Mixed Workplace N&A): 8 multi-step problems across pay, ratios, percentages, budgets and unit pricing.
45–50 min	Review top 3 most common errors.

Week 12 — 3D Shapes and Volume

Topic Focus: Measurement and Space — 3D shapes and volume in workplace
Workbook Chapter: Chapter 11 — 3D Shapes and Volume

Lesson 45 — 3D Shapes: Recognition and Properties

Duration: 50 minutes

Lesson Objectives:

1. Identify prisms and cylinders
2. Describe properties using faces, edges and vertices
3. Recognise 3D shapes in the workplace

Resources:

- Student Workbook Chapter 11
- 3D shape models or images
- Workplace object photos

Lesson Outline:

Time	Activity
0–8 min	**Starter:** Show images of a cereal box, pipe, brick, tin can. "What shape is each one? How are they similar or different?"
8–22 min	**Direct instruction:** Prisms (rectangular, triangular, square — faces are rectangles + 2 matching end faces). Cylinders (2 circular faces + curved surface). Properties: number and shape of faces. Workbook p.2–5 worked examples.
22–40 min	Students complete Workbook Ch.11 Exercises 11.1–11.2: identifying 3D shapes, counting faces, matching to everyday objects.
40–48 min	**Workplace object sort:** Students sort 12 workplace objects by their 3D shape (box = rectangular prism, pipe = cylinder, etc.).
48–50 min	**Exit question:** "Name one difference between a prism and a cylinder."

Lesson 46 — Uses of 3D Shapes in the Workplace

Duration: 50 minutes

Lesson Objectives:

1. Identify and name 3D shapes used in workplace settings
2. Explain why particular shapes are used for particular purposes
3. Describe 3D shapes using mathematical language

Resources:

- Student Workbook Chapter 11
- Workplace image cards
- Rulers

Lesson Outline:

Time	Activity
0–8 min	**Starter:** "Why are pipes cylindrical rather than rectangular? Why are bricks rectangular prisms?" Class discussion.
8–20 min	**Direct instruction:** Purpose of 3D shapes in industry (packaging, construction, storage). Describing shapes to colleagues. Workbook p.6–7 worked examples.
20–40 min	Students complete Workbook Ch.11 Exercise 11.3: match workplace objects to their 3D shape; describe each using correct mathematical language.
40–48 min	**Design activity:** Students sketch a workplace storage solution using named 3D shapes and explain why each shape was chosen.
48–50 min	Share designs.

Lesson 47 — Volume of Prisms

Duration: 50 minutes

Lesson Objectives:

1. Calculate volume of prisms using $V = A \times H$
2. Calculate the base area first, then apply the formula
3. Apply volume calculations to workplace problems

Resources:

- Student Workbook Chapter 11
- Calculators
- Formula book (volume formula)

Lesson Outline:

Time	Activity
0–8 min	**Starter:** "A storage room is 5m long, 4m wide and 3m high. How much space is inside it?" Students attempt.
8–22 min	**Direct instruction:** Volume = Area of base × Height ($V = A \times H$). For rectangular prisms: $V = L \times W \times H$. Step 1: find base area. Step 2: multiply by height. Units: m^3, cm^3. Workbook p.8–11 worked examples.
22–40 min	Students complete Workbook Ch.11 Exercises 11.4–11.5: volume calculations for rectangular and triangular prisms in workplace contexts (shipping containers, concrete slabs, storage boxes).
40–48 min	**Practical problem:** "How many boxes (30 cm × 20 cm × 15 cm) will fit in a storage room that is 3 m × 2 m × 2.4 m?"
48–50 min	**Exit ticket:** "What is $V = A \times H$? What does A represent?"

Lesson 48 — Practical Volume Calculations

Duration: 50 minutes

Lesson Objectives:

1. Apply volume formula to workplace scenarios
2. Convert between volume units where needed
3. Solve multi-step volume problems

Resources:

- Student Workbook Chapter 11
- Calculators
- Formula book

Lesson Outline:

Time	Activity
0–5 min	**Warm-up:** "Calculate the volume of a box 40 cm × 25 cm × 20 cm."
5–15 min	**Revision:** Review any misconceptions from Lesson 47. Introduce compound shapes (briefly): add volumes of two prisms.
15–40 min	Students complete Workbook Ch.11 Exercise 11.6 (Extended Volume Problems): 6 multi-step workplace problems.
40–48 min	**Estimation challenge:** Students estimate then calculate the volume of the classroom (model as a rectangular prism).
48–50 min	**Discussion:** "Why is volume important in trades?"

Week 13 — Mass, Capacity and Temperature

Topic Focus: Measurement and Space — Mass, capacity and temperature in workplace
Workbook Chapter: Chapter 12 — Mass, Capacity and Temperature

Lesson 49 — Mass Units and Conversions

Duration: 50 minutes

Lesson Objectives:

1. Identify and use metric mass units (mg, g, kg, t)
2. Convert between metric mass units
3. Estimate and measure mass in workplace contexts

Resources:

- Student Workbook Chapter 12
- Kitchen scales (if available)
- Calculators
- Conversion chart

Lesson Outline:

Time	Activity
0–8 min	**Starter:** "What unit would you use to weigh: a tablet? a bag of flour? a car?"
8–22 min	**Direct instruction:** mg, g, kg, t — relationships. 1000 mg = 1 g. 1000 g = 1 kg. 1000 kg = 1 t. Converting: ×1000 (going smaller), ÷1000 (going larger). Workbook p.2–5 worked examples.
22–40 min	Students complete Workbook Ch.12 Exercises 12.1–12.2: mass conversions and estimation in workplace contexts (pharmacy, kitchen, freight).
40–48 min	**Weighing activity:** Students weigh several classroom objects and record in grams and kilograms.
48–50 min	**Exit ticket:** "Convert 2.5 kg to grams and milligrams."

Lesson 50 — Capacity Units and Conversions

Duration: 50 minutes

Lesson Objectives:

1. Identify and use metric capacity units (mL, L, kL, ML)
2. Convert between capacity units
3. Apply capacity in workplace contexts (catering, plumbing, agriculture)

Resources:

- Student Workbook Chapter 12
- Measuring jugs / bottles (if available)
- Calculators

Lesson Outline:

Time	Activity
0–8 min	**Starter:** "A swimming pool holds 1,500,000 litres. Write this in kL and ML."
8–22 min	**Direct instruction:** mL, L, kL, ML. 1000 mL = 1 L. 1000 L = 1 kL. 1000 kL = 1 ML. Conversions. Practical contexts. Workbook p.6–9 worked examples.
22–40 min	Students complete Workbook Ch.12 Exercises 12.3–12.4: capacity conversions in workplace contexts (cooking, chemical mixing, irrigation, tank capacity).
40–48 min	**Practical scenario:** "A café makes 350 coffees per day, each needing 200 mL of milk. How many litres of milk are needed? How many 2 L bottles?"
48–50 min	Class review.

Lesson 51 — Temperature in the Workplace

Duration: 50 minutes

Lesson Objectives:

1. Read and interpret temperature in Celsius
2. Compare temperatures across contexts
3. Apply temperature knowledge to workplace health and safety

Resources:

- Student Workbook Chapter 12
- Thermometers (if available)
- Workplace temperature guidelines (food safety, etc.)

Lesson Outline:

Time	Activity
0–8 min	**Starter:** "At what temperature does water boil? Freeze? What temperature is a healthy human body?"
8–20 min	**Direct instruction:** Reading temperature (°C). Normal ranges: body temperature (37°C), food safety guidelines (hot food >60°C, cold food <5°C), comfortable workplace (18–24°C). Workbook p.10–12 worked examples.
20–38 min	Students complete Workbook Ch.12 Exercises 12.5–12.6: reading temperatures, comparing temperatures, temperature in workplace scenarios (food handling, outdoor work, storage).
38–48 min	**Case study:** "A food worker checks fridge temperatures over a week. Identify any days where temperatures indicate a food safety risk."
48–50 min	**Discussion:** "Why is temperature important in the workplace?"

Lesson 52 — Measuring in the Workplace

Duration: 50 minutes

Lesson Objectives:

1. Select and use appropriate measuring instruments
2. Take accurate measurements in workplace simulations
3. Record measurements in the correct units

Resources:

- Student Workbook Chapter 12
- Rulers, scales, measuring jugs, thermometers
- Workplace measurement activity

Lesson Outline:

Time	Activity
0–5 min	**Recap:** all measurement skills from Weeks 12–13.
5–20 min	**Direct instruction:** Choosing the right instrument for the job. Reading measurement instruments accurately. Recording measurements with correct units. Workbook p.13–14.
20–45 min	**Measurement stations activity:** Students rotate through 4 stations, each simulating a workplace measurement task: (1) mass of items, (2) capacity of containers, (3) temperature of samples, (4) length/area of a surface. Record all measurements in workbook.
45–50 min	**Debrief:** "Which measurement was hardest to take accurately? Why?"

Week 14 — Volume Units, Maps and Mixed M&S

Topic Focus: Measurement and Space — Volume units, workplace maps, review
Workbook Chapter: Chapter 13 — Workplace Measurement Mixed

Lesson 53 — Volume Unit Conversions

Duration: 50 minutes

Lesson Objectives:

1. Convert between volume units (mm^3, cm^3, m^3, km^3)
2. Apply volume unit conversions to workplace problems
3. Choose appropriate units for given situations

Resources:

- Student Workbook Chapter 13
- Calculators
- Formula book

Lesson Outline:

Time	Activity
0–8 min	**Starter:** "A concrete slab is 0.005 m^3. Express this in cm^3."
8–22 min	**Direct instruction:** Volume unit conversions: 1 cm^3 = 1000 mm^3. 1 m^3 = 1,000,000 cm^3. 1 km^3 = 1,000,000,000 m^3. When to use each unit. Workbook p.2–4 worked examples.
22–40 min	Students complete Workbook Ch.13 Exercises 13.1–13.2: volume unit conversions in industry contexts (mining, construction, engineering).
40–48 min	**Comparison problem:** "A storage tank holds 5,000,000 cm^3. Convert to m^3. How many 1m × 1m × 1m boxes would it hold?"
48–50 min	Class review.

Lesson 54 — Following Instructions and Measuring Materials

Duration: 50 minutes

Lesson Objectives:

1. Follow written instructions to complete a measurement task
2. Measure materials accurately
3. Record and report measurements

Resources:

- Student Workbook Chapter 13
- Rulers, measuring tapes
- "Assembly instructions" activity sheet

Lesson Outline:

Time	Activity
0–5 min	**Starter:** "Think of a time you followed instructions at home or work. What happened if you missed a step?"
5–15 min	**Direct instruction:** Reading and following sequential instructions. Importance of accuracy in workplace measurement. Workbook p.5–6.
15–40 min	**Practical activity:** Students receive a set of simple workplace instructions (e.g. measure and mark out a space for furniture, or measure ingredients for a recipe to serve a group). They follow each step, record measurements, and check their work.
40–48 min	**Discussion:** "What would happen in a real workplace if your measurements were wrong?"
48–50 min	**Self-assessment:** "Did I follow every step? Were my measurements accurate?"

Lesson 55 — Maps and Plans in the Workplace

Duration: 50 minutes

Lesson Objectives:

1. Read a workplace map or floor plan
2. Use a scale to calculate distances
3. Describe a delivery route or work route using direction

Resources:

- Student Workbook Chapter 13
- Printed workplace floor plan / delivery map
- Rulers, calculators

Lesson Outline:

Time	Activity
0–8 min	**Starter:** Show a warehouse floor plan. "Where is the exit? Where are the toilets? How far is the loading dock from the break room?"
8–20 min	**Direct instruction:** Reading workplace maps and floor plans. Scales in workplace contexts (1:50 means 1 cm = 50 cm). Using direction in delivery routes. Workbook p.7–10 worked examples.
20–40 min	Students complete Workbook Ch.13 Exercises 13.3–13.4: floor plan reading, scale calculations, delivery route planning.
40–48 min	**Route planning:** Given a local area delivery map, students plan the most efficient route for 5 deliveries and calculate total distance.
48–50 min	Share routes. Which is most efficient?

Lesson 56 — Mixed M&S Workplace Review

Duration: 50 minutes

Lesson Objectives:

1. Apply all M&S skills in combined workplace scenarios
2. Select appropriate formulas and units
3. Demonstrate understanding of M&S skills across workplace contexts

Resources:

- Student Workbook Chapter 13
- Calculators
- Formula book

Lesson Outline:

Time	Activity
0–5 min	**Quick quiz:** 10 rapid-fire M&S questions (unit conversions, shape identification, area formula).
5–45 min	Students complete Workbook Ch.13 Exercise 13.5 (Mixed M&S Workplace Review): 8 multi-step problems combining shapes, measurement, volume, maps.
45–50 min	Peer-marking and class discussion of answers.

Week 15 — Workplace Data and Graphs

Topic Focus: Statistics and Probability — Data in workplace contexts
Workbook Chapter: Chapter 14 — Data at Work

Lesson 57 — Recording Data in Workplace Tables

Duration: 50 minutes

Lesson Objectives:

1. Record workplace data accurately in tables
2. Organise data into columns and rows
3. Interpret simple tables to answer questions

Resources:

- Student Workbook Chapter 14
- Printed stock tally / sales record templates
- Calculators

Lesson Outline:

Time	Activity
0–8 min	**Starter:** "In a shop, why would you keep a record of how many items are sold each day?"
8–20 min	**Direct instruction:** Purpose of workplace data. Types of workplace records: stock lists, sales records, timesheets, temperature logs. Reading two-column tables. Workbook p.2–5 worked examples.
20–40 min	Students complete Workbook Ch.14 Exercises 14.1–14.2: filling in tables from given information; answering questions using two-column tables.
40–48 min	**Data recording activity:** Students receive a "day's worth" of sales information (described verbally or on cards) and must enter it into a table accurately.
48–50 min	Check tables for accuracy.

Lesson 58 — Constructing Workplace Charts and Graphs

Duration: 50 minutes

Lesson Objectives:

1. Construct a bar graph or chart from workplace data
2. Label graphs correctly
3. Use graphs to present information to others

Resources:

- Student Workbook Chapter 14
- Graph paper / digital tools
- Sales data from Lesson 57

Lesson Outline:

Time	Activity
0–5 min	**Recap:** how to draw a bar graph.
5–15 min	**Direct instruction:** Using data from a table to construct a bar chart. Choosing appropriate scale for workplace data. Workbook p.6–8 worked examples.
15–40 min	Students construct a bar graph from their workplace sales data (Lesson 57). Must be labelled with title, axis labels, scale. Workbook Ch.14 Exercise 14.3.
40–48 min	**Interpretation:** Students write 3 conclusions from their bar graph ("Tuesday had the highest sales of…").
48–50 min	Share graphs. "If you were the manager, what would this graph tell you?"

Lesson 59 — Two-Column Tables and Comparing Information

Duration: 50 minutes

Lesson Objectives:

1. Access information from two-column tables
2. Compare data across two or more tables
3. Identify the higher/lower value and explain the difference

Resources:

- Student Workbook Chapter 14
- Printed workplace tables (price lists, pay rates, specifications)

Lesson Outline:

Time	Activity
0–8 min	**Starter:** Display two supplier price lists. "Which supplier offers better value for 50 kg of flour?"
8–20 min	**Direct instruction:** Two-column tables — reading across rows and down columns. Comparing two tables. Identifying differences and drawing conclusions. Workbook p.9–11 worked examples.
20–40 min	Students complete Workbook Ch.14 Exercises 14.4–14.5: accessing and comparing two-column table information in workplace contexts (comparing job applicants, supplier prices, staff schedules).
40–48 min	**Decision-making task:** Given two supplier tables, students must choose the better supplier for a set of items and justify their choice using data.
48–50 min	Class share: "Which supplier did you choose? Why?"

Lesson 60 — Spreadsheets and Digital Data Entry

Duration: 50 minutes

Lesson Objectives:

1. Enter data into a spreadsheet or digital table
2. Use SUM or basic formula to calculate totals
3. Understand how digital data recording supports workplace decisions

Resources:

- Student Workbook Chapter 14
- Computers/tablets with spreadsheet software (Excel, Google Sheets) OR hand-held devices
- Sales/stock data

Lesson Outline:

Time	Activity
0–5 min	**Starter:** "What are the advantages of a spreadsheet over a paper table?"
5–18 min	**Direct instruction:** Opening a spreadsheet. Entering data in cells. Column headers. SUM function. Formatting as a table. Workbook p.12–13.
18–40 min	Students enter a week of stock data into a spreadsheet, add column headers, and calculate daily and weekly totals using SUM. Workbook Ch.14 Exercise 14.6.
40–48 min	**Extension:** Students add a "Total cost" column (quantity × unit price) and calculate total stock value.
48–50 min	**Discussion:** "How does a spreadsheet make the manager's job easier?"

Week 16 — Probability and Data Trends in the Workplace

Topic Focus: Statistics and Probability — Probability and interpretation in workplace contexts
Workbook Chapter: Chapter 15 — Probability and Trends at Work

Lesson 61 — Probability in the Workplace

Duration: 50 minutes

Lesson Objectives:

1. Identify situations in the workplace where probability is relevant
2. Calculate probability for simple workplace events
3. Use probability language in a professional context

Resources:

- Student Workbook Chapter 15
- Calculators
- Workplace probability scenarios

Lesson Outline:

Time	Activity
0–8 min	**Starter:** "A factory makes 500 items per day. Quality control finds that 1 in 50 items is defective. How many defective items are expected per day?"
8–20 min	**Direct instruction:** Probability in the workplace — quality control, risk assessment, chance events (weather delays, machinery failure). Applying $\Pr(E)$ formula. Workbook p.2–5 worked examples.
20–40 min	Students complete Workbook Ch.15 Exercises 15.1–15.2: probability calculations in workplace contexts (defective products, customer wait times, random selection from staff roster).
40–48 min	**Discussion scenario:** "If a restaurant has 8 chefs and randomly assigns 2 to a weekend shift, what is the probability that the most experienced chef is selected?"
48–50 min	**Exit ticket:** "Express Pr(defective item) as a fraction, decimal and percentage."

Lesson 62 — Data Trends and Workplace Decisions

Duration: 50 minutes

Lesson Objectives:

1. Identify trends in workplace data (line graphs)
2. Draw simple conclusions from trends
3. Use trend data to make basic workplace decisions

Resources:

- Student Workbook Chapter 15
- Printed line graphs (sales trends, website traffic, production output)

Lesson Outline:

Time	Activity
0–8 min	**Starter:** Display a line graph of a shop's weekly sales for 6 months. "What is the trend? When were sales highest? What might explain this?"
8–20 min	**Direct instruction:** Identifying trends in data (increasing, decreasing, stable, seasonal). Making basic predictions. Using trend information to make decisions (e.g. order more stock, increase staffing). Workbook p.6–8 worked examples.
20–40 min	Students complete Workbook Ch.15 Exercises 15.3–15.4: interpreting trend graphs in workplace contexts; writing one-paragraph conclusions.
40–48 min	**Case study:** "A café's monthly sales data is presented as a line graph. Write a short report (3–5 sentences) summarising the trend and recommending actions for next quarter."
48–50 min	Share reports.

Lesson 63 — Mixed S&P Workplace Practice

Duration: 50 minutes

Lesson Objectives:

1. Apply all S&P skills in combined workplace scenarios
2. Construct, read and interpret data displays
3. Calculate and express probability

Resources:

- Student Workbook Chapter 15
- Calculators
- Graph paper

Lesson Outline:

Time	Activity
0–5 min	**Recap:** all S&P skills.
5–45 min	Students complete Workbook Ch.15 Exercise 15.5 (Mixed S&P Workplace Practice): 8 problems covering data tables, bar graphs, line graphs, probability calculations and trend analysis in workplace contexts.
45–50 min	Peer-review answers.

Lesson 64 — Topic 2 Full Review

Duration: 50 minutes

Lesson Objectives:

1. Review all Topic 2 subject matter (N&A, M&S, S&P in workplace contexts)
2. Identify areas of strength and weakness
3. Prepare a study plan for the examination

Resources:

- Student Workbook (all chapters)
- Formula book
- Student self-assessment checklist

Lesson Outline:

Time	Activity
0–5 min	**Overview:** exam is in 2 weeks. What is covered? What format?
5–20 min	Students complete a self-assessment checklist: rate each skill 1–4 (1=not confident, 4=very confident). Identify 3 priority areas.
20–45 min	Students work independently on their weakest areas — teacher circulates and provides targeted support. Workbook revision exercises.
45–50 min	Set homework: students revise one topic area before next class.

Week 17 — Examination Preparation

Topic Focus: All skills — exam technique, practice, formula book
Workbook Chapter: All previous chapters (revision)

Lesson 65 — Exam Technique and Strategy

Duration: 50 minutes

Lesson Objectives:

1. Understand examination structure and conditions
2. Apply strategies for reading and answering exam questions
3. Use the formula book effectively

Resources:

- QCAA Formula Book
- Student Workbook
- Sample examination questions

Lesson Outline:

Time	Activity
0–10 min	Teacher explains exam structure: open-book, 5 min perusal + 45 min working. Real context, short response. Students may bring workbook, notes, formula book.
10–25 min	**Exam strategy:** Reading questions carefully. Identifying the mathematical task. Showing all working. Checking answers. Using the formula book. Teacher models approach with 2 sample questions.
25–45 min	Students answer 5 practice questions (from both topics). Teacher circulates.
45–50 min	Brief debrief. Most common mistakes discussed.

Lesson 66 — Practice Examination (Topic 1 Focus)

Duration: 50 minutes

Lesson Objectives:

1. Apply numeracy skills in an examination-style format (Topic 1 content)
2. Practise time management in exam conditions
3. Identify areas requiring further revision

Resources:

- Practice exam paper (Topic 1: Personal Identity and Community)
- QCAA Formula Book
- Student Workbook (open book)
- Calculators

Lesson Outline:

Time	Activity
0–5 min	Distribute practice exam. 5 minutes perusal (read through, no writing).
5–45 min	Students complete practice exam under exam-like conditions (open book).
45–50 min	Collect papers. Brief discussion of first impressions.

Note: Mark and return at start of Lesson 67.

Lesson 67 — Practice Examination (Topic 2 Focus)

Duration: 50 minutes

Lesson Objectives:

1. Apply numeracy skills in an examination-style format (Topic 2 content)

2. Practise time management in exam conditions
3. Identify and address remaining gaps

Resources:

- Practice exam paper (Topic 2: Workplace and Employment)
- QCAA Formula Book
- Student Workbook
- Calculators

Lesson Outline:

Time	Activity
0–5 min	Return Lesson 66 practice exam with marks. Students review their results.
5–10 min	Distribute Topic 2 practice exam. 5 minutes perusal.
10–45 min	Students complete practice exam under exam-like conditions.
45–50 min	Collect papers. Remind students to review formula book location before exam.

Lesson 68 — Review and Formula Book Familiarisation

Duration: 50 minutes

Lesson Objectives:

1. Review errors from practice exams
2. Know where to find key formulas in the formula book
3. Feel confident and prepared for the examination

Resources:

- Marked practice exams
- QCAA Formula Book
- Student Workbook

Lesson Outline:

Time	Activity
0–5 min	Return Lesson 67 practice exam.
5–25 min	Teacher goes through the most commonly missed questions from both practice exams. Students correct their work.
25–40 min	**Formula book tour:** Teacher leads students through the QCAA formula book — where is the area formula? Volume? Probability? Students annotate their workbook's formula reference page.
40–48 min	Students write a personal "exam checklist": equipment to bring, formula locations, key formulas to remember.
48–50 min	Exam logistics: time, room, what to bring. Encourage students.

Week 18 — Examination and Course Wrap-Up

Topic Focus: Assessment Task 2 + Course Completion

Lesson 69 — Final Revision and Q&A

Duration: 50 minutes

Lesson Objectives:

1. Answer any remaining questions before the examination
2. Final revision of key skills
3. Ensure all students are prepared and confident

Resources:

- Student Workbook
- Formula Book
- Student question list

Lesson Outline:

Time	Activity
0–10 min	Students write their top 3 questions on sticky notes. Collect and sort.
10–40 min	Teacher addresses questions in priority order. Students revise in workbook for any questions remaining.
40–48 min	**Final rapid-fire review:** teacher asks 10 quick questions covering one from each key skill area. Students answer on mini whiteboards or paper.
48–50 min	Encouragement and exam logistics reminder.

Lesson 70 — Assessment Task 2: Examination

Duration: 50 minutes (5 min perusal + 45 min working)

Conditions:

- Individual, supervised, open-book
- Students may use: QCAA Formula Book, student workbook, exercise books, notes, calculator
- No discussion or assistance from teacher during working time
- Questions sample subject matter from both Topic 1 and Topic 2
- Short-response, real context questions in simple familiar situations

Lesson Outline:

Time	Activity
0–5 min	Distribute exam. Perusal time — read questions, no writing on answer booklet.
5–50 min	Students complete examination. Teacher supervises.

Note: Collect all exam papers at end of lesson.

Lesson 71 — Post-Exam Debrief and Course Reflection

Duration: 50 minutes

Lesson Outline:

Time	Activity
0–10 min	**Brief debrief:** "How did you feel about the exam? Which questions felt confident? Which were tricky?" (Do not discuss specific answers.)
10–30 min	**Course reflection activity:** Students complete a written reflection — (1) What was the most useful thing you learned? (2) Where will you use numeracy in your future? (3) What would you do differently if you did this course again?
30–48 min	Students create a personal "Numeracy in My Life" summary page in their workbook — a visual summary of key formulas, skills, and one real-world application of each.
48–50 min	Preview: "How will you use these skills next year, in work, and in life?"

Lesson 72 — Course Celebration and Future Pathways

Duration: 50 minutes

Lesson Outline:

Time	Activity
0–15 min	Return marked practice exams (or preview results if assessment has been marked). Discuss overall performance trends.
15–35 min	**Future pathways discussion:** What careers use numeracy? What other maths courses are available? What supports are available if you want to develop further?
35–45 min	Certificate of completion / celebration activity (at teacher's discretion).
45–50 min	**Closing:** "You have completed Numeracy Short Course. You are now better equipped to manage maths in real life, in the community, and in the workplace."

A. Skill Coverage Summary

Numeracy Skill	Weeks Covered	Topic
Whole numbers, fractions, decimals, percentages	1–3	T1
Technology calculations	1, 10, 39	Both
Rates and ratios	3, 10–11	Both
Length conversions	4	T1
2D shapes (recognition + properties)	5	T1
Area of 2D shapes	4–5	T1
Area unit conversions	5	T1
Time (12-hr, 24-hr, dates, timetables)	6	T1
Maps, direction, coordinates, scales	5, 14	Both
Angles (360°, 180°, 90°)	6	T1
3D shapes (recognition + properties)	12	T2
Volume of prisms	12	T2
Volume unit conversions	14	T2
Mass conversions	13	T2
Capacity conversions	13	T2
Temperature	13	T2
Data tables, bar graphs, line graphs	7, 15	Both
Probability language	8, 16	Both
Probability calculation	8, 16	Both
Expressing probability	8, 16	Both

B. Resource Requirements

Resource	Notes
Student Workbook	One per student — also used in open-book exam
QCAA Formula Book	One per student for Task 2 exam (provided by teacher)
Calculators	One per student (phone calculator acceptable)
Rulers (30 cm)	One per student
Graph paper	Small supply per class
Spreadsheet software	Desirable for Lesson 60; Excel or Google Sheets
Printed materials	Timetables, maps, catalogues, receipts — sourced by teacher

C. Formula Reference Sheet

Key Formulas Reference Sheet

Topic 1 — Personal Identity and Community (Chapters 1–8)

Chapter 1 — Numbers in Everyday Life

BODMAS: Brackets → Orders → Division/-Multiplication (L to R) → Addition/Subtraction (L to R)

Rounding: Look at the digit to the *right*. 5 or more → round up. 4 or less → round down.

Length: mm $\xrightarrow{\div 10}$ cm $\xrightarrow{\div 100}$ m $\xrightarrow{\div 1000}$ km

Chapter 2 — Fractions, Decimals and Percentages

Fraction → decimal: divide top by bottom
Decimal → %: $\times 100$ % → decimal: $\div 100$

$$\text{Percentage of amount} = \text{Amount} \times \frac{\%}{100}$$

$$\text{Sale price} = \text{Original} \times \left(1 - \frac{\text{Disc. }\%}{100}\right)$$

$$P_{\text{rect}} = 2L + 2W \quad P_{\text{sq}} = 4S \quad P_{\triangle} = a + b + c$$

Chapter 3 — Rates, Ratios and Comparisons

$$\text{Rate} = \frac{\text{Quantity}}{\text{Unit}} \qquad \text{Unit price} = \frac{\text{Total cost}}{\text{Qty}}$$

$$D = S \times T \qquad S = \frac{D}{T} \qquad T = \frac{D}{S}$$

Simplify ratio: Divide both parts by HCF.

Share in a ratio: (1) Find total parts. (2) Total ÷ parts = value of 1 part. (3) Multiply each share.

Chapter 4 — Length, Perimeter and Area

$$A_{\text{rect}} = L \times W \quad A_{\text{sq}} = S^2 \quad A_{\triangle} = \frac{B \times H}{2}$$

Chapter 5 — Shapes, Area and Maps

Area conversions:
$1\,\text{cm}^2 = 100\,\text{mm}^2 \quad 1\,\text{m}^2 = 10{,}000\,\text{cm}^2$

Scale / maps:

$$\text{Actual} = \text{map length} \times \text{scale factor}$$

Chapter 6 — Time and Angles

12-hr → 24-hr: am = same (4 digits); pm = add 12; midnight = 00:00
Elapsed time = End time − Start time

Angle types:
Acute 0°–90° Right = 90° Obtuse 90°–180°
Straight = 180° Reflex 180°–360°

Chapter 7 — Data and Graphs

Graph checklist — SALT:
Scale **A**xes labelled **L**egend **T**itle

Count tallies → frequency → total must equal all items.

Chapter 8 — Chance and Probability

$$\Pr(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$$

$$\Pr(E) = \frac{\text{times event occurred}}{\text{total trials}}$$

$0 \le \Pr(E) \le 1 \quad \Pr(\text{even chance}) = 0.5 = 50\%$

Topic 2 — Workplace and Employment (Chapters 9–15)

Chapter 9 — Numbers at Work

$$\text{Gross pay} = \text{hours} \times \text{hourly rate}$$

$$\text{Net pay} = \text{Gross pay} - \text{Total deductions}$$

$$\text{Fortnightly pay} = 2 \times \text{weekly pay}$$

Key deductions (2025): Super = 11.5% of gross; Casual loading = 25% on base rate.

Chapter 10 — Workplace Numbers

GST: incl. = price × 1.10 excl. = total ÷ 1.10

Markup/discount:

$$\text{New price} = \text{original} \times (1 \pm r)$$

where r is the rate as a decimal.

Ratio mixing: $\text{qty/part} = \dfrac{\text{total volume}}{\text{total parts}}$

Chapter 11 — 3D Shapes and Volume

$$V_{\text{prism}} = A_{\text{base}} \times H \qquad V_{\text{rect}} = L \times W \times H$$

$$V_{\triangle\text{-prism}} = \tfrac{b \times h}{2} \times L$$

Conversions: $1\,\text{cm}^3 = 1000\,\text{mm}^3$ $1\,\text{m}^3 = 10^6\,\text{cm}^3$
$1\,\text{cm}^3 = 1\,\text{mL}$ $1\,\text{m}^3 = 1000\,\text{L}$

Chapter 12 — Mass, Capacity and Temperature

Mass: $\text{mg} \xrightarrow{\times 1000} \text{g} \xrightarrow{\times 1000} \text{kg} \xrightarrow{\times 1000} \text{t}$ (reverse: ÷1000)

Capacity: $\text{mL} \xrightarrow{\times 1000} \text{L} \xrightarrow{\times 1000} \text{kL} \xrightarrow{\times 1000} \text{ML}$

Food safety: Cold < 5°C Danger 5°–60°C Hot > 60°C

Chapter 13 — Workplace Measurement

$$\text{Real length} = \text{plan length} \times \text{scale factor}$$

Common scales: 1:50 = 1 cm → 0.5 m; 1:100 = 1 cm → 1 m; 1:200 = 1 cm → 2 m

Chapter 14 — Data at Work

SALT: **S**cale **A**xes labelled **L**egend **T**itle

Spreadsheet: `=B2*C2` multiply `=SUM(D2:D8)` add range

Compare tables: Extract values → calculate totals/differences → write conclusion.

Chapter 15 — Probability and Trends at Work

$$\Pr(\text{defect}) = \frac{\text{defective items}}{\text{total items checked}}$$

$$\text{Expected defects} = \Pr(\text{defect}) \times \text{total produced}$$

Trend reading: Compare start and end points. Identify highest/lowest. Describe overall direction (increasing, decreasing, stable).

Conclusion stem: *"The data shows that... Between [period] and [period], [value] increased/decreased by..."*

D. Practice Examinations

Two full practice examinations with marking guides are reproduced below for use during Week 17 (Exam Preparation).

Source Files

- Practice Exam 1 Student Paper: `Stage3_PracticeExam1_Student.tex`
- Practice Exam 1 Marking Guide: `Stage3_PracticeExam1_MarkingGuide.tex`
- Practice Exam 2 Student Paper: `Stage3_PracticeExam2_Student.tex`
- Practice Exam 2 Marking Guide: `Stage3_PracticeExam2_MarkingGuide.tex`

Practice Examination 1

PRACTICE EXAMINATION 1

Topic 1: Personal Identity and Community

Year 10 Numeracy Short Course | QCAA 2025 v1.2

Student Name: _____ **Class:** _____

Teacher: _____ **Date:** _____

Examination Conditions

- **Time allowed:** 5 minutes perusal + 45 minutes working time
- This is an **OPEN-BOOK** examination
- **You may use:** your student workbook, notes, exercise books, QCAA Formula Book, calculator
- Write your answers in the spaces provided
- Show **ALL** working — answers without working may not receive full marks
- Where a question asks you to show working, marks **ARE** awarded for correct working even if the final answer is wrong

Marks Summary

Total Marks: 30	Number of Questions: 20	Suggested time per mark: approx. 1.5 minutes

Question 1 **[1 mark]**

A music festival sold 1,348,250 tickets last year. Write this number in words.

2pt 6pt

Question 2 **[1 mark]**

What is the place value of the digit 6 in the number 2,467,381?

2pt 6pt

Question 3 **[1 mark]**

Order these numbers from smallest to largest:

12,504 12,045 12,450 12,540

2pt 6pt

Question 4 **[2 marks]**

You earn \$1,248 per fortnight. You spend \$387 on rent and \$156 on food each fortnight. How much money do you have remaining after paying rent and food? *Show all working.*

2pt 6pt

Question 5 **[1 mark]**

Simplify the fraction $\frac{18}{24}$ to its simplest form.

2pt 6pt

Question 6 **[1 mark]**

Convert 0.65 to a percentage.

2pt 6pt

Question 7 **[2 marks]**

A jacket costs \$180. It is on sale for 35% off.

(a) Calculate the discount amount.
(b) Calculate the sale price.

2pt 6pt

Question 8 **[2 marks]**

Juice is sold in two sizes:

- Size A: 600 mL for \$3.60
- Size B: 1 L for \$5.40

Which size is better value for money? Show your working by calculating the unit price (cost per mL or per L) for each size.

2pt 6pt

Question 9 **[2 marks]**

A concrete mix uses cement and sand in the ratio 1:3. If you need 20 kg of mix in total:

(a) How many kg of cement is needed?
(b) How many kg of sand is needed?

2pt 6pt

Question 10 **[1 mark]**

Convert 3.2 km to metres.

2pt 6pt

Question 11 **[2 marks]**

A rectangular room is 6.5 m long and 4.2 m wide.
Calculate the perimeter of the room. Include the correct unit in your answer.

2pt 6pt

Question 12 **[2 marks]**

A triangular garden bed has a base of 8 m and a perpendicular height of 5 m.
Calculate the area of the garden bed using the formula $A = \dfrac{B \times H}{2}$.

2pt 6pt

Question 13 **[1 mark]**

Convert $3.5\,\text{m}^2$ to cm^2.

2pt 6pt

Question 14 **[1 mark]**

A map has a scale of 1:50,000. A road measures 4.6 cm on the map.
What is the actual length of the road in kilometres?

2pt 6pt

Question 15 **[1 mark]**

Write 6:45 pm in 24-hour time.

2pt 6pt

Question 16 **[2 marks]**

A train departs at 07:52 and arrives at 09:38.
How long is the journey? Give your answer in hours and minutes.

2pt 6pt

Question 17 **[1 mark]**

A bus departs at 14:15. A passenger arrives at the stop at 2:00 pm.
How many minutes does the passenger wait for the bus?

2pt 6pt

Question 18 **[2 marks]**

A survey of 28 students asked about their favourite activity.

Activity	**Votes**
Swimming	14
Football	8
Cycling	6
Total	28

(a) What percentage of students chose swimming?

(b) Which activity received exactly $\frac{2}{7}$ of the votes?

2pt 6pt

Question 19 **[1 mark]**

A spinner has 8 equal sections numbered 1 to 8.
What is the probability of spinning an even number?
Write your answer as a simplified fraction.

2pt 6pt

Question 20 **[2 marks]**

A bag contains 5 red, 3 blue and 2 green marbles. One marble is drawn at random.

(a) What is the probability of drawing a red marble? Write as a simplified fraction.
(b) Express this probability as a decimal and as a percentage.

2pt 10pt

— END OF EXAMINATION —

Check your work if time permits.

Extra Working Space

Practice Examination 1 — Marking Guide

PRACTICE EXAMINATION 1 — MARKING GUIDE

Topic 1: Personal Identity and Community

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Year 10 Numeracy Short Course | QCAA 2025 v1.2

Total: 30 marks

Mark Allocation Summary

Question	Topic / Skill	Marks
1	Place value — reading numbers	1
2	Place value — digit value	1
3	Ordering whole numbers	1
4	Operations — multi-step problem	2
5	Fractions — simplifying	1
6	Fractions/decimals/percentages — converting	1
7	Percentages — discount	2
8	Rates — unit pricing / best buy	2
9	Ratios — dividing in a ratio	2
10	Length — unit conversion	1
11	Measurement — perimeter	2
12	Measurement — area of triangle	2
13	Measurement — area unit conversion	1
14	Maps — scale calculation	1
15	Time — 24-hour conversion	1
16	Time — elapsed time	2
17	Time — timetable reading	1
18	Data — percentage from table	2
19	Probability — calculating Pr(E)	1
20	Probability — all three forms	2
TOTAL		**30**

Question 1 **[1 mark]**

A music festival sold 1,348,250 tickets last year. Write this number in words.

Answer & Mark Scheme

Answer: One million, three hundred and forty-eight thousand, two hundred and fifty.

Award: 1 mark for a fully correct answer in words.

Question 2 **[1 mark]**

What is the place value of the digit 6 in the number 2,467,381?

Answer & Mark Scheme

Answer: The digit 6 is in the **ten-thousands** place. Its value is **60,000**.

Award: 1 mark for "sixty thousand" or "60,000" or "ten-thousands place" (accept either the place name *or* the value).

Question 3 **[1 mark]**

Order these numbers from smallest to largest: 12,504 12,045 12,450 12,540

Answer & Mark Scheme

Answer: 12,045 ; 12,450 ; 12,504 ; 12,540

Award: 1 mark for all four numbers in the correct order.

Question 4 **[2 marks]**

You earn \$1,248 per fortnight. You spend \$387 on rent and \$156 on food each fortnight. How much money do you have remaining after paying rent and food?

Answer & Mark Scheme

Working (Method 1):
\$1,248 − \$387 = \$861
\$861 − \$156 = \$705

Working (Method 2):
\$387 + \$156 = \$543
\$1,248 − \$543 = \$705

Answer: \$705 remaining

Award: 1 mark for correct method/working shown; 1 mark for correct final answer (\$705).

Question 5 **[1 mark]**

Simplify the fraction $\frac{18}{24}$ to its simplest form.

Answer & Mark Scheme

Working: $18 \div 6 = 3; \quad 24 \div 6 = 4$

Answer: $\frac{3}{4}$

Award: 1 mark for $\frac{3}{4}$.

Question 6 **[1 mark]**

Convert 0.65 to a percentage.

Answer & Mark Scheme

Working: $0.65 \times 100 = 65$

Answer: 65%

Award: 1 mark for 65%.

Question 7 **[2 marks]**

A jacket costs \$180. It is on sale for 35% off.
(a) Calculate the discount amount. (b) Calculate the sale price.

Answer & Mark Scheme

(a) Discount:
35% of \$180 = $0.35 \times$ \$180 = **\$63**

(b) Sale price:
\$180 − \$63 = **\$117**

Alternative method: \$180 × 0.65 = \$117 (direct calculation of sale price)

Award: 1 mark for correct discount (\$63); 1 mark for correct sale price (\$117).
Note: If a student calculates the sale price directly (\$180 × 0.65 = \$117) without separately finding the discount, award both marks.

Question 8 **[2 marks]**

Juice: Size A = 600 mL for \$3.60; Size B = 1 L for \$5.40. Which size is better value? Show working by calculating unit price.

Answer & Mark Scheme

Size A:
$\$3.60 \div 600 = \$0.006$ per mL (equivalent to \$6.00 per litre)

Size B:
$\$5.40 \div 1000 = \$0.0054$ per mL (equivalent to \$5.40 per litre)

Conclusion: Size B (1 L) is **better value** as it costs less per mL / per litre.

Award: 1 mark for correct unit price calculation for *both* sizes; 1 mark for correct conclusion (Size B / 1 L is better value).
Note: Accept calculations in either mL or L as long as both sizes are compared in the same unit.

Question 9 **[2 marks]**

Cement : sand ratio = 1:3. Total mix = 20 kg.
(a) How many kg of cement? (b) How many kg of sand?

Answer & Mark Scheme

Working:
Total parts $= 1 + 3 = 4$
One part $= 20 \div 4 = 5$ kg

(a) Cement: $1 \times 5 =$ **5 kg**
(b) Sand: $3 \times 5 =$ **15 kg**

Check: $5 + 15 = 20$ kg ✓

Award: 1 mark for correct cement (5 kg); 1 mark for correct sand (15 kg). Both must be shown to award 2 marks.

Question 10 **[1 mark]**

Convert 3.2 km to metres.

Answer & Mark Scheme

Working: $3.2 \times 1{,}000 = 3{,}200$

Answer: 3,200 m

Award: 1 mark for 3,200 m (accept 3200 m).

Question 11 **[2 marks]**

Rectangular room: length = 6.5 m, width = 4.2 m. Calculate the perimeter.

Answer & Mark Scheme

Working (Method 1):
$P = 2L + 2W = 2(6.5) + 2(4.2) = 13 + 8.4 = 21.4\,\text{m}$

Working (Method 2):
$P = 2(L + W) = 2(6.5 + 4.2) = 2 \times 10.7 = 21.4\,\text{m}$

Answer: $21.4\,\text{m}$

Award: 1 mark for correct formula/method; 1 mark for correct answer ($21.4\,\text{m}$). Unit must be included.

Question 12 **[2 marks]**

Triangular garden bed: base $= 8\,\text{m}$, perpendicular height $= 5\,\text{m}$. Calculate area using $A = \dfrac{B \times H}{2}$.

Answer & Mark Scheme

Working:

$$A = \frac{B \times H}{2} = \frac{8 \times 5}{2} = \frac{40}{2} = 20\,\text{m}^2$$

Answer: $20\,\text{m}^2$

Award: 1 mark for correct substitution into formula; 1 mark for correct answer ($20\,\text{m}^2$). Unit must be included.

Question 13 **[1 mark]**

Convert $3.5\,\text{m}^2$ to cm^2.

Answer & Mark Scheme

Working: $1\,\text{m}^2 = 10{,}000\,\text{cm}^2$
$3.5 \times 10{,}000 = 35{,}000\,\text{cm}^2$

Answer: $35{,}000\,\text{cm}^2$

Award: 1 mark for $35{,}000\,\text{cm}^2$.

Question 14 **[1 mark]**

Scale 1:50,000. Road measures $4.6\,\text{cm}$ on map. What is the actual length in km?

Answer & Mark Scheme

Working:
Actual length (cm) $= 4.6 \times 50{,}000 = 230{,}000\,\text{cm}$
$230{,}000\,\text{cm} = 2{,}300\,\text{m} = 2.3\,\text{km}$

Answer: 2.3 km

Award: 1 mark for 2.3 km (must convert to km).

Question 15 **[1 mark]**
Write 6:45 pm in 24-hour time.

Answer & Mark Scheme

Working: $6 + 12 = 18 \quad \Rightarrow \quad 18{:}45$

Answer: 18:45

Award: 1 mark for 18:45.

Question 16 **[2 marks]**
Train departs 07:52, arrives 09:38. How long is the journey? Give answer in hours and minutes.

Answer & Mark Scheme

Working:
From 07:52 to 08:52 = 1 hour
From 08:52 to 09:38 = 46 minutes

Answer: 1 hour 46 minutes

Award: 1 mark for correct method; 1 mark for 1 hour 46 minutes.
Note: Accept any valid method of calculating elapsed time.

Question 17 **[1 mark]**
Bus departs 14:15. Passenger arrives at stop at 2:00 pm. How many minutes does the passenger wait?

Answer & Mark Scheme

Working:
Bus departs $14:15 = 2:15$ pm. Passenger arrives $2:00$ pm.
Wait = 15 minutes.

Answer: 15 minutes

Award: 1 mark for 15 minutes.

Question 18 **[2 marks]**

Survey of 28 students (Swimming: 14, Football: 8, Cycling: 6).
(a) Percentage who chose swimming? (b) Which activity received exactly $\frac{2}{7}$ of the votes?

Answer & Mark Scheme

(a) Swimming percentage:
$\frac{14}{28} \times 100 = 50\%$

(b) Activity with $\frac{2}{7}$ of votes:
$\frac{2}{7} \times 28 = 8$ votes $\Rightarrow$ **Football** (8 votes)

Verification: $\frac{8}{28} = \frac{2}{7}$ ✓

Award: 1 mark for 50%; 1 mark for Football.

Question 19 **[1 mark]**

Spinner with 8 equal sections numbered 1 to 8. Probability of spinning an even number (as a simplified fraction)?

Answer & Mark Scheme

Working:
Even numbers on spinner: 2, 4, 6, 8 $\Rightarrow$ 4 favourable outcomes out of 8 total.

$\Pr(\text{even}) = \frac{4}{8} = \frac{1}{2}$

Answer: $\frac{1}{2}$

Award: 1 mark for $\frac{1}{2}$ (accept $\frac{4}{8}$).

Question 20 **[2 marks]**

Bag: 5 red, 3 blue, 2 green marbles. One marble drawn at random.
(a) Pr(red) as a simplified fraction. (b) Express as decimal AND percentage.

Answer & Mark Scheme

Working:
Total marbles $= 5 + 3 + 2 = 10$

(a) $\Pr(\text{red}) = \dfrac{5}{10} = \mathbf{\dfrac{1}{2}}$

(b) Decimal: **0.5** Percentage: **50%**

Award: 1 mark for $\dfrac{1}{2}$ (part a); 1 mark for *both* 0.5 *and* 50% (part b — student must provide both for the mark).

Grade Boundaries

Grade	Mark Range	Percentage
A	25–30	$> 80\%$
B	19–24	$> 60\%$
C	13–18	$> 40\%$
D	7–12	$> 20\%$
E	0–6	$\geq 0\%$

Teacher Notes

- Accept equivalent forms unless the question specifies a particular form.
- **Q7:** If a student calculates the sale price directly ($\$180 \times 0.65 = \$117$) without separately finding the discount, award both marks.
- **Q8:** Accept calculations in either mL or L as long as both sizes are compared in the same unit.
- **Q16:** Accept any valid method of calculating elapsed time.
- **Q20(b):** The student must provide **BOTH** the decimal and the percentage to receive the mark.

PRACTICE EXAMINATION 2

Topic 2: Workplace and Employment

Year 10 Numeracy Short Course | QCAA 2025 v1.2

Student name: _____

Class: _____

Teacher: _____

Date: _____

Examination Conditions

- This is an **open-book** examination. You may use your textbook, notes, and a calculator.
- **Perusal time:** 5 minutes (you may *not* write during perusal).
- **Working time:** 45 minutes.
- Show **all working** in the space provided. Answers without working may not receive full marks.
- Write your answers clearly. Where units are required, include them in your response.
- If you make a mistake, cross it out neatly and rewrite — do not use correction fluid.

Total marks	Number of questions	Topics covered
30	20	Topic 2 (Chapters 9–15)

Question 1 **[3 marks]**

Casey works as a café assistant and earns $22.50 per hour. Last Tuesday she worked from 9:00 am to 3:30 pm (with no unpaid breaks).

(a) How many hours did Casey work on Tuesday?

(b) Calculate her gross pay for that day. Show all working.

Question 2 **[1 mark]**

Casey's timesheet for the week shows:

Day	Hours worked
Monday	4.0
Tuesday	5.5
Wednesday	0.0
Thursday	3.0
Friday	6.0

What are Casey's total hours worked for the week?

Question 3 **[1 mark]**

Using your answer from Question 2 and Casey's hourly rate of $22.50 per hour, calculate Casey's weekly gross pay.

Question 4 **[1 mark]**

A plumber charges $420 for a job (excluding GST).
Calculate the GST (10%) on this amount.

Question 5 **[2 marks]**

An electrician's invoice lists:

- Labour: \$380
- Parts: \$175

(a) Calculate the subtotal (before GST).

(b) Calculate the total invoice amount including 10% GST.

Question 6 **[2 marks]**

A cleaning product is mixed in the ratio of 1 part concentrate to 9 parts water. A cleaner needs to make 5 litres of solution.

(a) How many litres of concentrate are needed?

(b) How many litres of water are needed?

Question 7 **[1 mark]**

A storage manager has a budget of \$1,200 to purchase shelving units. Each unit costs \$185.
What is the maximum number of units the manager can buy within budget?

Question 8 **[2 marks]**

A rectangular storage room is 8 m long, 5 m wide and 3 m high.
Calculate the volume of the room using the formula $V = L \times W \times H$.
Include the correct unit in your answer.

Question 9 **[1 mark]**

Name the 3D shape that has exactly 2 circular faces and 1 curved surface.

Question 10 **[2 marks]**

A delivery truck can carry a maximum load of 2.5 tonnes. A shipment of boxes each weighing 80 kg needs to be transported.

(a) Convert 2.5 tonnes to kilograms.

(b) How many boxes can fit in one truck load?

Question 11 **[1 mark]**

Convert 3,500 mL to litres.

Question 12 **[2 marks]**

A café uses 250 mL of milk per coffee. On a busy Saturday the café makes 280 coffees.

(a) How many litres of milk are needed?

(b) The café orders milk in 2 L bottles. How many bottles must be ordered?

Question 13 **[1 mark]**

A food worker checks the temperature of hot soup and records 58°C. Food safety guidelines require hot food to be kept above 60°C.
Is the soup within food safety guidelines? Circle your answer and explain.

YES / **NO** (circle one)

Reason:

Question 14 **[1 mark]**

Convert 0.004 m^3 to cm^3.

Question 15 **[2 marks]**

A warehouse floor plan has a scale of 1:200. A storage bay measures 3.5 cm × 2 cm on the plan.
Calculate the actual dimensions of the storage bay in metres.

Question 16 **[1 mark]**

The table below shows a shop's daily sales for one week.

Day	Sales
Monday	\$1,240
Tuesday	\$980
Wednesday	\$1,150
Thursday	\$870
Friday	\$1,560

On which day were sales the highest?

Question 17 **[2 marks]**

Using the sales data from Question 16:

(a) Calculate the total sales for the week.

(b) Calculate the average (mean) daily sales.

Question 18 **[1 mark]**

A quality control check of 400 items finds 12 defective items.
Express the probability of a defective item as a simplified fraction.

Question 19 **[1 mark]**

Using the probability from Question 18, how many defective items would you expect in a production run of 1,000 items?

Question 20 **[2 marks]**

A line graph shows a shop's monthly revenue. Revenue is low in January (\$8,200), increases through the year, peaks in December (\$24,600), with a noticeable dip in July (\$11,400).

(a) Describe the overall trend in the data.

(b) What action would you recommend the shop owner take to prepare for July? Give a reason based on the data.

END OF EXAMINATION

Additional Working Space

(Work written here will only be marked if you clearly indicate which question it belongs to.)

PRACTICE EXAMINATION 2 — MARKING GUIDE

Topic 2: Workplace and Employment

Year 10 Numeracy Short Course | QCAA 2025 v1.2

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Total: 30 marks | 20 Questions

Mark Allocation Summary

Q	Topic / Skill	Marks
1	Gross pay calculation	3
2	Timesheet — total hours	1
3	Weekly gross pay	1
4	GST calculation	1
5	Invoice — subtotal and GST total	2
6	Ratio mixing	2
7	Budget — maximum units	1
8	Volume of rectangular prism	2
9	3D shape identification	1
10	Mass conversion + max load	2
11	Capacity conversion	1
12	Capacity word problem	2
13	Temperature — food safety	1
14	Volume unit conversion	1
15	Floor plan scale	2
16	Table reading — maximum	1
17	Total and average sales	2
18	Probability — defect rate	1
19	Expected frequency	1
20	Trend analysis and recommendation	2
TOTAl		**30**

Worked Answers

Correct answers are highlighted. Award marks as specified for each question.

Question 1 [3 marks] Gross pay calculation

Worked solution:

(a) 9:00 am to 3:30 pm = **6.5** hours

(b) Gross pay = hours × rate = 6.5 × \$22.50 = \$**146.25**

1 mark for **6.5 hours**; 1 mark for correct method (hours × rate); 1 mark for **\$146.25**.

Question 2 [1 mark] Timesheet — total hours

Worked solution:

$$4.0 + 5.5 + 0.0 + 3.0 + 6.0 = \mathbf{18.5} \text{ hours}$$

1 mark for **18.5 hours**.

Question 3 [1 mark] Weekly gross pay

Worked solution:

$$18.5 \times \$22.50 = \$\mathbf{416.25}$$

1 mark for **\$416.25**. *Follow-through:* if the student used an incorrect answer from Q2, award the mark if their method (hours × \$22.50) is correct.

Question 4 [1 mark] GST calculation

Worked solution:

$$\text{GST} = \$420 \times 0.10 = \$\mathbf{42}$$

1 mark for **\$42**.

Question 5 [2 marks] Invoice — subtotal and GST total

Worked solution:

(a) Subtotal $= \$380 + \$175 = \$\mathbf{555}$

(b) Total including GST $= \$555 \times 1.10 = \$\mathbf{610.50}$

1 mark for **\$555**; 1 mark for **\$610.50**.

Question 6 [2 marks] Ratio mixing

Worked solution:

Total parts $= 1 + 9 = 10$
One part $= 5 \div 10 = 0.5\,\text{L}$

(a) Concentrate $= 1 \times 0.5 = \mathbf{0.5}\,\text{L}$

(b) Water $= 9 \times 0.5 = \mathbf{4.5}\,\text{L}$

Check: $0.5 + 4.5 = 5\,\text{L}$ ✓

1 mark for **0.5 L** (concentrate); 1 mark for **4.5 L** (water).

Question 7 [1 mark] Budget — maximum units

Worked solution:

$$\$1{,}200 \div \$185 = 6.486\ldots \quad \Rightarrow \quad \text{maximum } \mathbf{6} \text{ units}$$

(Cannot purchase a fraction of a unit; must round *down*.)

1 mark for **6 units**. Do **NOT** award a mark for 7.

Question 8 [2 marks] Volume of rectangular prism

Worked solution:

$$V = L \times W \times H = 8 \times 5 \times 3 = \mathbf{120}\,\text{m}^{\mathbf{3}}$$

1 mark for correct method; 1 mark for $\mathbf{120\,m^3}$ (unit required — do not award second mark if unit is missing or incorrect).

Question 9 [1 mark] 3D shape identification

Answer: Cylinder

1 mark for **cylinder**. Accept "cylinder" only; do not accept "tube", "pipe", or other informal names.

Question 10 [2 marks] Mass conversion + max load

Worked solution:

(a) $2.5\,\text{t} = 2.5 \times 1{,}000 = \mathbf{2{,}500}\,\text{kg}$

(b) $2{,}500 \div 80 = 31.25 \quad \Rightarrow \quad$ maximum **31** boxes
(Cannot exceed the load limit; must round *down*.)

1 mark for **2,500 kg**; 1 mark for **31 boxes**. Do **NOT** award a mark for 32.

Question 11 [1 mark] Capacity conversion

Worked solution:

$$3{,}500 \div 1{,}000 = \mathbf{3.5}\,\text{L}$$

1 mark for **3.5 L**.

Question 12 [2 marks] Capacity word problem

Worked solution:

(a) $280 \times 250\,\text{mL} = 70{,}000\,\text{mL} = \mathbf{70}\,\text{L}$

(b) $70 \div 2 = \mathbf{35}$ bottles

1 mark for **70 L**; 1 mark for **35 bottles**.

Question 13 [1 mark] Temperature — food safety

Answer: NO

Reason: 58°C is below the required minimum of 60°C, so the soup is *not* within food safety guidelines.

1 mark for **NO** with a correct reason that references 60°C. Do **NOT** award for "Yes". Students must provide both the answer *and* a reason; "No" alone without explanation does not receive the mark.

Question 14 [1 mark] Volume unit conversion

Worked solution:

$$0.004\,\text{m}^3 \times 1{,}000{,}000 = \mathbf{4{,}000}\,\text{cm}^{\mathbf{3}}$$

(1 m^3 = 1,000,000 cm^3; multiply by 100^3)

1 mark for **4,000 cm^3**.

Question 15 [2 marks] Floor plan scale

Worked solution (scale 1:200):

$$\begin{aligned} \text{Length:} \quad & 3.5\,\text{cm} \times 200 = 700\,\text{cm} = \mathbf{7}\,\text{m} \\ \text{Width:} \quad & 2\,\text{cm} \times 200 = 400\,\text{cm} = \mathbf{4}\,\text{m} \end{aligned}$$

Actual dimensions: **7 m** × **4 m**

1 mark for **7 m**; 1 mark for **4 m**. Both measurements must be converted to metres to receive full marks.

Question 16 [1 mark] Table reading — maximum

Answer: Friday ($1,560)

1 mark for **Friday**.

Question 17 [2 marks] Total and average sales

Worked solution:

(a) $\$1{,}240 + \$980 + \$1{,}150 + \$870 + \$1{,}560 = \$\mathbf{5{,}800}$

(b) Mean $= \$5{,}800 \div 5 = \$\mathbf{1{,}160}$

1 mark for **\$5,800**; 1 mark for **\$1,160**.

Question 18 [1 mark] Probability — defect rate

Worked solution:

$$P(\text{defective}) = \frac{12}{400} = \frac{\mathbf{3}}{\mathbf{100}}$$

1 mark for $\frac{\mathbf{3}}{\mathbf{100}}$. Accept $\frac{12}{400}$ if the question is interpreted as not requiring simplification, however the question explicitly states "simplified fraction" so prefer $\frac{3}{100}$.

Question 19 [1 mark] Expected frequency

Worked solution:

$$\frac{3}{100} \times 1{,}000 = \mathbf{30} \text{ defective items}$$

1 mark for **30**. *Follow-through:* if the student used an incorrect Q18 fraction but applied it correctly to 1,000, award the mark.

Question 20 [2 marks] Trend analysis and recommendation

Acceptable answers:

(a) **Trend description** — any of the following (or equivalent):

- Overall increasing trend through the year.
- Revenue increases from January to December with seasonal variation.
- Revenue is lowest in winter (July) and peaks in summer/December.
- Sales generally rise throughout the year but dip in July.

(b) **Recommendation** — any sensible, data-referenced response, for example:

- Reduce stock orders in July to avoid overstocking during the slow period.
- Reduce casual staff hours in July to lower labour costs.
- Run winter promotions or specials to increase revenue in July.
- Build up cash reserves before July to manage the expected revenue dip.
- Plan for reduced revenue in July by cutting discretionary spending beforehand.

1 mark for a correct trend description (must mention increasing trend and/or seasonal variation); 1 mark for a reasonable recommendation with a data-based reason referencing July or low sales.

Grade Boundaries

Grade	Mark range	Percentage
A (Very High)	27–30	90–100%
B (High)	23–26	77–89%
C (Sound)	18–22	60–76%
D (Limited)	12–17	40–59%
E (Very Limited)	0–11	0–39%

Grade boundaries are indicative only. Teachers should use professional judgement when assigning final grades, particularly near boundaries.

Teacher Notes

- **Q3 follow-through:** If a student used an incorrect answer from Q2, award the Q3 mark if their method (hours × \$22.50) is correct and applied consistently.
- **Q7 and Q10(b):** Both require rounding *down* to a whole number because fractions of a unit or box are not possible in context. Do not award marks if the student rounds up in either case.
- **Q8:** The unit m^3 must be present in the final answer to receive the second mark. A correct numerical answer of 120 without a unit receives only 1 mark.
- **Q13:** Students must write **NO** and provide a reason. Writing only "No" without a reason does **not** receive the mark. The reason must reference the 60°C guideline or state that 58°C is below the required temperature.
- **Q15:** Answers left in centimetres (700 cm × 400 cm) do not receive full marks — both dimensions must be converted to metres.
- **Q19 follow-through:** If the student used an incorrect Q18 fraction but applied it correctly to 1,000 items, award the Q19 mark.
- **Q20(a):** Accept any reasonable trend description. Do not penalise students who describe it simply as "increasing" without explicitly mentioning seasonal variation, provided the answer is not factually incorrect.
- **Q20(b):** Accept any sensible workplace recommendation. The key requirement is that it references the data (July, low sales, etc.). Generic advice not tied to the data does **not** receive the mark.

END OF MARKING GUIDE — TEACHER COPY ONLY